

Introduction to Quantitative Finance

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Introduction

Since Kiyoshi Itô began developing Itô calculus in the mid-twentieth century [1], stochastic calculus has become increasingly important in modeling a wide range of phenomena, with applications ranging from the evolution of financial asset prices to physical processes [2, 3]. In particular, stochastic differential equations (SDEs) have become a fundamental tool in financial market modeling for describing the dynamics of interest rates, derivative prices, and hedging strategies under uncertainty [4].

As with ordinary differential equations, few stochastic differential equations can be solved analytically. Exceptions include linear models and those that can be reduced to more tractable forms through changes of variables [5]. Numerical methods therefore play a crucial role in obtaining approximate solutions, enabling both theoretical analysis and the practical implementation of financial strategies based on stochastic models. These notes aim to provide a systematic introduction to the theory of numerical methods for stochastic differential equations, with an emphasis on applications in quantitative finance, introducing the essential tools and presenting simulations that illustrate the theoretical results.

These notes are organized as follows:

Chapter 1 introduces the fundamental concepts of stochastic processes, including martingales, Markov processes, and Wiener processes, which form the basis of stochastic calculus.

Chapter 2 presents stochastic integrals, with particular emphasis on the Itô integral and the notion of a stochastic differential, key elements in the development of stochastic differential equations.

Chapter 3 studies stochastic differential equations (SDEs), beginning with linear cases and considering examples such as the Langevin equation and geometric Brownian motion, which is widely used in quantitative finance.

Chapter 4 is devoted to the Fokker-Planck equation, which describes the evolution of probability distributions in stochastic systems and is fundamental to financial market modeling.

Chapter 5 develops Itô-Taylor numerical methods for solving SDEs. It introduces concepts such as multi-indices, multiple Itô integrals, and approximation coefficients, together with various strong and weak numerical methods and their stability.

Chapter 6 presents numerical simulations to validate the methods introduced, including the stability of Euler-Maruyama methods and the analysis of strong and weak errors.

Chapter 7 introduces financial derivatives, covering derivatives markets, forward contracts, financial options, and investment strategies involving derivatives. It presents continuous-time models and the pricing of options and forwards, with a detailed analysis of the Greeks and their use in hedging strategies.

Chapter 8 studies local volatility models, including calibration of the local volatility surface, implied volatility dynamics, and the pricing and hedging of forward-start options.

Chapter 9 introduces variance swaps, their relation to log contracts, forward variances, and weighted variance swaps.

Chapter 10 discusses trading volatility with options and the risks associated with delta hedging, dividends, and financing.

Chapter 11 studies fixed-income models, including the term structure of interest rates, the yield curve, and models such as Vasicek and CIR, which are fundamental to interest rate modeling.

Appendix A reviews probability preliminaries, including random variables, expectation, convergence, conditional expectation, and change of measure. Appendix B provides Python code for option strategies and simulations of option prices, delta, and hedging portfolios.

Several directions for further work are proposed: first, a more detailed study of advanced numerical methods, such as stochastic Runge-Kutta methods or adaptive schemes for equations with jumps (jump processes); second, the application of the methods studied to specific financial problems, such as portfolio risk management or the calibration of stochastic volatility models; and finally, an exploration of the relationship between stochastic differential equations and other approaches in quantitative finance, such as stochastic control theory and dynamic programming.

Chapter 1

Stochastic processes

Definition 1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and I a nonempty index set. A real d -dimensional **stochastic process** is defined as a family $\{X_t : t \in I\}$ of random vectors taking values in the state space $\mathbb{R}^d$ over a parameter set I .

Alternatively, a stochastic process can be viewed as a random vector¹ such that we have a random vector

$$\begin{aligned} X_t(\cdot) &: \Omega \rightarrow \mathbb{R}^d \\ \omega &\mapsto X_t(\omega) \end{aligned}$$

for each fixed $t \in I$, and

$$\begin{aligned} X(\omega) &: I \rightarrow \mathbb{R}^d \\ t &\mapsto X_t(\omega) \end{aligned}$$

for each $\omega \in \Omega$. The set $\{X_t(\omega) : t \in I\}$ with fixed $\omega \in \Omega$ is known as a *path of the process*² or *realization*. Note that there are only four distinct possibilities, determined by the countability of I and the continuity of $X_t(\cdot)$. Throughout the remainder of these notes, we take $I = [t_0, T]$ to represent a time interval, where $t_0 < T$ (possibly taking infinite values), unless otherwise stated. We will also focus on stochastic processes with continuous paths.

Definition 1.2. Two stochastic processes X_t and Y_t defined on the same probability space are **stochastically equivalent** if, for every $t \in [t_0, T]$, $X_t = Y_t$ almost surely.³

Introductory probability courses establish that a random variable, whether continuous or discrete, is determined by its distribution function. We may similarly ask which tools are needed to determine a stochastic process completely. To do so, we must specify its joint distribution function

$$\mathbb{P}(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n) \equiv F_{t_1, \dots, t_n}(x_1, \dots, x_n)$$

where $t_i \in I$, $x \in \mathbb{R}^d$, and $n \in \mathbb{N}$, which can be quite restrictive. However, we will focus on certain types of stochastic processes for which this requirement can be relaxed.

¹In fact, the range of $X_t(\cdot)$ can be any measurable space. However, these notes will primarily consider $\mathbb{R}^d$ or $\mathbb{R}^{d \times m}$.

²*Sample path.*

³We retain the abbreviation *c.t.p.* (almost surely) to indicate that a property holds with probability 1; in this case, this means $\mathbb{P}(\{X_t = Y_t\}) = \mathbb{P}(\{\omega \in \Omega : X_t(\omega) = Y_t(\omega)\}) = 1$ for every $t \in [t_0, T]$. Equivalently, the property fails at most on a set of measure zero.

Definition 1.3. Let $\{X_t : t \in I\}$ be a stochastic process. Given $s, t \in I$ such that $s \leq t$, the random vector $X_t - X_s$ is called an **increment** of the process.

Definition 1.4. A stochastic process $\{X_t : t \in I\}$ has **independent increments** if the random vectors $X_t - X_s$ are independent⁴ of $\mathcal{F}_s^X = \sigma(X_r, t_0 \leq r \leq s)$ for all $s, t \in I$ such that $s < t$.

The preceding definition can also be understood as requiring the random vectors $X_{t_0}, X_{t_1} - X_{t_0}, \dots, X_{t_n} - X_{t_{n-1}}$ to be independent for any $t_0 < t_1 < t_2 < \dots < t_n < T$.

Definition 1.5. A stochastic process $\{X_t : t \in I\}$ has **stationary increments** if, for all $s, t \in I$ such that $s < t$,

$$X_t - X_s \sim X_{t+t_0-s} - X_{t_0}$$

Recall that $\sim$ means precisely that the two random variables have the same probability distribution. In particular, if $t_0 = 0$, $X_t - X_s \sim X_{t-s} - X_0$.

Although requiring a continuous stochastic process to have independent and stationary increments can be quite restrictive, if both conditions hold, knowing the distributions of X_{t_0} and $X_t - X_{t_0}$ for every $t \in [t_0, T]$ is sufficient to characterize the process completely.

Having defined a stochastic process, we may ask on which probability space it is defined. As shown in Appendix A.2, random variables are defined on a measure space; in probability theory, we work specifically with probability spaces, see A.3. In general, Ω represents the space of possible outcomes of an experiment. Here, we want the outcomes of our experiment to be functions defined on I representing realizations of the stochastic process, such as the evolution of the value of a portfolio of stocks or the amount of water available in a reservoir over a given time period I . We therefore define⁵

$$\Omega = (\mathbb{R}^d)^I := \{\omega : I \rightarrow \mathbb{R}^d\}$$

Now, since $X(\omega) : I \rightarrow \mathbb{R}^d$, we want the realizations of the stochastic process for outcome $\omega \in \Omega$ to coincide with the function $\omega(t)$. We must therefore define

$$X_t(\omega) = \omega(t)$$

To construct the measure space, it remains only to specify a σ -algebra, which we denote by $\mathcal{B}(\mathbb{R}^d)^I$. We want the random vectors $X_t(\cdot) : \Omega \rightarrow \mathbb{R}^d$ to be measurable with respect to this σ -algebra, that is, analogously to (A.1),

$$X_t^{-1}(B) = \{\omega \in \Omega : \omega(t) \in B\} \in \mathcal{B}(\mathbb{R}^d)^I$$

for every $B \in \mathcal{B}(\mathbb{R}^d)$. We therefore take this σ -algebra to be the one generated by the cylinder sets in Ω given by

$$A_{t_1, \dots, t_n} = \{\omega \in \Omega : \omega(t_1) \in B_1, \dots, \omega(t_n) \in B_n\}$$

⁴In the sense of independence of σ -algebras, i.e., $\sigma(X_t - X_s)$ is independent of $\mathcal{F}_s^X$.

⁵This reasoning is the idea underlying the proof of Kolmogorov's theorem. See [6].

where $t_1, \dots, t_n \in I$ and $B_1, \dots, B_n \in \mathcal{B}(\mathbb{R}^d)$. Thus, the probability $\mathbb{P}$ that we define on this space will not be a product of probabilities as in A.17, but will be defined as

$$\mathbb{P}(\{\omega \in \Omega : \omega(t_1) \in B_1, \dots, \omega(t_n) \in B_n\})$$

From now on, we will always work on this probability space, which we denote simply by $(\Omega, \mathcal{F}, \mathbb{P})$.

Martingales are a particularly useful class of stochastic processes in stochastic calculus. We next introduce several concepts that will be central throughout these notes.

1.1 Martingales

Throughout this section, we take $(\Omega, \mathcal{F}, \mathbb{P})$ to be the probability space on which the following concepts are defined.

Definition 1.6. A **filtration** $\{\mathcal{F}_t, t \in I\}$ is a family of sub- σ -algebras of $\mathcal{F}$ such that $\mathcal{F}_s \subset \mathcal{F}_t$ for every $s, t \in I$, where $s \leq t$.

We begin by introducing the concept of a *filtration*, which represents the information available immediately after *time* t . That is, the possible values of $Z(\omega)$ for all $\mathcal{F}_t$ -measurable random variables. The 4-tuple $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in I}, \mathbb{P})$ is usually called a *filtered probability space*.

In this context, it may be useful to consider the information *generated* by a stochastic process at time $t \in I$, motivating the following definition.

Definition 1.7. The **natural filtration** of the stochastic process $\{X_t : t \in I\}$ is defined as $\mathcal{F}_t^X = \sigma(X_s, t_0 \leq s \leq t)$ for every $t \in I$.

Definition 1.8. A stochastic process $\{X_t : t \in I\}$ is said to be **adapted** to a filtration $\{\mathcal{F}_t : t \in I\}$ if X_t is⁶ $\mathcal{F}_t$ -measurable.

A stochastic process being adapted to a filtration means that, at a fixed *time* $t \in I$, the filtration $\mathcal{F}_t$ provides sufficient *information* to determine⁷ the value of $X_t(\omega)$ for $\omega \in \Omega$. A natural filtration is simply the *smallest*⁸ filtration with respect to which X_t is $\mathcal{F}_t$ -measurable.

Note that any process $\{X_t : t \in I\}$ is adapted to its natural filtration. This follows immediately because $\sigma(X_t) \subset \mathcal{F}_t^X = \sigma(X_s, t_0 \leq s \leq t)$ and X_t is $\sigma(X_t)$ -measurable by definition.

Definition 1.9. A stochastic process $\{X_t : t \in I\}$ is a **martingale** (with respect to a filtration $\{\mathcal{F}_t\}_{t \in I}$) if

- (a) The process $\{X_t : t \in I\}$ is adapted to the filtration $\{\mathcal{F}_t\}_{t \in I}$.
- (b) $X_t \in L^1(\Omega, \mathcal{F}_t, \mathbb{P})$ for every $t \in I$, i.e., $\mathbb{E}[|X_t|] < \infty$.

⁶As a random vector.

⁷See Example A.2, where the outcome of an experiment cannot be fully determined.

⁸The intersection of all sub- σ -algebras on which X_t is σ -measurable.

(c) $\mathbb{E}[X_t|\mathcal{F}_s] = X_s$ almost surely, where $s, t \in I$ satisfy $s \leq t$.

We now examine some properties that follow from this definition.

Proposition 1.10. *Let $\{X_t : t \in I\}$ be a martingale with respect to $\{\mathcal{F}_t\}_{t \in I}$. Then*

(a) $\mathbb{E}[X_t] = \mathbb{E}[X_s]$ for every $s, t \in I$.

(b) $\mathbb{E}[X_t - X_s|\mathcal{F}_s] = 0$ for all $s, t \in I$ such that $s \leq t$.

(c) If $I \subset \mathbb{R}_+$, then $\mathbb{E}[X_{t+s}|\mathcal{F}_s] = X_s$.

Proof. (a) First, we prove that

$$\mathbb{E}[X_t] = \mathbb{E}[\mathbb{E}[X_t|\mathcal{F}_s]]$$

It suffices to apply property (i) of Proposition A.27 with $\mathcal{G}_1 = \{\emptyset, \Omega\} \subset \mathcal{F}_t = \mathcal{G}_2$. Then $\mathbb{E}[\mathbb{E}[X_t|\mathcal{G}_2]|\mathcal{G}_1] = \mathbb{E}[X_t|\mathcal{G}_1]$, and property (a) of the same proposition gives $\mathbb{E}[X_t|\mathcal{G}_1] = \mathbb{E}[X_t]$.

Therefore, $\mathbb{E}[X_t] = \mathbb{E}[\mathbb{E}[X_t|\mathcal{F}_s]]$, but since the process $\{X_t : t \in I\}$ is a martingale with respect to $\{\mathcal{F}_t\}_{t \in I}$, we have $\mathbb{E}[X_t|\mathcal{F}_s] = X_s$ for all $s, t \in I$ such that $s \leq t$.

(b) Since $\{X_t : t \in I\}$ is a martingale, we have $X_t \in L^1(\Omega, \mathcal{F}_t, \mathbb{P})$ and $X_s \in L^1(\Omega, \mathcal{F}_s, \mathbb{P})$. Since $\mathcal{F}_s \subset \mathcal{F}_t$ because $\{\mathcal{F}_t\}_{t \in I}$ is a filtration, we have $L^1(\Omega, \mathcal{F}_s, \mathbb{P}) \subset L^1(\Omega, \mathcal{F}_t, \mathbb{P})$ and hence $X_s \in L^1(\Omega, \mathcal{F}_t, \mathbb{P})$. In this case, property (e) of Proposition A.27 gives

$$\mathbb{E}[X_t - X_s|\mathcal{F}_s] = \mathbb{E}[X_t|\mathcal{F}_s] - \mathbb{E}[X_s|\mathcal{F}_s]$$

Since X_s is $\mathcal{F}_s$ -measurable, applying property (c) gives $\mathbb{E}[X_s|\mathcal{F}_s] = X_s$. On the other hand, the definition of a martingale gives $\mathbb{E}[X_t|\mathcal{F}_s] = X_s$, yielding the desired result.

(c) The proof follows directly from the definition of a martingale and the fact that $s \leq s + t$ whenever $s, t \in I \subset \mathbb{R}_+$. $\square$

Thus, martingales are an *abstract* representation of the concept of a *fair game*, since the expected value of the process remains constant over time.

Relaxing condition (c) in Definition 1.9 gives the concepts of a **supermartingale**, if $\mathbb{E}[X_t|\mathcal{F}_s] \leq X_s$, and a **submartingale**, if $\mathbb{E}[X_t|\mathcal{F}_s] \geq X_s$. Furthermore, if $\{X_t : t \in I\}$ and $\{Y_t : t \in I\}$ are two martingales (defined on the same probability space) with respect to the same filtration, then $AX_t + BY_t$, where $A, B \in M_{p \times d}(\mathbb{R})$, and $X_t - X_{t_0}$ are martingales.

We next study a type of stochastic process widely used in a variety of applications. In particular, the process on which we will construct stochastic calculus is a Markov process.

1.2 Markov processes

In this chapter, we consider a stochastic process $\{X_t : t \in I\}$ with $I = [t_0, T] \subset \mathbb{R}_+$ and, for $s, t \in I$, define the sub- σ -algebra of $\mathcal{F}$ given by

$$\mathcal{F}([t_0, s]) := \sigma(X_t, t_0 \leq t \leq s)$$

that is, the smallest sub- σ -algebra of $\mathcal{F}$ with respect to which the random vector X_t is measurable for any $t_0 \leq t \leq s$.

Definition 1.11. A stochastic process $\{X_t : t \in I\}$ defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **Markov process** if it satisfies the **Markov property**: for $t_0 \leq s \leq t \leq T$ and every $B \in \mathcal{B}(\mathbb{R}^d)$, the equality

$$\mathbb{P}(X_t \in B | \mathcal{F}([t_0, s])) = \mathbb{P}(X_t \in B | X_s)$$

holds with probability 1.

The following theorem gives meaning to the familiar statement about Markov processes: “In a Markov process, the past and the future are statistically independent when the present is known”.

Theorem 1.12. *The Markov property is equivalent to any of the following conditions:*

(a) For $t_0 \leq s \leq t \leq T$ and $A \in \mathcal{F}([t_0, s])$,

$$\mathbb{P}(A | \mathcal{F}([t_0, s])) = \mathbb{P}(A | X_s)$$

(b) For $t_0 \leq s \leq t \leq T$ and an integrable, $\mathcal{F}([t_0, s])$ -measurable random variable Y ,

$$\mathbb{E}[Y | \mathcal{F}([t_0, s])] = \mathbb{E}[Y | X_s]$$

(c) For $t_0 \leq t_1 \leq t \leq t_2 \leq T$, $A_1 \in \mathcal{F}([t_0, t_1])$, and $A_2 \in \mathcal{F}([t_1, T])$,

$$\mathbb{P}(A_1 \cap A_2 | X_t) = \mathbb{P}(A_1 | X_t) \mathbb{P}(A_2 | X_t)$$

(d) For $n \geq 1$, $t_0 \leq t_1 < \dots < t_n < t \leq T$, and $B \in \mathcal{B}(\mathbb{R}^d)$,

$$\mathbb{P}(X_t \in B | X_{t_1}, \dots, X_{t_n}) = \mathbb{P}(X_t \in B | X_{t_n})$$

Note that the function $P(s, X_s, t, B)$ corresponding to $\mathbb{P}(X_t \in B | X_s)$ is a function where $s, t \in I$, with $s \leq t$, $x \in \mathbb{R}^d$, and $B \in \mathcal{B}(\mathbb{R}^d)$, satisfying

(a) For fixed $s \leq t$ and $B \in \mathcal{B}(\mathbb{R}^d)$, the following holds with probability 1:

$$P(s, X_s, t, B) = \mathbb{P}(X_t \in B | X_s)$$

(b) $P(s, x, t, \cdot)$ defines a probability (measure) on $\mathcal{B}(\mathbb{R}^d)$ for fixed $s \leq t$ and $x \in \mathbb{R}^d$.

(c) $P(s, \cdot, t, B)$ is a $\mathcal{B}(\mathbb{R}^d)$ -measurable function for fixed $s \leq t$ and $B \in \mathcal{B}(\mathbb{R}^d)$.

(d) For $t_0 \leq s \leq u \leq t \leq T$ and $B \in \mathcal{B}(\mathbb{R}^d)$, and for every $x \in \mathbb{R}^d$ with $N \subset \mathbb{R}^d$ such that $\mathbb{P}(X_s \in N) = 0$, it satisfies the Chapman-Kolmogorov equation

$$P(s, x, t, B) = \int_{\mathbb{R}^d} P(u, y, t, B) P(s, x, u, dy)$$

We call this function the **transition probability** or transition function and use the notation

$$P(s, x, t, B) = \mathbb{P}(X_t \in B | X_s = x)$$

1.3 Wiener processes and white noise

Much of the literature uses the terms Wiener process and Brownian motion interchangeably. To distinguish them, however, we will see that a Wiener process is actually a particular case of Brownian motion, also known as standard Brownian motion. In essence, a Wiener process is a homogeneous Markov process defined for $I = \mathbb{R}_+$.

Definition 1.13. Let $W = \{W_t, t \in \mathbb{R}_+\}$ be a stochastic process defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let $\{\mathcal{F}_t\}_{t \in \mathbb{R}_+}$ be a filtration. Then W is a **Wiener process** if:

- (a) W is adapted to the filtration $\{\mathcal{F}_t\}_{t \in \mathbb{R}_+}$.
- (b) $W_0 = 0$ and the paths of the process, $W(\omega)$, are continuous for almost every $\omega \in \Omega$.
- (c) W has independent increments.
- (d) The distribution of $W_t - W_s$ is⁹ $\mathcal{N}(0, (t-s)\mathcal{I})$, where $\mathcal{I}$ is the identity matrix, and hence its increments are stationary.

Note that, for any given filtration $\{\mathcal{F}_t\}_{t \in \mathbb{R}_+}$, if W is a Wiener process adapted to $\{\mathcal{F}_t\}_{t \in \mathbb{R}_+}$, then W is also a Wiener process adapted to the natural filtration of W , $\{\mathcal{F}_t^W\}_{t \in \mathbb{R}_+}$. Furthermore, the continuity condition in (b) is redundant, since it follows from Kolmogorov's criterion, see [7]. Indeed, since $\mathbb{E}[|W_t - W_s|^4] = (d^2 + 2d)(t-s)^2$, where d is the dimension of the Wiener process (as a vector consisting of d independent Wiener processes), Kolmogorov's criterion holds with $a = 4, b = 1$ and $c = d(d+2)$, ensuring continuity of the paths.

An important property of Wiener processes is that any realization, $W(\omega) = \omega(\cdot)$, is nowhere differentiable.

Theorem 1.14. *With probability 1, the paths of a Wiener process are nowhere differentiable.*

We next present a result that will be useful in showing that the paths of a Wiener process are unbounded.

Lemma 1.15. *Let $\{W_t : t \in \mathbb{R}_+\}$ be a d -dimensional Wiener process and let $s = t_0^{(n)} < t_1^{(n)} < \dots < t_n^{(n)} = t$ be a sequence of partitions of the interval $[s, t] \subset \mathbb{R}_+$ such that $\delta_n = \max_{k=1, \dots, n} (t_k^{(n)} - t_{k-1}^{(n)})$. Then*

$$\text{qm} \lim_{\delta_n \rightarrow 0} \sum_{k=1}^n \left(W_{t_k^{(n)}} - W_{t_{k-1}^{(n)}} \right) \left(W_{t_k^{(n)}} - W_{t_{k-1}^{(n)}} \right)' = (t-s)\mathcal{I} \quad (1.1)$$

where convergence is understood entrywise for the matrix. In particular,

$$\text{qm} \lim_{\delta_n \rightarrow 0} \sum_{k=1}^n \left| W_{t_k^{(n)}} - W_{t_{k-1}^{(n)}} \right|^2 = d(t-s) \quad (1.2)$$

Furthermore, if $\sum_{n=0}^{\infty} \delta_n < \infty$, convergence in the preceding expressions holds with probability 1.

⁹The letter $\mathcal{N}$ will be used for both random variables and random vectors with a normal distribution. In the one-dimensional case, clearly $\mathcal{I} = 1$.

Proof. Let W_t^i be the i th component of the vector W_t , with $i = 1, \dots, d$. Writing

$$S_n^{ij} = \sum_{k=1}^n \left(W_{t_k^{(n)}}^i - W_{t_{k-1}^{(n)}}^i \right) \left(W_{t_k^{(n)}}^j - W_{t_{k-1}^{(n)}}^j \right)$$

the properties of expectation imply that, if $i \neq j$,

$$\mathbb{E} [S_n^{ij}] = \sum_{k=1}^n \mathbb{E} \left[\left(W_{t_k^{(n)}}^i - W_{t_{k-1}^{(n)}}^i \right) \right] \mathbb{E} \left[\left(W_{t_k^{(n)}}^j - W_{t_{k-1}^{(n)}}^j \right) \right] = 0$$

by the definition of a Wiener process and because the components i and j of the vector W_t are independent. On the other hand, if $i = j$, then

$$\mathbb{E} [S_n^{ii}] = \sum_{k=1}^n \mathbb{E} \left[\left(W_{t_k^{(n)}}^i - W_{t_{k-1}^{(n)}}^i \right)^2 \right] = \sum_{k=1}^n \text{Var} \left[\left(W_{t_k^{(n)}}^i - W_{t_{k-1}^{(n)}}^i \right) \right] = \sum_{k=1}^n \left(t_k^{(n)} - t_{k-1}^{(n)} \right)$$

and the definition of the partition of $[s, t]$ gives $\mathbb{E} [S_n^{ii}] = (t - s)$. That is,

$$\mathbb{E} [S_n^{ij}] = (t - s) \delta_{ij}$$

On the other hand, for $i = j$, the definition of variance gives

$$\begin{aligned} \text{Var} [S_n^{ii}] &= \sum_{k=1}^n \left(\mathbb{E} \left[\left(W_{t_k^{(n)}}^i - W_{t_{k-1}^{(n)}}^i \right)^4 \right] - \left(t_k^{(n)} - t_{k-1}^{(n)} \right)^2 \right) \\ &= 2 \sum_{k=1}^n \left(t_k^{(n)} - t_{k-1}^{(n)} \right)^2 \end{aligned}$$

and for $i \neq j$, since the components of the random vector are mutually independent, we have

$$\begin{aligned} \text{Var} [S_n^{ij}] &= \sum_{k=1}^n \mathbb{E} \left[\left(W_{t_k^{(n)}}^i - W_{t_{k-1}^{(n)}}^i \right)^2 \left(W_{t_k^{(n)}}^j - W_{t_{k-1}^{(n)}}^j \right)^2 \right] \\ &= \sum_{k=1}^n \mathbb{E} \left[\left(W_{t_k^{(n)}}^i - W_{t_{k-1}^{(n)}}^i \right)^2 \right] \mathbb{E} \left[\left(W_{t_k^{(n)}}^j - W_{t_{k-1}^{(n)}}^j \right)^2 \right] \\ &= \sum_{k=1}^n \left(t_k^{(n)} - t_{k-1}^{(n)} \right)^2 \end{aligned}$$

That is,

$$\begin{aligned} \text{Var} [S_n^{ij}] &= (1 + \delta_{ij}) \sum_{k=1}^n \left(t_k^{(n)} - t_{k-1}^{(n)} \right)^2 \\ &\leq 2 \max_{k=1, \dots, n} \left(t_k^{(n)} - t_{k-1}^{(n)} \right) \sum_{k=1}^n \left(t_k^{(n)} - t_{k-1}^{(n)} \right) \\ &= 2\delta_n(t - s) \end{aligned}$$

and consequently

$$\lim_{\delta_n \rightarrow 0} \text{Var} [S_n^{ij}] = 0$$

which proves mean-square convergence, since $\mathbb{E} [|S_n^{ij}|^2] = \text{Var} [S_n^{ij}] + \mathbb{E} [S_n^{ij}]^2$. Applying the trace operator to expression (1.1) gives expression (1.2). To prove convergence with probability 1, it suffices to observe that, if $\sum_{n=0}^{\infty} \delta_n < \infty$, then $\sum_{n=0}^{\infty} \text{Var} [S_n^{ij}] \leq 2(t-s) \sum_{n=0}^{\infty} \delta_n < \infty$ and consequently $\sum_{n=1}^{\infty} \mathbb{E} [|S_n^{ij}|^2] < \infty$. Applying (A.20) therefore establishes convergence with probability 1. $\square$

In the particular case $t_k^{(n)} = s + (t-s)/2^n$, we have $\delta_n = (t-s)/2^n$ and, since $\sum_{n=0}^{\infty} \delta_n < \infty$, expression (1.2) converges with probability 1. Furthermore, note that

$$0 < t-s = \sum_{k=1}^{2^n} \left| W_{t_k}^{(n)} - W_{t_{k-1}}^{(n)} \right|^2 \leq \max_{k=1, \dots, 2^n} \left| W_{t_k}^{(n)} - W_{t_{k-1}}^{(n)} \right| \sum_{k=1}^{2^n} \left| W_{t_k}^{(n)} - W_{t_{k-1}}^{(n)} \right|$$

and continuity of the realizations of the Wiener process (by Kolmogorov's criterion) gives

$$\text{ac} \lim_{n \rightarrow \infty} \max_{k=1, \dots, 2^n} \left| W_{t_k}^{(n)} - W_{t_{k-1}}^{(n)} \right| = 0$$

This necessarily implies that

$$\text{ac} \lim_{n \rightarrow \infty} \sum_{k=1}^{2^n} \left| W_{t_k}^{(n)} - W_{t_{k-1}}^{(n)} \right| = \infty$$

that is, with probability 1, the paths of a Wiener process have unbounded variation on a finite time interval.

Notation 1.16. Expressions (1.1) and (1.2) motivate the following notation, which we will use throughout these notes.

$$(dW_t)(dW_t)' = \mathcal{I}dt$$

In particular, if $d = 1$, we write

$$dW_t^2 = dt \tag{1.3}$$

Spectral density function

To give a heuristic definition of the white noise process, we first introduce the concept of a spectral density function.

Definition 1.17. A stochastic process $\{X_t : t \in I\}$ is said to be **(strictly) stationary** if, given $t_i, t_i + t \in I$, for every $i = 1, \dots, n$ we have

$$F_{t_1+t, \dots, t_n+t}(x_1, \dots, x_n) = F_{t_1, \dots, t_n}(x_1, \dots, x_n)$$

Definition 1.18. A stochastic process $\{X_t : t \in I\}$ is said to be **(weakly) stationary** if it satisfies:

$$(a) \quad \mathbb{E} [|X_t|^2] < \infty.$$

- (b) $\mathbb{E}[X_t] = m$ for every $t \in I$, where m may be a constant vector if the process is d -dimensional.
- (c) $\gamma_X(t, s) = \gamma_X(t + h, s + h)$ for every $s, t \in I$ and $h \in \mathbb{R}$ such that $s + h, t + h \in I$, where $\gamma_X(t, s) = \text{Cov}(X_t, X_s)$

In this case, we write $\gamma_X(h) = \gamma_X(t + h, t)$.

If a stochastic process is (weakly) stationary, Bochner's theorem [8] ensures that there exists a unique measure F defined on $\mathbb{R}$ such that

$$\gamma_X(h) = \int_{-\infty}^{+\infty} e^{ihu} dF(u)$$

for $h \in \mathbb{R}$, where F is called the **spectral distribution function** of X_t .

The term *spectral* is used because F gives the frequency distribution of the harmonic oscillators that *contribute to generating* X_t . If F also has a density function f ,

$$\gamma_X(h) = \int_{-\infty}^{+\infty} e^{ihu} f(u) du$$

this is called the **spectral density function**. Furthermore, if

$$\int_{-\infty}^{+\infty} |\gamma_X(h)| dh < \infty,$$

that is, γ_X is integrable, then f can be obtained using the Fourier inversion formula

$$f(u) = \int_{-\infty}^{+\infty} e^{-2\pi i h u} \gamma_X(h) dh$$

Note that $f(0) = \int_{\mathbb{R}} \gamma_X(h) dh$.

White noise

To construct a stochastic process $\{\xi_t : t \in \mathbb{R}\}$ in which all frequencies *contribute* with equal intensity¹⁰ (uniformly) and which also satisfies $\mathbb{E}[\xi_t] = 0$ for every $t \in \mathbb{R}$, we must have

$$f(u) = 1$$

for every $u \in \mathbb{R}$. Thus, the preceding relationship between the spectral density function and the autocovariance function $\gamma_\xi(h)$ gives $\gamma_\xi(h) = \delta(h)$, where δ denotes the Dirac delta. However, this choice leads to

$$\text{Var}[\xi_t] = \gamma_\xi(0) = \int_{-\infty}^{+\infty} \underbrace{f(u)}_1 du = \infty$$

and therefore this process has infinite variance.

¹⁰As with white light, hence the name white noise.

Nevertheless, white noise can be defined using distribution theory, see [7], as a stochastic process defined on the space of tempered distributions. The white noise process can thus be viewed as the distributional derivative of a Wiener process, leading to the notation $\xi_t = \frac{dW_t}{dt}$.

From the preceding discussion, we can define a d -dimensional white noise process as a stochastic process with mean vector $\mathbb{E}[\xi_t] = 0$ for every $t \in \mathbb{R}$ and covariance matrix $\gamma_X(h) = \mathcal{I}\delta(h)$.

The aim of these notes is to study differential equations in which one component is a multiple of white noise, that is,

$$dX_t = a(t, X_t)dt + b(t, X_t)\xi_t dt$$

where the term multiplying ξ_t models the *variable* or *random* component of the process. In the notation just introduced, we can write this process as

$$dX_t = a(t, X_t)dt + b(t, X_t)dW_t$$

Thus, when solving these equations,

$$X_t = \int_I a(s, X_s)ds + \int_I b(s, X_s)dW_s$$

we must properly define the *integral*

$$\int_I b(s, X_s)dW_s$$

known as a stochastic integral. In particular, this integral must be constructed in a completely different way because of the earlier result showing that the paths of a Wiener process are unbounded. The next chapter, which forms a central part of these notes, therefore aims to introduce the concept of a stochastic integral formally.

Chapter 2

Stochastic integrals

To study the foundations of stochastic differential equations, we need to define a new type of integral, known as a stochastic integral. This chapter introduces the stochastic integral naturally and then considers several generalizations. In general, we assume that $\{W_t : t \in I\}$ is a Wiener process defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 2.1. Let $\{\mathcal{F}_s\}_{s \in I}$ be a filtration such that the m -dimensional Wiener process $\{W_s : s \in I\}$ is adapted to $\mathcal{F}_s$ and, for every $t_0 \leq s \leq t$, $W_t - W_s$ is independent¹ of $\mathcal{F}_s$. We then define the space $L^2_{\text{ad}}(I \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mu \times \mathbb{P})$, where $I = [t_0, T]$ and μ denotes Lebesgue measure, as the space of functions

$$\begin{aligned} b &: (I \times \Omega, \mathcal{B}(I) \times \mathcal{F}) \rightarrow (\mathbb{R}^{d \times m}, \mathcal{B}(\mathbb{R}^{d \times m})) \\ (s, \omega) &\mapsto b(s, \omega) \end{aligned}$$

such that:

- (a) b is $\mathcal{B}(I) \times \mathcal{F}$ -measurable; recall that this means

$$b^{-1}(B) = \{(s, \omega) \in I \times \Omega : b(s, \omega) \in B\} \in \mathcal{B}(I) \times \mathcal{F}$$

for every $B \in \mathcal{B}(\mathbb{R}^{d \times m})$. Here, $\mathcal{B}(I) \times \mathcal{F}$ denotes the smallest σ -algebra containing elements of the form $B \times F$ with $B \in \mathcal{B}(I)$ and $F \in \mathcal{F}$.

- (b) b is adapted to the filtration $\{\mathcal{F}_s\}_{s \in I}$. That is, for each $s \in I$, the function $b(s, \cdot)$ is $\mathcal{F}_s$ -measurable. Thus, $b(s, \omega)$ can be viewed as a stochastic process² with paths $b(\cdot, \omega)$.

- (c) The functions $b(\cdot, \omega)$ are square-integrable for every $\omega \in \Omega$, meaning that $b(\cdot, \omega) \in L^2(I, \mathcal{B}(I), \mu)$, and hence

$$\int_{t_0}^T |b(s, \omega)|^2 ds < \infty$$

where $|b| = \text{tr}(bb')^{1/2}$ is the Frobenius norm³ and the integral is a Lebesgue integral; that is, we are simplifying the notation $\int_I |b(s, \omega)|^2 d\mu(s)$.

¹As σ -algebras, i.e., $\sigma(W_t - W_s : s \leq t < \infty)$ is independent of $\mathcal{F}_s$. See independence of σ -algebras in A.18.

²Nevertheless, we will refer to b as a function, although the literature commonly refers to b as a stochastic process in L^2_{ad} . That is, it is a stochastic process defined on I , adapted to the filtration $\{\mathcal{F}_s\}_{s \in I}$, and with paths in $L^2(I, \mathcal{B}(I), \mu)$.

³Note that $b^* = b' \equiv b^t$ because the matrix is real.

The space $L_{\text{ad}}^2(I \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mu \times \mathbb{P})$ is a Banach space with norm

$$\|b\|_{L_{\text{ad}}^2} = \left[\int_{t_0}^T \mathbb{E} [|b|^2] ds \right]^{1/2}$$

For simplicity, we denote $L_{\text{ad}}^2(I \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mu \times \mathbb{P})$ by L_{ad}^2 . We first define the stochastic integral on a subset of functions in L_{ad}^2 .

Definition 2.2. Let $b \in L_{\text{ad}}^2$. We say that it is a step function if there exist $t_0 < t_1 < \dots < t_n = T$ such that $b(s, \omega) = b(t_{i-1}, \omega)$ for every $s \in [t_{i-1}, t_i)$ with $i = 1, \dots, n$ and⁴ every $\omega \in \Omega$. We denote by $L_{\text{step}}^2(I \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mathbb{P} \times \mu)$ the subspace of L_{ad}^2 consisting of step functions, abbreviated as L_{step}^2 .

Definition 2.3. Let $b \in L_{\text{step}}^2$. The **stochastic integral** of b with respect to W_t is defined as the random vector

$$I[b] = \int_{t_0}^T b dW = \int_{t_0}^T b(s) dW_s = \sum_{i=1}^n b(t_{i-1}) (W_{t_i} - W_{t_{i-1}}) \quad (2.1)$$

Remark 2.4. We again use an abuse of notation that will be common throughout these notes for simplicity. Denoting $I[b] = \int_{t_0}^T b dW$ as in expression (2.1), we have $I[b] = I[b](\omega) = \int_{t_0}^T b(s, \omega) dW_s(\omega) = \sum_{i=1}^n b(t_{i-1}, \omega) (W_{t_i}(\omega) - W_{t_{i-1}}(\omega))$ for $\omega \in \Omega$. Also note that $I[b]$ is a d -dimensional vector, since we take W_t to be an m -dimensional Wiener process and b takes values in (a subset of) $\mathbb{R}^{d \times m}$.

Note that the function b is evaluated at t_{i-1} , so we may ask what would happen if we evaluated b at another point $\tau_i \in [t_{i-1}, t_i]$. Unlike the definition of the Riemann integral, the result does depend on the point in the interval at which the function is evaluated.

We will prove that the choice $\tau_i = t_{i-1}$ gives the resulting integral several useful properties. The generalization of this case is known as the **Itô integral**, introduced by Kiyoshi Itô in 1944, see [1], and is what we call the stochastic integral throughout these notes. However, another choice is widely used in fields such as physics because of its properties: choosing the midpoint of the subinterval $\tau_i = (t_i + t_{i-1})/2$. The corresponding integral is known as the **Stratonovich integral**. These notes focus entirely on the Itô integral, which we will define fully below.

Having defined the stochastic integral for functions in L_{step}^2 , we study its properties before generalizing these results to a larger class of functions.

Theorem 2.5. Let $b_1, b_2 \in L_{\text{step}}^2$ with $I = [t_0, T]$, $t_0 \leq T$. Then:

(a) Given $c_1, c_2 \in \mathbb{R}$,

$$\int_{t_0}^T (c_1 b_1 + c_2 b_2) dW = c_1 \int_{t_0}^T b_1 dW + c_2 \int_{t_0}^T b_2 dW. \quad (2.2)$$

⁴From now on, we omit the argument ω in b whenever there is no risk of ambiguity.

(b) We have

$$\int_{t_0}^T b dW = \begin{pmatrix} \sum_{k=1}^m \int_{t_0}^T b_{1,k}(s) dW_s^k \\ \vdots \\ \sum_{k=1}^m \int_{t_0}^T b_{d,k}(s) dW_s^k \end{pmatrix} \quad (2.3)$$

with $W_t = (W_t^1, \dots, W_t^m)'$.

(c) If $\mathbb{E}[|b(s)|] < \infty$ for every $s \in I$, then

$$\mathbb{E} \left[\int_{t_0}^T b dW \right] = 0 \quad (2.4)$$

Recall that $\mathbb{E}[|b(s)|] < \infty$ means $\mathbb{E}[|b(s, \omega)|] = \int_{\Omega} |b(s, \omega)| d\mathbb{P} < \infty$ for each $s \in I$.

(d) If $\mathbb{E}[|b(s)|^2] < \infty$ for every $s \in I$, then the covariance matrix of the stochastic matrix is given by

$$\mathbb{E} \left[\left(\int_{t_0}^T b dW \right) \left(\int_{t_0}^T b dW \right)' \right] = \int_{t_0}^T \mathbb{E}[b(s)b(s)'] ds \quad (2.5)$$

and in particular

$$\mathbb{E} \left[\left| \int_{t_0}^T b dW \right|^2 \right] = \int_{t_0}^T \mathbb{E}[|b|^2] ds \quad (2.6)$$

Proof. (a) Since L_{step}^2 is a vector space, $c_1 b_1 + c_2 b_2 \in L_{\text{step}}^2$, and applying definition (2.1) gives the desired result.

(b) This follows directly by expanding the matrix product and applying property (a).

(c) From the definition of the integral for functions in L_{step}^2 and the linearity of expectation,

$$\mathbb{E} \left[\int_{t_0}^T b dW \right] = \sum_{i=1}^n \mathbb{E}[b(t_{i-1})(W_{t_i} - W_{t_{i-1}})] = \sum_{i=1}^n \mathbb{E}[b(t_{i-1})] \mathbb{E}[W_{t_i} - W_{t_{i-1}}]$$

since $b(t_{i-1})$ is independent of $W_{t_i} - W_{t_{i-1}}$ by construction of L_{ad}^2 . The result follows directly from the definition of a Wiener process, since $\mathbb{E}[W_{t_i} - W_{t_{i-1}}] = 0$.

(d) Expanding expression (2.5),

$$\begin{aligned} \mathbb{E} \left[\left(\int_{t_0}^T b dW \right) \left(\int_{t_0}^T b dW \right)' \right] &= \\ &= \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[b(t_{i-1})(W_{t_i} - W_{t_{i-1}})(W_{t_j} - W_{t_{j-1}})'b(t_{j-1})'] \\ &= \sum_{i=1}^n \mathbb{E}[b(t_{i-1})(W_{t_i} - W_{t_{i-1}})(W_{t_i} - W_{t_{i-1}})'b(t_{i-1})'] \\ &\quad + 2 \sum_{i < j} \mathbb{E}[b(t_{i-1})(W_{t_i} - W_{t_{i-1}})(W_{t_j} - W_{t_{j-1}})'b(t_{j-1})'] \\ &= S_1 + S_2 \end{aligned}$$

where the sum has been decomposed using the symmetry of the matrix. First, note that, for $i < j$, $W_{t_j}^q - W_{t_{j-1}}^q$ is independent of $b_{k,p}(t_{i-1})(W_{t_i}^p - W_{t_{i-1}}^p)b_{h,q}(t_{j-1})$ for any $k, h = 1, \dots, d$ with $p, q = 1, \dots, m$. Arguing as in (c), we therefore obtain $S_2 = 0$. We thus compute S_1 , denoting by $S_1^{k,h}(i)$ the entry of the matrix S_1 in row k and column h at t_i . Thus,

$$\begin{aligned} S_1^{k,h}(i) &= \sum_{p=1}^m \sum_{q=1}^m \mathbb{E} \left[b_{k,p}(t_{i-1})(W_{t_i}^p - W_{t_{i-1}}^p)(W_{t_j}^q - W_{t_{j-1}}^q)b_{h,q}(t_{j-1}) \right] \\ &= \sum_{p=1}^m \sum_{q=1}^m \mathbb{E} [b_{k,p}(t_{i-1})b_{h,q}(t_{j-1})] \mathbb{E} [(W_{t_i}^p - W_{t_{i-1}}^p)(W_{t_j}^q - W_{t_{j-1}}^q)] \end{aligned}$$

since $b \in L_{ad}^2$. Furthermore, $W_{t_i} - W_{t_{i-1}} \sim \mathcal{N}(0, (t_i - t_{i-1})\mathcal{I})$, and hence

$$S_1^{k,h}(i) = \sum_{p=1}^m \mathbb{E} [b_{k,p}(t_{i-1})b_{h,p}(t_{j-1})] (t_i - t_{i-1})$$

and it follows directly that

$$S_1 = \sum_{i=1}^n \mathbb{E} [b(t_{i-1})b(t_{j-1})'] (t_i - t_{i-1}) = \int_{t_0}^T \mathbb{E} [b(s)b(s)'] ds$$

and, in particular, applying the trace operator, where $\text{tr}(bb') = |b|^2$, gives expression (2.6). $\square$

We now generalize the definition of the stochastic integral to functions in L_{ad}^2 . To do so, we first establish an approximation result.

Theorem 2.6. *Given $b \in L_{ad}^2$, there exists a sequence of functions $\{b_n\}_{n \in \mathbb{N}} \subset L_{step}^2$ such that*

$$\lim_{n \rightarrow \infty} \|b_n - b\|_{L_{ad}^2} = 0$$

that is, L_{step}^2 is dense in L_{ad}^2 .

Proof. We prove this by first considering two particular cases.

Case 1. Suppose that $\mathbb{E}[b(s)b(t)]$ is a continuous function on $I \times I$. Let $t_0 < t_1 < \dots < t_n = T$ be a partition of I . Defining $b_n(s, \omega) = b(t_{i-1}, \omega)$ for $t_{i-1} \leq s < t_i$, we have $\{b_n\}_{n \in \mathbb{N}} \subset L_{step}^2$, and continuity of $\mathbb{E}[b(s)b(t)]$ on $I \times I$ gives

$$\lim_{t \rightarrow s} \mathbb{E} [|b(s) - b(t)|^2] = 0$$

therefore, for every $s \in I$,

$$\lim_{n \rightarrow \infty} \mathbb{E} [|b(s) - b_n(s)|^2] = 0 \quad (2.7)$$

Furthermore, since $|f - g|^2 \leq 2(|f|^2 + |g|^2)$, we have

$$\mathbb{E} [|b(s) - b_n(s)|^2] \leq 2(\mathbb{E} [|b(s)|^2] + \mathbb{E} [|b_n(s)|^2]) \leq 4 \sup_{s \in I} \mathbb{E} [|b(s)|^2] \quad (2.8)$$

so expressions (2.7) and (2.8) allow us to apply the dominated convergence theorem. Thus,

$$\lim_{n \rightarrow \infty} \|b_n - b\|_{L^2_{ad}} = \lim_{n \rightarrow \infty} \mathbb{E} \left[\int_{t_0}^t |b_n(s) - b(s)|^2 ds \right] = \mathbb{E} \left[\lim_{n \rightarrow \infty} \int_{t_0}^t |b_n(s) - b(s)|^2 ds \right] = 0$$

Case 2. Suppose that b is bounded, and define the sequence of functions

$$H_n(s, \omega) = \int_0^{n(s-T)} e^{-t} b(s - t/n, \omega) dt$$

We want to prove that $\mathbb{E}[H_n(s)H_n(t)]$ is a continuous function on $I \times I$ so that we can apply case 1 to $H_n(s)$. We will also prove that $\lim_{n \rightarrow \infty} \|H_n - b\|_{L^2_{ad}} = 0$ to establish the result for bounded functions.

We first prove continuity. Note that $H_n(s)$ is adapted to $\mathcal{F}_s$ and that $\int_{t_0}^T \mathbb{E}[|H_n(s)|^2] ds < \infty$. Let $u = s - t/n$. Then

$$H_n(s, \omega) = n \int_{t_0}^s e^{n(u-s)} b(u, \omega) du$$

which satisfies

$$\lim_{s \rightarrow t} \mathbb{E}[|H_n(s) - H_n(t)|^2] = 0$$

and therefore $\mathbb{E}[H_n(s)H_n(t)]$ is a continuous function on $I \times I$. On the other hand, we have

$$b(s) - H_n(s) = \int_0^\infty e^{-t} (b(s) - b(s - t/n)) dt$$

where $b(s)$ has been extended so that $b(s) = 0$ for $s < t_0$. Since $e^{-t} dt$ is a probability measure on $[0, \infty)$, applying the Cauchy-Schwarz inequality, see (A.15), gives

$$|b(s) - H_n(s)|^2 \leq \int_0^\infty |b(s) - b(s - t/n)|^2 e^{-t} dt$$

Then

$$\begin{aligned} \int_{t_0}^T \mathbb{E}[|b(s) - H_n(s)|^2] ds &\leq \int_{t_0}^T \int_0^\infty e^{-t} \mathbb{E}[|b(s) - b(s - t/n)|^2] dt ds \\ &= \int_0^\infty e^{-t} \left(\int_{t_0}^T \mathbb{E}[|b(s) - b(s - t/n)|^2] ds \right) dt \\ &= \int_0^\infty e^{-t} \mathbb{E} \left[\int_{t_0}^T |b(s) - b(s - t/n)|^2 ds \right] dt \end{aligned} \quad (2.9)$$

and since we assumed that b is bounded, we have

$$\lim_{n \rightarrow \infty} \int_{t_0}^T |b(s, \omega) - b(s - t/n, \omega)|^2 ds = 0 \quad (2.10)$$

for almost every $\omega \in \Omega$. From (2.9) and (2.10), it follows that

$$\lim_{n \rightarrow \infty} \int_{t_0}^T \mathbb{E}[|b(s) - H_n(s)|^2] ds = 0 \quad (2.11)$$

By continuity of $\mathbb{E}[H_n(s)H_n(t)]$ on $I \times I$, we can apply the result obtained in *case 1* to $H_n(s, \omega)$ and choose a sequence of functions $\{b_n\}_{n \in \mathbb{N}} \subset L_{\text{step}}^2$ such that

$$\int_{t_0}^T \mathbb{E}[|H_n(s) - b_n(s)|^2] ds \leq \frac{1}{n} \quad (2.12)$$

and then applying (2.11) to the preceding expression gives

$$\lim_{n \rightarrow \infty} \int_{t_0}^T \mathbb{E}[|b(s) - b_n(s)|^2] ds = 0$$

which proves the second case.

General case. Let $b \in L_{\text{ad}}^2$. We define

$$H_n(s, \omega) = \begin{cases} b(s, \omega) & \text{if } |b(s, \omega)| \leq n \\ 0 & \text{otherwise.} \end{cases}$$

for every $n \in \mathbb{N}$. By the dominated convergence theorem,

$$\lim_{n \rightarrow \infty} \int_{t_0}^T \mathbb{E}[|b(s) - H_n(s)|^2] ds = \mathbb{E} \left[\lim_{n \rightarrow \infty} \int_{t_0}^T |b(s)|^2 \mathbb{1}_{\{|b(s)| > n\}} ds \right] = 0 \quad (2.13)$$

This proves the desired result. $\square$

This result enables us to define the stochastic integral for any function in L_{ad}^2 . If we have a sequence of functions $\{b_n(s, \omega)\}_{n \in \mathbb{N}} \subset L_{\text{step}}^2$ such that $\lim_{n \rightarrow \infty} \|b - b_n\|_{L_{\text{ad}}^2} = 0$ and apply Theorem 2.6, then

$$\lim_{n, m \rightarrow \infty} \mathbb{E}[|I[b_n] - I[b_m]|^2] = 0$$

and therefore $\{I[b_n]\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $L^2(\Omega, \mathcal{F}, \mathbb{P})$. We thus define the stochastic integral as

$$I[b] := \lim_{n \rightarrow \infty} I[b_n] \quad (2.14)$$

where $I[b] \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ (because $I[b_n]$ is a Cauchy sequence in the complete space L^2 , so its limit belongs to that space), and by definition it inherits linearity. To verify that the integral is well defined, we must check that the limit in equation (2.14) is independent of the chosen sequence $\{b_n\}_{n \in \mathbb{N}}$; see [9] for the proof. As with the stochastic integral for functions in L_{step}^2 , property (d) of Theorem 2.5 continues to hold.

Theorem 2.7. *Let $b \in L_{\text{ad}}^2(I \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mu \times \mathbb{P})$. Its stochastic integral given by $I[b](\omega) = \int_{t_0}^T b(s, \omega) dW_s(\omega)$ is a random variable with $\mathbb{E}[I[b]] = 0$ and satisfies*

$$\mathbb{E}[|I[b]|^2] = \int_{t_0}^T \mathbb{E}[|b(s)|^2] ds \quad (2.15)$$

According to this theorem, the stochastic integral $I : L_{\text{ad}}^2(I \times \Omega) \rightarrow L^2(\Omega)$ is an isometry. This identity is also known as the Itô isometry.

We next extend the concept of a stochastic integral to a more general class of functions than those in L_{ad}^2 . We relax condition (c) in Definition 2.1 by requiring the functions $b(\cdot, \omega)$ to be square-integrable with probability 1, that is,

$$\mathbb{P} \left(\left\{ \omega \in \Omega : \int_{t_0}^T |b(s, \omega)|^2 d\mu(s) < \infty \right\} \right) \equiv \mathbb{P} \left(\int_{t_0}^T |b(s)|^2 ds < \infty \right) = 1$$

thereby defining the space $\mathcal{L}_{\text{ad}}(I \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mu \times \mathbb{P})$ as the most general space on which we study the stochastic integral in these notes.

Remark 2.8. Note that $L_{\text{ad}}^2 \subset \mathcal{L}_{\text{ad}}$ since, given $b \in L_{\text{ad}}^2$,

$$\mathbb{E} \left[\int_{t_0}^T |b(s)|^2 ds \right] = \int_{t_0}^T \mathbb{E} [|b(s)|^2] ds < \infty$$

and therefore

$$\int_{t_0}^T |b(s)|^2 ds < \infty$$

for almost every $\omega \in \Omega$, where we have used Theorem 2.7 and Fubini's theorem.

We next show that functions in $\mathcal{L}_{\text{ad}}$ can be written as the limit (in probability) of a sequence of functions in L_{ad}^2 .

Lemma 2.9. *Given $b \in \mathcal{L}_{\text{ad}}$, there exists a sequence of functions $\{b_n\}_{n \in \mathbb{N}} \subset L_{\text{ad}}^2$ such that*

$$\lim_{n \rightarrow \infty} \int_{t_0}^T |b(s) - b_n(s)|^2 ds = 0$$

for almost every $\omega \in \Omega$, that is, it converges⁵ with probability 1.

Proof. Let $n \in \mathbb{N}$. We define

$$b_n(s, \omega) = \begin{cases} b(s, \omega) & \text{if } \int_{t_0}^s |b(t, \omega)|^2 dt \leq n \\ 0 & \text{otherwise.} \end{cases}$$

and since $b \in \mathcal{L}_{\text{ad}}$ and it is adapted to $\{\mathcal{F}_s\}_{s \in I}$, b_n is also adapted. Defining

$$\tau_n(\omega) = \sup \left\{ s \in I : \int_{t_0}^s |b(t, \omega)|^2 dt \leq n \right\}$$

we obtain

$$\int_{t_0}^T |b_n(s, \omega)|^2 ds = \int_{t_0}^{\tau_n(\omega)} |b(s, \omega)|^2 ds \leq n$$

and taking expectations on both sides gives

$$\int_{t_0}^T \mathbb{E} [|b_n(s, \omega)|^2] ds \leq n < \infty$$

⁵Convergence with probability 1 implies convergence in probability, see A.6.

so $b_n \in L^2_{\text{ad}}$, where we have used Fubini's theorem. We now show that this sequence indeed approximates b . By definition of $\mathcal{L}_{\text{ad}}$, there exists a null set $N \subset \Omega$ such that, for every $\omega \in \Omega \setminus N$,

$$\int_{t_0}^T |b(s, \omega)|^2 ds < \infty$$

since the paths of b belong to L^2 with probability 1, that is,

$$\mathbb{P} \left(\left\{ \omega \in \Omega : \int_{t_0}^T |b(s, \omega)|^2 d\mu(s) < \infty \right\} \right) = \mathbb{P}(N) = 0$$

Taking n sufficiently large for a fixed $\omega \in \Omega \setminus N$, we have

$$\int_{t_0}^T |b(s, \omega)|^2 ds \leq n$$

and therefore $b_n(s, \omega) = b(s, \omega)$ for every $s \in I$. Thus,

$$\lim_{n \rightarrow \infty} \int_{t_0}^T |b_n(s, \omega) - b(s, \omega)|^2 ds = 0$$

for every $\omega \in \Omega \setminus N$, that is, for almost every $\omega \in \Omega$. □

We now present a key result for defining the stochastic integral on $\mathcal{L}_{\text{ad}}$.

Lemma 2.10. *Let $b \in L^2_{\text{step}}$. Then, for every $N > 0$ and $c > 0$,*

$$\mathbb{P} \left(\left| \int_{t_0}^T b(s) dW_s \right| > c \right) \leq \frac{N}{c^2} + \mathbb{P} \left(\int_{t_0}^T |b(s)|^2 ds > N \right)$$

Proof. Let $t_0 < t_1 < \dots < t_n = T$ be a partition of I and suppose that $b(s) = b(t_{i-1})$ for $t_{i-1} \leq s < t_i$. Then the function

$$b_N(s) = \begin{cases} b(s) & \text{if } \int_{t_0}^{t_i} |b(s)|^2 ds \leq N, \\ 0 & \text{otherwise.} \end{cases}$$

belongs to L^2_{step} since $b \in L^2_{\text{step}}$. Since

$$\int_{t_0}^T |b_N(s)|^2 ds = \sum_{i=1}^n |b_N(t_{i-1})|^2 (t_i - t_{i-1}) \leq N$$

because b_N is a step function, we have $|b_N(t_{i-1})| \leq N/(t_i - t_{i-1})$ and therefore

$$\mathbb{E} [|b_N(s)|^2] < \infty$$

so we can apply part (d) of Theorem 2.5 to obtain

$$\mathbb{E} \left[\left| \int_{t_0}^T b_N(s) dW_s \right|^2 \right] = \int_{t_0}^T \mathbb{E} [|b_N(s)|^2] ds \leq N$$

Finally, the definition of $b_N(s)$ clearly implies that $b_N(s) \neq b(s)$ if and only if

$$\int_{t_0}^T |b(s)|^2 ds > N$$

which can be rewritten as

$$\mathbb{P} \left(\left\{ \omega \in \Omega : \sup_{t_0 \leq s \leq T} |b_N(s, \omega) - b(s, \omega)| > 0 \right\} \right) = \mathbb{P} \left(\left\{ \omega \in \Omega : \int_{t_0}^T |b(s, \omega)|^2 ds > N \right\} \right)$$

Note that the set of ω for which the absolute value of the integral exceeds c is contained in the union of the set of ω for which $b_N(s)$ satisfies the same property and the set of ω on which $b_N(s) \neq b(s)$, that is,

$$\begin{aligned} \left\{ \omega \in \Omega : \left| \int_{t_0}^T b(s, \omega) dW_s(\omega) \right| > c \right\} &\subset \left\{ \omega \in \Omega : \left| \int_{t_0}^T b_N(s, \omega) dW_s(\omega) \right| > c \right\} \\ &\cup \left\{ \omega \in \Omega : \left| \int_{t_0}^T (b_N(s, \omega) - b(s, \omega)) dW_s(\omega) \right| > 0 \right\} \end{aligned}$$

and since $\mathbb{P}$ is an additive measure,

$$\begin{aligned} \mathbb{P} \left(\left| \int_{t_0}^t b(s) dW_s \right| > c \right) &\leq \mathbb{P} \left(\left| \int_{t_0}^t b(s) dW_s \right| > c \right) + \mathbb{P} \left(\left| \int_{t_0}^t (b_N(s) - b(s)) dW_s \right| > 0 \right) \\ &\leq \frac{1}{c^2} \mathbb{E} \left[\left| \int_{t_0}^t b_N(s) dW_s \right|^2 \right] + \mathbb{P} \left(\int_{t_0}^t |b(s)|^2 ds > N \right) \\ &\leq \frac{N}{c^2} + \mathbb{P} \left(\int_{t_0}^t |b(s)|^2 ds > N \right) \end{aligned}$$

where we have used the preceding results and Chebyshev's inequality given in (A.17). $\square$

Finally, before defining the stochastic integral on this more general function space, we need the following result.

Lemma 2.11. *Let $b \in \mathcal{L}_{ad}$. Then there exists a sequence $\{b_n\}_{n \in \mathbb{N}} \subset L_{step}^2$ such that*

$$\lim_{n \rightarrow \infty} \int_{t_0}^T |b_n(s) - b(s)|^2 ds = 0$$

where convergence is in probability, that is,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left\{ \omega \in \Omega : \int_{t_0}^T |b_n(s, \omega) - b(s, \omega)|^2 ds > \epsilon \right\} \right) = 0$$

Proof. By Lemma 2.9, we can choose a sequence $\{H_n\}_{n \in \mathbb{N}} \subset L_{ad}^2$ such that

$$\lim_{n \rightarrow \infty} \int_{t_0}^T |H_n(s) - b(s)|^2 ds = 0$$

with convergence in probability, since convergence with probability 1 implies convergence in probability; see Appendix A.6. Applying Theorem 2.6, we can now find a sequence of functions $\{b_n\}_{n \in \mathbb{N}} \subset L_{ad}^2$ such that

$$\mathbb{E} \left[\int_{t_0}^T |b_n(s) - H_n(s)|^2 ds \right] \leq \frac{1}{n} \quad (2.16)$$

and using inequality $|a + b|^2 \leq 2(|a|^2 + |b|^2)$ again, where $a = b_n - H_n$ and $b = H_n - b$, we see that

$$\int_{t_0}^T |b_n(s, \omega) - b(s, \omega)|^2 ds \leq 2 \int_{t_0}^T |b_n(s, \omega) - H_n(s, \omega)|^2 ds + 2 \int_{t_0}^T |H_n(s, \omega) - b(s, \omega)|^2 ds$$

so, for every $\epsilon > 0$,

$$\begin{aligned} \left\{ \omega \in \Omega : \int_{t_0}^T |b_n(s, \omega) - b(s, \omega)|^2 ds > \epsilon \right\} &\subset \left\{ \omega \in \Omega : \int_{t_0}^T |b_n(s, \omega) - H_n(s, \omega)|^2 ds > \frac{\epsilon}{4} \right\} \\ &\cup \left\{ \omega \in \Omega : \int_{t_0}^T |H_n(s, \omega) - b(s, \omega)|^2 ds > \frac{\epsilon}{4} \right\} \end{aligned}$$

and since $\mathbb{P}$ is an additive measure,

$$\begin{aligned} \mathbb{P} \left(\int_{t_0}^T |b_n(s, \omega) - b(s, \omega)|^2 ds > \epsilon \right) &\leq \mathbb{P} \left(\int_{t_0}^T |b_n(s, \omega) - H_n(s, \omega)|^2 ds > \frac{\epsilon}{4} \right) \\ &\quad + \mathbb{P} \left(\int_{t_0}^T |H_n(s, \omega) - b(s, \omega)|^2 ds > \frac{\epsilon}{4} \right) \end{aligned}$$

Applying Chebyshev's inequality (A.17) and expression (2.16) to the first term on the right gives

$$\mathbb{P} \left(\int_{t_0}^T |b_n(s, \omega) - b(s, \omega)|^2 ds > \epsilon \right) \leq \frac{4}{\epsilon n} + \mathbb{P} \left(\int_{t_0}^T |H_n(s, \omega) - b(s, \omega)|^2 ds > \frac{\epsilon}{4} \right)$$

and it therefore suffices to take the limit as $n \rightarrow \infty$ to complete the proof: the first term clearly vanishes, and the second vanishes because H_n was chosen to ensure convergence in probability. $\square$

To ensure that the stochastic integral of functions in $\mathcal{L}_{ad}$ is well defined, we establish the following result.

Lemma 2.12. *Suppose that $b \in \mathcal{L}_{ad}$ and that there exists a sequence of functions $\{b_n\}_{n \in \mathbb{N}} \subset L_{step}^2$ approximating b in the sense of Lemma 2.11, that is,*

$$\lim_{n \rightarrow \infty} \int_{t_0}^T |b(s) - b_n(s)|^2 ds = 0$$

Then

$$\lim_{n \rightarrow \infty} \int_{t_0}^T b_n(s) dW_s = I[b]$$

where convergence is in probability and the integral is defined by expression (2.1). Furthermore, this limit is independent of the chosen sequence.

Proof. Applying inequality $|a + b|^2 \leq 2(|a|^2 + |b|^2)$ gives

$$\int_{t_0}^T |b_n(s) - b_m(s)|^2 ds \leq 2 \int_{t_0}^T |b(s) - b_n(s)|^2 ds + 2 \int_{t_0}^T |b(s) - b_m(s)|^2 ds$$

and by assumption we necessarily have

$$\lim_{n,m \rightarrow \infty} \int_{t_0}^T |b_n(s) - b_m(s)|^2 ds = 0$$

in probability. That is,

$$\lim_{n,m \rightarrow \infty} \mathbb{P} \left(\left\{ \omega \in \Omega : \int_{t_0}^T |b_n(s, \omega) - b_m(s, \omega)|^2 ds > \epsilon \right\} \right) = 0$$

for every $\epsilon > 0$. Applying Lemma 2.10 to $b_n - b_m$ gives

$$\begin{aligned} \lim_{n,m \rightarrow \infty} \sup \mathbb{P} \left(\left| \int_{t_0}^T b_n(s) dW_s - \int_{t_0}^T b_m(s) dW_s \right| > \delta \right) &\leq \frac{\epsilon}{\delta^2} \\ + \lim_{n,m \rightarrow \infty} \sup \mathbb{P} \left(\int_{t_0}^T |b_n(s) - b_m(s)|^2 dW_s > \delta \right) &= \frac{\epsilon}{\delta^2} \end{aligned}$$

and since $\epsilon > 0$ is arbitrary, we have

$$\lim_{n,m \rightarrow \infty} \mathbb{P} \left(\left| \int_{t_0}^T b_n(s) dW_s - \int_{t_0}^T b_m(s) dW_s \right| > \delta \right) = 0$$

We can therefore establish convergence in probability⁶. For verification that the limit is independent of the chosen sequence, see [9]. $\square$

We have now developed the tools needed to define the Itô integral properly.

Definition 2.13. Let $b \in \mathcal{L}_{\text{ad}}$. Its **stochastic integral** or **Itô integral** with respect to the Wiener process $\{W_t : t \in I\}$ over the interval $I = [t_0, T]$ is defined as the random variable $I[b]$ given by

$$I[b] = \int_{t_0}^T b dW = \int_{t_0}^T b(s) dW_s = \lim_{n \rightarrow \infty} \int_{t_0}^T b_n dW, \quad (2.17)$$

where convergence is in probability and $\{b_n\}_{n \in \mathbb{N}}$ is a sequence of functions in L^2_{step} approximating b in the sense of Lemma 2.11, that is,

$$\lim_{n \rightarrow \infty} \int_{t_0}^T |b_n(s) - b(s)|^2 ds = 0$$

in probability.

⁶More precisely, to prove convergence of this sequence, we note that these integrals are defined on a space $L^0(\Omega, \mathcal{F}, \mathbb{P}) = \{X : \Omega \rightarrow \mathbb{R}^d : \mathbb{P}(X < \infty) = 1\} / \sim$, where $\sim$ is the equivalence relation given by $X \sim Y$ if and only if $\mathbb{P}(X = Y) = 1$. This space is complete under the Ky Fan metric, so the Cauchy sequence converges to a random variable in that space.

Thus, the stochastic integral for functions in $\mathcal{L}_{ad}$ is well defined. Furthermore, if $b \in L^2_{ad}$, expression (2.14) follows directly. However, the properties that we were able to extend from step functions to functions in L^2_{ad} no longer generalize; for example, we can no longer guarantee an isometry for integrands in $\mathcal{L}_{ad}$. Indeed, recall that, for functions in L^2_{ad} , the corresponding integral was a function in $L^2(\Omega)$. However, $I[b]$ for $b \in \mathcal{L}_{ad}$ need not have finite expectation. We therefore present a result analogous to, but slightly different from, Theorem 2.5 for this new integral.

Theorem 2.14. *Let $b, b_1, b_2 \in \mathcal{L}_{ad}$ and let W_s be a Wiener process. Then the stochastic integral given in (2.17) satisfies the following properties:*

(a) *The stochastic integral is a linear map,*

$$\int_{t_0}^T (c_1 b_1 + c_2 b_2) dW = c_1 \int_{t_0}^T b_1 dW + c_2 \int_{t_0}^T b_2 dW$$

(b) *We have*

$$\int_{t_0}^T b dW = \begin{pmatrix} \sum_{k=1}^m \int_{t_0}^T b_{1,k}(s) dW_s^k \\ \vdots \\ \sum_{k=1}^m \int_{t_0}^T b_{d,k}(s) dW_s^k \end{pmatrix} \quad (2.18)$$

with $W_t = (W_t^1, \dots, W_t^m)'$.

(c) *If b is such that*

$$\int_{t_0}^T \mathbb{E} [|b(s)|^2] ds < \infty$$

then

$$\mathbb{E} \left[\int_{t_0}^T b dW \right] = 0$$

and the covariance matrix is therefore given by

$$\mathbb{E} \left[\left(\int_{t_0}^T b dW \right) \left(\int_{t_0}^T b dW \right)' \right] = \int_{t_0}^T \mathbb{E} [bb'] ds$$

In particular,

$$\mathbb{E} \left[\left| \int_{t_0}^T b dW \right|^2 \right] = \int_{t_0}^T \mathbb{E} [|b|^2] ds$$

Proof. (a) and (b) follow directly from parts (a) and (b) of Theorem 2.5. We therefore prove part (c). By assumption, we know that

$$\int_{t_0}^T \mathbb{E} [|b(s)|^2] ds < \infty$$

and from this condition one can prove, see [7], that there always exists a sequence of functions $\{b_n\}_{n \in \mathbb{N}} \subset L^2_{step}$ such that

$$\lim_{n \rightarrow \infty} \int_{t_0}^T \mathbb{E} [|b_n(s) - b(s)|^2] ds = 0$$

Applying parts (c) and (d) of Theorem 2.5 then gives the result directly. $\square$

Although we have defined the Itô integral for a class of functions $\mathcal{L}_{\text{ad}}$, this provides no information about how to compute stochastic integrals explicitly. This theoretical framework is essential for properly defining stochastic differential equations, whose solutions require evaluating such stochastic integrals. In fact, only *relatively few* stochastic differential equations can be solved explicitly: those that are linear or reducible to linear equations through a change of variables. Those known to admit explicit solutions are presented in [5].

2.1 Itô processes

The definition of the stochastic integral immediately shows that, if $J \subset I = [t_0, T]$ with $J \in \mathcal{B}(I)$, then

$$\int_J b dW = \int_{t_0}^T b \mathbb{1}_J dW$$

where the integral is well defined since $b \mathbb{1}_J \in \mathcal{L}_{\text{ad}}$ by the requirement $J \in \mathcal{B}(I)$. By Theorem 2.14(a), for two disjoint Borel subsets $J, K \in \mathcal{B}(I)$, we have

$$\int_{J \cup K} b dW = \int_I b \mathbb{1}_{J \cup K} dW = \int_I b (\mathbb{1}_J + \mathbb{1}_K) dW = \int_J b dW + \int_K b dW$$

For $b \in \mathcal{L}_{\text{ad}}$, we then define the random vector $X_t : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ as

$$X_t := \int_{t_0}^t b(s) dW_s = \int_{t_0}^T b(s) \mathbb{1}_{[0,t]} dW_s \quad (2.19)$$

such that

$$X_t(\omega) = \int_{t_0}^T b(s, \omega) \mathbb{1}_{[0,t]}(\omega) dW_s(\omega)$$

and thus $\{X_t : t \in I\}$ is a stochastic process satisfying

$$X_{t_0} = 0$$

for almost every $\omega \in \Omega$, and

$$X_t - X_s = \int_s^t b(u) dW_u$$

Furthermore, if $b \in \mathcal{L}_{\text{ad}}$ when $T \rightarrow \infty$, the process X_t is defined on the interval $[t_0, \infty)$. We examine some useful properties of X_t .

Theorem 2.15. *Let $b \in \mathcal{L}_{\text{ad}}$ and suppose that*

$$X_t = \int_{t_0}^t b(s) dW_s$$

for every $t \in I$. Then:

- (a) X_t is $\mathcal{F}_t$ -measurable⁷.

⁷Recall that $\mathcal{F}_t$ is the σ -algebra to which b is adapted, defined in 2.1.

(b) If

$$\int_{t_0}^t \mathbb{E} [|b(s)|^2] ds < \infty$$

for every $t \in I$, then $(X_t, \mathcal{F}_t)$ is a martingale. That is,

$$\mathbb{E} [X_t | \mathcal{F}_s] = X_s$$

Furthermore, for $s, t \in I$, we have $\mathbb{E} [X_t] = 0$, so the covariance matrix is

$$\mathbb{E} [X_t X_s'] = \int_{t_0}^{\min\{s, t\}} \mathbb{E} [b(u)b(u)'] du$$

In particular,

$$\mathbb{E} [|X_t|^2] = \int_{t_0}^t \mathbb{E} [|b(u)|^2] du$$

and for every $c > 0$ and $t_0 \leq a \leq t \leq b \leq T$,

$$\mathbb{P} \left(\left\{ \omega \in \Omega : \sup_{a \leq t \leq b} |X_t(\omega) - X_a(\omega)| > c \right\} \right) \leq \frac{1}{c^2} \int_a^b \mathbb{E} [|b(s)|^2] ds$$

and

$$\mathbb{E} \left[\sup_{a \leq t \leq b} |X_t(\omega) - X_a(\omega)| \right] \leq 4 \int_a^b \mathbb{E} [|b(s)|^2] ds$$

(c) The realizations of X_t , that is, $X_t(\omega)$, are continuous functions for almost every $\omega \in \Omega$.

(d) If there exists $k \in \mathbb{N}$ such that

$$\int_a^t \mathbb{E} [|b(s)|^{2k}] ds < \infty$$

for every $t_0 \leq a \leq t \leq T$, then

$$\mathbb{E} [|X_t - X_a|^{2k}] \leq (k(2k-1))^{k-1} (t-a)^{k-1} \int_a^t \mathbb{E} [|b(s)|^{2k}] ds$$

Proof. (a) By the definition of the stochastic integral,

$$X_t^{(n)} = \int_{t_0}^t b_n dW \rightarrow \int_{t_0}^t b dW = X_t$$

and $X_t^{(n)}$ is $\mathcal{F}_t$ -measurable because $b_n \in L_{\text{step}}^2$.

(b) Since X_t is adapted to the filtration $\mathcal{F}_t$ by part (a), and its first and second moments are finite by Theorem 2.14(e), it suffices to verify that $\mathbb{E} [X_t | \mathcal{F}_s] = X_s$, or equivalently,

$$\mathbb{E} [X_t - X_s | \mathcal{F}_s] = 0$$

Recall, however, that by definition of the space L_{ad}^2 , this equality holds for functions in L_{ad}^2 and therefore holds in general by taking the limit. The covariance matrix is obtained by carrying out the expansion for $b_n \in L_{\text{step}}^2$ and then taking the limit.

(c) and (d) For the proof of these parts, see [7]. $\square$

Having developed an integral calculus using Itô integrals, we now wish to study the definition and properties of a stochastic differential calculus built on these concepts.

2.2 Stochastic differentials

The definition of the random vector X_t given in (2.19) can also be expressed as⁸

$$dX_t = b(t)dW_t$$

This expression is called a **stochastic differential**. In this section, we study the differentials of more general stochastic processes, namely those that can be written in the form

$$X_t(\omega) = X_{t_0}(\omega) + \int_{t_0}^t a(s, \omega)ds + \int_{t_0}^t b(s, \omega)dW_s(\omega) \quad (2.20)$$

where, as in the preceding section, W_t is an m -dimensional Wiener process, $\mathcal{F}_s$ is a filtration independent of $W_t - W_s$ such that the Wiener process $\{W_s : s \in I\}$ is adapted to $\{\mathcal{F}_s\}_{s \in I}$, $b \in \mathcal{L}_{\text{ad}}$, and the stochastic integral is defined in (2.17) on $t \in I$. We assume that:

- (a) X_{t_0} is an $\mathcal{F}_{t_0}$ -measurable random variable independent of $W_t - W_{t_0}$. Note that these conditions hold if X_{t_0} is a constant.
- (b) $a(t, \omega)$ is an $\mathbb{R}^d$ -valued, $(\mathcal{B}(I) \times \mathcal{F})$ -measurable function. Furthermore, $a(t, \cdot)$ is $\mathcal{F}_t$ -measurable and

$$\int_{t_0}^T |a(s, \omega)|ds < \infty$$

with probability 1.

Since both integrals in (2.20) are continuous, the process X_t has continuous paths with probability 1. The following notation provides a simpler way to write expression (2.20).

Definition 2.16. We say that the stochastic process X_t defined in (2.20) has stochastic differential $a(t)dt + b(t)dW_t$ and write

$$dX_t = a(t)dt + b(t)dW_t \equiv fdt + b dW \quad (2.21)$$

We next state, without proof, one of the main results of stochastic calculus, known as Itô's lemma or Itô's formula, the stochastic counterpart of the chain rule in differential calculus. We consider only the one-dimensional case.

Theorem 2.17. Let $u = u(t, x)$ be a continuous function on $I \times \mathbb{R}$ with continuous partial derivatives u_t, u_x, u_{xx} . If X_t has a differential form defined on I such that

$$dX_t = a(t)dt + b(t)dW_t$$

where a, b and W_t are scalar functions in t , then $Y_t = u(t, X_t)$ has a differential form on I defined by

$$dY_t = \left(u_t(t, X_t) + u_x(t, X_t)a(t) + \frac{1}{2}u_{xx}(t, X_t)b(t)^2 \right) dt + u_x(t, X_t)b(t)dW_t$$

⁸Again, recall that dW_t is merely notation for the infinitesimal increment in W_t .

We consider an example illustrating the importance of the preceding theorem.

Example 2.18. If we take two stochastic processes $u(x_1, x_2) = x_1 x_2$ and $X_1(t), X_2(t)$ ⁹ with stochastic differentials

$$\begin{aligned} dX_1(t) &= a_1(t)dt + b_1(t)dW_t \\ dX_2(t) &= a_2(t)dt + b_2(t)dW_t \end{aligned}$$

then

$$\begin{aligned} d(X_1(t)X_2(t)) &= X_1(t)dX_2(t) + X_2(t)dX_1(t) + b_1(t)b_2(t)dt \\ &= (X_1a_2 + X_2a_1 + b_1b_2)dt + (X_1b_2 + X_2b_1)dW_t \end{aligned}$$

This equality is known as the integration-by-parts formula for Itô integrals, which in integral form is written as

$$X_1(t)X_2(t) = X_1(t_0)X_2(t_0) + \int_{t_0}^t X_1dX_2 + \int_{t_0}^t X_2dX_1 + \int_{t_0}^t b_1b_2ds$$

where $t_0 \leq t \leq T$.

Thus, although Itô calculus gives integrals important properties, such as the resulting process being a martingale, the Leibniz rule of ordinary differential calculus no longer holds. This is where the Stratonovich integral becomes relevant: the corresponding differentials defined using these integrals do satisfy the chain rule, but do not satisfy the martingale condition.

⁹For convenience, the notation X_t has been replaced by $X(t)$.

Chapter 3

Stochastic differential equations

Starting from the stochastic differential

$$dX_t = a(t, X_t)dt + b(t, X_t)dW_t \quad (3.1)$$

which, recall, was simply a way of writing the expression

$$X_t = c + \int_{t_0}^t a(s, X_s)ds + \int_{t_0}^t b(s, X_s)dW_s \quad (3.2)$$

where $X_{t_0} = c$ and $t_0 \leq t \leq T < \infty$, we may ask whether there exists a stochastic process X_t defined on $[t_0, T]$ whose differential coincides with (3.1) and which satisfies $X_{t_0} = c$.

Throughout this chapter, we assume that the functions a and b do not depend explicitly on ω , but only through X_t . That is, $a(t, X_t(\omega))$ and $b(t, X_t(\omega))$.

Definition 3.1. Equation (3.1) is called a **stochastic differential equation**. The random vector c is known as the **initial condition** or initial value. A stochastic process X_t is said to be a **solution** of this equation on $[t_0, T]$ if:

- (a) X_t is adapted to the filtration $\{\mathcal{F}_t\}_{t \in [t_0, T]}$. That is, X_t is $\mathcal{F}_t$ -measurable.
- (b) The functions $a(t, X_t(\omega))$ and $b(t, X_t(\omega))$ are such that

$$\int_{t_0}^T |a(s, X_t(\omega))|ds < \infty$$

and

$$\int_{t_0}^T |b(s, X_t(\omega))|^2 ds < \infty$$

with probability 1. That is, $b \in \mathcal{L}_{\text{ad}}$ and the paths of the process $a(t, X_t)$ belong to $L^1([t_0, T], \mathcal{B}([t_0, T]), \mu)$ with probability 1.

- (c) X_t satisfies equation (3.2) for every $t \in [t_0, T]$ with probability 1.

Note that we have not specified the filtration $\{\mathcal{F}\}_{t \in [t_0, T]}$ in the preceding definition.

Remark 3.2. To study stochastic differential equations, it suffices to take $\mathcal{F}_t$ to be the smallest σ -algebra with respect to which c and W_s are measurable, that is,

$$\mathcal{F}_t = \sigma(\{c\} \cup \{W_s, s \leq t\})$$

Note that, by definition, $\mathcal{F}_t$ is independent of $\sigma(W_s - W_t, s \geq t)$, so $\mathcal{F}_t$ satisfies the conditions imposed in the definition of the stochastic integral.

Before studying a result that establishes conditions under which a stochastic differential equation of type (3.1) has a unique solution, observe that, if $b \equiv 0$, we recover an ordinary differential equation, for which Picard's theorem ensures a unique solution. Similar conditions must therefore hold in the general case. We first state without proof two preliminary results that will be useful in proving the existence and uniqueness theorem.

Lemma 3.3 (Gronwall). *Let $\alpha(t)$ and $\beta(t)$ be two integrable functions on $[t_0, T]$, where $\alpha \geq 0$. If, for every $t \in [t_0, T]$,*

$$\alpha(t) \leq L \int_{t_0}^t \alpha(s) ds + \beta(t)$$

with $L > 0$, then

$$\alpha(t) \leq \beta(t) + L \int_{t_0}^t e^{L(t-s)} \beta(s) ds$$

for any $t \in [t_0, T]$.

Lemma 3.4. *Let $\{\theta_n\}_{n=1}^\infty$ be a sequence of functions in $L^1([t_0, T])$ such that*

$$\theta_{n+1}(t) \leq \Phi(t) + \beta \int_{t_0}^t \theta_n(s) ds$$

for every $t \in [t_0, T]$, with constant $\beta \geq 0$ and nonnegative $\Phi \in L^1([t_0, T])$. Then

$$\theta_{n+1}(t) \leq \Phi(t) + \beta \int_{t_0}^t \Phi(s) e^{\beta(t-s)} ds + \beta^n \int_{t_0}^t \frac{(t-s)^{n-1}}{(n-1)!} \theta_1(s) ds$$

holds for every $n \geq 2$. In particular, if $\Phi(t) = \alpha$ and $\theta_1(t) = c$ are constants, then, for every $n \geq 1$,

$$\theta_{n+1}(t) \leq \alpha e^{\beta(t-t_0)} + c \beta^n \frac{(t-t_0)^n}{n!}$$

We now present the existence and uniqueness theorem for stochastic differential equations.

Theorem 3.5. *Consider the stochastic differential equation*

$$dX_t = a(t, X_t)dt + b(t, X_t)dW_t \quad X_{t_0} = c, t_0 \leq t \leq T \quad (3.3)$$

where W_t is an m -dimensional Wiener process and c is a random variable independent of $W_t - W_{t_0}$ for $t \geq t_0$. Suppose that the functions $a : [t_0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $b : [t_0, T] \times \mathbb{R}^d \rightarrow \mathcal{M}_{d \times m}(\mathbb{R})$ are measurable on $[t_0, T] \times \mathbb{R}^d$ and satisfy the following properties: there exists $K > 0$ such that

- (a) For every $t \in [t_0, T]$, $x \in \mathbb{R}^d$ and $y \in \mathbb{R}^d$, both functions satisfy the Lipschitz condition, i.e.,

$$|a(t, x) - a(t, y)| + |b(t, x) - b(t, y)| \leq K|x - y| \quad (3.4)$$

- (b) For every $t \in [t_0, T]$ and $x \in \mathbb{R}^d$, both functions satisfy the growth condition

$$|a(t, x)|^2 + |b(t, x)|^2 \leq K^2 (1 + |x|^2) \quad (3.5)$$

Then (3.3) has a unique d -dimensional solution X_t on $[t_0, T]$, whose paths are continuous with probability 1 and which satisfies the initial condition $X_{t_0} = c$. Uniqueness means that, if X_t and Y_t are two continuous solutions of (3.3) with the same initial value c , then

$$\mathbb{P} \left(\left\{ \omega \in \Omega : \sup_{t_0 \leq t \leq T} |X_t(\omega) - Y_t(\omega)| > 0 \right\} \right) = 0$$

Proof. Uniqueness: The argument is analogous to the final part of the proof of Theorem 2.17: (i) we take solutions X_t and Y_t of the differential equation and verify that their bounded versions coincide with probability 1; (ii) we then verify that the set on which they are unbounded has measure zero.

(i) Let X_t and Y_t be two continuous solutions of (3.3). We want to prove that $\mathbb{E}[|X_t - Y_t|^2] = 0$ for every $t \in [t_0, T]$, since applying Chebyshev's inequality gives

$$\mathbb{P} \left(\left\{ |X - Y| \geq \frac{1}{n} \right\} \right) \leq n^2 \mathbb{E}[|X - Y|^2] = 0$$

and, using the countable union,

$$\mathbb{P}(\{|X - Y| > 0\}) = \mathbb{P} \left(\bigcup_{n \geq 1} \left\{ |X - Y| \geq \frac{1}{n} \right\} \right) = 0$$

which would prove $\mathbb{P}(\{X = Y\}) = 1$. However, since the moments of order 2 of the two solutions need not be bounded, we define

$$I_N(t) = \begin{cases} 1 & \text{if } |X_s| \leq N \text{ and } |Y_s| \leq N \text{ for all } t_0 \leq s \leq t \\ 0 & \text{otherwise} \end{cases}$$

and verify that $\mathbb{E}[I_N(t)|X_t - Y_t|^2] = 0$. Note that $I_N(t) = I_N(t)I_N(s)$ for $s \leq t$. Therefore, since both X_t and Y_t are solutions of (3.3) by assumption,

$$\begin{aligned} I_N(t)(X_t - Y_t) &= I_N(t) \left(\int_{t_0}^t I_N(s)(a(s, X_s) - a(s, Y_s))ds \right. \\ &\quad \left. + \int_{t_0}^t I_N(s)(b(s, X_s) - b(s, Y_s))dW_s \right) \end{aligned}$$

where the integral over dW_s is well defined since, if $b \in \mathcal{L}_{\text{ad}}$, then $g = g(t, b) \in \mathcal{L}_{\text{ad}}$ if g is a measurable function. Multiplying condition (3.4) by $I_N(t)$ gives

$$I_N(s) (|a(s, X_s) - a(s, Y_s)| + |b(s, X_s) - b(s, Y_s)|) \leq K I_N(s) |X_s - Y_s| \leq 2KN$$

since $I_N(t)|X_s| \leq N$ and $I_N(t)|Y_s| \leq N$. Thus, the squares of the integrals in the preceding expression have finite expectation. From inequality $(a + b)^2 \leq 2(a^2 + b^2)$,

$$\begin{aligned} \mathbb{E} [I_N(t)|X_t - Y_t|^2] &\leq 2\mathbb{E} \left[\left| \int_{t_0}^t I_N(s)(a(s, X_s) - a(s, Y_s))ds \right|^2 \right] \\ &\quad + 2\mathbb{E} \left[\left| \int_{t_0}^t I_N(s)(b(s, X_s) - b(s, Y_s))dW_s \right|^2 \right] \end{aligned}$$

Applying Cauchy's inequality to the first term and the Itô isometry to the second gives

$$\begin{aligned} \mathbb{E} [I_N(t)|X_t - Y_t|^2] &\leq 2(T - t_0) \int_{t_0}^t \mathbb{E} [I_N(s) |a(s, X_s) - a(s, Y_s)|^2] ds \\ &\quad + 2 \int_{t_0}^t \mathbb{E} [I_N(s) |b(s, X_s) - b(s, Y_s)|^2] ds \end{aligned}$$

Applying condition (3.4) now gives

$$\mathbb{E} [I_N(s)|X_t - Y_t|^2] \leq 2(T - t_0 + 1)K^2 \int_{t_0}^t \mathbb{E} [I_N(s)|X_s - Y_s|^2] ds$$

and using Gronwall's lemma, see Lemma 3.3, with $\alpha(t) = \mathbb{E} [I_N(s)|X_s - Y_s|^2]$, $\beta(t) = 0$, and $L = 2(T - t_0 + 1)K^2$ yields

$$\mathbb{E} [I_N(s)|X_s - Y_s|^2] \leq 0$$

and since $\mathbb{E} [I_N(s)|X_s - Y_s|^2] \geq 0$ by definition, we obtain the desired result. This means that, for every $t \in [t_0, T]$, $I_N(t)X_t = I_N(t)Y_t$ holds with probability 1. (ii) Thus, the set of $\omega \in \Omega$ where $I_N(t) \neq 1$, or in other words where the processes X_t and Y_t are unbounded, satisfies

$$\begin{aligned} \mathbb{P}(\{\omega \in \Omega : I_N(t) \neq 1 \text{ for } t \in [t_0, T]\}) &\leq \mathbb{P} \left(\left\{ \omega \in \Omega : \sup_{t_0 \leq t \leq T} |X_t(\omega)| > N \right\} \right) \\ &\quad + \mathbb{P} \left(\left\{ \omega \in \Omega : \sup_{t_0 \leq t \leq T} |Y_t(\omega)| > N \right\} \right) \end{aligned}$$

but both terms on the right tend to zero by continuity (and hence boundedness) of X_t and Y_t , respectively, so this set has measure zero. Thus, the two processes are stochastically equivalent (see Definition 1.2). Since both processes are continuous with probability 1,

$$\mathbb{P} \left(\left\{ \omega \in \Omega : \sup_{t_0 \leq t \leq T} |X_t - Y_t| > 0 \right\} \right) = 0$$

Existence: To prove existence of a solution, we define a sequence of stochastic processes $\{X_t^{(n)}\}_{n=1}^\infty$ by

$$X_t^{(n)} = c + \int_{t_0}^t a(s, X_s^{(n-1)})ds + \int_{t_0}^t b(s, X_s^{(n-1)})dW_s$$

and set $X_t^{(0)} = c$. The proof consists of 3 steps: (i) we prove that the sequence $\{X_t^{(n)}\}_{n=0}^\infty$ is contained in L_{ad}^2 ; (ii) we then verify that its limit exists; and (iii) we show that the limit satisfies the stochastic differential equation given in (3.3).

(i) We prove this by induction, assuming that c is $\mathcal{F}_{t_0}$ -measurable and $\mathbb{E}[|c|^2] < \infty$, so $c \in L_{ad}^2$ and its paths are clearly continuous. Now assume that $X_t^{(n-1)} \in L_{ad}^2$ for every $n \geq 1$ and prove that $X_t^{(n)} \in L_{ad}^2$. Assumption (b) of the theorem gives

$$\begin{aligned} \mathbb{E} \left[\int_{t_0}^T |b(s, X_s^{(n-1)})|^2 ds \right] &\leq \mathbb{E} \left[\int_{t_0}^T K^2(1 + |X_s^{(n-1)}|^2) ds \right] \\ &= K^2(T - t_0) + K^2 \int_{t_0}^T \mathbb{E} [|X_t^{(n-1)}|^2] < \infty \end{aligned}$$

by the assumption that $X_t^{(n-1)} \in L_{ad}^2$. Furthermore, $b(t, X_t^{(n-1)})$ is $\mathcal{F}_t$ -measurable, and therefore $b(t, X_t^{(n-1)}) \in L_{ad}^2$. Thus, the Itô integral is well defined, and by properties (a) and (c) of Theorem 2.15 it is $\mathcal{F}_t$ -measurable and has continuous paths with probability 1. On the other hand, applying assumption (b) of the theorem gives

$$\int_{t_0}^t |a(s, X_s^{(n-1)})| ds \leq \sqrt{K} \left(\int_{t_0}^t (1 + |X_s^{(n-1)}|^2) ds \right)^{1/2} < \infty$$

by continuity of the paths of $X_t^{(n-1)}$. Thus, the Lebesgue integral is continuous with probability 1 and is adapted to $\{\mathcal{F}_t\}$. Using $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$, we obtain

$$|X_t^{(n)}|^2 \leq 3 \left(c^2 + \left(\int_{t_0}^t a(s, X_s^{(n-1)}) ds \right)^2 + \left(\int_{t_0}^t b(s, X_s^{(n-1)}) dW_s \right)^2 \right)$$

Taking expectations on both sides and integrating over $[t_0, T]$ gives

$$\begin{aligned} \int_{t_0}^T \mathbb{E} [|X_s^{(n)}|^2] ds &\leq 3(T - t_0) \mathbb{E} [|c|^2] + 3 \int_{t_0}^T \mathbb{E} \left[\left(\int_{t_0}^t a(s, X_s^{(n-1)}) ds \right)^2 \right] dt \\ &\quad + 3 \int_{t_0}^T \mathbb{E} \left[\left(\int_{t_0}^t b(s, X_s^{(n-1)}) dW_s \right)^2 \right] dt \\ &\leq 3(T - t_0) \mathbb{E} [|c|^2] + 3 \int_{t_0}^T \mathbb{E} \left[\left(\int_{t_0}^t a(s, X_s^{(n-1)}) dW_s \right)^2 \right] dt \\ &\quad + 3 \int_{t_0}^T \int_{t_0}^t \mathbb{E} [|b(s, X_s^{(n-1)})|^2] ds dt < \infty \end{aligned}$$

by assumption (b) of the theorem and because $X_t^{(n-1)} \in L_{ad}^2$ by assumption. Thus, we have proved that $X_t^{(n)} \in L_{ad}^2$.

(ii) We now prove that the sequence $\{X_t^{(n)}\}_{n=0}^\infty$ converges uniformly with probability 1. Defining

$$Y_t^{(n)} = \int_{t_0}^t b(s, X_s^{(n-1)}) dW_s \quad Z_t^{(n)} = \int_{t_0}^t a(s, X_s^{(n-1)}) ds$$

we have $X_t^{(n)} = c + Y_t + Z_t$. Applying the Itô isometry gives

$$\begin{aligned} \mathbb{E} \left[\left| Y_t^{(n)} - Y_t^{(n-1)} \right|^2 \right] &= \mathbb{E} \left[\left(\int_{t_0}^t (b(s, X_s^{(n-1)}) - b(s, X_s^{(n-2)})) dW_s \right)^2 \right] \\ &= \int_{t_0}^t \mathbb{E} \left[|b(s, X_s^{(n-1)}) - b(s, X_s^{(n-2)})|^2 \right] ds \\ &\leq K^2 \int_{t_0}^t \mathbb{E} \left[|X_s^{(n-1)} - X_s^{(n-2)}|^2 \right] ds \end{aligned} \quad (3.6)$$

where assumption (a) of the theorem was applied in the final inequality. Similarly, now applying Hölder's inequality and assumption (a),

$$\begin{aligned} \mathbb{E} \left[\left| Z_t^{(n)} - Z_t^{(n-1)} \right|^2 \right] &= \mathbb{E} \left[\left(\int_{t_0}^t (a(s, X_s^{(n-1)}) - a(s, X_s^{(n-2)})) dW_s \right)^2 \right] \\ &\leq (T - t_0) K^2 \int_{t_0}^T |X_s^{(n-1)} - X_s^{(n-2)}|^2 ds \end{aligned} \quad (3.7)$$

Thus, from the two preceding inequalities and using $(a + b)^2 \leq 2(a^2 + b^2)$,

$$\begin{aligned} \mathbb{E} \left[\left| X_t^{(n)} - X_t^{(n-1)} \right|^2 \right] &= \mathbb{E} \left[\left| Y_t^{(n-1)} - Y_t^{(n-2)} + Z_t^{(n-1)} - Z_t^{(n-2)} \right|^2 \right] \\ &= \leq 2\mathbb{E} \left[\left| Y_t^{(n-1)} - Y_t^{(n-2)} \right|^2 \right] + 2\mathbb{E} \left[\left| Z_t^{(n-1)} - Z_t^{(n-2)} \right|^2 \right] \\ &\leq 2K^2(1 + T - t_0) \int_{t_0}^t \mathbb{E} \left[|X_s^{(n-1)} - X_s^{(n-2)}|^2 \right] ds \end{aligned} \quad (3.8)$$

We can now apply Lemma 3.4 to obtain

$$\mathbb{E} \left[\left| X_t^{(n)} - X_t^{(n-1)} \right|^2 \right] \leq (2K^2(1 + T - t_0))^{n-2} \int_{t_0}^t \frac{(t-s)^{n-2}}{(n-2)!} \mathbb{E} \left[|X_s^{(2)} - X_s^{(1)}|^2 \right] ds \quad (3.9)$$

Expression (3.8) directly implies that

$$\mathbb{E} \left[|X_s^{(2)} - X_s^{(1)}|^2 \right] \leq 2K^2(1 + T - t_0) (1 + \mathbb{E} [|c|^2]) s$$

Thus, applying this result to inequality (3.9) gives

$$\mathbb{E} \left[\left| X_t^{(n)} - X_t^{(n-1)} \right|^2 \right] \leq 2K^2(1 + T - t_0) (1 + \mathbb{E} [|c|^2]) \frac{(t - t_0)^n}{n!} \quad (3.10)$$

On the other hand, note that

$$\left| X_t^{(n)} - X_t^{(n-1)} \right| \leq \left| Y_t^{(n)} - Y_t^{(n-1)} \right| + \left| Z_t^{(n)} - Z_t^{(n-1)} \right|$$

which implies

$$\sup_{t \in [t_0, T]} \left| X_t^{(n)} - X_t^{(n-1)} \right| \leq \sup_{t \in [t_0, T]} \left| Y_t^{(n)} - Y_t^{(n-1)} \right| + \sup_{t \in [t_0, T]} \left| Z_t^{(n)} - Z_t^{(n-1)} \right|$$

or, equivalently, the set inclusion

$$\left\{ \omega \in \Omega : \sup_{t \in [t_0, T]} |X_t^{(n)} - X_t^{(n-1)}| > \frac{1}{n^2} \right\} \subset \left\{ \omega \in \Omega : \sup_{t \in [t_0, T]} |Y_t^{(n)} - Y_t^{(n-1)}| > \frac{1}{2n^2} \right\} \\ \cup \left\{ \omega \in \Omega : \sup_{t \in [t_0, T]} |Z_t^{(n)} - Z_t^{(n-1)}| > \frac{1}{2n^2} \right\}$$

Taking probabilities (and using the fact that a probability measure is, in particular, additive),

$$\mathbb{P} \left(\left\{ \sup_{t \in [t_0, T]} |X_t^{(n)} - X_t^{(n-1)}| > \frac{1}{n^2} \right\} \right) \leq \mathbb{P} \left(\left\{ \sup_{t \in [t_0, T]} |Y_t^{(n)} - Y_t^{(n-1)}| > \frac{1}{2n^2} \right\} \right) \\ + \mathbb{P} \left(\left\{ \sup_{t \in [t_0, T]} |Z_t^{(n)} - Z_t^{(n-1)}| > \frac{1}{2n^2} \right\} \right)$$

Applying Doob's inequality, see [10], and inequality (3.6) gives

$$\mathbb{P} \left(\left\{ \sup_{t \in [t_0, T]} |Y_t^{(n)} - Y_t^{(n-1)}| > \frac{1}{2n^2} \right\} \right) \leq 4n^4 \mathbb{E} \left[|Y_T^{(n)} - Y_T^{(n-1)}|^2 \right] \\ \leq 4n^4 K^2 \rho \beta^{n-1} \frac{(T - t_0)^n}{n!}$$

where $\rho = 1 + \mathbb{E}[|c|^2]$ and $\beta = 2K^2(1 + T - t_0)$. On the other hand, Chebyshev's inequality, see A.5, and inequality (3.7) give

$$\mathbb{P} \left(\left\{ \sup_{t \in [t_0, T]} |Z_t^{(n)} - Z_t^{(n-1)}| > \frac{1}{2n^2} \right\} \right) \leq 4n^4 \mathbb{E} \left[\sup_{t \in [t_0, T]} |Z_T^{(n)} - Z_T^{(n-1)}|^2 \right] \\ \leq 4n^4 K^2 (T - t_0) \rho \beta^{n-1} \frac{(T - t_0)^n}{n!}$$

Thus, from the last two inequalities and summing over $n \in \mathbb{N}$,

$$\sum_{n=1}^{\infty} \mathbb{P} \left(\left\{ \sup_{t \in [t_0, T]} |X_t^{(n)} - X_t^{(n-1)}| > \frac{1}{n^2} \right\} \right) \leq 4\rho \sum_{n=1}^{\infty} \frac{n^4 \beta^n (T - t_0)^n}{n!} < \infty$$

By the Borel-Cantelli lemma, see [9], for almost every $\omega \in \Omega$ there exists an $n_0 \in \mathbb{N}$ such that $\forall n \geq n_0$

$$\sup_{t \in [t_0, T]} |X_t^{(n)} - X_t^{(n-1)}| \leq \frac{1}{n^2}$$

and applying this result to the representation

$$X_t^{(n)} = c + \sum_{k=1}^{n-1} (X_t^{(k+1)} - X_t^{(k)}) \quad (3.11)$$

shows that the series converges uniformly for almost every $\omega \in \Omega$. That is, the limit

$$\lim_{n \rightarrow \infty} X_t^{(n)} := X_t$$

exists, with uniform convergence on $t \in [t_0, T]$ with probability 1.

(iii) It remains to verify that $\{X_t, t \in [t_0, T]\}$ is indeed a solution of the stochastic differential equation. The properties of $X_t^{(n)}$ and uniform convergence ensure that X_t is adapted to the filtration $\{\mathcal{F}_t\}_{t \in [t_0, T]}$ and has continuous paths for almost every $\omega \in \Omega$. Taking the L^2 norm in expression (3.11) gives

$$\|X_t\| \leq \|c\| + \sum_{n=1}^{\infty} \sqrt{\rho} \frac{\beta^{n/2} (T - t_0)^{n/2}}{\sqrt{n!}}$$

and consequently $\int_{t_0}^T \mathbb{E} [|X_t|^2] dt < \infty$, which implies that $X_t \in L_{ad}^2 \subset \mathcal{L}_{ad}$. It therefore suffices to prove that X_t satisfies the conditions for a solution given in Definition 3.1, which follow directly from assumptions (a) and (b) of the theorem. $\square$

3.1 Linear stochastic differential equations

Having defined stochastic differential equations and established the corresponding existence and uniqueness theorem, we focus on linear stochastic differential equations.

Definition 3.6. A stochastic differential equation of the form (3.3) is said to be **linear** if

$$a(t, x) = A(t)x + f(t)$$

where $A : [t_0, T] \rightarrow \mathcal{M}_{d \times d}(\mathbb{R})$ and $f : [t_0, T] \rightarrow \mathbb{R}^d$, and also

$$b(t, x) = (B_1(t)x + g_1(t), \dots, B_m(t)x + g_m(t))$$

where $B_i : [t_0, T] \rightarrow \mathcal{M}_{d \times d}(\mathbb{R})$, $g_i : [t_0, T] \rightarrow \mathbb{R}^d$, and $W_t = (W_t^1, \dots, W_t^m)'$. Thus, a linear differential equation has the form

$$dX_t = (A(t)X_t + f(t))dt + \sum_{i=1}^m (B_i(t)X_t + g_i(t))dW_t^i$$

If $f(t) = g_1(t) = \dots = g_m(t) = 0$ for every $t \in [t_0, T]$, the equation is said to be **homogeneous**. If $B_1(t) = \dots = B_m(t) = 0$ for every $t \in [t_0, T]$, the equation is said to be **linear in the strict sense**.

Remark 3.7. Note that $b(t, x)$ is well defined since, as $x \in \mathbb{R}^d$, we have $B_i(t)x + g_i(t) \in \mathbb{R}^d$ and hence $b(t, x) \in \mathcal{M}_{d \times m}(\mathbb{R})$, so $b(t, x)dW_t \in \mathbb{R}^d$.

We first study stochastic differential equations that are linear in the strict sense. In this case,

$$b(t, x) = (g_1(t), \dots, g_m(t)) =: B(t)$$

and the stochastic differential is

$$dX_t = (A(t)X_t + f(t))dt + B(t)dW_t \tag{3.12}$$

Theorem 3.8. *A stochastic differential equation that is linear in the strict sense, of the form (3.12) with $X_{t_0} = c$, has solution*

$$X_t = \Phi(t) \left(c + \int_{t_0}^t \Phi^{-1}(s) f(s) ds + \int_{t_0}^t \Phi^{-1}(s) B(s) dW_s \right) \quad (3.13)$$

defined on $[t_0, T]$, where $\Phi(t)$ is the fundamental matrix of the ordinary differential equation

$$\frac{dX_t}{dt} \equiv \dot{X}_t = A(t)X_t$$

Proof. Taking

$$Y_t = c + \int_{t_0}^t \Phi^{-1}(s) f(s) ds + \int_{t_0}^t \Phi^{-1}(s) B(s) dW_s$$

and computing its stochastic differential gives

$$dY_t = \Phi^{-1}(t) (f(t)dt + B(t)dW_t)$$

Thus, applying Itô's formula, the process $X_t = \Phi(t)Y_t$ has stochastic differential

$$dX_t = \dot{\Phi}(t)Y_t dt + \Phi(t)dY_t \quad (3.14)$$

$$= A(t)\Phi(t)Y_t dt + f(t)dt + B(t)dW_t \quad (3.15)$$

$$= (A(t)X_t + f(t)) dt + B(t)dW_t \quad (3.16)$$

□

3.1.1 The Langevin equation and the Ornstein-Uhlenbeck process

To model the velocity of a dust particle suspended in a fluid, Langevin proposed the following differential equation in 1908:

$$\frac{dX_t}{dt} = -\alpha X_t + \sigma \xi_t$$

where X_t represents the particle's velocity, the term $-\alpha X_t$ represents the frictional force, and $\sigma \xi_t$ is a force caused by collisions of the dust particle with the molecules of the fluid. Here, X_t denotes one of the three components of the particle's velocity, $\alpha > 0$ and σ are constants, and ξ_t is 1-dimensional white noise. We can then rewrite the preceding equation as the stochastic differential equation

$$dX_t = -\alpha X_t dt + \sigma dW_t$$

which is linear in the strict sense, so its solution is given by expression (3.13), where $\Phi(t)$ is obtained from the fundamental solution of the ordinary differential equation

$$\frac{dX_t}{dt} = -\alpha X_t$$

Thus, $\Phi = e^{-\alpha t}$, taking $t_0 = 0$, and hence

$$X_t = e^{-\alpha t}c + \sigma \int_0^t e^{-\alpha(t-s)} dW_s$$

where, recall, $c = X_0$. Furthermore, one can prove that

$$\mathbb{E}[X_t] = e^{-\alpha t} \mathbb{E}[c]$$

since the expectation of an Itô integral is zero; see part (e) of Theorem 2.14. On the other hand, the covariance between X_s and X_t is

$$\text{Cov}(X_t, X_s) = e^{-\alpha(t+s)} \left(\text{Var}[c] + \frac{1}{2\alpha} \sigma^2 (e^{2\alpha \min\{t,s\}} - 1) \right)$$

Therefore,

$$\text{Var}[X_t] = e^{-2\alpha t} \text{Var}[c] + \frac{1}{2\alpha} \sigma^2 (1 - e^{-2\alpha t})$$

and since $\lim_{t \rightarrow \infty} e^{-\alpha t}c = 0$ almost surely, the distribution of X_t approaches a normal distribution $\mathcal{N}(0, \sigma^2/2\alpha)$ as $t \rightarrow \infty$. If the initial condition c has a normal distribution, then X_t is a Gaussian process called an **Ornstein-Uhlenbeck process**. If, in addition, $c \sim \mathcal{N}(0, \sigma^2/2\alpha)$, then X_t is a stationary Gaussian process, called **colored noise**, such that $\mathbb{E}[X_t] = 0$, with covariance

$$\mathbb{E}[X_s X_t] = \frac{\sigma^2}{2\alpha} e^{-\alpha|t-s|}$$

3.1.2 Geometric Brownian motion

Geometric Brownian motion is one of the most important stochastic processes in quantitative finance because of its ability to model the evolution of financial asset prices.

As we will see later, the Black-Scholes-Merton model, a cornerstone of financial derivatives pricing theory, assumes that underlying asset prices follow geometric Brownian motion. This assumption yields a partial differential equation whose analysis has transformed the pricing of options and other derivatives. However, geometric Brownian motion is also relevant beyond this model, providing a foundation for exploring more complex variants, such as those incorporating stochastic volatility or jumps.

First, we show that linear stochastic differential equations have a unique solution with continuous paths.

Theorem 3.9. *The linear stochastic differential equation*

$$dX_t = (A(t)X_t + f(t))dt + \sum_{i=1}^m (B_i(t)X_t + g_i(t))dW_t^i$$

has a unique solution on the interval $[t_0, t]$ with continuous paths for each initial condition $X_{t_0} = c$ independent of $W_t - W_{t_0}$, given bounded measurable functions $A(t)$, $f(t)$, $B_i(t)$ and $g_i(t)$. Furthermore, if these assumptions hold on any interval $[t_0, +\infty)$, then a unique global solution exists.

In particular, geometric Brownian motion is modeled by the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (3.17)$$

Theorem 3.10. *The stochastic differential equation given in equation (3.17) has the unique solution*

$$S_t = S_{t_0} e^{(\mu - \frac{1}{2}\sigma^2)(t-t_0) + \sigma(W_t - W_{t_0})} \quad (3.18)$$

Proof. Applying the one-dimensional Itô formula gives

$$d(\ln S_t) = \frac{1}{S_t} dS_t - \frac{1}{2} \frac{1}{S_t^2} [dS, dS]_t$$

where $[dS, dS]_t$ is the quadratic variation of dS_t . Computing the quadratic variation gives

$$[dS, dS]_t = \sigma^2 S_t^2 dt$$

and finally,

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t$$

which can be integrated directly to obtain the desired result. □

Remark 3.11. Applying Itô's theorem 2.17 to expression (3.18) verifies that it is a solution of the stochastic differential equation.

Chapter 4

The Fokker-Planck equation

This chapter introduces the Fokker-Planck equation, a partial differential equation that determines the time evolution of the probability density function for a class of Markov processes called diffusion processes. We first define the following operators.

Definition 4.1. Let X_t be a Markov process defined on $t \in [t_0, T]$ with transition probability $P(s, x, t, B)$. We define the **transition operator** as

$$T_t g(s, x) = \mathbb{E}_{s,x} [g(s+t, X_{t+s})] = \int_{\mathbb{R}^d} g(s+t, y) P(s, x, t+s, dy)$$

for $t \in [0, T-s]$ and a bounded measurable function $g : [t_0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ on $[t_0, T] \times \mathbb{R}^d$. We also define the **infinitesimal generator** as

$$Ag(s, x) = \lim_{t \rightarrow 0^+} \frac{T_t g(s, x) - g(s, x)}{t}$$

Remark 4.2. When X_t is a homogeneous Markov process, the definitions reduce to

$$T_t g(x) = \mathbb{E}_x [g(X_t)] = \int_{\mathbb{R}^d} g(y) P(t, x, dy)$$

and

$$Ag(x) = \lim_{t \rightarrow 0^+} \frac{T_t g(x) - g(x)}{t} \tag{4.1}$$

4.1 Diffusion processes

Diffusion processes play an important role in the theory of stochastic differential equations because one can prove, see [7], that if the functions a and b in the stochastic differential equation (3.3) are independent of t , the solution is a homogeneous Markov process. In particular, if these functions are continuous with respect to t , the solution is a diffusion process.

Definition 4.3. Let X_t be a Markov process on $t \in [t_0, T]$ taking values in $\mathbb{R}^d$, whose paths $X(\omega)$ are continuous for almost every $\omega \in \Omega$. We say that X_t is a diffusion process if its transition probability $P(s, x, t, B)$ satisfies the following properties for every $s \in [t_0, T)$, $x \in \mathbb{R}^d$, and $\epsilon > 0$:

$$\begin{aligned}
 (a) \quad & \lim_{t \rightarrow s^+} \frac{1}{t-s} \int_{|y-x| > \epsilon} P(s, x, t, dy) = 0. \\
 (b) \quad & \lim_{t \rightarrow s^+} \frac{1}{t-s} \int_{|y-x| \leq \epsilon} (y-x) P(s, x, t, dy) = a(s, x). \\
 (c) \quad & \lim_{t \rightarrow s^+} \frac{1}{t-s} \int_{|y-x| \leq \epsilon} (y-x)(y-x)' P(s, x, t, dy) = b(s, x).
 \end{aligned}$$

where a and b are called the **diffusion coefficients**, a is called the drift vector (or coefficient if $d = 1$), and b is called the diffusion matrix (or coefficient if $d = 1$). Furthermore, $b(s, x)$ is symmetric and positive semidefinite.

We now explain the preceding conditions and how restrictive they are. Condition (a) implies that the probability of large changes in X_t over a short time interval is small. That is,

$$\mathbb{P}(\{\omega \in \Omega : |X_t(\omega) - X_s(\omega)| \leq \epsilon | X_s = x\}) = 1 - \mathcal{O}(t-s)$$

To analyze conditions (b) and (c), suppose that X_t has finite first and second moments. Then the preceding definition gives

$$\mathbb{E}_{s,x}[X_t - X_s] = a(s, x)(t-s) + \mathcal{O}(t-s)$$

and

$$\mathbb{E}[(X_t - X_s)(X_t - X_s)'] = b(s, x)(t-s) + \mathcal{O}(t-s)$$

Thus, the covariance matrix of the random vector $X_t - X_s$ (with respect to the probability measure $P(s, x, t, \cdot)$) is

$$\text{Cov}_{s,x}[X_t - X_s] = b(s, x)(t-s) + \mathcal{O}(t-s)$$

We can therefore interpret $a(s, x)$ as the *mean velocity* of the random motion described by X_t given $X_s = x$. On the other hand, $b(s, x)$ measures the *fluctuation* of $X_t - X_s$ around its mean. In particular, we can write

$$X_t - X_s \approx a(s, X_s)(t-s) + b(s, X_s)\xi$$

where $\mathbb{E}_{s,x}[\xi] = 0$, $\text{Cov}_{s,x}[\xi] = (t-s)\mathcal{I}$, and $b \in \mathcal{M}_{d \times d}(\mathbb{R})$ such that $bb' = B$.

Since Definition 1.13 of a Wiener process shows that $W_t - W_s$ has this distribution, we can write

$$X_t - X_s \approx a(s, X_s)(t-s) + b(s, X_s)(W_t - W_s)$$

or, in differential form,

$$dX_t = a(t, X_t)dt + b(t, X_t)dW_t$$

Thus, solutions of stochastic differential equations and diffusion processes are essentially the same class of stochastic processes. For a definition of diffusion processes in terms of stochastic differential equations, see [7].

Example 4.4. A Wiener process is a diffusion process with $a = 0$ and $B = \mathcal{I}$.

4.2 Kolmogorov's *backward* and *forward* equations

We next show that a diffusion process can be determined solely from its diffusion coefficients a and b . This is quite remarkable, since these coefficients are related to the first and second moments, and in general these moments alone do not determine a distribution. To each diffusion process, we associate the second-order differential operator

$$\mathfrak{D} = \sum_{i=1}^d a_i(s, x) \frac{\partial}{\partial x_i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d b_{ij}(s, x) \frac{\partial^2}{\partial x_i \partial x_j} \quad (4.2)$$

where $b_{ij} = (b)_{ij}$. One can prove that every Markov process, and in particular every diffusion process, is uniquely determined by its infinitesimal generator A , which, expanding expression (4.1), takes the form

$$Ag(s, x) = \lim_{t \rightarrow 0^+} \frac{1}{t} \int_{\mathbb{R}^d} (g(s+t, y) - g(s, x)) P(s, x, t+s, dy)$$

Assume that $g : [t_0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a bounded function of class $\mathcal{C}^1(\mathbb{R})$ with respect¹ to x_i and of class $\mathcal{C}^1(\mathbb{R})$ with respect to s . Expanding $g(s+t, y)$ in a Taylor series around (s, x) and applying conditions (b) and (c) of Definition 4.3 then shows that, for all functions g described above for which the limit in (4.1) exists, the following relation holds:

$$A = \frac{\partial}{\partial s} + \mathfrak{D}$$

and for time-independent functions in the homogeneous case, $A = \mathfrak{D}$. That is, since $\mathfrak{D}$ depends only on a and b , a diffusion process is uniquely determined by the coefficients a and b .

We next state three important results without proof; their proofs can be found in [11].

Theorem 4.5. *Let X_t be a d -dimensional diffusion process defined on $[t_0, T]$ with continuous diffusion coefficients $a(s, x)$ and $b(s, x)$. Let $g(x)$ be a bounded continuous scalar function such that*

$$u(s, x) = \mathbb{E}_{s,x} [g(X_t)] = \int_{\mathbb{R}^d} g(y) P(s, x, t, dy)$$

where $t \in (s, T]$ is fixed and $x \in \mathbb{R}^d$ is continuous and bounded, as are $\partial u / \partial x_i$ and $\partial^2 u / \partial x_i \partial x_j$ for $i, j = 1, \dots, d$. Then $u(s, x)$ is differentiable with respect to s and satisfies Kolmogorov's backward equation:

$$\frac{\partial u}{\partial s} + \mathfrak{D}u = 0, \quad (4.3)$$

where $\mathfrak{D}$ is defined in (4.2) and $u(s, x)$ satisfies the condition

$$\lim_{s \rightarrow t^-} u(s, x) = g(x)$$

¹That is, the function $g_i(x_i) = g(s, (x_1, \dots, x_i, \dots, x_d)) \in \mathcal{C}^2(\mathbb{R})$ for every $i = 1, \dots, d$, where $(s, (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_d)) \in [t_0, T] \times \mathbb{R}^{d-1}$ are fixed.

The term *backward* refers to differentiation with respect to the initial arguments s and x , rather than with respect to t and y as in the *forward* equation.

Theorem 4.6. *Under the assumptions of Theorem 4.5, if $P(s, x, t, \cdot)$ has a transition density function $p(s, x, t, y)$ continuous with respect to s , and its derivatives $\partial p / \partial x_i$ and $\partial^2 p / \partial x_i \partial x_j$ exist and are continuous with respect to s , then p is said to be a fundamental solution of Kolmogorov's backward equation*

$$\frac{\partial p}{\partial s} + \mathfrak{D}p = 0$$

where p satisfies (as a fundamental solution)

$$\lim_{s \rightarrow t^-} p(s, x, t, y) = \delta(x - y)$$

where δ denotes the Dirac delta.

Remark 4.7. If X_t is a homogeneous diffusion process, then $a(s, x) = a(x)$ and $b(s, x) = b(x)$, so $\mathfrak{D}$ is independent of s . Since we defined $P(s, x, t, y) = P(t - s, x, y)$, the sign of $\partial p / \partial s$ must change in this case, giving the equation

$$-\frac{\partial p}{\partial s} + \mathfrak{D}p = 0$$

Theorem 4.8. *Let X_t be a d -dimensional diffusion process on $[t_0, T]$ satisfying the properties in 4.3 uniformly in s and x , with transition density $p(s, x, t, y)$. If the derivatives $\partial p / \partial t$, $\partial(a_i(t, y)p) / \partial y_i$, and $\partial^2(b_{ij}(t, y)p) / \partial y_i \partial y_j$ exist and are continuous functions, then, for fixed s and x such that $s \leq t$, the density $p(s, x, t, y)$ is a fundamental solution of Kolmogorov's forward equation, also known as the **Fokker-Planck equation***

$$\frac{\partial p}{\partial t} + \sum_{i=1}^d \frac{\partial}{\partial y_i} (a_i(t, y)p) - \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \frac{\partial^2}{\partial y_i \partial y_j} (b_{ij}(t, y)p) = 0 \quad (4.4)$$

Example 4.9. Recall from Example 4.4 that the Wiener process is a homogeneous diffusion process with $a_i = 0$ for every $i = 1, \dots, n$ and $b_{ij} = \delta_{ij}$. Substituting into 4.4 gives the partial differential equation

$$\frac{\partial p}{\partial t} = \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 p}{\partial y_i^2}$$

whose solution, the homogeneous transition density function, is given by

$$p(t, x, y) = \frac{1}{(2\pi t)^{-d/2}} e^{-\frac{|y-x|^2}{2t}}$$

The Fokker-Planck equation is used in problems where the initial probability distribution is known. In physics, it has applications in areas such as plasma physics and laser optics. Furthermore, the Fokker-Planck equation is a particular case of the *master equation*, which, in its most general form, is used in physics to describe open quantum systems (quantum systems that interact with another external quantum system), with wide-ranging applications in areas such as quantum optics and quantum information.

Chapter 5

Itô-Taylor methods

We begin by briefly recalling the construction and form of Taylor methods for ordinary differential equations¹. Starting from an ordinary differential equation given by

$$\frac{dX_t}{dt} = a(X_t)$$

defined on $[t_0, T]$ with initial value X_{t_0} , if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function, the chain rule gives

$$\frac{df(X_t)}{dt} = a(X_t) \frac{\partial f(X_t)}{\partial x};$$

therefore, defining the operator $L = a \frac{\partial}{\partial x}$, we obtain, in integral form,

$$f(X_t) = f(X_{t_0}) + \int_{t_0}^t a(X_s) \frac{\partial f(X_s)}{\partial x} ds = f(X_{t_0}) + \int_{t_0}^t Lf(X_s) ds$$

Note that, if $f(x) = x$, then $Lf = a$, and we recover the original differential equation in integral form:

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds. \quad (5.1)$$

Applying the same reasoning to $a(X_s)$ gives

$$a(X_s) = a(X_{t_0}) + \int_{t_0}^s La(X_u) du$$

so expression (5.1) can be written as

$$X_t = X_{t_0} + \int_{t_0}^t \left(a(X_{t_0}) + \int_{t_0}^s La(X_u) du \right) ds = X_{t_0} + a(X_{t_0})(t - t_0) + R_2$$

with $R_2 = \int_{t_0}^t \int_{t_0}^s La(X_u) du ds$. Repeating the same procedure for $La(X_u)$ gives

$$X_t = X_{t_0} + a(X_{t_0})(t - t_0) + La(X_{t_0}) \frac{(t - t_0)^2}{2} + \underbrace{\int_{t_0}^t \int_{t_0}^s \int_{t_0}^u L^2 a(X_v) dv du ds}_{R_3}$$

¹We consider autonomous ordinary differential equations.

where we have written $L^2 = L \circ L$. Thus, assuming that $a = a(x)$ is a $r + 1$ -times differentiable function, we obtain

$$X_t = \sum_{l=0}^r \frac{(t-t_0)^l}{l!} L^l a(X_{t_0}) + \int_{t_0}^t \cdots \int_{t_0}^{s_2} L^{r+1} a(X_{s_1}) ds_1 \dots ds_{r+1} \quad (5.2)$$

where $L^0 a(X_{t_0}) = X_{t_0}$. In general, taking $f : \mathbb{R} \rightarrow \mathbb{R}$ to be $r + 1$ -times differentiable gives Taylor's formula

$$f(X_t) = f(X_{t_0}) + \sum_{l=1}^r \frac{(t-t_0)^l}{l!} L^l f(X_{t_0}) + \int_{t_0}^t \cdots \int_{t_0}^{s_2} L^{r+1} f(X_{s_1}) ds_1 \dots ds_{r+1} \quad (5.3)$$

The main idea of these notes is to extend this type of approximation to stochastic differential equations. Given a stochastic differential equation in integral form (3.2),

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds + \int_{t_0}^t b(X_s) dW_s \quad (5.4)$$

defined on $[t_0, T]$, if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a twice differentiable function, applying Itô's formula² gives

$$f(X_t) = f(X_{t_0}) + \int_{t_0}^t \left(a(X_s) \frac{\partial f}{\partial x}(X_s) + \frac{1}{2} b^2(X_s) \frac{\partial^2 f}{\partial x^2}(X_s) \right) ds + \int_{t_0}^t b(X_s) dW_s$$

Thus, similarly to the preceding case, we define the operators

$$L^0 = a \frac{\partial}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2}{\partial x^2} \quad \text{and} \quad L^1 = b \frac{\partial}{\partial x}$$

so that

$$f(X_t) = f(X_{t_0}) + \int_{t_0}^t L^0 f(X_s) ds + \int_{t_0}^t L^1 f(X_s) dW_s,$$

Taking $f(x) = x$ recovers the stochastic integral equation given in (5.4). Applying Itô's formula to $f = a$ and $f = b$ and using the notation introduced gives

$$\begin{aligned} X_t &= X_{t_0} + \int_{t_0}^t \left(a(X_{t_0}) + \int_{t_0}^s L^0 a(X_u) du + \int_{t_0}^s L^1 a(X_u) dW_u \right) ds \\ &\quad + \int_{t_0}^t \left(b(X_{t_0}) + \int_{t_0}^s L^0 b(X_u) du + \int_{t_0}^s L^1 b(X_u) dW_u \right) dW_s \\ &= X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t dW_s + R_2 \end{aligned}$$

with remainder

$$\begin{aligned} R_2 &= \int_{t_0}^t \int_{t_0}^s L^0 a(X_z) du ds + \int_{t_0}^t \int_{t_0}^s L^1 a(X_z) dW_u ds \\ &\quad + \int_{t_0}^t \int_{t_0}^s L^0 b(X_z) du dW_s + \int_{t_0}^t \int_{t_0}^s L^1 b(X_z) dW_u dW_s \end{aligned}$$

²Recall that the chain rule does not hold in Itô calculus, as we saw in 2.17.

Applying Itô's formula again, for example to $f = L^1 b$, gives

$$X_t = X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t dW_s + L^1 b(X_{t_0}) \int_{t_0}^t \int_{t_0}^s dW_u dW_s + R_3$$

where R_3 contains triple iterated integrals with respect to dW_s and ds . These expansions can be generalized to the d -dimensional, nonhomogeneous case, that is, when (5.4) has the form

$$X_t = X_{t_0} + \int_{t_0}^t a(s, X_s) ds + \sum_{j=1}^m \int_{t_0}^t b^j(s, X_s) dW_s^j \quad (5.5)$$

where m is the dimension of the Wiener process.

We therefore need to define several preliminary concepts to generalize Taylor approximations to the stochastic case.

5.1 Itô-Taylor expansions

5.1.1 Multi-indices

We begin this section by recalling the definition of a multi-index.

Definition 5.1. We say that α is a multi-index of length $l := l(\alpha)$ if α is a vector of the form

$$\alpha = (j_1, j_2, \dots, j_l) \quad (5.6)$$

where $j_i \in \{0, 1, \dots, m\}$ for $i = 1, \dots, l$ and $m \in \mathbb{N}$.

We will later see that m denotes the number of components of the Wiener process under consideration. For example, for a one-dimensional Wiener process, $m = 1$, and we consider multi-indices of the form $(j_1, \dots, j_l)$, where $j_i \in \{0, 1\}$ for every $i = 1, \dots, l$. On the other hand, in the trivial case with no Wiener process, $m = 0$, and $\alpha = (0, \dots, 0)$ is the only possible multi-index under the preceding definition.

We denote by v the multi-index of length zero, $l(v) = 0$. We also define $n(\alpha)$ as the number of zero components in a multi-index. In the trivial case discussed above, clearly $n(\alpha) = l(\alpha)$.

Definition 5.2. For fixed $m \in \mathbb{N}$, we define the set of all multi-indices as

$$\mathcal{M} = \{(j_1, j_2, \dots, j_l) : j_i \in \{0, 1, \dots, m\} \text{ with } i \in \{1, 2, \dots, l\}; l \in \mathbb{N}\} \cup \{v\} \quad (5.7)$$

Note that this set depends on the fixed value of $m \in \mathbb{N}$. Since the one-dimensional case is particularly relevant, observe that, if $m = 1$, then

$$\mathcal{M} = \{(j_1, j_2, \dots, j_l) : j_i \in \{0, 1\} \text{ with } i \in \{1, 2, \dots, l\}; l \in \mathbb{N}\} \cup \{v\}$$

To conclude this section, we introduce some useful operations on $\mathcal{M}$. Let $\alpha \in \mathcal{M}$ with $l(\alpha) \geq 1$. We define the multi-indices $-\alpha$ and $\alpha-$ as those obtained by deleting the first or last component of α , respectively. Note that, if $l(\alpha) = 1$, then $-\alpha = \alpha- = v$.

Finally, we define the concatenation of two multi-indices $\alpha, \beta \in \mathcal{M}$ with components $\alpha = (j_1, \dots, j_r)$ and $\beta = (i_1, \dots, i_s)$ as

$$\alpha * \beta = (j_1, \dots, j_r, i_1, \dots, i_s)$$

5.1.2 Multiple Itô integrals

Before describing the function space $\mathcal{H}_\alpha$ to which the integrands below must belong, we define the multiple Itô integral and use this definition to construct the space $\mathcal{H}_\alpha$.

Definition 5.3. Let $\alpha \in \mathcal{M}$ be a multi-index and $f \in \mathcal{H}_\alpha$ a stochastic process. We define its multiple Itô integral recursively. That is, taking $l = l(\alpha)$,

$$I_\alpha[f]_{t_0, T} := \begin{cases} f(T) & \text{if } l = 0 \\ \int_{t_0}^T I_{\alpha-}[f]_{t_0, s} ds & \text{if } l \geq 1 \text{ and } j_l = 0 \\ \int_{t_0}^T I_{\alpha-}[f]_{t_0, s} dW_s^{j_l} & \text{if } l \geq 1 \text{ and } j_l \geq 1 \end{cases} \quad (5.8)$$

Thus, given $\alpha = (j_1, j_2, \dots, j_l) \in \mathcal{M}$, the *order of integration* is from left to right, where $j_i = 0$ denotes a Lebesgue integral with respect to the parameter t , and $j_i \geq 1$ denotes the Itô integral defined in (2.17) with respect to the j_i th component of the m -dimensional Wiener process.

Examples that will be useful later include the multiple integrals associated with the multi-indices v , (0) , (1) , and $(0, 1)$:

$$\begin{aligned} I_v[f]_{t_0, T} &= f(T) \\ I_{(0)}[f]_{t_0, T} &= \int_{t_0}^T f(t) dt \\ I_{(1)}[f]_{t_0, T} &= \int_{t_0}^T f(t) dW_t^1 \\ I_{(0,1)}[f]_{t_0, T} &= \int_{t_0}^T \int_{t_0}^t f(s) ds dW_t^1 \end{aligned}$$

Remark 5.4. Recall that these notes frequently use an abuse of notation: f is not merely a function of t , but shorthand for $f(t, \omega)$ with $\omega \in \Omega$. Thus, the corresponding multiple integrals are themselves random variables. Therefore, given $\omega \in \Omega$, the full notation is

$$I_{(0,1)}[f]_{t_0, T}(\omega) = \int_{t_0}^T \int_{t_0}^t f(s, \omega) ds dW_t^1(\omega).$$

We now construct the space $\mathcal{H}_\alpha$ to ensure that the multiple integrals above are well defined. We construct this space recursively, defining three sets of right-continuous adapted³ stochastic processes $\{f(t) : t \in [t_0, T]\}$.

- $\mathcal{H}_v$ is the set of such processes satisfying

$$|f(t, \omega)| < \infty$$

- $\mathcal{H}_{(0)}$ is the set of processes satisfying

$$\int_{t_0}^t |f(s, \omega)| ds < \infty$$

³By adapted, we mean that condition (b) of Definition 2.1 is satisfied.

- $\mathcal{H}_{(j)}$ with $j > 1$ is the set of all processes satisfying

$$\int_{t_0}^t |f(t, \omega)|^2 < \infty$$

where these inequalities hold for almost every $\omega \in \Omega$ and every $t \in [t_0, T]$.

Thus, given $\alpha = (j_1, \dots, j_l) \in \mathcal{M}$ and following the preceding construction, we define $\mathcal{H}_\alpha$ as the right-continuous adapted stochastic processes for which the integrals $\{I_{\alpha-}[f]_{t_0,t}, t \in [t_0, T]\}$ depending on t satisfy

$$I_{\alpha-}[f]_{t_0,t} \in \mathcal{H}_{(j_l)}.$$

In this way, all integrals appearing in each multiple integral associated with α are well defined, both the Lebesgue integrals and the Itô integrals. For example:

$$I_{(1,0,2)}[f]_{t_0,T} = \underbrace{\int_{t_0}^T \underbrace{\int_{t_0}^t \underbrace{\int_{t_0}^s \underbrace{f(u)}_{\in \mathcal{H}_{(1)}} dW_u^1}_{\in \mathcal{H}_{(0)}} ds}_{\in \mathcal{H}_{(2)}} dW_t^2$$

Thus, $\mathcal{H}_{(1,0,2)}$ consists of the processes such that $f \in \mathcal{H}_{(1)}$, $I_{(1)}[f]_{t_0,s} \in \mathcal{H}_{(0)}$, and $I_{(2)}[I_{(1)}[f]_{t_0,s}]_{t_0,t} \in \mathcal{H}_{(2)}$.

A particular case of these multiple integrals occurs when $f(t, \omega) \equiv 1$, for which we use the notation $I_{\alpha, t_0, T}$. If the time interval is unambiguous, we further simplify the notation to I_α . These are known as multiple integrals of the unit function. As examples, consider the multi-indices $\alpha = (0)$, $\alpha = (1)$, $\alpha = (0, 0)$, and $\alpha = (1, 1)$ with $t_0 = t_n$ and $T = t_{n+1}$:

$$\begin{aligned} I_{(0)} &= \int_{t_n}^{t_{n+1}} dt = t_{n+1} - t_n := \Delta \\ I_{(1)} &= \int_{t_n}^{t_{n+1}} dW_t = W_{t_{n+1}} - W_{t_n} := \Delta W \\ I_{(0,0)} &= \int_{t_n}^{t_{n+1}} \int_{t_n}^t ds dt = \Delta^2/2 \\ I_{(1,1)} &= \int_{t_n}^{t_{n+1}} \int_{t_n}^t dW_s dW_t \end{aligned} \tag{5.9}$$

In particular, if $\alpha = (0, 0, \dots, 0)$, then $I_\alpha = \Delta^{l(\alpha)}/l(\alpha)!$.

5.1.3 Properties of multiple Itô integrals

In this section, we study several properties of, and relationships between, multiple integrals of the unit function. For simplicity, we denote $I_\alpha[1]_{0,t}$ by $I_{\alpha,t}$ for $\alpha \in \mathcal{M}$, use $W_t^0 = t$ for $t \geq 0$, and let 1_A be the indicator function of the set A .

Proposition 5.5. *Let $l = l(\alpha)$ with $\alpha = (j_1, j_2, \dots, j_l) \in \mathcal{M}$, where $j_i \in \{0, 1, \dots, m\}$. Then*

$$\begin{aligned} I_{(j),t} I_{\alpha,t} &= \sum_{i=0}^l I_{(j_1, \dots, j_i, j, j_{i+1}, \dots, j_l), t} \\ &\quad + \sum_{i=1}^l 1_{\{j_i \neq 0\}} I_{(j_1, \dots, j_{i-1}, 0, j_{i+1}, \dots, j_l), t} \end{aligned} \quad (5.10)$$

for every $t \geq 0$ and $j \in \{0, 1, \dots, m\}$.

Proof. Let $\{X_t, t \geq 0\}$ be a multidimensional Itô process whose components are multiple integrals of the unit function:

$$X_t = \begin{pmatrix} X_t^{(0)} \\ \vdots \\ X_t^{(m)} \\ X_t^{(j_1, j_2)} \\ \vdots \\ X_t^{(j_1, j_2, \dots, j_l)} \end{pmatrix} = \begin{pmatrix} I_{(0),t} \\ \vdots \\ I_{(m),t} \\ I_{(j_1, j_2),t} \\ \vdots \\ I_{(j_1, j_2, \dots, j_l),t} \end{pmatrix}$$

Let $\beta = (i_1, i_2, \dots, i_r)$ with $1 \leq r \leq l$, and define

$$a^\beta = \begin{cases} x^{\beta-} & \text{if } i_r = 0, \\ 0 & \text{if } i_r \neq 0. \end{cases} \quad b^{\beta,i} = \begin{cases} x^{\beta-} & \text{if } i = i_r \in \{1, \dots, m\}, \\ 0 & \text{otherwise.} \end{cases}$$

By definition of $I_{\beta,t}$, clearly $I_{\beta,0} = 0$, so X_0 is the zero vector. Applying Itô's formula gives

$$d(I_{(j),t} I_{\alpha,t}) = I_{\alpha,t} dI_{(j),t} + I_{(j),t} dI_{\alpha,t}$$

which, in integral form and using the recursive definition of multiple integrals, gives

$$I_{(j),t} I_{\alpha,t} = \int_0^t I_{\alpha,s} dI_{(j),s} + \int_0^t I_{(j),s} I_{\alpha-,s} dW_s^{j_l} + 1_{\{j_l \neq 0\}} \int_0^t I_{\alpha-,s} ds \quad (5.11)$$

We first consider the case $l(\alpha) = 1$. The first term in (5.11) is

$$\int_0^t I_{\alpha,s} dI_{(j),s} = \int_0^t \int_0^s dW_u^{j_1} dW_s^j = I_{(j_1, j), t},$$

while the second term is

$$\int_0^t I_{(j),s} I_{\alpha-,s} dW_s^{j_1} = \int_0^t \int_0^s dW_u^j dW_s^{j_1} = I_{(j, j_1), t},$$

and the final term in (5.11) is $1_{\{j_1 \neq 0\}} I_{(0),t}$. Thus, for $l = 1$, (5.11) coincides with the desired expression (5.10). We proceed by induction on $l(\alpha)$. If $l(\alpha) \geq 2$, an analogous argument gives

$$I_{(j),t} I_{\alpha,t} = I_{(j_1, \dots, j_l, j), t} + \int_0^t I_{(j),s} I_{\alpha-,s} dW_s^{j_l} + 1_{\{j_l \neq 0\}} I_{(j_1, \dots, j_l, 0), t}$$

and since $l(\alpha-) < l(\alpha)$, applying induction to the product $I_{(j),s}I_{\alpha-,s}$ yields

$$I_{(j),s}I_{\alpha-,s} = I_{(j_1,\dots,j_{l-1}),s} + \int_0^s I_{(j),u}I_{(\alpha-)-,u}dW_s^{l_{j-1}} + 1_{\{j_{l-1}=j \neq 0\}}I_{(j_1,\dots,j_{l-1},0),t}$$

and applying this procedure recursively gives the desired result. $\square$

We next consider examples applying this proposition. Fix $j = 0$. If $\alpha = (1)$, then (5.10) gives

$$I_{(0)}I_{(1)} = I_{(0,1)} + I_{(1,0)} \quad (5.12)$$

and similarly, taking $\alpha = (1, 1)$ gives

$$I_{(0)}I_{(1,1)} = I_{(0,1,1)} + I_{(1,0,1)} + I_{(1,1,0)}. \quad (5.13)$$

Corollary 5.6. *If $\alpha = (j, \dots, j) \in \mathcal{M}$ with $l(\alpha) = l \geq 2$ and $j \in \{0, 1, \dots, m\}$, then, for every $t \geq 0$,*

$$I_{\alpha,t} = \begin{cases} \frac{1}{l!}t^l & \text{if } j = 0, \\ \frac{1}{l}(W_t^j I_{\alpha-,t} - tI_{(\alpha-)-,t}) & \text{if } j \geq 1. \end{cases} \quad (5.14)$$

Proof. Let $\alpha = (j_1, j_2, \dots, j_l) = (j, j, \dots, j)$. We first analyze the case $j \neq 0$. Applying Proposition 5.5 to the multi-index $(\alpha-)-$ and to $I_{(0),t} = t$ gives

$$\begin{aligned} I_{(0),t}I_{(\alpha-)-,t} &= \sum_{i=0}^{l-2} I_{(j_1,\dots,j_i,0,j_{i+1},\dots,j_{l-2})} + \sum_{i=1}^{l-2} \underbrace{1_{\{0=j \neq 0\}}}_0 I_{(j_1,\dots,j_{i-1},0,j_{i+1},\dots,j_{l-2})} \\ &= \sum_{i=0}^{l-2} I_{(j_1,\dots,j_i,0,j_{i+1},\dots,j_{l-2})} = \sum_{i=1}^{l-1} I_{(j_1,\dots,j_{i-1},0,j_{i+1},\dots,j_{l-1})} \end{aligned}$$

since $l((\alpha-)-) = l - 2$. Similarly, taking $\alpha-$ and $I_{(j),t}$ gives

$$\begin{aligned} I_{(j),t}I_{\alpha-,t} &= \sum_{i=0}^{l-1} I_{(j_1,\dots,j_i,j,j_{i+1},\dots,j_{l-1}),t} + \sum_{i=1}^{l-1} \underbrace{1_{\{j=j \neq 0\}}}_1 I_{(j_1,\dots,j_{i-1},0,j_{i+1},\dots,j_{l-1})} \\ &= \sum_{i=0}^{l-1} I_{(j_1,\dots,j_i,j,j_{i+1},\dots,j_{l-1}),t} + \sum_{i=1}^{l-1} I_{(j_1,\dots,j_{i-1},0,j_{i+1},\dots,j_{l-1})} \\ &= lI_{\alpha,t} + \sum_{i=1}^{l-1} I_{(j_1,\dots,j_{i-1},0,j_{i+1},\dots,j_{l-1})}. \end{aligned}$$

Thus, subtracting the two equalities gives

$$I_{(j),t}I_{\alpha-,t} - tI_{(\alpha-)-,t} = I_{(j),t}I_{\alpha-,t} - I_{(0),t}I_{(\alpha-)-,t} = lI_{\alpha,t}$$

which proves the case $j \geq 1$. If $j = 0$, then $\alpha = (0, 0, \dots, 0)$, so the integrals are ordinary integrals and, as shown earlier, $I_{\alpha,t} = t^l/l!$. $\square$

For example, taking $m = 1$ and $\alpha = (1, 1)$ gives

$$I_{(1,1),t_n,t_{n+1}} = \frac{1}{2} (\Delta W \cdot I_{(1),t_n,t_{n+1}} - \Delta \cdot I_{v,t_n,t_{n+1}}) = \frac{1}{2} (\Delta W^2 - \Delta).$$

Thus, Corollary 5.6 allows us to reduce the *degree*⁴ of multiple integrals of the unit function.

Because of its relevance to the numerical methods studied later, we next describe the random variable $I_{(1,0)}$.

⁴In the sense of expressing the integral in terms of integrals associated with a multi-index of shorter length.

Proposition 5.7. *The random variable $I_{(1,0),t_n,t_{n+1}} \equiv I_{(1,0)} = \Delta Z$ is normally distributed, with mean, variance, and covariance*

$$\mathbb{E}[I_{(1,0)}] = 0, \quad \mathbb{E}[I_{(1,0)}^2] = \frac{1}{3}\Delta^3, \quad \mathbb{E}[I_{(1,0)}I_{(1)}] = \frac{1}{2}\Delta^2.$$

Proof. The normality of ΔZ follows from the integrals being independent of ω ; see [7]. Applying Fubini's theorem and the definition of a Wiener process gives

$$\mathbb{E}[I_{(1,0)}] = \mathbb{E}\left[\int_{t_n}^{t_{n+1}} (W_s - W_{t_n}) ds\right] = 0.$$

In general, if there exists $j \geq 1$ such that $j \in \alpha$, then $\mathbb{E}[I_\alpha] = 0$ because the expectation of an Itô integral is zero. On the other hand, Proposition 5.5 gives

$$I_{(1)}I_{(1,0)} = I_{(1,1,0)} + I_{(1,1,0)} + I_{(1,0,1)} + I_{(0,0)}$$

and taking expectations gives $\mathbb{E}[I_{(1)}I_{(1,0)}] = \mathbb{E}[I_{(0,0)}] = \Delta^2/2$. Finally, relation (5.12) gives

$$\mathbb{E}[I_{(1,0)}^2] = \mathbb{E}[(I_{(0)}I_{(1)} - I_{(0,1)})^2] = \Delta^2\mathbb{E}[I_{(1)}^2] - 2\Delta\mathbb{E}[I_{(1)}I_{(0,1)}] + \mathbb{E}[I_{(0,1)}^2]$$

where $\mathbb{E}[I_{(1)}^2] = \Delta$ is the variance of the Itô integral with unit integrand. Again, Proposition 5.5 gives

$$I_{(1)}I_{(0,1)} = I_{(1,0,1)} + I_{(0,1,1)} + I_{(0,1,1)} + I_{(0,0)}$$

and taking expectations gives $\mathbb{E}[I_{(1)}I_{(0,1)}] = \mathbb{E}[I_{(0,0)}] = \Delta^2/2$. Applying the Itô isometry,

$$\mathbb{E}[I_{(0,1)}^2] = \mathbb{E}\left[\left(\int_{t_n}^{t_{n+1}} s dW_s\right)^2\right] = \int_{t_n}^{t_{n+1}} \mathbb{E}[s^2] ds = \int_{t_n}^{t_{n+1}} s^2 ds = \frac{\Delta^3}{3}.$$

Thus, $\mathbb{E}[I_{(1,0)}^2] = \Delta^2\Delta - 2\Delta\Delta^2/2 + \Delta^3/3 = \Delta^3/3$, as required. $\square$

5.1.4 Coefficients of the Itô-Taylor approximation

Recall that the stochastic integral equation in the d -dimensional case is given by

$$X_t = X_{t_0} + \int_{t_0}^t a(s, X_s) ds + \sum_{j=1}^m \int_{t_0}^t b^j(s, X_s) dW_s^j$$

where $a : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $b : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$. Following the development in the introduction and Itô's formula, we define the operators for the d -dimensional case:

$$L^0 = \frac{\partial}{\partial t} + \sum_{k=1}^d a^k \frac{\partial}{\partial x^k} + \frac{1}{2} \sum_{k,l=1}^d \sum_{j=1}^m b^{k,j} b^{l,j} \frac{\partial^2}{\partial x^k \partial x^l} \quad (5.15)$$

and for $j \in \{1, 2, \dots, m\}$,

$$L^j = \sum_{k=1}^d b^{k,j} \frac{\partial}{\partial x^k} \quad (5.16)$$

where W^j is the j th component of an m -dimensional Wiener process and $b^{j,k}$ is the entry of the matrix b in row j and column k . We require f to be a $l(\alpha) + n(\alpha)$ -times continuously differentiable function⁵. Given the multi-index $\alpha = (j_1, j_2, \dots, j_l)$, we denote

⁵Unless otherwise stated, we take $f(t, x) = x$, as at the beginning of this chapter.

the composition $L^{j_1} \circ \dots \circ L^{j_l}$ by L^α and call $L^\alpha f$ an Itô coefficient. We take $L^v f = f$.

We consider some examples. In the simplest case, we consider an autonomous equation, $d = m = 1$ and $f(t, x) = x$. Then

$$L^{(0)} f = a, \quad L^{(1)} f = b \quad \text{and} \quad L^{(0,1)} f = ab' + \frac{1}{2} b^2 b''$$

where a' and b' denote derivatives with respect to x . We compute the case $\alpha = (1, 0)$:

$$L^{(1,0)} f = L^1(L^0 f) = L^1 \left(\frac{\partial f}{\partial t} + a \frac{\partial f}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} \right) = L^1 a = ba'$$

since $f(t, x) = x$. Similarly, if $\alpha = (1, 1, 1)$, then

$$L^{(1,1,1)} = L^{(1,1)}(L^1 f) = L^{(1,1)} \left(b \frac{\partial f}{\partial x} \right) = L^1(L^1 b) = L^1(bb') = b^2 b'' + b(b')^2$$

In general, the coefficients for the one-dimensional autonomous case with $l(\alpha) \leq 3$ can be found in [5].

5.1.5 Hierarchical sets

The multiple integrals appearing in the Itô-Taylor expansion are not indexed by arbitrary sets of multi-indices; only multi-indices belonging to specific sets, called hierarchical sets, occur.

Definition 5.8. Given a subset $\mathcal{A} \neq \emptyset$ of $\mathcal{M}$, we say that $\mathcal{A}$ is a hierarchical set if:

- (a) $\sup_{\alpha \in \mathcal{A}} l(\alpha) < \infty$,
- (b) $-\alpha \in \mathcal{A}$ for every $\alpha \in \mathcal{A} \setminus \{v\}$.

Clearly, $v \in \mathcal{A}$ because condition 2 of the preceding definition implies that, whenever there exists $\alpha \in \mathcal{A}$ such that $l(\alpha) \geq 1$, we have $-(\dots(-\alpha)) = v \in \mathcal{A}$. For example, in the case $m = 1$, the sets $\{v\}$, $\{v, (0), (1)\}$, and $\{v, (0), (1, 0), (0, 1, 0)\}$ are hierarchical sets.

As seen in the introduction to this chapter, Itô-Taylor approximations of different orders leave a remainder involving multiple integrals associated with multi-indices in a particular set, which we call the remainder set.

Definition 5.9. Given a hierarchical set $\mathcal{A}$, we denote the remainder set of $\mathcal{A}$ by $\mathcal{R}(\mathcal{A})$, defined as

$$\mathcal{R}(\mathcal{A}) = \{\alpha \in \mathcal{M} \setminus \mathcal{A} : -\alpha \in \mathcal{A}\}$$

Continuing the preceding example, we have

$$\begin{aligned} \mathcal{R}(\{v\}) &= \{(0), (1)\} \\ \mathcal{R}(\{v, (0), (1)\}) &= \{(0, 0), (0, 1), (1, 0), (1, 1)\} \\ \mathcal{R}(\{v, (0), (1, 0), (0, 1, 0)\}) &= \{(1), (0, 0), (1, 1, 0), (0, 0, 1, 0), (1, 0, 1, 0)\} \end{aligned} \tag{5.17}$$

We next study two hierarchical sets that will be central to describing Itô-Taylor methods.

Proposition 5.10. *The set*

$$\Gamma_r = \{\alpha \in \mathcal{M} : l(\alpha) \leq r\} \quad (5.18)$$

is a hierarchical set for any $r \in \mathbb{N}$.

Proof. The first condition clearly holds, since

$$\sup_{\alpha \in \mathcal{A}} l(\alpha) \leq \sup_{\alpha \in \mathcal{A}} r = r < \infty$$

On the other hand, given $\alpha \in \mathcal{A} \setminus \{v\}$, we have $l(-\alpha) = l(\alpha) - 1 < r$ and therefore $-\alpha \in \mathcal{A}$. $\square$

Proposition 5.11. *The set*

$$\Lambda_r = \{\alpha \in \mathcal{M} : l(\alpha) + n(\alpha) \leq r\} \quad (5.19)$$

is a hierarchical set for any $r \in \mathbb{N}$.

Proof. Since $0 \leq n(\alpha) \leq l(\alpha)$, we have $l(\alpha) \leq l(\alpha) + n(\alpha) \leq r$ and consequently

$$\sup_{\alpha \in \mathcal{A}} l(\alpha) \leq r < \infty$$

It remains to check that $-\alpha \in \mathcal{A}$ for any $\alpha \in \mathcal{A} \setminus \{v\}$. Let $\alpha \in \mathcal{A} \setminus \{v\}$. Then $l(-\alpha) = l(\alpha) - 1 < r$ and $n(-\alpha) \leq n(\alpha)$, so

$$l(-\alpha) + n(-\alpha) < l(\alpha) + n(\alpha) \leq r$$

as required. $\square$

We can also examine the elements of the set $\mathcal{R}(\Gamma_r)$. Given $\alpha \in \mathcal{R}(\Gamma_r)$, since $-\alpha \in \mathcal{A}$, we have $l(-\alpha) \leq r$ and consequently $l(\alpha) \leq r + 1$. However, necessarily $l(\alpha) > r$; otherwise, $\alpha \in \mathcal{A}$, contradicting the definition of $\mathcal{R}$. Thus, the remainder set of Γ_r is

$$\mathcal{R}(\Gamma_r) = \{\alpha \in \mathcal{M} : l(\alpha) = r + 1\} \quad (5.20)$$

With these preliminary concepts in place, we can now define and study Itô-Taylor expansions.

5.1.6 Itô-Taylor expansions

We assume that X_t is a d -dimensional Itô process, see (2.20), determined by

$$X_t = X_{t_0} + \int_{t_0}^t a(s, X_s) ds + \sum_{j=1}^m \int_{t_0}^t b^j(s, X_s) dW_s^j \quad (5.21)$$

where $t \in [t_0, T]$. Before defining Itô-Taylor expansions, we establish some preliminary results useful in their proof.

Lemma 5.12. *Let $f : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}$ be of class⁶ $\mathcal{C}^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$. Then*

$$f(t, X_t) = f(t_0, X_{t_0}) + \sum_{j=0}^m I_{(j)} [L^j f]_{t_0, t} \quad (5.22)$$

Lemma 5.13. *Let $f : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}$ and let $\alpha, \beta \in \mathcal{M}$ be such that $l(\beta) \geq 1$. Then*

$$I_\alpha [L^\beta f]_{t_0, t} = (L^\beta f)(t_0, X_{t_0}) I_\alpha + \sum_{j=0}^m I_{(j)*\alpha} [L^{(j)*\beta} f]_{t_0, t}. \quad (5.23)$$

We are now ready to introduce Itô-Taylor expansions, which play a fundamental role in these notes.

Theorem 5.14. *Given a function $f : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}$ and a hierarchical set $\mathcal{A}$, we have*

$$f(t, X_t) = \sum_{\alpha \in \mathcal{A}} (L^\alpha f)(t_0, X_{t_0}) I_\alpha + \sum_{\alpha \in \mathcal{R}(\mathcal{A})} I_\alpha [L^\alpha f]_{t_0, t} \quad (5.24)$$

where X_t is described by (5.21), assuming that the derivatives of f , a , and b and the multiple Itô integrals appearing in (5.24) exist. Expression (5.24) is known as the Itô-Taylor expansion.

Proof. Define $l_\infty(\mathcal{A}) = \sup_{\alpha \in \mathcal{A}} l(\alpha)$. We proceed by induction on $l_\infty(\mathcal{A})$. If $l_\infty(\mathcal{A}) = 0$, then $l(\alpha) = 0$ for every $\alpha \in \mathcal{A}$, and necessarily $\mathcal{A} = \{v\}$, so $\mathcal{R}(\mathcal{A}) = \{(0), (1), \dots, (m)\}$. By Lemma 5.12,

$$f(t, X_t) = f(t_0, X_{t_0}) + \sum_{j=0}^m I_{(j)} [L^{(j)} f]_{t_0, t} = \sum_{\alpha \in \mathcal{A}} (L^\alpha f)(t_0, X_{t_0}) I_\alpha + \sum_{\alpha \in \mathcal{R}(\mathcal{A})} I_\alpha [L^\alpha f]_{t_0, t}$$

and therefore (5.24) holds for $l_\infty(\mathcal{A}) = 0$. Now suppose that the expansion holds for every hierarchical set of length $k - 1$ with $k > 1$. Let $l_\infty(\mathcal{A}) = k \geq 1$ and define $\mathcal{E} = \{\alpha \in \mathcal{A} : l(\alpha) \leq k - 1\}$. Since $l_\infty(\mathcal{E}) = k - 1$, the induction hypothesis gives

$$f(t, X_t) = \sum_{\alpha \in \mathcal{E}} (L^\alpha f)(t_0, X_{t_0}) I_\alpha + \sum_{\alpha \in \mathcal{R}(\mathcal{E})} I_\alpha [L^\alpha f]_{t_0, t}$$

We show that $\mathcal{A} \setminus \mathcal{E} \subseteq \mathcal{R}(\mathcal{E})$. Given $\alpha \in \mathcal{A} \setminus \mathcal{E}$, we have $l(\alpha) > k - 1$. In particular, $\alpha \in \mathcal{A}$ and therefore $l(\alpha) \leq l_\infty(\mathcal{A}) = k$. Thus, $l(\alpha) = k$ and $l(-\alpha) = k - 1$, so $-\alpha \in \mathcal{E}$. Using the definition of the remainder set in 5.9 proves that $\alpha \in \mathcal{R}(\mathcal{E})$.

Since $\mathcal{R}(\mathcal{E}) = (\mathcal{A} \setminus \mathcal{E}) \cup (\mathcal{R}(\mathcal{E}) \setminus (\mathcal{A} \setminus \mathcal{E}))$ is a disjoint union, we can rewrite the preceding equality as

$$f(t, X_t) = \sum_{\alpha \in \mathcal{E}} (L^\alpha f)(t_0, X_{t_0}) I_\alpha + \sum_{\alpha \in \mathcal{A} \setminus \mathcal{E}} I_\alpha [L^\alpha f]_{t_0, t} + \sum_{\alpha \in \mathcal{R}(\mathcal{E}) \setminus (\mathcal{A} \setminus \mathcal{E})} I_\alpha [L^\alpha f]_{t_0, t}$$

⁶That is, f is of class $\mathcal{C}^1$ with respect to the first component and of class $\mathcal{C}^2$ with respect to the second, on their respective domains.

and applying Lemma 5.13 with $\alpha \in \mathcal{A} \setminus \mathcal{E}$ and $\beta = \alpha$ gives

$$\begin{aligned}
 f(t, X_t) &= \sum_{\alpha \in \mathcal{E}} (L^\alpha f)(t_0, X_{t_0}) I_\alpha + \sum_{\alpha \in \mathcal{A} \setminus \mathcal{E}} \left((L^\alpha f)(t_0, X_{t_0}) I_\alpha + \sum_{j=0}^m I_{(j)*\alpha} [L^{(j)*\alpha} f]_{t_0, t} \right) \\
 &\quad + \sum_{\alpha \in \mathcal{R}(\mathcal{E}) \setminus (\mathcal{A} \setminus \mathcal{E})} I_\alpha [L^\alpha f]_{t_0, t} \\
 &= \sum_{\alpha \in \mathcal{A}} (L^\alpha f)(t_0, X_{t_0}) I_\alpha + \sum_{\alpha \in \mathcal{A} \setminus \mathcal{E}} \sum_{j=0}^m I_{(j)*\alpha} [L^{(j)*\alpha} f]_{t_0, t} \\
 &\quad + \sum_{\alpha \in \mathcal{R}(\mathcal{E}) \setminus (\mathcal{A} \setminus \mathcal{E})} I_\alpha [L^\alpha f]_{t_0, t} \\
 &= \sum_{\alpha \in \mathcal{A}} (L^\alpha f)(t_0, X_{t_0}) I_\alpha + \sum_{\alpha \in \mathcal{R}_1} I_\alpha [L^\alpha f]_{t_0, t}
 \end{aligned}$$

where

$$\begin{aligned}
 \mathcal{R}_1 &= \left(\bigcup_{j=0}^m \{(j)*\alpha : \alpha \in \mathcal{A} \setminus \mathcal{E}\} \right) \cup (\mathcal{R}(\mathcal{E}) \setminus (\mathcal{A} \setminus \mathcal{E})) \\
 &= \{\alpha \in \mathcal{M} \setminus \mathcal{E} : -\alpha \in \mathcal{A} \setminus \mathcal{E}\} \cup (\{\alpha \in \mathcal{M} \setminus \mathcal{E} : -\alpha \in \mathcal{E}\} \setminus \{\alpha \in \mathcal{M} \setminus \mathcal{E} : \alpha \in \mathcal{A} \setminus \mathcal{E}\}) \\
 &= \{\alpha \in \mathcal{M} \setminus \mathcal{A} : -\alpha \in \mathcal{A} \setminus \mathcal{E}\} \cup \{\alpha \in \mathcal{M} \setminus \mathcal{A} : -\alpha \in \mathcal{E}\} \\
 &= \{\alpha \in \mathcal{M} \setminus \mathcal{A} : -\alpha \in \mathcal{A}\} \\
 &= \mathcal{R}(\mathcal{A})
 \end{aligned}$$

which establishes (5.24). $\square$

We next consider some relevant examples. Taking the trivial hierarchical set, $\mathcal{A} = \{v\}$, and generalizing the first equality in (5.17) to an m -dimensional Wiener process gives $\mathcal{R}(\{v\}) = \{(0), (1), \dots, (m)\}$. Applying Theorem 5.14 then yields

$$f(t, X_t) = f(t_0, X_{t_0}) + \sum_{j=1}^m I_{(j)} [L^{(j)} f]_{t_0, t}$$

since $I_v = 1$, and the definition of the multiple Itô integral in 5.3 gives

$$f(t, X_t) = f(t_0, X_{t_0}) + \int_{t_0}^t L^0 f(s, X_s) ds + \sum_{j=1}^m \int_{t_0}^t L^j f(s, X_s) dW_s^j$$

which is Itô's formula in integral form. In particular, taking $d = m = 1$ gives

$$f(t, X_t) = f(t_0, X_{t_0}) + \int_{t_0}^t \left(\frac{\partial f}{\partial s} + a \frac{\partial f}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} \right) ds + \int_{t_0}^t b \frac{\partial f}{\partial x} dW_s,$$

where we have used the definitions of L^0 and L^j in (5.15) and (5.16) for $d = m = 1$. We can write this expression in differential form as

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + a \frac{\partial f}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} \right) dt + b \frac{\partial f}{\partial x} dW_t$$

which coincides with the one-dimensional Itô formula presented in 2.17. Thus, the Itô-Taylor expansion can be viewed as a generalized version of Itô's formula.

Now consider the case $f(t, x) = f(x)$ with $d = m = 1$, $a = 1$, and $b = 0$; for simplicity, take $t_0 = 0$. These choices imply that $dX_t = dt$ or, equivalently, $X_t = t$ with $t \in [0, T]$. Note that $L^1 f = 0$ by the choice of b , while $L^0 f = f' \equiv \partial f / \partial x$ since we took f to be independent of t . In general⁷, for a multi-index $\alpha = (0, 0, \dots, 0)$ of length $l(\alpha) = r$, we have $L^\alpha f = f^{(r)}$.

Now take the hierarchical set Γ_r defined in Proposition 5.10 for $r = 1, 2, \dots$, and let $\mathcal{R}(\Gamma_r)$ be its remainder set, computed in 5.20. Applying the Itô-Taylor expansion (5.24) gives

$$\begin{aligned} f(X_t) &= \sum_{\alpha \in \Gamma_r} (L^\alpha f)(X_{t_0}) I_\alpha + \sum_{\alpha \in \mathcal{R}(\Gamma_r)} I_\alpha [L^\alpha f]_{t_0, t} \\ &= f(X_{t_0}) + f'(X_{t_0}) \int_{t_0}^t ds + \dots + f^{(r)}(X_{t_0}) \int_{t_0}^t \dots \int_{t_0}^{s_2} ds_1 \dots ds_r \\ &\quad + \int_{t_0}^t \dots \int_{t_0}^{s_2} f^{(r+1)}(X_{s_1}) ds_1 \dots ds_{r+1} \end{aligned}$$

and since $X_t \equiv t$, we have

$$f(t) = f(t_0) + f'(t_0)(t - t_0) + \dots + f^{(r)} \frac{(t - t_0)^r}{r!} + \int_{t_0}^t \dots \int_{t_0}^{s_2} f^{(r+1)}(s_1) ds_1 \dots ds_{r+1}$$

which is the ordinary Taylor expansion. Thus, the Itô-Taylor expansion generalizes the deterministic Taylor expansion.

As a first illustration of the usefulness of this expansion for numerical methods, consider the hierarchical set $\Gamma_2 = \{\alpha \in \mathcal{M} : l(\alpha) \leq 2\}$ and the function $f(t, x) = x$. Applying the truncated Itô-Taylor expansion (5.24) in the one-dimensional case $d = m = 1$ gives

$$\begin{aligned} X_t &\approx \sum_{\alpha \in \Gamma_2} (L^\alpha X_t)(X_{t_0}) I_\alpha \\ &= X_{t_0} + a(X_{t_0}) I_{(0)} + b(X_{t_0}) I_{(1)} + \left(a(X_{t_0}) a'(X_{t_0}) + \frac{1}{2} b^2(X_{t_0}) a''(X_{t_0}) \right) I_{(0,0)} \\ &\quad + \left(a(X_{t_0}) b'(X_{t_0}) + \frac{1}{2} b^2(X_{t_0}) b''(X_{t_0}) \right) I_{(0,1)} + b(X_{t_0}) a'(X_{t_0}) I_{(1,0)} \\ &\quad + b(X_{t_0}) b'(X_{t_0}) I_{(1,1)} \end{aligned}$$

where we assume that $a = a(x)$ and $b = b(x)$. Taking $t_0 = t_n$, $t = t_{n+1}$, and $\Delta = t_{n+1} - t_n$ gives

$$\begin{aligned} X_{t_{n+1}} &\approx X_{t_n} + a(X_{t_n}) \Delta + b(X_{t_n}) \Delta W + \left(a(X_{t_n}) a'(X_{t_n}) + \frac{1}{2} b^2(X_{t_n}) a''(X_{t_n}) \right) \frac{\Delta^2}{2} \\ &\quad + b(X_{t_n}) b'(X_{t_n}) \frac{(\Delta W^2 - \Delta)}{2} + \left(a(X_{t_0}) b'(X_{t_0}) + \frac{1}{2} b^2(X_{t_0}) b''(X_{t_0}) \right) I_{(0,1)} \\ &\quad + b(X_{t_0}) a'(X_{t_0}) I_{(1,0)} \end{aligned}$$

⁷We assume that $f \in \mathcal{C}^{r+1}(\mathbb{R})$ and denote the r th derivative of f with respect to x by $f^{(r)}$, or by f' if $r = 1$.

with $\Delta W = W_{t_{n+1}} - W_{t_n}$.

As we have seen, the Itô-Taylor expansion allows considerable freedom in choosing the hierarchical set. In the next section, we examine two particular cases of these expansions for two given hierarchical sets and study their respective properties.

5.2 Strong Itô-Taylor methods

We begin by defining strong convergence, which characterizes the methods presented in this section. From now on, we take $t_0 = 0$.

Definition 5.15. Let Y_t^δ be a discrete-time approximation, with maximum step size δ , of the stochastic process X_t . We say that Y_t^δ converges strongly to X_t with order $\gamma > 0$ at time T if there exist $C > 0$ independent of δ and $\delta_0 > 0$ such that

$$\mathbb{E} [|X_T - Y_T^\delta|] \leq C\delta^\gamma$$

for every $0 < \delta < \delta_0$.

In general, we assume a uniform partition of the time interval $[t_0, T]$ and hence have $t_n = t_0 + n\Delta$ (in which case $\delta = \Delta$), and write Y_t^δ as Y_t .

The Euler-Maruyama method

The simplest case is known as the Euler-Maruyama method. In the one-dimensional case, $d = m = 1$, taking the truncated Itô-Taylor expansion (5.24) for the hierarchical set $\mathcal{A} = \{v, (0), (1)\}$ gives

$$Y_{n+1} = Y_n + a(t_n, Y_n)\Delta + b(t_n, Y_n)\Delta W \quad (5.25)$$

where we have taken a uniform partition of the interval $[0, T]$ and $Y_n := Y_{t_n}$. We have also used the earlier notation, where $\Delta = t_{n+1} - t_n = I_{(0)}$ and $\Delta W = W_{t_{n+1}} - W_{t_n} = I_{(1)}$. By the definition of a Wiener process, we know that $\Delta W \sim N(0, \Delta)$. From now on, we use the abbreviations $a_n = a(t_n, Y_n)$ and $b_n = b(t_n, Y_n)$.

More generally, taking $\mathcal{A} = \{v, (0), (1), \dots, (m)\}$ as the hierarchical set, if the stochastic process is d -dimensional and W_t is an m -dimensional Wiener process, then

$$Y_{n+1}^k = Y_n^k + a_n^k \Delta + \sum_{j=1}^m b_n^{k,j} \Delta W^j \quad (5.26)$$

denotes the k th component of the Euler-Maruyama method. Recall that the Wiener increments $\Delta W^j = I_{(j)}$ are independent for different values of $j = 1, \dots, m$.

We next show that, under certain assumptions on the functions $a(t, X_t)$ and $b(t, X_t)$, the Euler-Maruyama method converges strongly with order $\gamma = 0.5$. We consider only the autonomous case where X_0 is a known constant and therefore take $Y_0 = X_0$. Before stating the theorem, we present a preliminary result.

Lemma 5.16. (Gronwall) *Let $\alpha : [0, T] \rightarrow \mathbb{R}_+$ be an integrable function. If there exist $A > 0$ and $B > 0$ such that*

$$0 \leq \alpha(t) \leq A + B \int_0^t \alpha(s) ds$$

for every $t \in [0, T]$, then $\alpha(t) \leq Ae^{Bt}$. In particular, $\alpha(t) \leq Ae^{BT}$.

Theorem 5.17. *Let X_t be the d -dimensional Itô process defined on the interval $[0, T]$ that solves (3.2). Suppose that the functions $a = a(x)$ and $b = b(x)$ satisfy the Lipschitz condition, that is,*

$$|a(x) - a(y)| + |b(x) - b(y)| \leq C_1 |x - y| \quad (5.27)$$

and the linear growth condition

$$|a(x)|^2 + |b(x)|^2 \leq C_2 (1 + |x|^2) \quad (5.28)$$

for any $s, t \in [0, T]$, $x, y \in \mathbb{R}^d$. Then the Euler-Maruyama method Y_n obtained with a uniform partition of step size Δ converges strongly with order $\gamma = 0.5$; that is, there exists $C_3 > 0$ such that

$$\mathbb{E}[|X_T - Y_T|] \leq C_3 \Delta^{1/2}. \quad (5.29)$$

Proof. Let $\{t_n\}_{n=0}^N$ be a uniform partition of the interval $[0, T]$ such that $t_n = n\Delta$ with $n = 0, 1, \dots, N$ and $T = N\Delta$. Define the function

$$Y(t) = \sum_{n=0}^{N-1} 1_{[t_n, t_{n+1}]}(t) Y_n$$

for every $t \in [0, T)$ with $Y(T) = Y_N$. Thus, $Y(t) = Y_n$ for every $t \in [t_n, t_{n+1})$. Define

$$\alpha(s) = \sup_{r \in [0, s]} \mathbb{E}[|Y(r) - X_r|^2]$$

The Euler-Maruyama method (5.25) gives

$$\begin{aligned} Y_{k+1} - Y_k &= a_k \Delta + b_k \Delta W = \int_{t_k}^{t_{k+1}} a_k ds + \int_{t_k}^{t_{k+1}} b_k dW_s \\ &= \int_{t_k}^{t_{k+1}} a(Y(s)) ds + \int_{t_k}^{t_{k+1}} b(Y(s)) dW_s \end{aligned}$$

From this equality, for fixed $t \in [0, T]$ and the index n such that $t \in [t_n, t_{n+1})$, we see that

$$\begin{aligned} Y(t) &= Y_n = Y_0 + \sum_{k=0}^{n-1} (Y_{k+1} - Y_k) \\ &= Y_0 + \sum_{k=0}^{n-1} \left(\int_{t_k}^{t_{k+1}} a(Y(s)) ds + \int_{t_k}^{t_{k+1}} b(Y(s)) dW_s \right) \\ &= Y_0 + \int_0^{t_n} a(Y(s)) ds + \int_0^{t_n} b(Y(s)) dW_s. \end{aligned}$$

On the other hand, writing the original differential equation in integral form (5.21) gives

$$X_t = X_0 + \int_0^t a(X_s)ds + \int_0^t b(X_s)dW_s$$

so, subtracting the two equalities and recalling that we consider the case $Y(0) = Y_0 = X_0$, we obtain

$$\begin{aligned} Y(t) - X_t &= \underbrace{\int_0^{t_n} (a(Y(s)) - a(X_s)) ds}_{I_1} + \underbrace{\int_0^{t_n} (b(Y(s)) - b(X_s)) dW_s}_{I_2} \\ &\quad - \underbrace{\int_{t_n}^t a(X_s) ds}_{I_3} - \underbrace{\int_{t_n}^t b(X_s) dW_s}_{I_4} \end{aligned}$$

since we previously fixed $t \in [t_n, t_{n+1})$ and hence $t \geq t_n$. The Cauchy-Schwarz inequality gives $(I_1 + I_2 - I_3 - I_4)^2 \leq 4(I_1^2 + I_2^2 + I_3^2 + I_4^2)$, so taking the expectation of the squared norm yields

$$\mathbb{E}[|Y(t) - X_t|^2] \leq 4\mathbb{E}[I_1^2 + I_2^2 + I_3^2 + I_4^2].$$

By linearity of expectation, we analyze each term separately. First, recall that, for any function $u \in L^2([0, T])$, the Cauchy-Schwarz inequality implies

$$\left(\int_0^{t_n} u(s) ds \right)^2 \leq t_n \int_0^{t_n} |u(s)|^2 ds$$

therefore, applying this inequality and the Lipschitz condition (5.27) gives

$$\begin{aligned} \mathbb{E}[I_1^2] &= \mathbb{E}\left[\left(\int_0^{t_n} (a(Y(s)) - a(X_s)) ds\right)^2\right] \\ &\leq t_n \mathbb{E}\left[\int_0^{t_n} |a(Y(s)) - a(X_s)|^2 ds\right] \\ &\leq t_n C_1^2 \int_0^{t_n} \mathbb{E}[|Y(s) - X_s|^2] ds \\ &\leq TC_1^2 \int_0^{t_n} \mathbb{E}\left[\sup_{r \in [0, s]} |Y(r) - X_r|^2\right] ds = TC_1^2 \int_0^{t_n} \alpha(s) ds \\ &\leq TC_1^2 \int_0^t \alpha(s) ds. \end{aligned}$$

On the other hand, applying the Itô isometry and again the Lipschitz condition gives

$$\begin{aligned} \mathbb{E}[I_2^2] &= \mathbb{E}\left[\left(\int_0^{t_n} (b(Y(s)) - b(X_s)) dW_s\right)^2\right] \\ &= \int_0^{t_n} \mathbb{E}[|b(Y(s)) - b(X_s)|^2] ds \\ &\leq C_1^2 \int_0^{t_n} \mathbb{E}[|Y(s) - X_s|^2] ds \\ &\leq C_1^2 \int_0^t \alpha(s) ds. \end{aligned}$$

Applying the linear growth condition (5.28) and the Cauchy-Schwarz inequality gives

$$\begin{aligned}
 \mathbb{E} [I_3^2] &= \mathbb{E} \left[\left(\int_{t_n}^t a(X_s) ds \right)^2 \right] \\
 &\leq (t - t_n) \mathbb{E} \left[\int_{t_n}^t |a(X_s)|^2 ds \right] \\
 &\leq \Delta C_2 \mathbb{E} \left[\int_{t_n}^t (1 + |X_s|^2) ds \right] \\
 &\leq C\Delta
 \end{aligned}$$

since

$$\sup_{s \in [0, T]} (1 + |X_s|^2) \leq K$$

by the existence and uniqueness theorem 3.3, where we have taken $C = KC_2T$. Finally, an analogous argument using the Itô isometry gives

$$\begin{aligned}
 \mathbb{E} [I_4^2] &= \mathbb{E} \left[\left(\int_{t_n}^t b(X_s) dW_s \right)^2 \right] \\
 &= \int_{t_n}^t \mathbb{E} [|b(X_s)|^2] ds \\
 &\leq C_2 \mathbb{E} \left[\int_{t_n}^t (1 + |X_s|^2) ds \right] \\
 &\leq C\Delta
 \end{aligned}$$

Thus, we finally obtain

$$0 \leq \mathbb{E} [|Y(t) - X_t|^2] \leq \alpha(t) \leq 4(T+1)C_1^2 \int_0^t \alpha(s) ds + 8C\Delta$$

and applying Gronwall's lemma 5.16 with $B = 4(T+1)C_1^2$ and $A = 8C\Delta$ gives

$$\alpha(t) \leq C_3^2 \Delta$$

where $C_3^2 = 8Ce^{BT}$ for every $t \in [0, T]$. Since any random variable satisfies $\mathbb{E} [Z] \leq \sqrt{\mathbb{E} [Z^2]}$, we have

$$\begin{aligned}
 \max_{n=0, \dots, N} \mathbb{E} [|Y_n - X_{t_n}|] &\leq \sup_{t \in [0, T]} \mathbb{E} [|Y(t) - X_t|] \\
 &\leq \sup_{t \in [0, T]} \sqrt{\mathbb{E} [|Y(t) - X_t|^2]} \\
 &= \sqrt{\alpha(t)} \\
 &\leq C_3 \Delta^{1/2}.
 \end{aligned}$$

In particular, this holds for $n = N$ and therefore $t_n = T$, yielding expression (5.29). $\square$

General case

We define the strong Itô-Taylor method of order $\gamma = 0.5, 1, 1.5, \dots$ in the general case by taking the set of multi-indices

$$\mathcal{A}_\gamma = \left\{ \alpha \in \mathcal{M} : l(\alpha) + n(\alpha) \leq 2\gamma \text{ or } l(\alpha) = n(\alpha) = \gamma + \frac{1}{2} \right\}. \quad (5.30)$$

Proposition 5.18. *The set $\mathcal{A}_\gamma$ described in (5.30) is a hierarchical set.*

Proof. Rewriting (5.30) as

$$\mathcal{A}_\gamma = \Lambda_{2\gamma} \cup \mathcal{B}$$

where $\Lambda_{2\gamma}$ is defined in 5.11 and $\mathcal{B} = \{\alpha \in \mathcal{M} : l(\alpha) = n(\alpha) = \gamma + 1/2\}$, we have

$$\sup_{\alpha \in \mathcal{B}} l(\alpha) = \gamma + \frac{1}{2} < \infty$$

and $\sup_{\alpha \in \Lambda_{2\gamma}} l(\alpha) < \infty$ because $\Lambda_{2\gamma}$ is a hierarchical set. Thus, $\sup_{\alpha \in \mathcal{A}_\gamma} l(\alpha) < \infty$. On the other hand, given $\alpha \in \mathcal{A}_\gamma$, we have

1. If $\alpha \in \Lambda_{2\gamma}$, then $-\alpha \in \Lambda_{2\gamma} \subset \mathcal{A}_\gamma$ because $\Lambda_{2\gamma}$ is a hierarchical set.
2. If $\alpha \in \mathcal{B}$, then $l(-\alpha) = \gamma - 1/2$ and $n(-\alpha) \leq \gamma + 1/2$. That is, $l(-\alpha) + n(-\alpha) \leq 2\gamma$ and therefore $-\alpha \in \Lambda_{2\gamma} \subset \mathcal{A}_\gamma$.

□

Note that, if $\gamma \in \mathbb{N}$, then $\mathcal{A}_\gamma = \Lambda_{2\gamma}$. We also define the approximation coefficients as

$$L^\alpha f(t, x) = L^{\alpha^-} b^j(t, x) \quad (5.31)$$

for every $(t, x) \in \mathbb{R} \times \mathbb{R}^d$ and $\alpha = (j_1, j_2, \dots, j_l) \in \mathcal{M}$, where we denote the function $a(t, x)$ by $b^0(t, x)$.

Thus, given a time discretization $t_0 < t_1 < \dots < t_n < \dots$, we define the strong method of order γ as the truncated Itô-Taylor expansion

$$Y_{n+1} = \sum_{\alpha \in \mathcal{A}_\gamma} (L^\alpha f)(t_n, Y_n) I_\alpha \quad (5.32)$$

where $\mathcal{A}_\gamma$ is the hierarchical set in (5.11) and I_α are the multiple Itô integrals from Definition 5.3, computed over $[t_n, t_{n+1}]$. In particular, taking $\mathcal{A}_{0.5} = \{v, (0), (1)\}$ gives the Euler-Maruyama method in (5.25) with order $\gamma = 0.5$. An important property of these strong methods is that the paths of the generated approximations are continuous and exhibit irregular behavior, as does the Wiener process. Analogously to Theorem 5.17, a convergence result exists for the general case, which we state without proof.

Theorem 5.19. *Let Y^δ be the Itô-Taylor approximation associated with the hierarchical set $\mathcal{A}_\gamma$ in (5.11), defined in (5.32) for a time discretization with $0 < \delta < 1$ and $\gamma = 0.5, 1, 1.5, \dots$. Suppose that the approximation coefficients defined in (5.31) satisfy the Lipschitz condition*

$$|L^\alpha f(t, x) - L^\alpha f(t, y)| \leq C_1 |x - y|$$

for every $\alpha \in \mathcal{A}_\gamma$, $t \in [0, T]$, $x, y \in \mathbb{R}^d$; $L^{-\alpha} f \in \mathcal{C}^{1,2}$ and $L^\alpha f \in \mathcal{H}_\alpha$ for every $\alpha \in \mathcal{A}_\gamma \cup \mathcal{R}(\mathcal{A}_\gamma)$; and the coefficients satisfy the growth condition

$$|L^\alpha f(t, x)| \leq C_2 (1 + |x|)$$

for every $\alpha \in \mathcal{A}_\gamma \cup \mathcal{R}(\mathcal{A}_\gamma)$, $t \in [0, T]$, and $x \in \mathbb{R}^d$. If we take $Y_0^\delta = X_0$, then

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t - Y_t^\delta|^2 \right] \leq C_3 \delta^\gamma$$

where the constants are independent of δ .

Thus, strong Itô-Taylor approximations ensure strong convergence not only at the final time T , but uniformly over the entire interval $[0, T]$.

The Euler-Maruyama method introduced above is the strong method with the lowest convergence order; in practice, its simplicity limits its usefulness. We therefore consider several examples derived from the general definition that ensure a higher convergence order, although they are computationally more expensive than the Euler-Maruyama method.

The Milstein method

The Milstein method is the strong Itô-Taylor method of order $\gamma = 1$. For the one-dimensional autonomous case, we therefore take $\mathcal{A}_1 = \{v, (0), (1), (1, 1)\}$, since if $\alpha = (j_1, j_2)$ and any of its components is zero, then $l(\alpha) + n(\alpha) > 2$; furthermore, $l(\alpha) = n(\alpha) = 3/2$ does not hold for any $\alpha \in \mathcal{M}$. It therefore suffices to add the term $\alpha = (1, 1)$ to the Euler-Maruyama method in (5.25), giving

$$Y_{n+1} = Y_n + a_n \Delta + b_n \Delta W + \frac{1}{2} b_n b'_n (\Delta W^2 - \Delta) \quad (5.33)$$

since, by (5.16) and (5.31),

$$L^{(1,1)} f = L^1 b(t, x) = b \frac{\partial b}{\partial x} := b(x) b'(x)$$

and finally, $I_{(1,1)}$ was obtained in (5.9) by applying Itô's formula. Extending expression (5.33) to the general multidimensional case gives

$$Y_{n+1}^k = Y_n^k + a_n^k \Delta + \sum_{j=1}^m b^{k,j} \Delta W^j + \sum_{j_1, j_2=1}^m L^{j_1} b^{k, j_2} I_{(j_1, j_2)} \quad (5.34)$$

as the k th component of the Milstein method. To simplify notation, $L^{j_1} b^{k, j_2}$ denotes the function $L^{j_1} b^{k, j_2}$ evaluated at (t_n, Y_n) , just as $b^{k, j}$ denotes $b^{k, j}(t_n, Y_n)$. Clearly, this expression is analogous to the multidimensional Euler-Maruyama method (5.26), with the addition of the terms for $\alpha = (j_1, j_2)$ where neither component is zero.

The Taylor method of strong order 1.5

As a final example of the general case, we consider the Itô-Taylor method of strong order $\gamma = 1.5$. Beginning with the one-dimensional autonomous case, we must take $\mathcal{A}_{1.5} = \{v, (0), (1), (0, 0), (1, 0), (0, 1), (1, 1), (1, 1, 1)\}$. Thus, the strong Itô-Taylor approximation of order $\gamma = 1.5$ is

$$\begin{aligned} Y_{n+1} = & Y_n + a_n \Delta + b_n \Delta W + \frac{1}{2} b_n b'_n (\Delta W^2 - \Delta) + a'_n b_n \Delta Z \\ & + \frac{1}{2} \left(a_n a'_n + \frac{1}{2} b_n^2 a''_n \right) \Delta^2 + \left(a_n b'_n + \frac{1}{2} b_n^2 b''_n \right) (\Delta W \Delta - \Delta Z) \\ & + \frac{1}{2} b_n (b_n b''_n + (b'_n)^2) \left(\frac{1}{3} \Delta W^2 - \Delta \right) \Delta W \end{aligned} \quad (5.35)$$

which results from adding the terms $(1, 0), (0, 1), (1, 1), (1, 1, 1)$ to the Milstein method. Note that

$$L^{(1,0)} f = L^1 b^0 = L^1 a = b(x) a'(x)$$

with $I_{(1,0)} = \Delta Z$ obtained in Proposition 5.7. On the other hand,

$$L^{(0,0)} f = L^0 b^0 = L^0 a = a(x) a'(x) + \frac{1}{2} b^2(x) a''(x)$$

and $I_{(0,0)} = \Delta^2/2$, generalizing the result obtained in (5.9). Furthermore,

$$L^{(0,1)} f = L^0 b = a(x) b'(x) + \frac{1}{2} b^2(x) b''(x)$$

since (5.12) gives $I_{(0,1)} = \Delta W \Delta - \Delta Z$. Finally, observe that

$$L^{(1,1,1)} f = L^{(1,1)} b = L^1 (b(x) b'(x)) = b(x) ((b'(x))^2 + b(x) b''(x))$$

where $I_{(1,1,1)}$ can be computed from (5.14), since

$$\begin{aligned} I_{(1,1,1)} &= \frac{1}{3} (I_{(1)} I_{(1,1)} - I_{(0)} I_{(1)}) \\ &= \frac{1}{3} \left(\frac{1}{2} I_{(1)} (I_{(1)} I_{(1)} - I_{(0)}) - I_{(0)} I_{(1)} \right) \\ &= (\Delta W^2/3 - \Delta) \Delta W/2. \end{aligned}$$

Note that, in the deterministic case⁸, we recover the Taylor method of order 2. In the general (nonautonomous) multidimensional case, the Taylor method of order 1.5 is determined by

$$\begin{aligned} Y_{n+1}^k = & Y_n^k + a_n^k \Delta + \frac{1}{2} L^0 a^k \Delta^2 \\ & + \sum_{j=1}^m (b^{k,j} \Delta W^j + L^0 b^{k,j} I_{(0,j)} + L^j a^k I_{(j,0)}) \\ & + \sum_{j_1, j_2=1}^m L^{j_1} b^{k, j_2} I_{(j_1, j_2)} + \sum_{j_1, j_2, j_3=1}^m L^{(j_1, j_2)} b^{k, j_3} I_{(j_1, j_2, j_3)} \end{aligned} \quad (5.36)$$

To complete this section, we study the families of θ -methods⁹ defined from the explicit

⁸That is, $b(x) = 0$.

⁹Or implicit methods.

methods presented here, giving a brief introduction for the Euler-Maruyama, Milstein, and strong Taylor methods of order 1.5.

5.2.1 Strong implicit Itô-Taylor methods

Starting from the one-dimensional Euler-Maruyama method in (5.25), we can define the family of implicit methods, or family of θ -methods, for Euler-Maruyama as

$$Y_{n+1} = Y_n + (\theta a(t_{n+1}, Y_{n+1}) + (1 - \theta)a_n) \Delta + b_n \Delta W \quad (5.37)$$

Note that, if $\theta = 0$, we recover the explicit method (5.25), while $\theta = 1$ gives the fully implicit Euler method. For the general multidimensional case, we similarly have

$$Y_{n+1}^k = Y_n^k + (\theta_k a^k(t_{n+1}, Y_{n+1}) + (1 - \theta)a_n^k) \Delta + \sum_{j=1}^m b^{k,j} \Delta W^j \quad (5.38)$$

as the k th component of the family of θ -methods for Euler-Maruyama.

The general case for the Milstein method is analogous, giving

$$Y_{n+1}^k = Y_n^k + (\theta_k a^k(t_{n+1}, Y_{n+1}) + (1 - \theta)a_n^k) \Delta + \sum_{j=1}^m b^{k,j} \Delta W^j \quad (5.39)$$

$$+ \sum_{j_1, j_2=1}^m L^{j_1} b^{k, j_2} I_{(j_1, j_2)} \quad (5.40)$$

Note that we take the convex linear combination only in the terms that are not multiplied by a Wiener increment. This is because Wiener increments are unbounded over a finite time interval, which can cause problems in defining the methods.

Finally, we study the Itô-Taylor method of strong order $\gamma = 1.5$. In the multidimensional case, we have the family of θ -methods

$$\begin{aligned} Y_{n+1}^k = & Y_n^k + (\theta_k a^k(t_{n+1}, Y_{n+1}) + (1 - \theta_k)a_n^k) \Delta \\ & + \left(\frac{1}{2} - \theta_k\right) (\eta_k L^0 a^k(t_{n+1}, Y_{n+1}) + (1 - \eta_k)L^0 a_n^k) \Delta^2 \\ & + \sum_{j=1}^m (b^{k,j} \Delta W^j + L^0 b^{k,j} I_{(0,j)} + L^j a_n^k (I_{(j,0)} - \theta_k \Delta W^j \Delta)) \\ & + \sum_{j_1, j_2=1}^m L^{j_1} b^{k, j_2} I_{(j_1, j_2)} + \sum_{j_1, j_2, j_3=1}^m L^{(j_1, j_2)} b^{k, j_3} I_{(j_1, j_2, j_3)} \end{aligned} \quad (5.41)$$

where $\theta_k, \eta_k \in [0, 1]$.

These families of implicit methods will become relevant later, when we study the linear stability of numerical methods, analogously to the existing theory for numerical methods for ordinary differential equations.

5.3 Weak Itô-Taylor methods

Before studying weak methods, we introduce the concept of weak convergence. Let $\mathcal{C}^l(\mathbb{R}^d, \mathbb{R})$ be the space of l -times continuously differentiable functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$. We denote by $\mathcal{C}_P^l(\mathbb{R}^d, \mathbb{R})$ the subspace consisting of $f \in \mathcal{C}^l(\mathbb{R}^d, \mathbb{R})$ whose partial derivatives up to order l all satisfy a polynomial growth condition; that is, there exist constants $C > 0$ and $r \in \{1, 2, \dots\}$ such that

$$|\partial^k f(x)| \leq C(1 + |x|^{2r})$$

for every $x \in \mathbb{R}^d$, where $\partial^k f(x)$ denotes any partial derivative of order $k \leq l$.

Definition 5.20. Let Y_t^δ be a discrete-time approximation, with maximum step size δ , of the stochastic process X_t . We say that Y_t^δ converges weakly with order $\beta > 0$ at time T if, for every $g \in \mathcal{C}_P^l(\mathbb{R}^d, \mathbb{R})$, there exist $C > 0$ independent of δ and $\delta_0 > 0$ such that

$$|\mathbb{E}[g(X_T)] - \mathbb{E}[g(Y_T^\delta)]| \leq C\delta^\beta$$

for every $0 < \delta < \delta_0$.

Remark 5.21. Strong convergence implies convergence of paths, whereas weak convergence implies convergence of moments (if $g(x) = x^k$) or probabilities (if $g(x) = 1_{[a,b]}(x)$). In general, strong convergence of order γ implies weak convergence of order at least γ .

Here, we begin with the general case.

General case

Analogously to strong Itô-Taylor methods, we can give a general expression for the case of weak convergence. Taking the hierarchical set Γ_β defined in 5.10 and applying the truncated Itô-Taylor expansion (5.24), we define Itô-Taylor methods with weak convergence order β as

$$Y_{n+1} = \sum_{\alpha \in \Gamma_\beta} (L^\alpha f)(t_n, Y_n) I_\alpha \quad (5.42)$$

where the coefficients $L^\alpha f$ are defined in (5.31) and I_α denotes the multiple Itô integral of the unit function associated with the multi-index α over the interval $[t_n, t_{n+1}]$. As before, a theorem ensures weak convergence of these methods with order β . We state the theorem only for the autonomous case, that is, for $a = a(x)$ and $b = b(x)$.

Theorem 5.22. Let Y^δ be the weak Itô-Taylor approximation defined in (5.42) for a time discretization with $0 < \delta < 1$ and $\beta \in \{1, 2, \dots\}$. Suppose that $a = a(x)$ and $b = b(x)$ are Lipschitz continuous functions whose components a^k and $b^{k,j}$ belong to $\mathcal{C}_P^{2(\beta+1)}(\mathbb{R}^d, \mathbb{R})$ for any $k, j = 1, 2, \dots, d$. Suppose also that $L^\alpha f$, where $f(t, x) = x$, satisfies the linear growth condition

$$|L^\alpha f| \leq C(1 + |x|) \quad (5.43)$$

for every $x \in \mathbb{R}^d$, $t \in [0, T]$, and $\alpha \in \Gamma_\beta$. Then, for each $g \in \mathcal{C}_P^{2(\beta+1)}(\mathbb{R}^d, \mathbb{R})$, there exists $C_g > 0$ independent of δ such that

$$|\mathbb{E}[g(X_T)] - \mathbb{E}[g(Y_T^\delta)]| \leq C_g \delta^\beta, \quad (5.44)$$

that is, Y^δ converges weakly with order β to X at time T as $\delta \rightarrow 0$.

Remark 5.23. In particular, polynomials of degree at most $2(\beta + 1)$ belong to $\mathcal{C}_P^{2(\beta+1)}$. Thus, (5.44) gives weak convergence of the moments of the approximation up to that order, as already noted in 5.21.

We next consider examples of methods with a specified weak convergence order.

The Euler-Maruyama method

The Euler-Maruyama method introduced in (5.25) is also the simplest Itô-Taylor method with weak convergence order $\beta = 1$. In the one-dimensional case, this Euler method corresponds to the truncated Itô-Taylor expansion for the hierarchical set $\Gamma_1 = \{v, (0), (1)\}$, which coincides with the hierarchical set $\mathcal{A}_{0.5}$ of the strong method.

Remark 5.24. The weak order of these methods is not reduced if we replace ΔW by another random variable with the same moments. For example, in (5.25), we can replace ΔW by a random variable $\Delta \hat{W}$ such that

$$\mathbb{P}(\Delta \hat{W} = \pm \sqrt{\Delta}) = \frac{1}{2} \quad (5.45)$$

Theorem 5.22 shows that weak convergence of order $\beta = 1$ for the Euler-Maruyama method requires the coefficients to be 4 times continuously differentiable. However, a result specific to this method allows this condition to be relaxed.

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}^l$ is said to be Hölder continuous with exponent $\nu \in (0, 1)$ if there exists a constant $C > 0$ such that

$$|f(x) - f(y)| \leq C|x - y|^\nu$$

for every $x, y \in \mathbb{R}^d$. We then define $\mathcal{H}_T^{(l)}$ for $l \in \mathcal{L} := (0, 1) \cup (1, 2) \cup (2, 3)$ as the set of functions $f : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ whose partial derivatives are Hölder continuous with exponent $l - [l]$ on x and exponent $(l - 2r - s)/2$ on t , where r is the order of differentiation with respect to t and s the order with respect to x , satisfying $2r + s \leq l$. In other words, there exists $C_1 > 0$ such that, for every $x, x' \in \mathbb{R}^d$,

$$|\partial_t^r \partial_x^s f(t, x) - \partial_t^r \partial_x^s f(t, x')| \leq C_1 |x - x'|^{l - [l]},$$

and there exists $C_2 > 0$ such that, for every $t, t' \in [0, T]$,

$$|\partial_t^r \partial_x^s f(t, x) - \partial_t^r \partial_x^s f(t', x)| \leq C_2 |t - t'|^{l - 2r - s}$$

where ∂_t^r denotes the partial derivative of order r with respect to t , and ∂_x^s any partial derivative of order s with respect to $x = (x_1, \dots, x_d)$. If the function f is independent of t , we denote this space by $\mathcal{H}^{(l)}$.

For example, if $l = 1/2$, then $2r + s \leq l$ with $r, s \in \mathbb{N} \cup \{0\}$ holds only if $r = s = 0$. Thus, $\mathcal{H}_T^{(1/2)}$ is the space of functions $u : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$|u(t, x) - u(t, x')| \leq C_1 |x - x'|^{1/2}$$

for any $x, x' \in \mathbb{R}$ and

$$|u(t, x) - u(t', x)| \leq C_2 |t - t'|^{1/4}$$

for every $t, t' \in [0, T]$.

On the other hand, if $l \in (2, 3)$, then $2r + s \leq l$ if $(r, s) \in \{(0, 0), (0, 1), (1, 0), (0, 2)\}$. Thus, in the one-dimensional case, $\mathcal{H}_T^{(l)}$ consists of functions $u : \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$\begin{aligned} |u(t, x) - u(t, x')| &\leq C_1 |x - x'|^{l-[l]} \\ \left| \frac{\partial u}{\partial x^j}(t, x) - \frac{\partial u}{\partial x^j}(t, x') \right| &\leq C_2 |x - x'|^{l-[l]} \\ \left| \frac{\partial u}{\partial t}(t, x) - \frac{\partial u}{\partial t}(t', x) \right| &\leq C_3 |t - t'|^{l-[l]} \\ \left| \frac{\partial^2 u}{\partial x^j \partial x^i}(t, x) - \frac{\partial^2 u}{\partial x^j \partial x^i}(t, x') \right| &\leq C_4 |x - x'|^{l-[l]} \end{aligned} \quad (5.46)$$

In the autonomous case, $\mathcal{H}^{(l)}$ with $l \in (2, 3)$ consists of twice differentiable functions with Hölder continuity. Having introduced these concepts, we can state the theorem ensuring weak convergence of the Euler-Maruyama method. We give it without proof; the proof can be found in [5].

Theorem 5.25. *Let X_t be a d -dimensional Itô process (we assume $m = 1$) defined on the interval $[0, T]$ such that all moments of X_T exist and*

$$B(t, x)\zeta \cdot \zeta \geq \lambda |\zeta|^2$$

for every $t \in [0, T]$ and $x, \zeta \in \mathbb{R}^d$, with constant $\lambda > 0$, where $B(t, x) = b(t, x)b(t, x)^t$. Suppose that

$$a, B \in \mathcal{H}_T^{(l)}$$

for some $l \in \mathcal{L}$. Let Y^δ also be a discrete approximation of X_t and let $g \in \mathcal{H}^{(l+2)}$ be a function. Then

$$|\mathbb{E}[g(X_T)] - \mathbb{E}[g(Y_T^\delta)]| \leq C\delta^{\chi(l)}$$

where

$$\chi(l) = \begin{cases} l/2 & , l \in (0, 1) \\ 1/(3-l) & , l \in (1, 2) \\ 1 & , l \in (2, 3) \end{cases}$$

with C independent of δ .

Thus, by the preceding theorem, to obtain weak convergence $\beta = 1$ for the Euler-Maruyama method, we must require $a(t, x)$ and $b(t, x)$ to satisfy the conditions in (5.46). That is, they must be twice continuously differentiable in x and Hölder continuous on x with exponent $l - [l]$. Recall that every Lipschitz continuous function is Hölder continuous. Thus, in the autonomous case, expression (5.46) shows that, if the functions $a = a(x)$ and $b = b(x)$ and their derivatives up to second order are Lipschitz continuous, then the Euler-Maruyama method converges with weak order $\beta = 1$. Unlike the conditions of Theorem 5.22, the coefficients need not be 4 times continuously differentiable. In particular, taking $\delta = \Delta$ gives the desired result.

The Taylor method of weak order 2

Starting with the autonomous one-dimensional case, we take the hierarchical set $\Gamma_2 = \{\alpha \in \mathcal{M} : l(\alpha) \leq 2\} = \{v, (0), (1), (0, 0), (1, 0), (0, 1), (1, 1)\}$. As we can see, this method has more terms than the Milstein method (and therefore has strong order 1), but fewer than the Taylor method of strong order 1.5 (so it does not have strong order 1.5). Thus, from expression (5.42), the Itô-Taylor method of weak order 2 is given by

$$\begin{aligned} Y_{n+1} = & Y_n + a_n \Delta + b_n \Delta W + \frac{1}{2} b_n b'_n (\Delta W^2 - \Delta) + a'_n b \Delta Z \\ & + \frac{1}{2} \left(a_n a'_n + \frac{1}{2} b_n^2 a''_n \right) \Delta^2 + \left(a_n b'_n + \frac{1}{2} b_n^2 b''_n \right) (\Delta W \Delta - \Delta Z) \end{aligned} \quad (5.47)$$

which clearly results from removing the final term in (5.35).

Remark 5.26. Analogously to Remark 5.24, in (5.47) we can replace ΔW by a random variable $\Delta \hat{W}$ such that

$$\mathbb{P}(\Delta \hat{W} = \pm \sqrt{3\Delta}) = \frac{1}{6}, \quad \mathbb{P}(\Delta \hat{W} = 0) = \frac{2}{3} \quad (5.48)$$

For the multidimensional (not necessarily autonomous) case, the Taylor method of weak order 2 can be generalized, with

$$\begin{aligned} Y_{n+1}^k = & Y_n^k + a_n^k \Delta + \frac{1}{2} L^0 a^k \Delta^2 + \sum_{j_1, j_2=1}^m L^{j_1} b^{k, j_2} I_{(j_1, j_2)} \\ & + \sum_{j=1}^m (b^{k, j} \Delta W^j + L^0 b^{k, j} I_{(0, j)} + L^j a^k I_{(j, 0)}) \end{aligned} \quad (5.49)$$

determining the k th component of the method. Note that this expression corresponds to the general strong method of order 1.5 in (5.36) with the sum over the integrals $I_{(j_1, j_2, j_3)}$ removed, that is, the terms corresponding to multi-indices of length 3 that do not belong to Γ_2 .

In relation to Remark 5.21, the Itô-Taylor method of strong order 1.5 in (5.35) contains all terms of the weak order 2 method (5.47). Therefore, the strong method of order 1.5 also has weak order at least 2.

5.3.1 Weak implicit Itô-Taylor methods

Just as we defined families of θ -methods for strong methods, we now present families of implicit methods for the weak methods studied in this section.

Following Remark 5.24, we can write the implicit Euler-Maruyama method in the multidimensional case as

$$Y_{n+1}^k = Y_n^k + a^k(t_{n+1}, Y_{n+1}) \Delta + \sum_{j=1}^m b_n^{k, j} \Delta \hat{W}^j \quad (5.50)$$

where $\Delta\hat{W}^j$ is a random variable given by

$$\mathbb{P}\left(\Delta\hat{W}^j = \pm\sqrt{\Delta}\right) = \frac{1}{2}$$

for every $j = 1, 2, \dots, d$ and $n = 1, 2, \dots$. Thus, we define the family of θ -methods for Euler-Maruyama as

$$Y_{n+1}^k = Y_n^k + (\theta_k a^k(t_{n+1}, Y_{n+1}) + (1 - \theta) a_n^k) \Delta + \sum_{j=1}^m b^{k,j} \Delta\hat{W}^j \quad (5.51)$$

where k denotes the k th component of the method.

On the other hand, for the autonomous case, the Itô-Taylor method of weak order 2 gives the family of θ -methods

$$\begin{aligned} Y_{n+1} &= Y_n + a(Y_{n+1}) + b_n \Delta\hat{W} + \frac{1}{2} b_n b'_n (\Delta\hat{W}^2 - \Delta) \\ &\quad - \frac{1}{2} \left(a(Y_{n+1}) a'(Y_{n+1}) + \frac{1}{2} b^2(Y_{n+1}) a''(Y_{n+1}) \right) \Delta^2 \\ &\quad + \frac{1}{2} \left(-a'_n b_n + a_n b'_n + \frac{1}{2} b''_n b_n^2 \right) \Delta\hat{W} \Delta \end{aligned} \quad (5.52)$$

in the one-dimensional case, where $\Delta\hat{W}$ is defined as in 5.26. For the general multidimensional case, we construct the family of θ -methods given by

$$\begin{aligned} Y_{n+1} &= Y_n + (\theta a(Y_{n+1}) + (1 - \theta) a_n) \\ &\quad + \frac{1}{2} \sum_{j_1, j_2=1}^m L^{j_1} b^{j_2} \left(\Delta\hat{W}^{j_1} \Delta\hat{W}^{j_2} + V_{j_1, j_2} \right) \\ &\quad + \sum_{j=1}^m \left(b_n^j + \frac{1}{2} (L^0 b^j + (1 - 2\theta) L^j a) \Delta \right) \Delta\hat{W}^j \\ &\quad + \frac{1}{2} (1 - 2\theta) (\eta L^0 a + (1 - \eta) L^0 a(t_{n+1}, Y_{n+1})) \Delta^2 \end{aligned} \quad (5.53)$$

where $\theta, \eta \in [0, 1]$, $\Delta\hat{W}$ is defined in 5.26, and V_{j_1, j_2} is a random variable such that

$$\mathbb{P}(V_{j_1, j_2} = \pm\Delta) = \frac{1}{2}$$

for $j_2 = 1, \dots, j_1 - 1$, $V_{j_1, j_1} = -\Delta$, and $V_{j_1, j_2} = -V_{j_2, j_1}$.

5.4 Linear stability of Itô-Taylor methods

In this section, we introduce linear stability for Itô-Taylor methods [12], [13]. As in the study of numerical methods for ordinary differential equations, we must choose a stochastic test equation for the analysis. We consider only the one-dimensional case $d = m = 1$. Following [14], consider the linear stochastic differential equation

$$dX_t = \lambda X_t dt + \mu X_t dW_t \quad (5.54)$$

defined on $t \in [0, T]$ with constant initial condition X_0 . Although the analysis can be extended to the complex case, we assume that λ and μ are real numbers such that $\lambda < 0$. Differential equation (5.54) has the exact solution

$$X_t = X_0 \exp \left(\left(\lambda - \frac{1}{2} \mu^2 \right) t + \mu W_t \right). \quad (5.55)$$

Observe that, if $X_0 = 0$, the unique solution is the trivial solution $X_t = 0$. We want to determine the values of λ and μ for which the condition

$$\lim_{t \rightarrow \infty} \mathbb{E} [|X_t|^2] = 0; \quad (5.56)$$

holds; in this case, we speak of linear mean-square stability (or stability of the second moment).

Lemma 5.27. *Condition (5.56) holds for stochastic differential equation (5.54) if and only if $\lambda + \frac{1}{2} \mu^2 < 0$.*

Proof. Taking $Y_t = \mathbb{E} [|X_t|^2]$, we find that Y_t solves the ordinary differential equation

$$dY_t = (2\lambda + \mu^2) Y_t dt$$

with initial condition $Y_0 = 1$. Since the solution of this differential equation is $Y_t = \exp((2\lambda + \mu^2)t)$, we have

$$\lim_{t \rightarrow \infty} Y_t = \lim_{t \rightarrow \infty} e^{(2\lambda + \mu^2)t} = 0$$

if and only if $2\lambda + \mu^2 < 0$. □

We aim to perform a similar analysis for the numerical methods¹⁰ introduced above. That is, if Y_n is a discrete approximation of X_t as a solution of (5.55), we want to determine under what conditions

$$\lim_{n \rightarrow \infty} \mathbb{E} [|Y_n|^2] = 0 \quad (5.57)$$

Define $\bar{Y}_n = \mathbb{E} [|Y_n|^2]$. Applying a numerical method to (5.54) and taking the mean square allows us to simplify¹¹ the resulting expression to

$$\bar{Y}_{n+1} = R(h, k) \bar{Y}_n \quad (5.58)$$

where $h = \Delta\lambda$ and $k = \mu^2\Delta$. We call this function the stability function of the numerical method. Thus, (5.57) holds if

$$|R(h, k)| < 1. \quad (5.59)$$

Definition 5.28. We say that a numerical method M applied with step size Δ is *MS-stable* (or mean-square stable) for the set of values (h, k) for which (5.59) holds. We define the *MS-stability domain* of a method as the set $\mathcal{S}$ given by

$$\mathcal{S}_M = \{(h, k) : |R(h, k)| < 1\}. \quad (5.60)$$

We next study the linear stability regions of some methods from the preceding sections. We compare $\mathcal{S}_M$ with the stability region of equation $\mathcal{S} = \{(h, k) : 2h + k < 0\}$.

¹⁰We consider only uniformly spaced time discretizations.

¹¹This always holds for stochastic differential equations of the form (5.54).

Euler-Maruyama

From the Euler-Maruyama method in (5.25), since $a(t, x) = \lambda x$ and $b(t, x) = \mu x$, we have

$$Y_{n+1} = Y_n + \lambda Y_n \Delta + \mu Y_n \Delta W$$

so, taking the expectation of the square,

$$\mathbb{E} [|Y_{n+1}|^2] = \mathbb{E} [|Y_n + \lambda Y_n \Delta + \mu Y_n \Delta W|^2] = \mathbb{E} [|Y_n|^2] (1 + 2\lambda\Delta + \mu^2\Delta + \lambda^2\Delta^2)$$

and simplifying and applying the definition of $\bar{Y}_n$ gives

$$R(h, k) = |1 + h|^2 + |k| \quad (5.61)$$

whose MS -stability region can be plotted as shown in Figure 5.1.

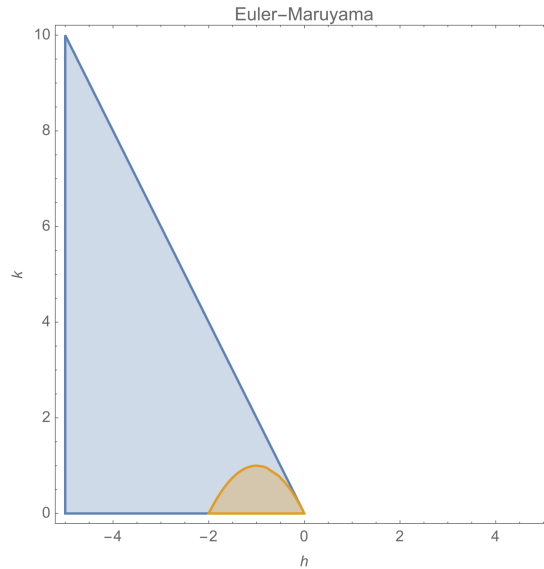


Figure 5.1: MS -stability region of the explicit Euler-Maruyama method. The stability region of the equation is shown in blue, and that of the numerical method in orange.

We can apply the same procedure to the family of Euler-Maruyama θ -methods in (5.37). Taking the expectation of the square gives

$$\bar{Y}_{n+1} |1 - \lambda\Delta\theta|^2 = \bar{Y}_n (|1 + (1 - \theta)\lambda\Delta|^2 + |k|)$$

so

$$R(h, k) = \frac{|1 + (1 - \theta)h|^2 + |k|}{|1 - h\theta|^2} \quad (5.62)$$

For example, Figure 5.2 shows the stability region of the θ -methods (5.37) for different values of θ .

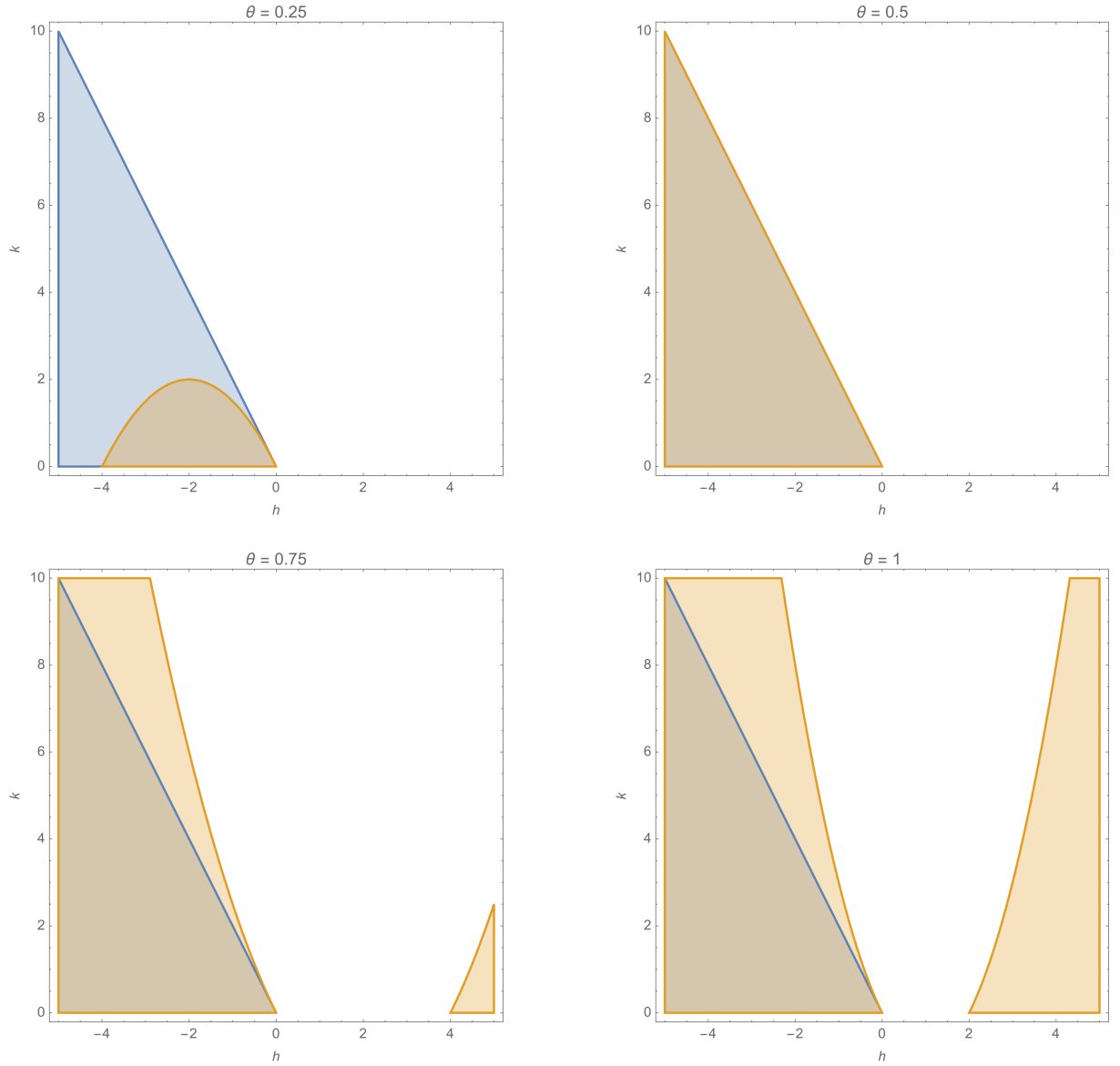


Figure 5.2: *MS*-stability region for different values of θ in the family of Euler-Maruyama θ -methods.

As expected, increasing θ enlarges the stability region of the methods, with the stability regions of the method and the equation coinciding when $\theta = \frac{1}{2}$. However, for $\theta > \frac{1}{2}$, the method's stability region is larger than that of the equation, implying that, for certain values of λ and μ , the method is stable even though the theoretical solution is unstable.

Milstein

Applying the Milstein method described in (5.33) to stochastic differential equation (5.54) gives

$$Y_{n+1} = Y_n + \lambda Y_n \Delta + \mu Y_n \Delta W + \frac{1}{2} \mu^2 Y_n (\Delta W^2 - \Delta)$$

and taking the expectation of the square therefore gives

$$R(h, k) = |1 + h|^2 + |k| + \frac{1}{2}|k|^2 \quad (5.63)$$

whose stability domain is shown in Figure 5.3.

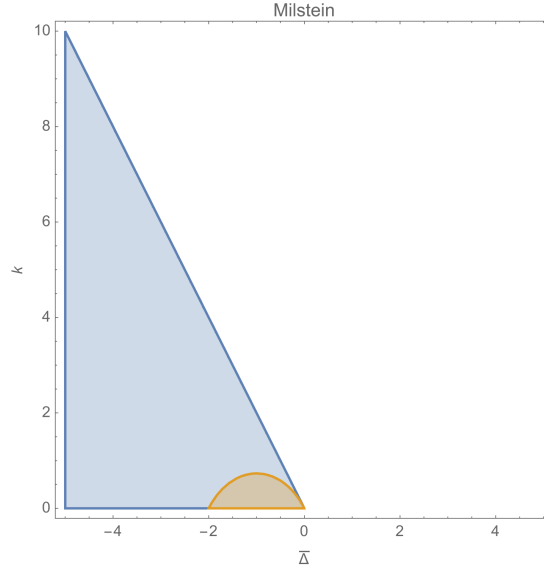


Figure 5.3: *MS*-stability region of the Milstein method.

Since $R(h, k)$ is the stability function of the explicit Euler method with an additional quadratic term in k , the smaller stability region of the Milstein method is to be expected.

Taylor method of strong order 1.5

An analogous argument gives the stability function

$$R(h, k) = \left| 1 + h + \frac{1}{2}h^2 \right|^2 + |1 + h|^2|k| + \frac{1}{2}|k|^2 + \frac{1}{6}|k|^3 \quad (5.64)$$

whose stability region is shown in 5.4a.

Taylor method of weak order 2

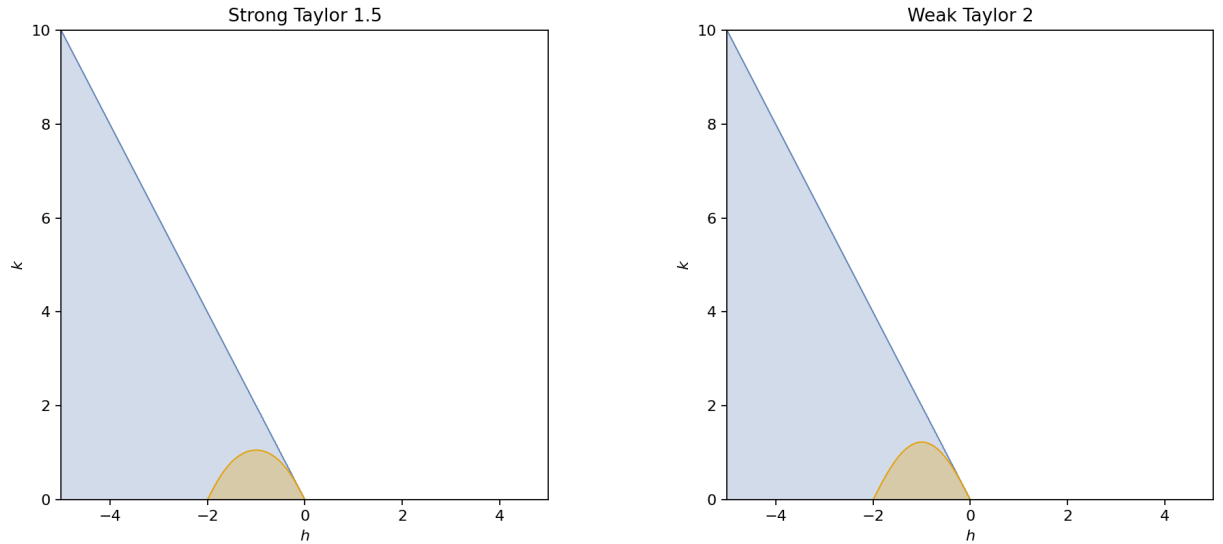
Stability regions can similarly be obtained for weak Itô-Taylor methods. In particular, for the method in (5.47), with $a(t, x) = \lambda x$ and $b(t, x) = \mu x$, we have

$$Y_{n+1} = Y_n \mathbb{E} \left[\left(1 + \lambda \Delta + \mu \Delta W + \frac{1}{2} \mu^2 (\Delta W^2 - \Delta) + \lambda \mu \Delta Z + \frac{1}{2} \lambda^2 \Delta^2 + \lambda \mu (\Delta W \Delta - \Delta Z) \right)^2 \right]$$

and using the properties of ΔW , Proposition 5.7, and the equalities $\mathbb{E}[\Delta W^2 \Delta Z] = 0$ and $\mathbb{E}[\Delta W^3 \Delta Z] = \frac{3}{2} \Delta^3$, we finally obtain

$$R(h, k) = \left| 1 + h + \frac{1}{2}h^2 \right|^2 + |1 + h|^2 |k| + \frac{1}{2}|k|^2 \quad (5.65)$$

with stability region shown in Figure 5.4b.



(a) Itô-Taylor method of strong order 1.5.

(b) Itô-Taylor method of weak order 2.

Figure 5.4: MS -stability region for other explicit Itô-Taylor methods.

Comparing (5.64) and (5.65) shows that the stability region of the Taylor method of strong order $\gamma = 1.5$ is contained in that of the Taylor method of weak order $\beta = 2$. Thus, the stability region is largely determined by the value of θ , with explicit methods having the smallest stability regions. Figure 5.5 compares the stability regions of different methods.

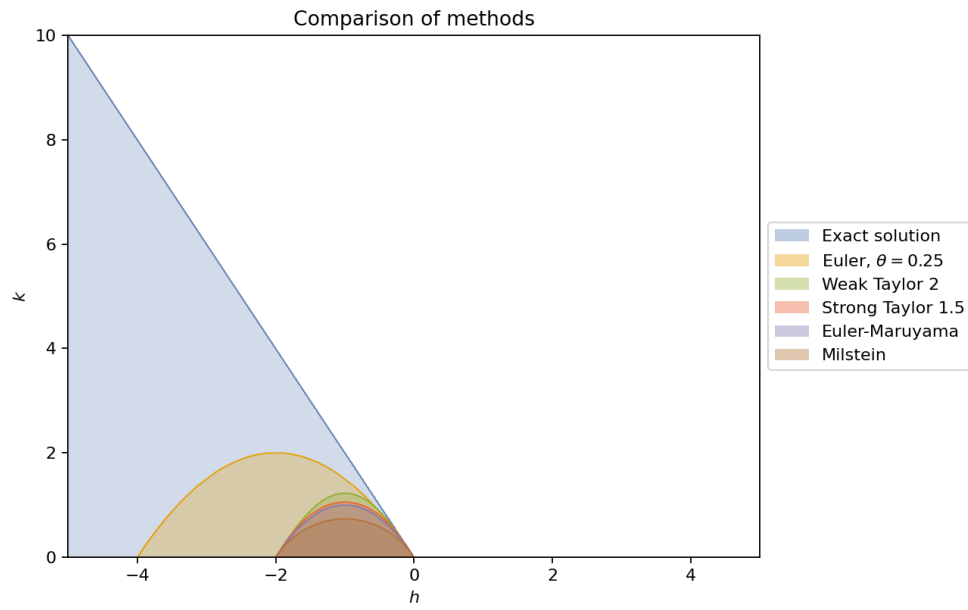


Figure 5.5: Comparison of the MS -stability regions of different methods.

Chapter 6

Numerical simulations

6.1 Stability of Euler-Maruyama θ -methods

In this chapter, we numerically simulate mean-square stability and estimate the strong and weak convergence orders of various numerical methods applied to the stochastic differential equation in (5.54). This allows us to compare the numerical results with the theoretical predictions.

In particular, we focus on mean-square stability of the family of Euler-Maruyama θ -methods in (5.37). Figure 5.2 shows that the stability region grows as θ increases. We therefore fix λ and k in the stability region of the exact solution, that is, such that $\lambda + \mu^2/2 < 0$, and simulate the mean square both of the exact solution from Lemma 5.27 and of the paths obtained for different values of θ .

For each θ , we simulate 1000 paths using the method, square each path, and average these squares at each time.

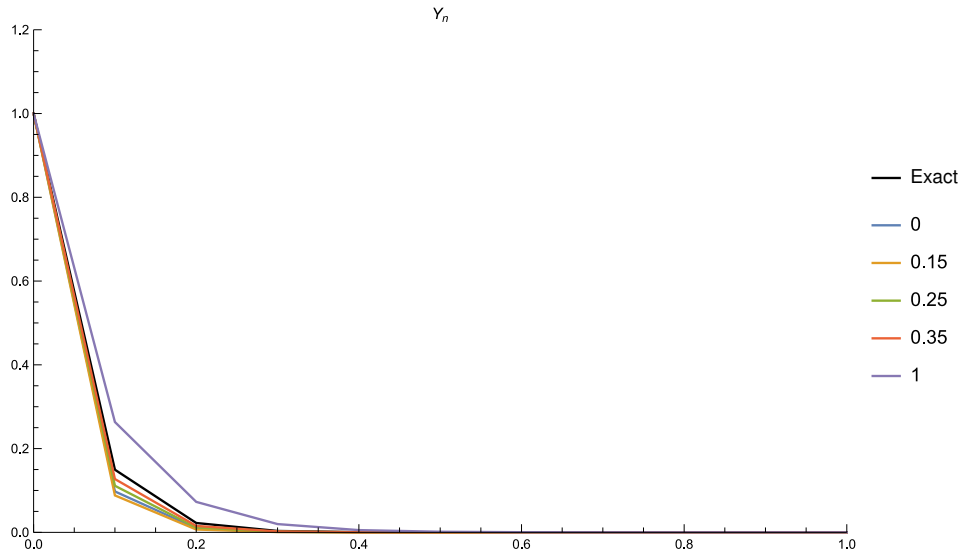


Figure 6.1: Stability of Euler-Maruyama θ -methods for different values of θ with $\lambda = -10$, $\mu = 1$, and $\Delta = 0.1$.

If $\lambda = -10$, $\mu = 1$, and $\Delta = 0.1$, Figures 5.1 and 5.2 show that the chosen point $h = -1$ and $k = 0.1$ lies in the stability region of the exact solution, see Figure 6.1, and of the numerical methods for any $\theta \in [0, 1]$. Thus, the simulation clearly shows mean-square convergence of the exact solution and the methods for every value of θ . Figure 6.2 gives an example in which some methods are no longer stable.

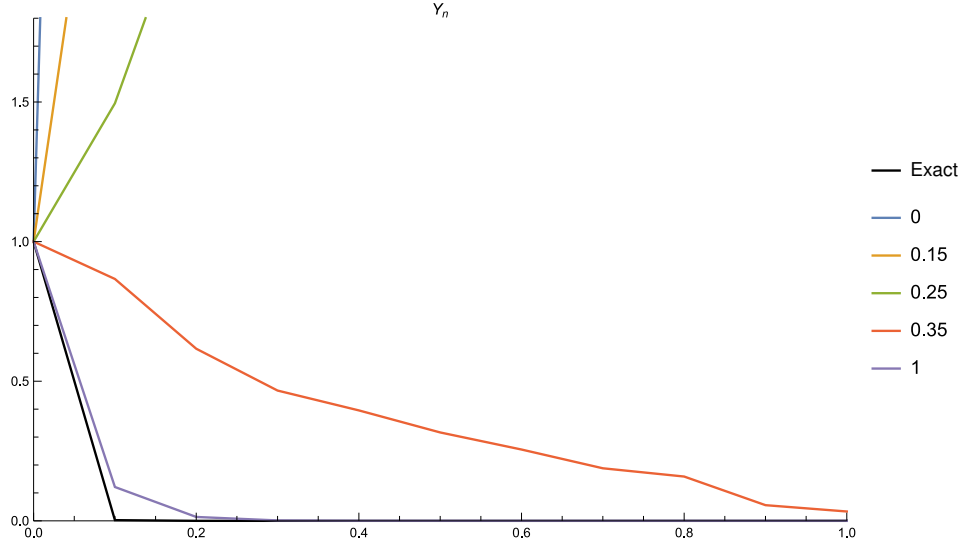


Figure 6.2: Stability of Euler-Maruyama θ -methods for different values of θ with $\lambda = -40$, $\mu = \sqrt{20}$, and $\Delta = 0.1$.

If $\lambda = -40$, $\mu = \sqrt{20}$, and $\Delta = 0.1$, the point $(h, k) = (-4, 2)$ lies in the stability region of the exact solution Y_t , but not, for example, for $\theta \in \{0, 0.15, 0.25\}$. Hence, for these values of θ , mean-square stability does not hold.

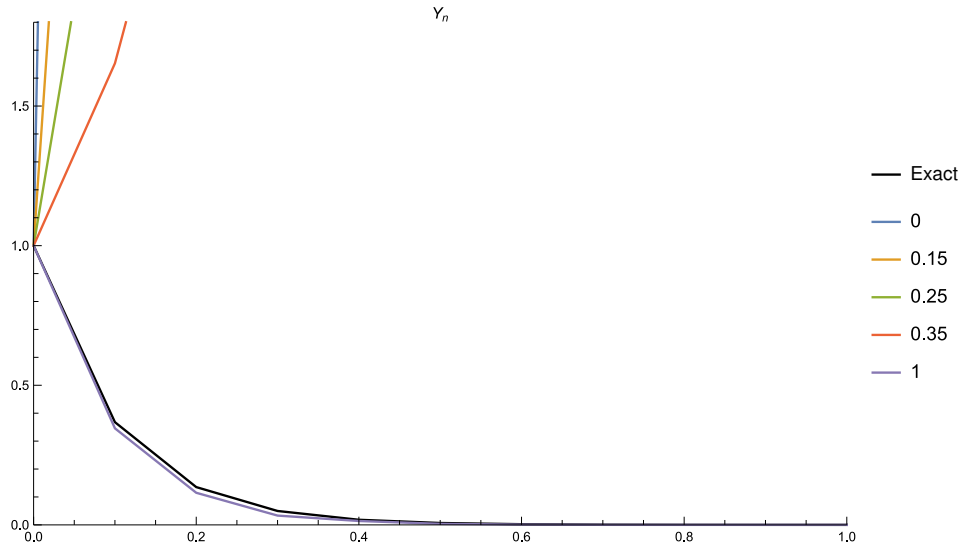


Figure 6.3: Stability of Euler-Maruyama θ -methods for different values of θ with $\lambda = -40$, $\mu = \sqrt{70}$, and $\Delta = 0.1$.

In Figure 6.3, taking $\lambda = -40$, $\mu = \sqrt{70}$, and $\Delta = 0.1$ gives values of $h = -7$

and $k = 4$ for which only the fully implicit method is stable among those simulated. Examining the stability region in 5.2 shows that, for any pair (h, k) for which the exact solution is stable, the implicit method is also stable. However, there are regions in which the implicit method is stable while the exact solution is not, as shown in Figure 6.4.

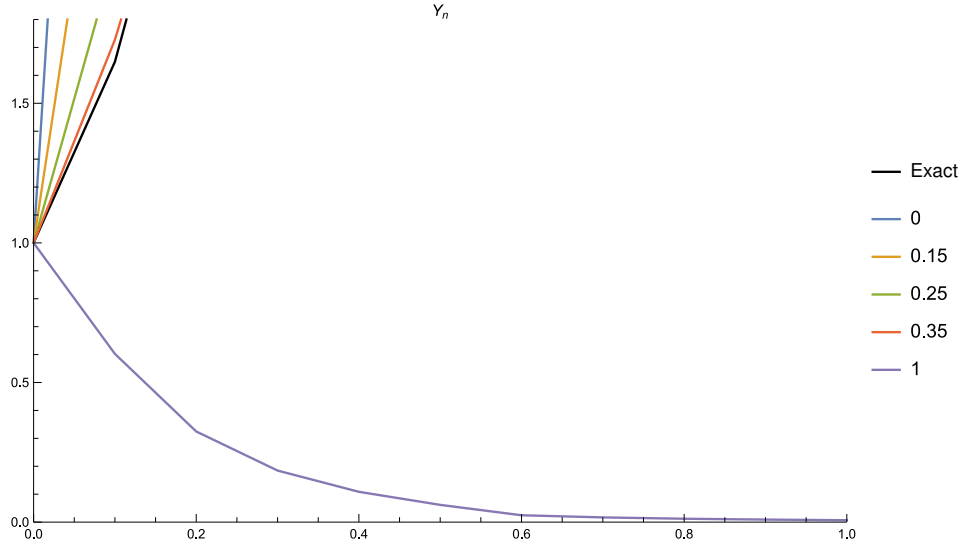


Figure 6.4: Stability of Euler-Maruyama θ -methods for different values of θ with $\lambda = -20$, $\mu = \sqrt{45}$, and $\Delta = 0.1$.

6.2 Simulation of strong and weak errors

In this section, we numerically reproduce the results presented in Sections 5.2 and 5.3. We simulate 1000 paths of the exact solution (5.55) and of each numerical method studied.

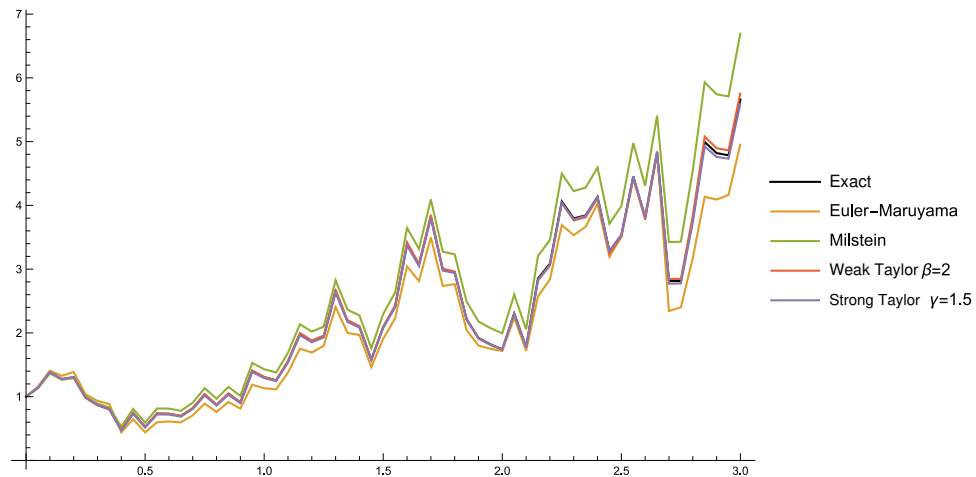


Figure 6.5: Simulation of a path using the different methods with $\lambda = 1.5$ and $\mu = 1$.

Figure 6.5 illustrates a simulated path obtained both from the exact solution and using all numerical methods studied in the preceding chapter.

To study the strong error, we take, for each simulated path, the absolute difference from the path obtained for the exact solution. We then average the 1000 simulations to estimate the theoretical expectation. That is,

$$e_f = \frac{1}{N} \sum_{i=1}^N |X_T^i - Y_T^i|$$

where N is the number of simulations and i is the i th simulation.

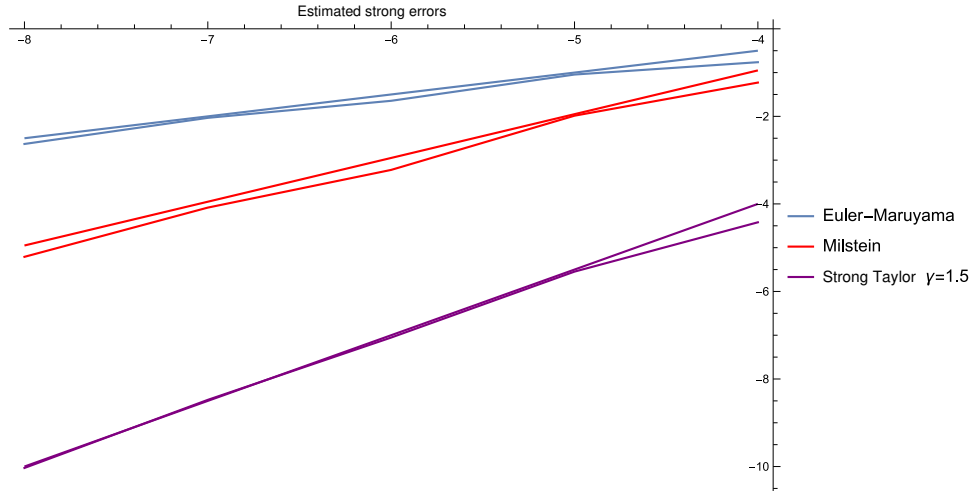


Figure 6.6: $\log_2(e_f)$ as a function of $\log_2 \Delta$ for $\Delta = 2^{-4}, 2^{-5}, 2^{-6}, 2^{-7}, 2^{-8}$. Dashed lines have slopes equal to the theoretical convergence order of each method.

Figure 6.6 plots simulated $\log_2 e_f$ as a function of $\log_2 \Delta$, showing a linear trend with slope γ . Thus,

$$e_f = C \Delta^\gamma$$

so all methods exhibit linear behavior with an order (slope) corresponding to the theoretical strong orders presented in 5.2.

For the weak error, we compute the absolute difference between the expectation of the exact solution and the estimated expectation for each method; that is, we take $g(x) = x$:

$$e_d = \left| \mathbb{E}[X_T] - \frac{1}{N} \sum_{i=1}^N Y_T^i \right| = \left| e^{\lambda T} - \frac{1}{N} \sum_{i=1}^N Y_T^i \right|$$

Simulating this error for different values of the step size Δ gives the following figure:

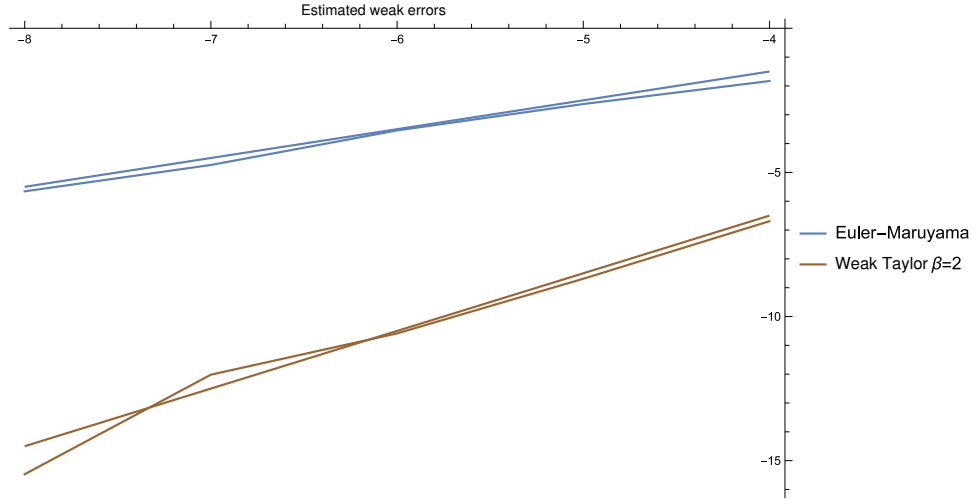


Figure 6.7: $\log_2(e_d)$ as a function of $\log_2 \Delta$ for $\Delta = 2^{-4}, 2^{-5}, 2^{-6}, 2^{-7}, 2^{-8}$. Dashed lines have slopes equal to the theoretical convergence order of each method.

Figure 6.7 shows that the numerical methods have the weak orders predicted by the theory in 5.3, since

$$e_d = C\Delta^\beta$$

Chapter 7

Introduction to financial derivatives

A derivative instrument is a financial product that, as its name suggests, derives from more basic financial products. It is generally a contract between two parties that confers rights and/or obligations according to its structure. In particular, the value of these instruments is usually determined by the price of one or more underlying financial assets [26].

Derivatives have a history extending back several centuries, although their modern development is relatively recent. The earliest examples date to ancient Mesopotamia, where farmers and merchants used rudimentary futures contracts to fix crop prices before planting. Contracts allowing agricultural products to be bought or sold at an agreed price on a future date also existed in ancient Greece and Rome [25].

The modern concept of derivatives began to take shape in the seventeenth century in commodity markets in Japan and Europe. In Japan, rice merchants used contracts called rice futures to protect themselves against price fluctuations. At the same time, the exchange established in Amsterdam in 1602 became a center for trading stocks and futures.

The most significant development occurred in the United States in the nineteenth century, with the establishment of the Chicago Board of Trade (CBOT) in 1848, where farmers and merchants traded futures on products such as corn and wheat. These contracts provided protection against price fluctuations, acting as a form of insurance against uncertainty.

In the final decades of the twentieth century, financial derivatives underwent a radical transformation with the introduction of more sophisticated contracts, such as options and swaps. This development accelerated through the work of economists such as Fischer Black and Myron Scholes, who published the celebrated Black-Scholes option pricing model in 1973, laying the foundations for quantitative analysis in financial markets [27].

There are three main types of participants in financial markets: (i) hedgers use derivatives to reduce the risk associated with possible future movements in a market variable, protecting themselves against adverse fluctuations; (ii) speculators use derivatives to profit from anticipating the future direction of a market variable, accepting greater risk in exchange for a potential reward; (iii) arbitrageurs seek risk-free profit opportunities by taking offsetting positions in two or more financial instruments, exploiting price discrepancies to secure a gain.

These notes are intended as a basic guide to the knowledge essential to a quant's

training. What is a quant, or quantitative analyst? What do they do? Quantitative analysts, known as quants, play a fundamental role in financial markets by applying advanced mathematical methods, statistics, and programming to solve complex problems in derivatives pricing, quantitative portfolio management, risk analysis, and related areas.

In derivatives pricing, they use probabilistic models such as Black-Scholes, binomial trees, or numerical methods (Monte Carlo, finite differences) to calculate fair prices for options, futures, and other financial products.

They also apply machine learning and time series analysis to predict market movements or develop derivatives trading strategies, identifying patterns in asset prices and designing algorithmic investment strategies.

For risk assessment, they develop statistical measurement models such as Value at Risk (VaR) and Expected Shortfall to quantify portfolio exposure to market fluctuations.

Based on Mark Joshi's classification¹, there are six types of quants according to their specialization:

- Front office/desk quant. Implements pricing models designed by other quants for use by traders. A trader executes transactions and makes market decisions, focusing more on execution, market dynamics, and P&L (profit and loss).
- Model validation quant. Independently implements pricing models to verify the correctness of those used in the front office. The work involves reproducing models from scratch, comparing results, assessing stability, and ensuring compliance with regulations and internal standards. Unlike front office quants, these quants audit and verify models rather than developing them for trading.
- Research quant. Develops new pricing approaches and, in some cases, conducts theoretical research without an immediate application. The work focuses on mathematical and computational innovation, exploring more efficient or accurate models for pricing and risk management.
- Quant developer. Essentially a programmer focused on computational optimization of pricing models, developing infrastructure for quantitative models, or debugging code in complex systems. Although quant developers do not generally design mathematical models, their work is crucial to efficient implementation in production environments.
- Statistical arbitrage quant. Develops trading strategies based on statistical patterns in data to identify arbitrage opportunities. Unlike quants working in derivatives pricing, the approach is more empirical and relies on large-scale data analysis, machine learning, and optimization techniques.
- Capital quant. Develops models to estimate credit risk exposure and determine how much capital a bank must set aside to absorb potential losses. The work involves simulating adverse scenarios, calculating default probabilities, and modeling correlations between assets. These quants also work with risk and compliance teams to ensure that the metrics used accurately reflect the institution's financial health.

¹See [On becoming a quant](#).

Unlike trading quants, their work focuses more on the bank's long-term stability and solvency.

Financial derivatives encompass a wide range of products whose values are linked to different types of underlying assets. These include:

- **Equities.** Derivatives on stocks and stock indices. Pricing these products relies on mathematical models that incorporate expectations about the evolution of underlying stock prices. These derivatives allow investors to hedge risks or speculate on future asset price movements.
- **FX.** Short for foreign exchange, this refers to currency trading. Contracts are generally short-dated, with high trading volumes and relatively simple specifications. Consequently, the main focus in this market is on execution speed and modeling the volatility smile, which captures the nonlinear structure of implied volatility² in currency options.
- **Fixed income.** Products based on interest rates, such as swaps, interest rate options, and structured bonds. This is probably the largest market by notional value. Modeling is more complex than for other assets because interest rates depend on multiple maturities, introducing a multidimensional structure. Advanced stochastic methods are used for pricing.
- **Credit derivatives.** Products whose payments depend on corporate default, that is, credit events such as defaults or changes in credit ratings.
- **Commodities.** Raw materials traded in financial markets, including energy (e.g., oil, gas), metals (e.g., gold, copper), and agricultural products (e.g., wheat, coffee). Trading may involve physical delivery or derivatives. Unlike other financial assets, commodity prices are strongly influenced by external factors such as global supply and demand, weather conditions, and geopolitical events. These characteristics require specific modeling approaches to capture the seasonality and volatility of each market.
- **Structured products.** Tailored financial instruments whose payments depend on the performance of one or more markets or underlying assets, combining, for example, interest rates, currencies, credit, equities, or commodities. Their complexity ranges from relatively simple to highly sophisticated, as some require integrating the dynamics of multiple risk factors. They are used for both investment and hedging strategies and are generally issued by banks or other financial institutions to provide specific risk-return profiles tailored to investors' objectives.

7.1 Derivatives markets

Derivatives markets fall into two broad categories: exchange-traded markets (ETM) and over-the-counter (OTC) markets.

²These concepts will be defined rigorously later.

ETM offer standardized contracts defined by the exchange and use clearing houses to reduce counterparty risk. Examples include the Chicago Mercantile Exchange (CME) and Eurex. For a more detailed explanation of how a clearing house operates, see [26].

OTC markets allow parties to negotiate customized contracts without an exchange intermediary, providing greater flexibility. Financial institutions act as market makers³ for the most common financial instruments.

Both markets offer derivatives, since both exchange-traded and OTC markets allow trading in contracts whose values depend on underlying assets. However, OTC markets are considerably larger in terms of trading volume and the notional value of contracts. According to data from the Bank for International Settlements (BIS), the notional value of OTC derivatives has historically been several times that of exchange-traded derivatives. This reflects the flexibility of OTC contracts, which can be customized by maturity, size, and structure, making them attractive to financial institutions and large companies with specific hedging and speculative needs. By contrast, ETM offer standardized contracts that, although highly liquid, limit customization.

As noted above, the main advantage of OTC markets over exchange-traded markets is that contracts need not follow exchange specifications. However, they carry greater credit risk (Lehman Brothers provides a clear example). Exchange-traded markets, on the other hand, have virtually eliminated credit risk.

7.2 Forward contracts

One of the simplest derivatives is a forward contract: an agreement between two parties to buy or sell an asset on a specified future date at a price fixed when the contract is entered into. Forward contracts are not traded on organized exchanges; they are OTC agreements between the parties, allowing greater flexibility in volume, maturity, and specific terms.

The agreed price, known as the forward price, is determined by market expectations about the evolution of the underlying asset price, together with factors such as the interest rate and the risk premium. At maturity, one party will have a gain and the other a loss, depending on the difference between the underlying asset's market price and the contract price.

In a forward contract, the party agreeing to buy the underlying asset on the specified date is said to hold a long position. The other party, agreeing to sell the asset at the agreed price, holds a short position. Figure 7.1 shows the possible payoff functions of a forward contract for both positions.

Mathematically, in the simplest model, the forward price at time t for maturity T on an underlying asset S , denoted by $\text{For}_S(t, T)$, can be expressed as⁴

$$\text{For}_S(t, T) = S_t e^{(r-q)(T-t)}, \quad (7.1)$$

³Market makers provide liquidity to a market, i.e., ensure that there is always supply and demand for an asset. Through these transactions, they earn money from the spread, defined as the difference between an asset's ask and bid prices.

⁴The forward price is derived later in the text. In particular, equation (7.1) gives the price obtained when the underlying asset (the asset on which the contract is written) follows geometric Brownian motion; see 3.1.2.

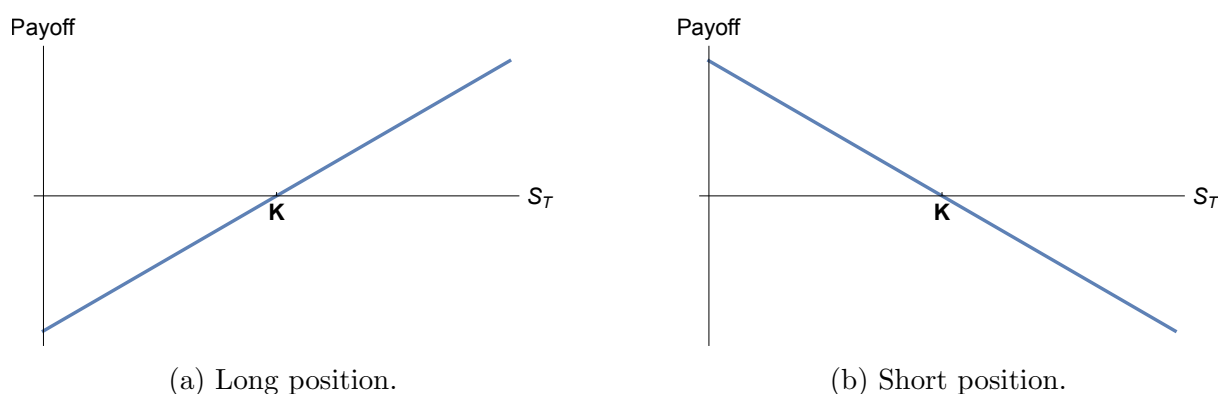


Figure 7.1: Payoff functions for long and short forward contracts.

where S_t is the price of the underlying asset (e.g., equity, commodities) at time t , r is the risk-free interest rate⁵, q represents continuous dividend payments⁶ on the underlying asset (if any), and T is the time to maturity. Note that $\text{For}_S(t, T)$ is not the price paid to acquire the forward contract, but the delivery price at which the parties agree to exchange the underlying asset at maturity. At inception, the contract is set up to have zero value to both parties, but its value may subsequently change as the market evolves. For assets generating cash flows, such as dividend-paying stocks or coupon bonds, the formula must be adjusted to reflect these payments, as we will see later.

Forward contracts are used primarily for hedging and speculation. Hedgers seek to eliminate the risk of changes in the underlying asset price, while speculators seek to profit from anticipated price movements. These derivatives are common in commodity, FX, and interest rate markets.

Despite their advantages, forwards have limitations such as illiquidity and default risk. Futures contracts are often used to mitigate these problems, offering greater security and transparency through a clearing house. Nevertheless, forward contracts remain an essential tool in financial markets, particularly for transactions requiring customized and flexible contractual terms.

7.3 Futures contracts

A futures contract is a standardized agreement imposing an obligation to buy or sell an underlying asset on a future date at a specified price. Unlike forward contracts, futures trade on organized exchanges, providing greater liquidity and transparency. They are also overseen by a clearing house, which acts as an intermediary and reduces default risk through margin requirements and daily settlement.

Standardization of futures contracts covers aspects such as contract size, maturity dates, and delivery or settlement mechanisms. In many cases, contracts are cash-settled rather than settled through physical delivery, facilitating trading for investors and speculators. This is because the underlying assets are generally commodities (sugar, copper,

⁵The risk-free interest rate is the return expected from an investment with no default risk. Chapter 11 examines its definition in greater detail.

⁶In practice, discrete dividend payments are more common; this case is introduced later.

aluminum, gold, ...) or other financial assets (equities, FX, fixed income, etc.).

A distinctive feature of futures is daily settlement, known as marking to market. Each day, gains and losses on open positions are calculated using the market closing price, and participants must maintain a minimum margin to cover potential losses. This mechanism reduces credit risk but also introduces the possibility of margin calls, requiring traders to provide additional capital if losses exceed a specified threshold.

Mathematically, an arbitrage-free futures price is modeled similarly to a forward price, with adjustments for daily settlement and the costs of maintaining the position. In efficient markets, assuming constant interest rates and volatility, the futures price at time t for expiry T on underlying asset S , denoted by $\text{Fut}_S(t, T)$, is expressed as⁷

$$\text{Fut}_S(t, T) = S_t e^{(r-q)(T-t)}, \quad (7.2)$$

where r is the risk-free interest rate and q represents, for example, continuous dividend payments⁸.

Futures contracts are widely used for hedging and speculation in commodity, currency, stock index, and interest rate markets. Their liquidity and regulated structure make them attractive to institutional investors and for algorithmic trading.

Thus, futures offer advantages over forwards in security and efficiency, at the cost of reduced flexibility. They play a crucial role in risk management, price discovery, and liquidity provision in global markets.

7.4 Options

An option is a derivative contract giving the buyer the right, but not the obligation, to buy (a call option) or sell (a put option) an underlying asset at a fixed price K , called the strike price. According to when the contract can be exercised, the simplest (plain vanilla) options are classified as:

- European. Exercisable only at maturity T .
- American. Exercisable at any time up to T .

Unless otherwise stated, we use vanilla options to refer only to options with European-style payoffs.

Exotic options have more complex structures. They include Bermudan options, exercisable only on specified dates before maturity; Asian options, whose values depend on the average underlying price over a given period; and variants such as barrier, lookback, or chooser options. Exotic options are studied later in the text.

In the exchange-traded equity options market, a contract is generally an agreement to buy or sell 100 shares. This is known as the option's contract multiplier, with 10, 100, or 1000 shares per contract being the most common. Furthermore, unlike forwards

⁷In general, forward and futures prices coincide only if both the interest rate and volatility are constant and there are –at most– continuous dividend payments. For more sophisticated models, $\text{For}_S(t, T) = \text{Fut}_S(t, T)$ generally does not hold. For example, if the interest rate is stochastic and correlated with S_t , the prices differ. See Section 7.6.3.

⁸The case of discrete dividends is considered later.

and futures, which have no initial cost, acquiring an option has a cost, called the option premium.

There are four types of participants in vanilla options markets: (i) call buyers, (ii) call sellers, (iii) put buyers, and (iv) put sellers. Buyers are said to hold long positions, and sellers short positions.

We next analyze the structure of European call and put options.

7.4.1 European call option

A call option gives the buyer the right to buy a specified financial asset from the seller at maturity T for a strike price K . Denoting the underlying financial asset's value at time t by S_t , there are two possible scenarios depending on the underlying value:

- At maturity, the asset value exceeds the strike price. The buyer exercises the option, buying the shares at the strike price K and selling them at S_T . The resulting profit is $S_T - K$.
- If, at time T , the stock price is at or below the strike price, the buyer does not exercise the option, so the payoff is⁹ 0.

Thus, the payoff function of this option is

$$(S_T - K)^+ = \max\{0, S_T - K\}. \quad (7.3)$$

When selling a call option, the seller receives the premium immediately but assumes an obligation to sell the underlying asset to the buyer if the underlying price exceeds the strike price at maturity. Thus, while gains from selling a call are limited, losses can be very large if the underlying asset price ends up far above the strike. As discussed later, call sellers use hedging tools to minimize the risk of loss.

The payoff structure of a European call option is shown in Figure 7.2.

7.4.2 European put option

The contract gives the buyer the right to sell a specified financial asset at maturity T for a strike price K . Again, we distinguish two scenarios:

- At T , the asset value is at or above the strike price. The buyer does not exercise the option, so the profit is 0.
- At time T , the stock price is below the strike price. The buyer exercises the option, buying the financial asset in the market at S_T and selling it to the option seller at K . The resulting profit is $K - S_T$.

Thus, the payoff function of this option is

$$(K - S_T)^+ = \max\{0, K - S_T\}.$$

As with calls, put sellers can face large losses if the underlying asset price ends up far below the strike at maturity. Hedging strategies are also used to minimize this risk.

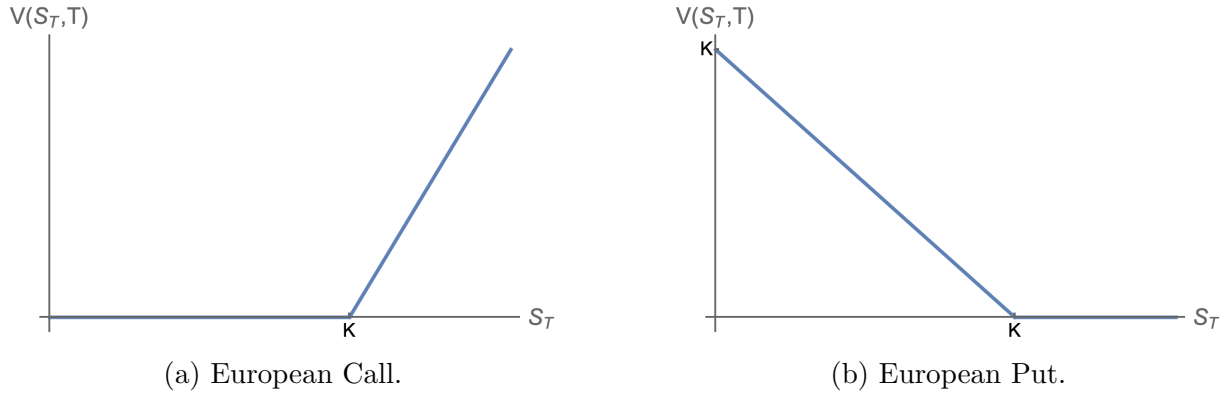


Figure 7.2: Payoff functions for European call and put options.

The payoff structure of a European put option is shown in Figure 7.2.

Example 7.1: European option

Suppose that a company A wants to purchase \$20,000 worth of shares in company B . At time $t = 0$, the share price of B is $S_0 = \$90$.

Company A does not want to spend more than \$90 per share and buys 200 call options with

$$K = \$90 \quad T = 6 \text{ months} \quad C_0 = \$500 \quad (7.4)$$

where C_0 is the price of each option, and each option gives the right to buy 100 shares in company B at \$90 per share.

If, after $T = 6$ months, the share price satisfies $S_T > \$90$, company A exercises the options and spends $90 \cdot 20,000 = \$1,800,000$ on shares and $200 \cdot C_0 = \$100,000$ on the options. If, for example, $S_T = \$97$, immediately selling the purchased shares yields a profit of $(97 - 90) \cdot 20,000 - 200 \cdot 500 = \$40,000$.

On the other hand, if $S_T \leq 90$, company A does not exercise the option and loses $200 \cdot C_0 = \$100,000$.

In general, we decompose an option's value at each time into its intrinsic value and its time value. That is,

$$\boxed{\text{Option value} = \text{Intrinsic value} + \text{Time value},} \quad (7.5)$$

where intrinsic value is defined as the amount of money that would be obtained if the option were exercised immediately. For example, the intrinsic value of a call at time t is $(S_t - K)^+$.

The time value of an option reflects the possibility that the underlying asset price may move favorably in the future, increasing the option's value before expiry. In general, time value decreases as maturity approaches and is exactly zero at maturity. Figure 7.3 illustrates this decomposition.

⁹In fact, the buyer loses money because options have a cost agreed in advance; see Example 7.1.

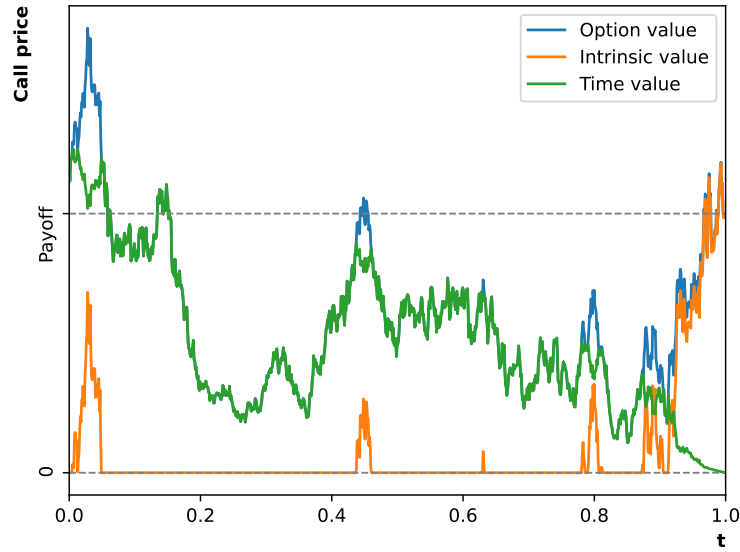


Figure 7.3: Evolution of a call option's value and its decomposition into intrinsic value and time value.

7.5 Continuous-time market models

A fundamental approach to financial market modeling treats price evolution as a continuous-time stochastic process. Unlike discrete-time models, in which transactions and price changes occur at separate times, continuous-time models allow infinitesimal price changes, enabling us to describe their dynamics using stochastic differential equations.

Continuous-time models are essential for derivatives pricing and the formulation of optimal investment strategies. In this setting, concepts such as self-financing portfolios and market completeness are central to determining whether any financial asset can be replicated using combinations of other instruments available in the market.

This section introduces the foundations of continuous-time models, providing the basis for studying the pricing of options and other financial derivatives.

First, suppose that we have $d + 1$ assets given by the components of the random vector

$$S_t = \left(S_t^{(0)}, S_t^{(1)}, \dots, S_t^{(d)} \right), \quad (7.6)$$

forming a stochastic process $S = \{S_t : t \in \mathbb{R}\} \subset \mathbb{R}^{d+1}$. We take $S_t^{(0)}$ to be a risk-free asset (for example, a savings account paying an interest rate r , initially assumed constant¹⁰), denoted by $S_t^{(0)} = B_t$. We can then model the evolution of B_t as

$$dB_t = rB_t dt, \quad (7.7)$$

with solution

$$B_t = B_0 e^{rt}, \quad (7.8)$$

¹⁰The concept of a money-market account is introduced more formally later.

where B_0 denotes the value of the risk-free asset at $t = 0$. The remaining assets $S_t^{(i)}$ with $i > 0$ are taken to be risky. We define the discount factor as

$$D_t = \frac{B_0}{B_t} \quad (7.9)$$

for $t \geq 0$ and the accumulation factor as $A_t = D_t^{-1}$. When the interest rate is constant, both D_t and A_t are deterministic processes. Thus, the discounted value of asset $S_t^{(i)}$ is $D_t S_t^{(i)}$.

Working with discounted asset prices removes the effect of exponential growth associated with interest rates. This allows us to focus on the asset's price fluctuations, making the mathematical models simpler and more tractable.

We also define a portfolio's investment strategy as the stochastic process $\xi_t = \{\xi_t^{(k)} : t \in \mathbb{R}\} \subset \mathbb{R}^{d+1}$, where $\xi_t^{(k)}$ represents the quantity (or portfolio fraction) of asset $S_t^{(k)}$ held at time $t \geq 0$.

The value of the portfolio with strategy ξ_t is then defined as

$$V_t = \xi_t \cdot S_t = \sum_{i=0}^d \xi_t^{(i)} S_t^{(i)}, \quad (7.10)$$

and its discounted value as $D_t V_t$.

7.5.1 Arbitrage

Arbitrage is a fundamental concept in financial markets: an opportunity to earn risk-free profits by exploiting price discrepancies for the same asset across markets. It rests on the idea that assets should be priced consistently across markets, since any price difference can be exploited for an immediate gain without risk exposure. Arbitrage is not only an investment strategy; it also plays a crucial role in economic and financial theory by ensuring market efficiency and contributing to price discovery. In this context, absence of arbitrage is a key condition in most financial models, ensuring consistent asset prices and ruling out opportunities for risk-free profits.

Definition 7.1. A portfolio strategy $\{\xi_t : t \in \mathbb{R}\}$ constitutes an arbitrage opportunity if:

- (i) $V_0 \leq 0$. That is, the portfolio starts at no cost or with debt.
- (ii) $V_T \geq 0$ for some $T > 0$. That is, the portfolio has positive value at a future time.
- (iii) $\mathbb{P}(V_T > 0) > 0$. That is, there is a nonzero probability of a profit.

We next define the risk-neutral probability measure, denoted by $\mathbb{P}^*$.

Definition 7.2. Let $(\Omega, \mathcal{F}, \mathbb{P}^*)$ be a probability space and $\{\mathcal{F}_t : t \in \mathbb{R}\}$ a filtration of $\mathcal{F}$ generated by a Wiener process W_t . That is,

$$\mathcal{F}_t = \sigma(W_u : 0 \leq u \leq t). \quad (7.11)$$

The probability measure $\mathbb{P}^*$ is said to be risk-neutral if

$$\mathbb{E}^* \left[S_t^{(k)} | \mathcal{F}_u \right] = e^{(t-u)r} S_u^{(k)} \quad (7.12)$$

for every $u \in [0, t]$ and $k = 1, 2, \dots, d$, where $\mathbb{E}^*$ denotes expectation under $\mathbb{P}^*$. That is, $\mathbb{P}^*$ is risk-neutral if $S_t^{(k)}$ is a martingale under this measure.

Risk-neutral probability implies that, on average, the gain associated with a risky asset is expected to equal that obtained from the risk-free asset.

Proposition 7.3. *Let $(\Omega, \mathcal{F}, \mathbb{P}^*)$ be a probability space. The probability measure $\mathbb{P}^*$ is risk-neutral if and only if the discounted risky asset price $D_t S_t^{(k)}$ is a martingale with respect to $\mathbb{P}^*$ for every $k = 1, 2, \dots, d$.*

Proof. $\Rightarrow$ Suppose that $\mathbb{P}^*$ is a risk-neutral probability measure and let $\mathbb{E}^*$ denote expectation under this measure. Then, using the properties of expectation,

$$\mathbb{E}^* \left[D_t S_t^{(k)} | \mathcal{F}_u \right] = D_t \mathbb{E}^* \left[S_t^{(k)} | \mathcal{F}_u \right]$$

and Definition 7.2 gives

$$D_t \mathbb{E}^* \left[S_t^{(k)} | \mathcal{F}_u \right] = e^{-rt} e^{(t-u)r} S_u = e^{-ur} S_u^{(k)} = D_u S_u^{(k)}$$

for every $k = 1, 2, \dots, d$.

$\Leftarrow$ If $D_t S_t^{(k)}$ is a martingale under $\mathbb{P}^*$ for every $k = 1, 2, \dots, d$, then

$$\begin{aligned} \mathbb{E}^* \left[S_t^{(k)} | \mathcal{F}_u \right] &= A_t \mathbb{E}^* \left[D_t S_t^{(k)} | \mathcal{F}_u \right] \\ &= A_t D_u S_u^{(k)} \\ &= e^{(t-u)r} S_u^{(k)} \end{aligned}$$

which proves that $\mathbb{P}^*$ is a *risk-neutral probability measure* by Definition 7.2. $\square$

Analogously to Definition 7.2, we can define a risk-premium probability measure, denoted by $\mathbb{P}^\sharp$, if it satisfies

$$\mathbb{E}^\sharp \left[S_t^{(k)} | \mathcal{F}_u \right] > e^{(t-u)r} \quad (7.13)$$

and a negative-risk-premium probability measure, denoted by $\mathbb{P}^\flat$, if it satisfies

$$\mathbb{E}^\flat \left[S_t^{(k)} | \mathcal{F}_u \right] < e^{(t-u)r} \quad (7.14)$$

In financial mathematics, probability measures such as the risk-neutral measure, the risk-premium measure, and the negative-risk-premium measure play a key role in modeling and pricing financial assets under different degrees of risk aversion. The risk-neutral measure transforms real-world market probabilities so that an asset's expected discounted value equals its current price, ensuring absence of arbitrage. This simplifies derivatives pricing and connects the underlying asset's stochastic dynamics with its valuation in models such as Black-Scholes.

The (negative) risk-premium measure introduces a bias in probabilities to reflect the compensation investors require for taking additional risk, adjusting prices to incorporate positive risk premia. By contrast, the negative-premium measure models situations in which investors accept lower expected returns to access safe-haven assets in volatile markets. These transformations, formalized through changes of measure based on Radon-Nikodym densities, capture complex market dynamics and differing preferences of economic agents, providing essential tools for consistent asset pricing in different settings.

Theorem 7.4: First fundamental theorem of asset pricing

A market has no arbitrage opportunities if and only if it admits at least one *risk-neutral probability measure* $\mathbb{P}^*$.

7.5.2 Self-financing strategies

7.5.3 Market completeness

Mathematically, a financial derivative has an associated payoff, modeled as a random variable Λ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P}^*)$. This payoff may depend on factors such as the underlying asset price S_t at a future time, the time at which a specific event occurs, or a combination of factors.

The most common approach to derivatives pricing is to discount the expected payoff. Consider a single¹¹ underlying asset S_t . If the derivative at time T depends only on the underlying asset price at maturity, that is, $\Lambda(S_T)$, then its price at time t depends only¹² on t and the current underlying price. For example, assuming a constant interest rate r , we will see that the present value of a financial derivative with payoff $\Lambda(S_T)$ under a risk-neutral probability measure $\mathbb{P}^*$ is calculated as

$$V(t, s) = e^{-r(T-t)} \mathbb{E}^* [\Lambda(S_T) | S_t = s], \quad (7.15)$$

where T is the maturity time.

In practice, this calculation may require numerical tools such as those introduced in Chapter 5.

Definition 7.5. A derivative with payoff $\Lambda(S_T)$ is said to be attainable (or replicable) if there exists a self-financing portfolio strategy ξ_t such that, at maturity T , the portfolio value satisfies

$$V_T = \xi_T \cdot S_T = \sum_{i=0}^d \xi_T^{(i)} S_T^{(i)} = \Lambda \quad (7.16)$$

with probability 1. That is, $\mathbb{P}(\{V_T = \Lambda\}) = 1$.

The ability to replicate a European call payoff exactly in a complete market has several important advantages. One is that it allows precise pricing using the no-arbitrage principle: the option price equals the cost of constructing a portfolio that exactly reproduces its behavior. For example, a call paying $\max(S_T - K, 0)$ can be replicated by holding an

¹¹Derivatives depending on more than one underlying asset are introduced later.

¹²This statement is established later.

adjusted number of shares and investing the remainder in a risk-free asset. The option price must therefore equal the initial cost of this strategy, ruling out unjustified prices that could generate risk-free profits (arbitrage).

Another advantage is the ability to hedge the option seller's risk. Suppose someone sells a call. If the stock price rises substantially, the seller must pay a larger amount to the option holder. Through replication, the seller can buy and sell small quantities of the stock at specified times, ensuring that the portfolio always has the funds needed to meet the payment at maturity. For example, if the stock price rises, the seller will already have bought some stock, so the gain offsets the amount owed to the option buyer.

This idea also improves investment efficiency. Suppose you want the same returns as a call option but cannot buy it directly because it is unavailable in the market. You can replicate it by buying stocks and bonds in dynamically adjusted proportions. For example, if a stock's price is S_t and you expect it to rise, you can replicate a call by buying a fraction of the stock and adjusting the investment as its price changes.

Finally, replication helps build confidence in financial markets. If an investor sees that an option's price equals its replication cost, this indicates an absence of distortions and supports confidence in the valuation. This facilitates derivatives trading and ensures that prices correctly reflect risk. However, this advantage holds only in complete markets, where all future cash flows can be replicated. In incomplete markets, such as those with illiquid assets, replication is not always possible and alternative methods are needed.

Definition 7.6. A market model is complete if every financial derivative is replicable.

Theorem 7.7

An arbitrage-free market model is complete if and only if it admits a unique^a *risk-neutral probability measure* $\mathbb{P}^*$.

^aUp to equivalence. That is, μ and ν are equivalent measures on the measurable space $(\Omega, \mathcal{F})$ if the sets $\{A \in \mathcal{F} \mid \mu(A) = 0\}$ and $\{A \in \mathcal{F} \mid \nu(A) = 0\}$ are equal.

We can now justify expression (7.15). In a complete, arbitrage-free market, any derivative can be replicated by constructing a portfolio ξ_t . Since the discounted value of each portfolio asset is a martingale under $\mathbb{P}^*$, the discounted portfolio value is also a martingale under $\mathbb{P}^*$ by construction. Thus, assuming that D_T is not stochastic, that is, that interest rates follow a deterministic process (initially assumed constant here), we have

$$\begin{aligned} D_t V_t &= \mathbb{E}^* [D_T V_T | \mathcal{F}_t] \\ &= D_T \mathbb{E}^* [\Lambda | \mathcal{F}_t]. \end{aligned}$$

In particular, if $\Lambda = \Lambda(S_T)$ and the underlying asset is modeled by geometric Brownian motion, one can show that

$$\begin{aligned} V_t &= e^{-r(T-t)} \mathbb{E}^* [\Lambda(S_T) | \mathcal{F}_t] \\ &= e^{-r(T-t)} \mathbb{E}^* [\Lambda(S_T) | S_t = s] \end{aligned}$$

since the two filtrations are equivalent. Note that, if interest rates are allowed to follow a stochastic process, then

$$V_t = D_t^{-1} \mathbb{E}^* [D_T V_T | \mathcal{F}_t] \quad (7.17)$$

Strictly speaking, real markets are neither complete nor arbitrage-free. One may therefore ask why this derivatives pricing approach is still used.

Arbitrage opportunities do exist, but they are short-lived or costly to exploit. Absence of arbitrage is therefore assumed as a theoretical basis for calculating fair prices, which are then adjusted using empirical data. With calibration and empirical adjustments, this useful simplification remains the foundation of derivatives pricing in practice.

7.6 Pricing forwards and futures

We assume that a unique risk-neutral measure exists, denoted by $\mathbb{P}^*$, so all assets satisfy the risk-neutral pricing formula in (7.17). We next study forward and futures pricing and discuss several relevant differences. The analysis applies to any non-dividend-paying asset, not only stocks.

7.6.1 Forward contracts

Let S_t be the stochastic process modeling the underlying asset price. Since the constant interest rate case is trivial, we assume that R_t is a stochastic process adapted to the filtration $\{\mathcal{F}_t\}_{t \in \mathbb{R}_+}$. The discount factor between times t_0 and t is then

$$D_t = \exp \left(- \int_{t_0}^t R_s ds \right), \quad (7.18)$$

and, by the risk-neutral pricing formula (7.17), the price of a zero-coupon bond¹³ is determined by

$$B(t, T) = D_t^{-1} \mathbb{E}^* [D_T | \mathcal{F}_t]. \quad (7.19)$$

Definition 7.8. A forward contract is an agreement between two parties in which the buyer undertakes to purchase, and the seller to deliver, an underlying asset whose price at time t is denoted by S_t , at an agreed price K on a future date T (the delivery time). The T -forward price of the asset at time t , denoted by $\text{For}_S(t, T)$, is the value of K for which the forward contract's present value is zero at t under the risk-neutral measure $\mathbb{P}^*$. That is, it is the price that prevents arbitrage opportunities.

It is essential to distinguish the forward price from the value of a forward contract. We next show that a forward contract must have zero value at inception to prevent arbitrage opportunities.

Lemma 7.9. *Let S_t be the spot price at time t of a non-dividend-paying underlying asset. If a forward contract's initial value at that time t is greater than zero¹⁴, that is, $V_t > 0$, then arbitrage opportunities exist.*

Proof. **To be added.** □

¹³A zero-coupon bond is a financial instrument promising to pay a fixed amount, called the face value, at a future time T , with no intermediate payments before maturity. To simplify notation, we take its face value to be 1. For a more detailed explanation, see Chapter 11.

¹⁴More precisely, this statement applies to contracts called on-market forwards. There are also forwards with nonzero initial value that do not generate arbitrage opportunities, known as off-market forwards.

Theorem 7.10

Suppose that zero-coupon bonds of any maturity can be traded. Then the forward price is given by

$$\text{For}_S(t, T) = \frac{S_t}{B(t, T)} \quad (7.20)$$

Proof. As seen in Section 7.2, the payoff of a forward contract with maturity T is $S_T - K$. Since we work in an arbitrage-free market where discounted asset prices are martingales under the risk-neutral probability measure $\mathbb{P}^*$, we can value the forward contract using the risk-neutral pricing formula

$$V_t = D_t^{-1} \mathbb{E}^* [D_T (S_T - K) \mid \mathcal{F}_t].$$

Expanding the conditional expectation gives

$$\begin{aligned} V_t &= D_t^{-1} \mathbb{E}^* [D_T S_T \mid \mathcal{F}_t] - K D_t^{-1} \mathbb{E}^* [D_T \mid \mathcal{F}_t] \\ &= S_t - K B(t, T), \end{aligned}$$

where we have used the martingale property of the discounted underlying asset price under $\mathbb{P}^*$, that is,

$$D_t^{-1} \mathbb{E}^* [D_T S_T \mid \mathcal{F}_t] = S_t,$$

and the zero-coupon bond price for maturity T given in expression (7.19).

By definition, the forward price $K = \text{For}_S(t, T)$ is chosen so that the forward contract has zero value at inception t , that is, $V_t = 0$. Imposing this condition in the preceding equation gives the desired expression. $\square$

7.6.2 Futures contracts

7.6.3 The spread between forwards and futures

7.7 Option pricing

The Black-Scholes model assumes a market consisting of a risk-free asset, denoted by B_t , and a risky asset, denoted by S_t , modeled by geometric Brownian motion. That is,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (7.21)$$

for $t \in \mathbb{R}_+$. Applying Girsanov's theorem, see [4], and assuming that a unique risk-neutral measure exists, denoted by $\mathbb{P}^*$, equation (7.21) can be written as

$$dS_t = r S_t dt + \sigma S_t dW_t \quad (7.22)$$

under this measure¹⁵.

¹⁵We use an abuse of notation: W_t is Brownian motion under $\mathbb{P}$ but not under $\mathbb{P}^*$. The explicit transformation of Brownian motion follows from Girsanov's theorem.

As seen in Section 3.1.2, this equation has the unique solution (with probability 1) given by the stochastic process

$$S_t = S_{t_0} \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) (t - t_0) + \sigma (W_t - W_{t_0}) \right). \quad (7.23)$$

The Black-Scholes model relies on several key assumptions for option pricing. In addition to assuming that the underlying asset price follows geometric Brownian motion with constant, known volatility, financial markets are taken to be perfect: there are no arbitrage opportunities, unrestricted continuous trading is possible, and a risk-free asset with a constant interest rate exists. Transaction costs and taxes are also assumed absent.

These assumptions yield an analytical formula for European option prices, although some may not hold fully in practice, motivating the development of more sophisticated models.

7.7.1 European call option

We compute the option price explicitly using expression (7.15). As seen earlier, the payoff of a European call at maturity is

$$\Lambda(S_T) = (S_T - K)^+ = \begin{cases} S_T - K & \text{if } S_T \geq K, \\ 0 & \text{if } S_T < K. \end{cases} \quad (7.24)$$

Thus, the option value at time t for underlying asset value $S_t = s$ is¹⁶

$$\begin{aligned} V(t, s) &= e^{-r(T-t)} \mathbb{E}^* [(S_T - K)^+ | S_t = s] \\ &= e^{-r(T-t)} \mathbb{E}^* [(S_T - K) \mathbb{1}_{\{S_T \geq K\}} | S_t = s] \\ &= e^{-r(T-t)} \mathbb{E}^* [S_T \mathbb{1}_{\{S_T \geq K\}} | S_t = s] - e^{-r(T-t)} K \mathbb{E}^* [\mathbb{1}_{\{S_T \geq K\}} | S_t = s]. \end{aligned}$$

Assuming that the underlying asset is modeled by geometric Brownian motion, we can write

$$S_T = S_t e^{(r - \frac{1}{2} \sigma^2)(T-t) + \sigma(W_T - W_t)},$$

with $W_T - W_t \sim N(0, T - t)$ independent of S_t by definition.

First, observe that

$$\begin{aligned} \{S_T \geq K\} &= \{S_t e^{(r - \frac{1}{2} \sigma^2)(T-t) + \sigma(W_T - W_t)} \geq K\} \\ &= \{e^{(r - \frac{1}{2} \sigma^2)(T-t) + \sigma(W_T - W_t)} \geq K/S_t\} \\ &= \left\{ \left(r - \frac{1}{2} \sigma^2 \right) (T - t) + \sigma(W_T - W_t) \geq \log(K/S_t) \right\} \\ &= \left\{ W_T - W_t \geq -\frac{\log(S_t/K) + (r - \frac{1}{2} \sigma^2)(T - t)}{\sigma} \right\} \\ &= \left\{ Z \geq -\frac{\log(S_t/K) + (r - \frac{1}{2} \sigma^2)(T - t)}{\sigma \sqrt{T - t}} \right\} \end{aligned}$$

¹⁶Conditioning on $S_t = s$ is equivalent to conditioning on $\mathcal{F}_t$, since the two filtrations are equivalent. That is, the filtration $\{\mathcal{F}_t : t \in I\}$ is generally chosen to be that generated by the underlying asset, so $\mathcal{F}_t = \sigma(S_t)$.

where, in the final step, the normally distributed random variable $W_T - W_t$ has been standardized by dividing by its standard deviation. Defining the function

$$d_-(x, \tau) := \frac{\log(x) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad (7.25)$$

we finally obtain

$$\{S_T \geq K\} = \{Z \geq -d_-(S_t/K, T-t)\}.$$

Thus, the option value becomes

$$\begin{aligned} V(t, s) &= e^{-\frac{1}{2}\sigma^2} \mathbb{E}^* \left[S_t e^{\sigma\sqrt{T-t}Z} \mathbb{1}_{\{Z \geq -d_-(S_t/K, T-t)\}} | S_t = s \right] \\ &\quad - e^{-r(T-t)} K \mathbb{E}^* \left[\mathbb{1}_{\{Z \geq -d_-(S_t/K, T-t)\}} | S_t = s \right] \\ &= e^{-\frac{1}{2}\sigma^2} s \mathbb{E}^* \left[e^{\sigma\sqrt{T-t}Z} \mathbb{1}_{\{Z \geq -d_-(s/K, T-t)\}} \right] - e^{-r(T-t)} K \mathbb{P}^*(Z \geq -d_-(s/K, T-t)) \end{aligned}$$

We now compute the first term. Since $Z \sim N(0, 1)$,

$$\begin{aligned} \mathbb{E}^* \left[e^{\sigma\sqrt{T-t}Z} \mathbb{1}_{\{Z \geq -d_-(s/K, T-t)\}} \right] &= \int_{-d_-(s/K, T-t)}^{+\infty} e^{\sigma\sqrt{T-t}z} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz \\ &= \int_{-d_-(s/K, T-t)}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{\sigma\sqrt{T-t}z - \frac{z^2}{2}} dz, \end{aligned}$$

and the change of variable $u = \sigma\sqrt{T-t} - z$ transforms the integral into

$$\begin{aligned} \mathbb{E}^* \left[e^{\sigma\sqrt{T-t}Z} \mathbb{1}_{\{Z \geq -d_-(s/K, T-t)\}} \right] &= e^{\frac{1}{2}\sigma^2(T-t)} \int_{-\infty}^{\sigma\sqrt{T-t} + d_-(s/K, T-t)} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du \\ &= e^{\frac{1}{2}\sigma^2(T-t)} \mathcal{N}(\sigma\sqrt{T-t} + d_-(s/K, T-t)) \\ &= e^{\frac{1}{2}\sigma^2(T-t)} \mathcal{N}(d_+(s/K, T-t)) \end{aligned}$$

where $\mathcal{N}$ denotes the standard normal cumulative distribution function and we have defined

$$d_+(x, \tau) := d_-(x, \tau) + \sigma\sqrt{\tau}. \quad (7.26)$$

The second term follows directly as

$$\mathbb{P}^*(Z \geq -d_-(s/K, T-t)) = \mathbb{P}^*(Z < d_-(s/K, T-t)) = \mathcal{N}(d_-(s/K, T-t)) \quad (7.27)$$

Finally, the value of a European call is

$$\boxed{V(t, s) = s \mathcal{N}(d_+(s/K, T-t)) - K e^{-r(T-t)} \mathcal{N}(d_-(s/K, T-t))} \quad (7.28)$$

7.7.2 European put option

The payoff function of a European put is

$$\Lambda(S_T) = (K - S_T)^+ = \begin{cases} 0 & \text{if } S_T > K, \\ K - S_T & \text{if } S_T \leq K, \end{cases} \quad (7.29)$$

and an argument analogous to the preceding section gives

$$\boxed{V(t, s) = K e^{-r(T-t)} \mathcal{N}(-d_-(s/K, T-t)) - s \mathcal{N}(-d_+(s/K, T-t))} \quad (7.30)$$

7.7.3 Binary (call) option

These options have a payoff function of the form

$$\Lambda(S_T) = \mathbb{1}_{\{S_T > K\}} = \begin{cases} 1 & \text{if } S_T \geq K, \\ 0 & \text{if } S_T < K. \end{cases} \quad (7.31)$$

The price of this option is the discounted expected payoff at maturity. That is,

$$V(t, s) = e^{-r(T-t)} \mathbb{E}^* [\Lambda(S_T) | S_t = s]. \quad (7.32)$$

Thus, using the payoff function in (7.31) gives

$$V(t, s) = e^{-r(T-t)} \mathbb{E}^* [\mathbb{1}_{\{S_T \geq K\}} | S_t = s], \quad (7.33)$$

where, assuming that the underlying asset is modeled by geometric Brownian motion, we can write

$$V(t, s) = e^{-r(T-t)} \mathcal{N}(d_-(s/K, T-t)) \quad (7.34)$$

7.7.4 Binary (put) option

Analogously to the preceding case, the payoff function is

$$\Lambda(S_T) = \mathbb{1}_{\{S_T < K\}} = \begin{cases} 1 & \text{if } S_T \leq K, \\ 0 & \text{if } S_T > K. \end{cases} \quad (7.35)$$

Thus, the risk-neutral pricing formula gives

$$V(t, s) = e^{-r(T-t)} \mathcal{N}(-d_-(s/K, T-t)) \quad (7.36)$$

7.7.5 Asset-or-nothing (call) option

These options have a payoff function of the form

$$\Lambda(S_T) = S_T \mathbb{1}_{\{S_T > K\}} = \begin{cases} S_T & \text{if } S_T \geq K, \\ 0 & \text{if } S_T < K. \end{cases} \quad (7.37)$$

Using the payoff function in (7.37) gives

$$V(t, s) = e^{-r(T-t)} \mathbb{E}^* [S_T \mathbb{1}_{\{S_T \geq K\}} | S_t = s], \quad (7.38)$$

where a calculation analogous to the preceding cases yields

$$V(t, s) = s \mathcal{N}(d_+(s/K, T-t)) \quad (7.39)$$

7.7.6 Asset-or-nothing (put) option

This is analogous to the preceding case, but with the structure of a put option. That is, the payoff is

$$\Lambda(S_T) = S_T \mathbb{1}_{\{S_T < K\}} = \begin{cases} S_T & \text{if } S_T \leq K, \\ 0 & \text{if } S_T > K, \end{cases} \quad (7.40)$$

whose price at time t is

$$V(t, s) = s \mathcal{N}(-d_+(s/K, T-t)) \quad (7.41)$$

7.7.7 Options with a general piecewise linear payoff

Here, we consider a payoff function given by

$$\Lambda(S_T) = \sum_{i=1}^n (A_i S_T + B_i) \mathbb{1}_{\{a_i \leq S_T < b_i\}}. \quad (7.42)$$

As we will see in the next section, these payoff functions are very common when designing investment strategies using options. Following the preceding argument, the value of this option is

$$\begin{aligned} V(t, s) = & \sum_{i=1}^n \{A_i s [\mathcal{N}(d_+(s/a_i, T-t)) - \mathcal{N}(d_+(s/b_i, T-t))] \\ & + e^{-r(T-t)} B_i [\mathcal{N}(d_-(s/a_i, T-t)) - \mathcal{N}(d_-(s/b_i, T-t))]\} \end{aligned} \quad (7.43)$$

Note that, for $n = 1$, setting $A_1 = 1$, $B_1 = -K$, $a_1 = K$, and $b_1 = +\infty$ recovers a call option. Taking $A_1 = 1$, $B_1 = -K$, $a_1 = -\infty$, and $b_1 = K$ instead recovers a put option.

7.8 Investment strategies using options

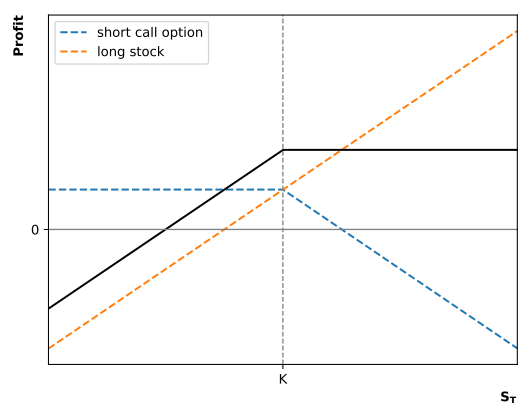
In this section, we study some of the most common investment strategies involving the purchase or sale of options.

7.8.1 Positions in options and the underlying asset

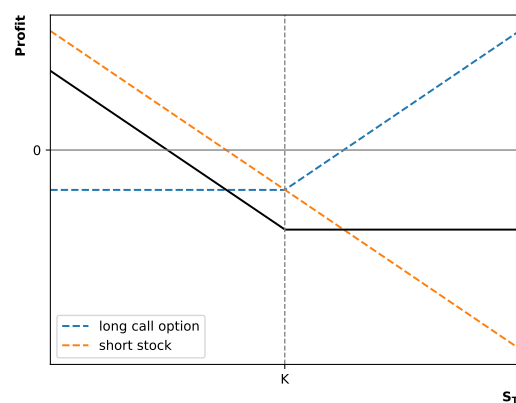
We generally assume that the underlying asset is a stock. We consider 4 possible strategies involving purchases or sales of options on an underlying asset S_t and purchases or sales of the asset itself. The possible *portfolios* are:

- **Short call + long stock.** That is, selling a European call and buying the underlying asset. This strategy is known as *writing a covered call*, since the long position in the underlying stock covers the investor's exposure to the *payoff* of the written call if the stock price rises substantially. The profit structure of this strategy is shown in Figure 7.4a.
- **Long call + short stock.** As shown in Figure 7.4b, the main objective is to profit from a decline in the stock price while providing protection if the price rises. If the stock price falls, the short stock position generates gains, while the call acts as insurance limiting losses if the stock price rises unexpectedly.
- **Long put + long stock.** The main objective is to benefit from an increase in the stock price while retaining protection against substantial declines, as shown in Figure 7.4c. This strategy is also known as a *protective put*.
- **Short put + short stock.** This is the opposite of the preceding strategy, as shown in Figure 7.4d.

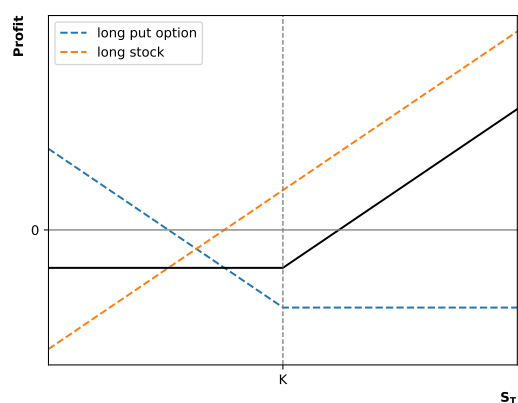
Figure 7.4 shows that these strategies replicate the structure of a call or put option. This follows from a property of options known as put-call parity (to be added).



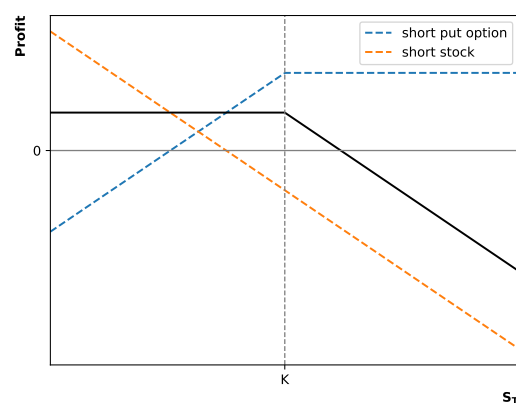
(a) Short call + long stock: Covered call.



(b) Long call + short stock.



(c) Long put + long stock: Protective put.



(d) Short put + short stock.

Figure 7.4: Profit diagrams for strategies combining options and the underlying asset.

7.8.2 Spreads

A *spread* is an investment strategy formed by buying or selling two or more options on the same underlying asset. Different types of *spreads* arise according to the properties of the contracts.

- **Bull spreads.** A bull spread is used when an investor expects the underlying stock price to rise. Its structure limits potential gains but also reduces risk exposure. It can be constructed in two ways:
 - Long call with strike K_1 + short call with strike $K_2 > K_1$.
 - Long put with strike K_1 + short put with strike $K_2 > K_1$.

For a bull spread constructed with calls, as in Figure 7.5a, since call prices decrease as the strike price increases¹⁷, the price of the option sold (income) is always lower than the price of the option bought (expense). Thus, since income < expense, a

¹⁷See, for example, the [Wolfram Demonstrations Project](#).

bull call spread always requires an initial investment. Spreads requiring an initial investment are called debit spreads.

By contrast, a bull spread constructed with puts (as in Figure 7.5b) generates a positive initial cash flow, since a put's price increases as its strike increases⁴. Spreads generating an initial cash inflow are called credit spreads.

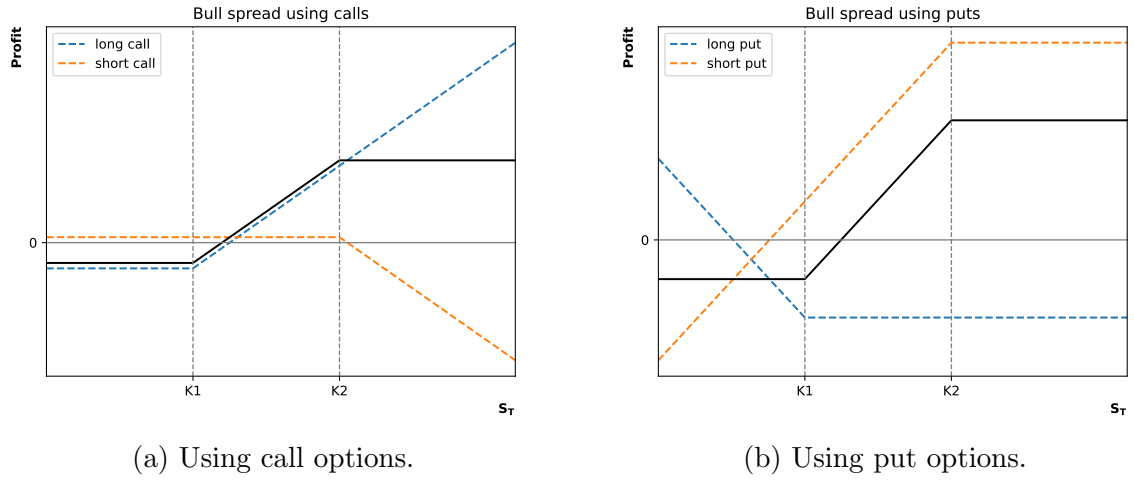


Figure 7.5: Profit diagram of a bull spread.

- **Bear spreads.** A bear spread is used when an investor expects the underlying stock price to fall. Like a bull spread, it limits both potential gains and risk. Again, it can be constructed in two ways:

- Short put with strike K_1 + long put with strike $K_2 > K_1$.
- Short call with strike K_1 + long call with strike $K_2 > K_1$.

A bear spread constructed with puts generates a negative initial cash flow because the short put price (income) is lower than the long put price (expense). Since income < expense, an initial investment is required (a debit spread).

Analogously, a bear spread constructed with calls generates a positive initial cash flow (a credit spread).

- **Box spreads.** A box spread combines a bull call spread and a bear put spread with the same strikes. As shown in Figure 7.7, its profit is constant and equal to $K_2 - K_1$. Thus, its present value is $e^{-r(T-t)}(K_2 - K_1)$. This instrument is particularly useful for detecting arbitrage opportunities. A box spread follows this structure only with European options, not American options.
- **Butterfly spreads.** As in the preceding cases, a butterfly spread can be constructed in two ways:
 - 1 long call with strike K_1 + 2 short calls with strike K_2 + 1 long call with strike $K_3 > K_2 > K_1$.

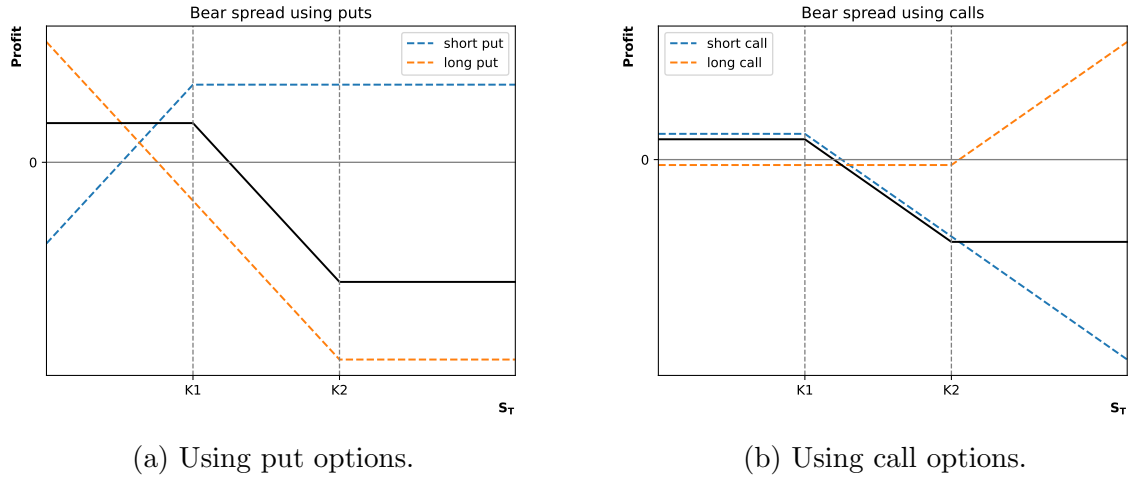


Figure 7.6: Profit diagram of a bear spread.

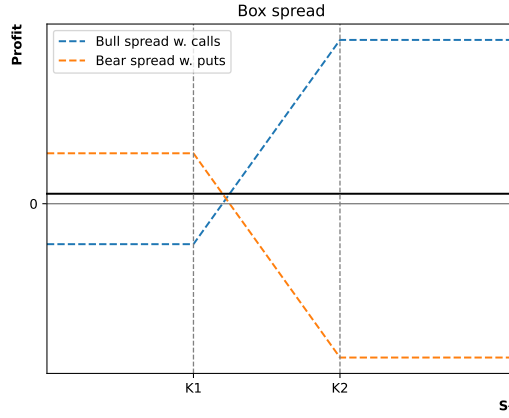


Figure 7.7: Profit diagram of a box spread.

- 1 long put with strike K_1 + 2 short puts with strike K_2 + 1 long put with strike $K_3 > K_2 > K_1$.

This spread generates a profit if the stock price remains close to the strike K_2 . Put-call parity shows that the initial investment is the same for both strategies. The result is shown in Figure 7.8.

An investor can also sell a butterfly spread by taking the opposite positions.

- **Calendar spreads.** Unlike the preceding cases, a calendar spread is constructed by buying or selling options with different maturities, $T_1 < T_2$. It can be formed through the following positions:
 - Short call with maturity T_1 + long call with maturity $T_2 > T_1$.
 - Short put with maturity T_1 + long put with maturity $T_2 > T_1$.

Since longer maturity implies a higher contract price, a calendar spread requires an initial investment. Its profit diagram is shown in Figure 7.9.

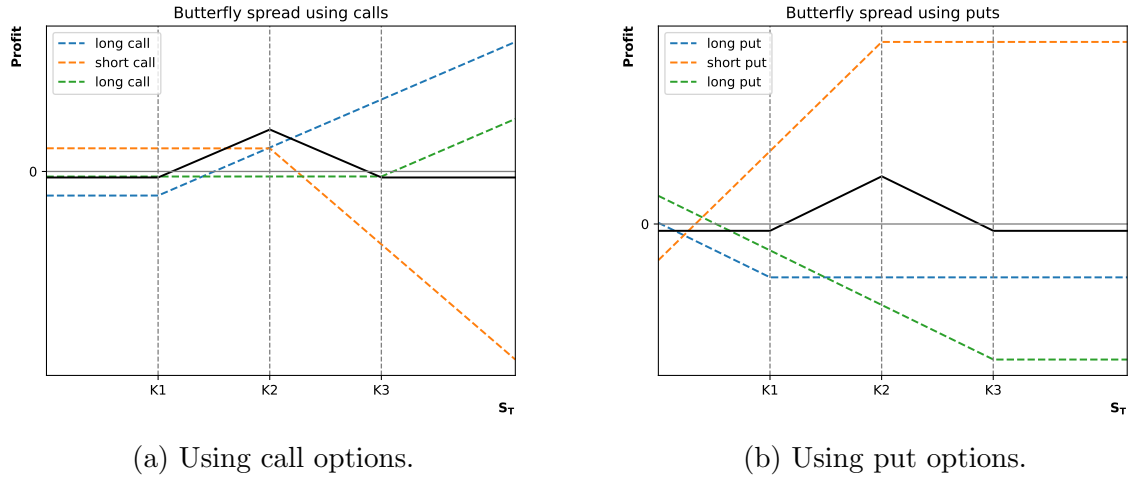
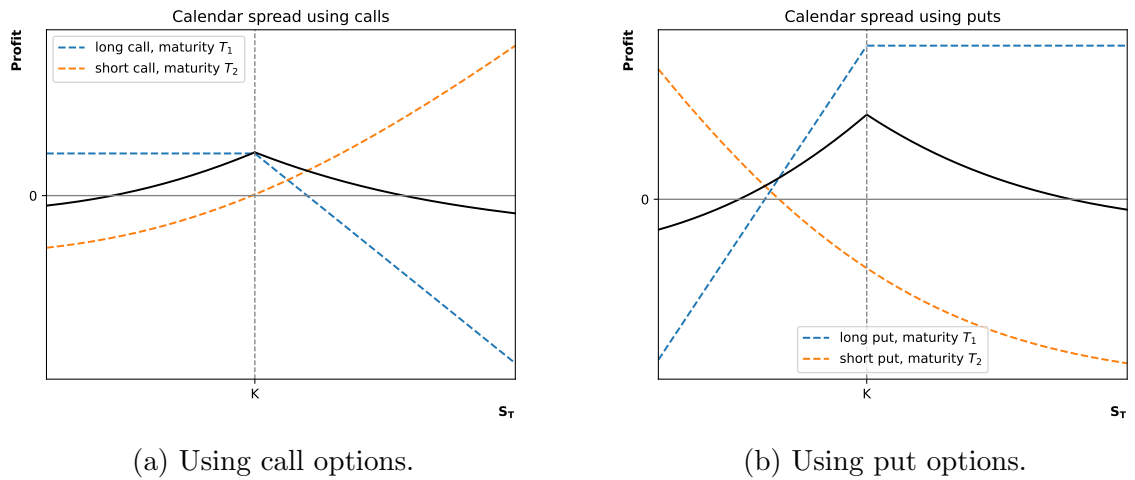


Figure 7.8: Profit diagram of a butterfly spread.

Depending on the difference between the strike and the stock price when the calendar spread is established, we distinguish three main types: (i) a *neutral calendar spread*, when the strike is close to the current stock price; (ii) a *bullish calendar spread*, when the strike exceeds the current price; and (iii) a *bearish calendar spread*, when the strike is below the current price.

Finally, a *reverse calendar spread* can be constructed by taking the opposite positions.


 Figure 7.9: Profit diagram of a calendar spread at $t = T_1$.

- **Diagonal spreads.** Here, both maturities and strikes differ, allowing a wide variety of spreads to be constructed.

7.8.3 Combinations

Combinations are options trading strategies involving both calls and puts on the same underlying asset. We distinguish four main types.

- **Straddle.** A straddle is an options strategy involving the purchase of a call and a put with the same strike and maturity (specifically, a *bottom straddle* or *straddle purchase*). It is used when the investor expects a substantial price movement but is uncertain about its direction. The objective is to profit from market volatility. The profit structure of a bottom straddle is shown in Figure 7.10a. A *top straddle* or *straddle write* is the opposite position (see Figure 7.10b), obtained by selling a call and a put with the same maturity and strike. It is used when the investor expects the underlying price to remain close to the strike, but involves substantial risk because losses are unbounded.

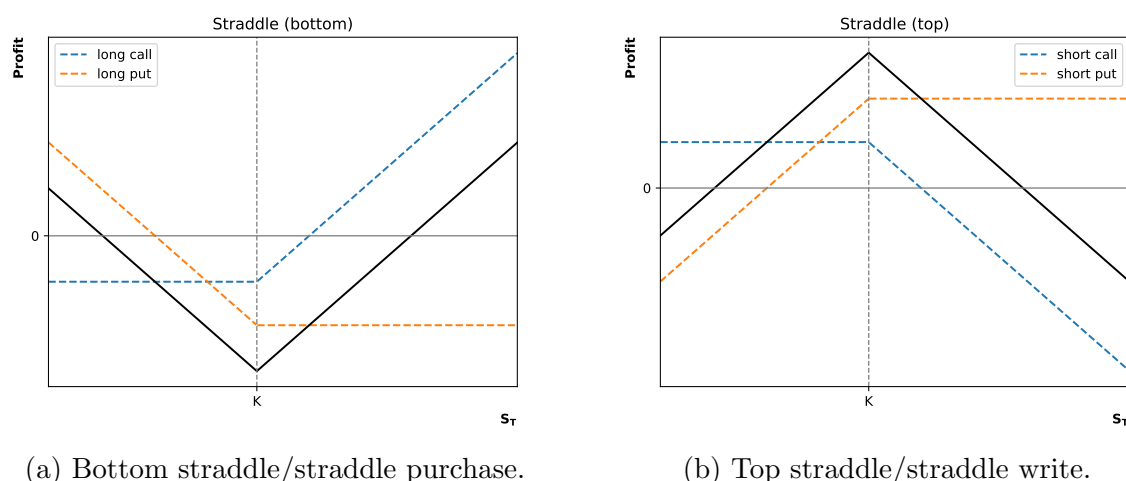


Figure 7.10: Profit diagram of a straddle.

- **Strip.** A strip consists of buying one call and two puts with the same strike and maturity. It is used similarly to a bottom straddle (the investor expects a large underlying price movement), but with the additional view that a price decline is more likely than a rise. Its profit diagram is shown in Figure 7.11.
- **Strap.** A strap consists of buying two calls and one put with the same strike and maturity. It is the counterpart of a strip: the investor expects a large underlying price movement but considers a rise more likely than a decline. Its profit diagram is shown in Figure 7.12.
- **Strangle.** Also known as a *bottom vertical combination*, a strangle is constructed by buying a put and a call with the same maturity but different strikes.

As shown in Figure 7.13a, a strangle is similar to a straddle, but the investor expects a substantial stock price movement without favoring either an upward or a downward move.

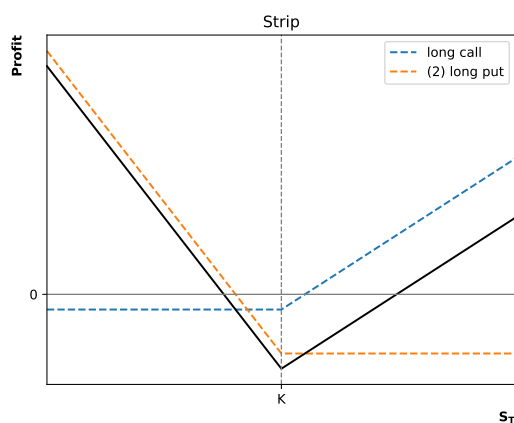


Figure 7.11: Profit diagram of a strip.

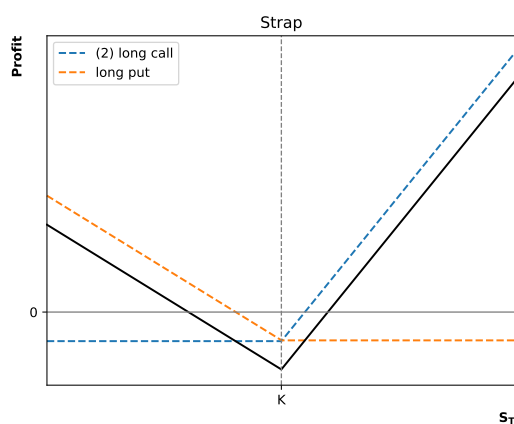


Figure 7.12: Profit diagram of a strap.

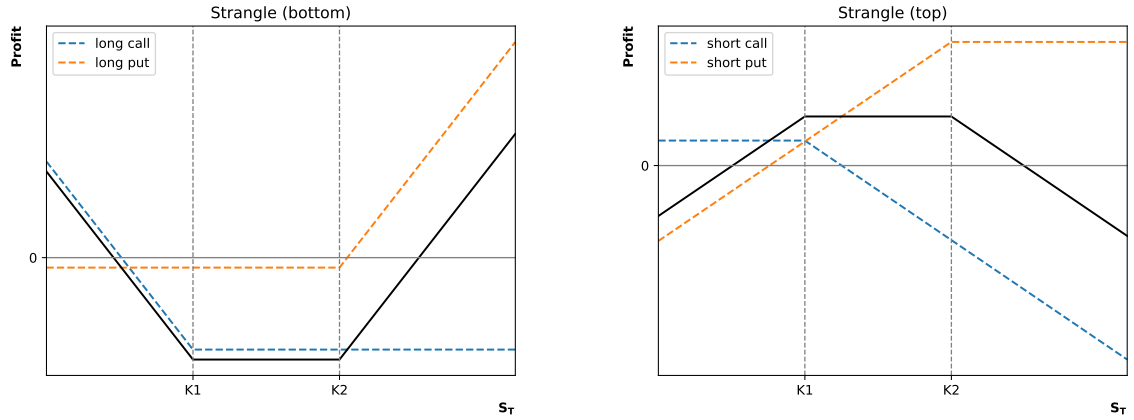
Like straddles, strangles can also be sold, a position known as a *top vertical combination*. This is used for the opposite view: the investor considers a large underlying price movement unlikely. Again, the strategy involves substantial risk because, as shown in Figure 7.13b, potential losses in extreme cases are unbounded.

The code used to generate the figures in this section can be found in Appendix ??.

7.8.4 Other strategies

A commonly used bearish strategy consists of buying a bear spread (a debit spread, with a short put at strike $K_1 = S_t$ and a long put at strike $K_2 > K_1$) and selling a call at strike K_3 whose premium equals the cost of the put spread. The strategy therefore has zero initial cost. Its payoff structure is shown in Figure 7.14.

Unlike a bear spread, this strategy again has unlimited risk because losses are unbounded. Its advantage, however, is protection against a market rise (recall that this is a bearish strategy). The available profit margin can also be adjusted at the cost of reducing the protection provided by selling the call.



(a) Bottom strangle/bottom vertical combination.

(b) Top strangle/top vertical combination.

Figure 7.13: Profit diagram of a strangle.

A similar strategy can be constructed using a credit call spread, that is, by selling and buying call options. The difference is that this is a credit strategy: the premium is received initially, and part of it must be repaid at maturity (possibly more than the premium, depending on the underlying price at T). This structure is shown in Figure 7.15.

7.8.5 Connection with partial differential equations

Proposition 7.11. *Let $\{(\eta_t, \Delta_t) : t \in \mathbb{R}_+\}$ be a self-financing portfolio strategy such that the portfolio value $V_t := \eta_t A_t + \Delta_t S_t$ can be written as a function of t and S_t , that is, $V_t = c(t, S_t)$ where $c \in \mathcal{C}^{(1,2)}(\mathbb{R}_+ \times \mathbb{R}_+)$. Then the function $c(t, x)$ satisfies the partial differential equation*

$$\begin{cases} rc(t, x) &= \frac{\partial c}{\partial t}(t, x) + rx \frac{\partial c}{\partial x}(t, x) + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 c}{\partial x^2}(t, x) \\ \Delta_t &= \frac{\partial c}{\partial x}(t, x) \end{cases} \quad (7.44)$$

for every $t \in \mathbb{R}_+$ and $x \in \mathbb{R}_+$.

A hedging portfolio starts with initial capital V_0 and invests in a risky financial asset S_t and a risk-free asset A_t so that, at each time $t \in [0, T]$, its value equals $c(t, S_t)$. This holds if and only if $e^{-rt} V_t = e^{-rt} c(t, S_t)$ for every t . One way to ensure this equality is to verify that

$$d(e^{-rt} V_t) = d(e^{-rt} c(t, S_t))$$

for every $t \in [0, T)$ and $V_0 = c(0, S_0)$. Integrating gives

$$e^{-rt} V_t - V_0 = e^{-rt} c(t, S_t) - c(0, S_0)$$

for every $t \in [0, T)$. Thus, if $V_0 = c(0, S_0)$, the desired equality follows.

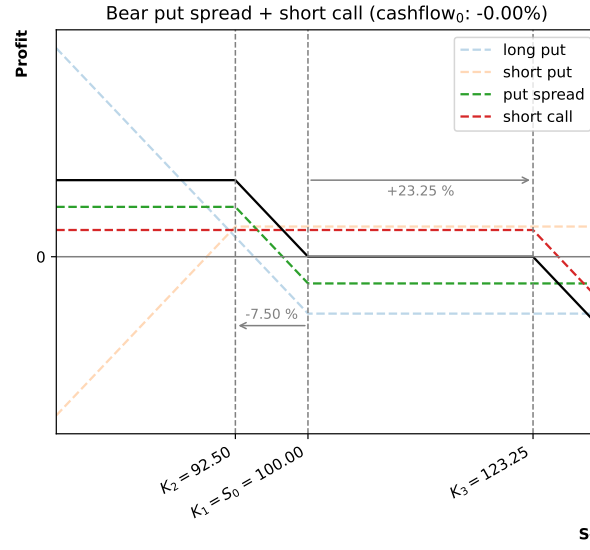


Figure 7.14: Profit diagram of a bear put spread + short call.

Proposition 7.12. *Under the assumptions of Proposition 7.11, the arbitrage-free price $\pi_t(C)$ at time $t \in [0, T]$ of the option with payoff $C = h(S_T)$ is given by $\pi_t(C) = g(t, S_t)$, and the quantity (or proportion) of the underlying asset in the portfolio is given by its partial derivative, where the function $g(t, x)$ solves the partial differential equation*

$$\begin{cases} rg(t, x) &= \frac{\partial g}{\partial t}(t, x) + rx \frac{\partial g}{\partial x}(t, x) + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 g}{\partial x^2}(t, x) \\ g(T, x) &= h(x) \end{cases} \quad (7.45)$$

for every $t \in [0, T]$ and $x \in \mathbb{R}_+$.

In particular, the payoff of a call is $h(x) = (x - K)^+$, where K is the strike price. The solution of partial differential equation (7.45) is then

$$g(t, x) = \text{Bl}(x, K, \sigma, r, T - t) = x\Phi(d_+(T - t)) - Ke^{-r(T-t)}\Phi(d_-(T - t)) \quad (7.46)$$

where

$$d_+(T - t) := \frac{\log(x/K) + (r + \sigma^2/2)(T - t)}{|\sigma|\sqrt{T - t}} \quad (7.47)$$

$$d_-(T - t) := \frac{\log(x/K) + (r - \sigma^2/2)(T - t)}{|\sigma|\sqrt{T - t}} \quad (7.48)$$

for every $t \in [0, T]$. Here, Φ denotes the cumulative distribution function of a random variable $Z \sim N(0, 1)$. The delta of a European call is also given by

$$\xi_t = \Phi(d_+(T - t)) \quad (7.49)$$

Identifying $x = S_t$ and $\xi_t = \Phi(d_+(T - t))$ in (7.46), and using assumption $V_t = g(t, S_t)$ of Proposition 7.11, we can decompose expression (7.46) as

$$g(t, S_t) = V_t = \underbrace{S_t \Phi(d_+(T - t))}_{\text{Risky investment (held)}} - \underbrace{Ke^{-r(T-t)} \Phi(d_-(T - t))}_{\text{Risk-free investment (borrowed)}} \quad (7.50)$$

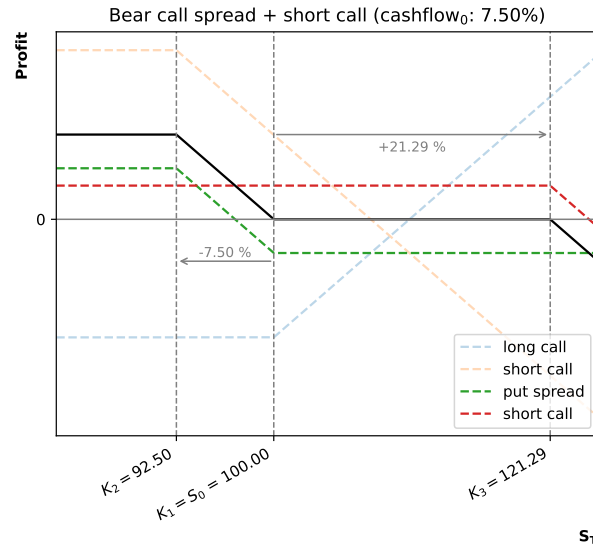


Figure 7.15: Profit diagram of a bear call spread + short call.

We can now implement `Python` code to price a European call. The result is shown in Figure 7.16.

Similarly, Figure 7.17 shows the delta of a particular European call.

Finally, Figure 7.18 shows the hedging strategy that replicates the option value at each time $t \in [0, T]$. The code generating these figures is provided in Appendix ??.

7.9 The Greeks

In risk management and option pricing, the Greeks quantify the sensitivity of an option's price to various market factors. These metrics help investors and portfolio managers understand how an option's value changes with the underlying asset price, volatility, time to maturity, and interest rates.

7.9.1 Delta

Delta is one of the most important Greeks in option pricing theory. It is defined as the change in the option price with respect to the underlying asset price, holding all other parameters constant. That is,

$$\Delta_t = \frac{\partial V}{\partial s}. \quad (7.51)$$

Delta measures the sensitivity of an option's price to small changes in the underlying asset price. It can be interpreted as the approximate number of units of the underlying asset a trader should hold to replicate the option through delta-neutral hedging. A portfolio with positive total delta benefits from increases in the asset price, while negative delta implies exposure to declines.

For a call, delta lies between 0 and 1. A delta of 0.5 means that an increase of 1 in the underlying asset price increases the option value by approximately 0.5 units. For a put, delta lies between -1 and 0, reflecting the decline in option value as the underlying

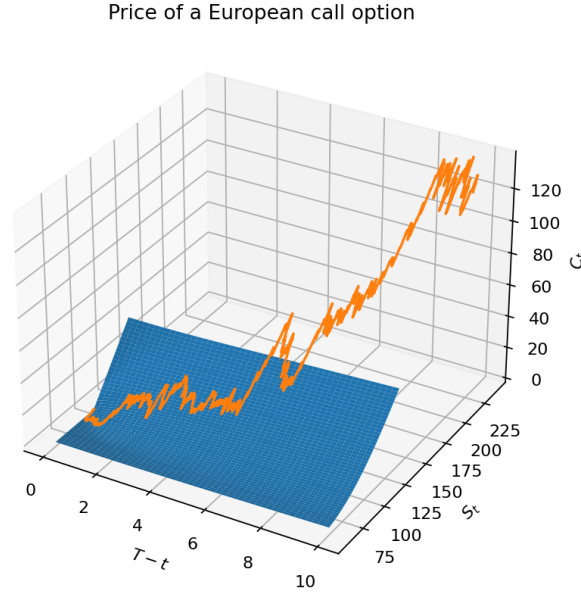


Figure 7.16: Price of a European call as a function of maturity and the underlying asset price.

price rises. As maturity approaches, the deltas of ITM and OTM options approach 1 and 0, respectively, which can be explained by analyzing Γ , introduced later.

Table 7.1 shows the expression for Δ_t for the options analyzed in Section 7.7.

Option type	$\mathcal{V}_t$
European call/asset-or-nothing call	$\mathcal{N}(d_+)$
European put/asset-or-nothing put	$-\mathcal{N}(-d_+)$
Binary call	$e^{-r(T-t)} \frac{\mathcal{N}(d_-)}{s \sigma \sqrt{T-t}}$
Binary put	$-e^{-r(T-t)} \frac{\mathcal{N}(d_-)}{s \sigma \sqrt{T-t}}$

Table 7.1: Expression for Δ_t for the options studied in Section 7.7.

Volatility also affects delta: as volatility increases, ATM option deltas change little, whereas ITM and OTM deltas move more slowly toward 1 or 0. High-volatility options tend to have deltas closer to 0.5 because there is greater uncertainty about whether they will finish ITM or OTM.

7.9.2 Theta

Note that both terms in the call price (7.28) are nonnegative. The highest option price therefore occurs when the first term reaches its maximum and the second its minimum. Thus, assuming that the underlying asset satisfies $S_t = s$, the option value is bounded above by the underlying price: $V(t, s) = s$.

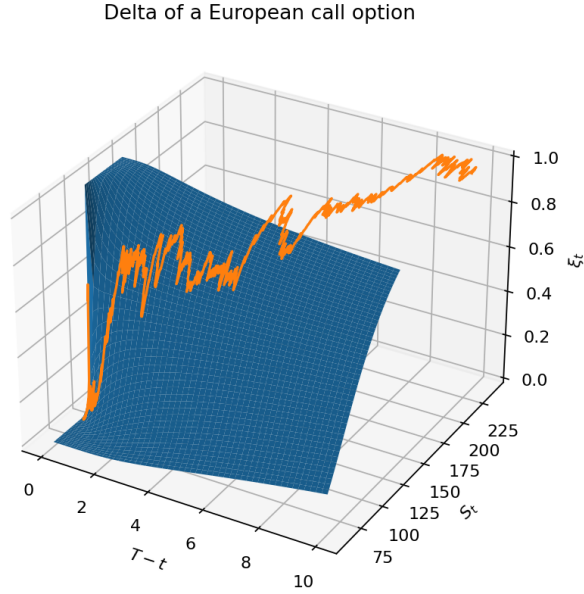


Figure 7.17: Delta of a European call as a function of maturity and the underlying asset price.

In particular, if the remaining time to maturity is very large, that is, $\tau = T - t \gg 1$, we can approximate

$$d_+(x, \tau) \approx \frac{r + \frac{1}{2}\sigma^2}{\sigma} \sqrt{\tau} \rightarrow +\infty$$

as $\tau \rightarrow +\infty$, so $\mathcal{N}(d_+) \rightarrow 1$. Although the behavior of $d_-(x, \tau)$ depends on the relationship between r and $\frac{1}{2}\sigma^2$, the exponential factor $e^{-r\tau}$ multiplying $\mathcal{N}(d_-)$ makes the second term tend to zero. Thus,

$$\lim_{T-t \rightarrow +\infty} V(t, s) = s \quad (7.52)$$

Figure 7.19 shows the value of a call as a function of time to maturity. As can be seen, larger τ implies a higher option price.

In this context, we define the stochastic process θ_t , adapted to the filtration $\{\mathcal{F}_t\}_{t \in I}$, as

$$\theta_t = \frac{\partial V_t}{\partial t}, \quad (7.53)$$

holding all other parameters constant.

Since $\tau = T - t$, we can write $\theta_t = -\frac{\partial V_t}{\partial \tau}$. Figure 7.19 therefore shows that $\theta_t < 0$ for a call, regardless of whether it is ITM, ATM, or OTM (recall that the underlying value S_t is held constant).

Figure 7.20 shows the evolution of a put's price as a function of time to maturity T . It indicates that a put's theta can be positive or negative depending on various factors.

Table 7.2 gives the expressions for theta for the options introduced above.

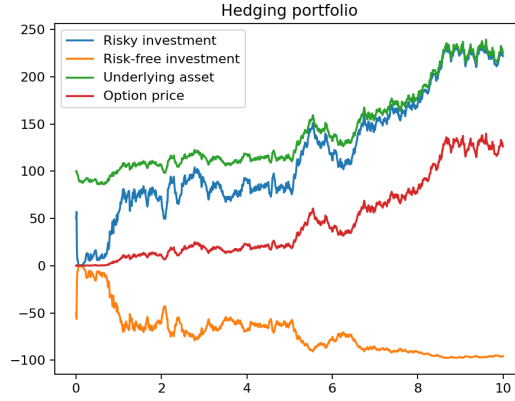


Figure 7.18: Evolution of a hedging portfolio for a call option.

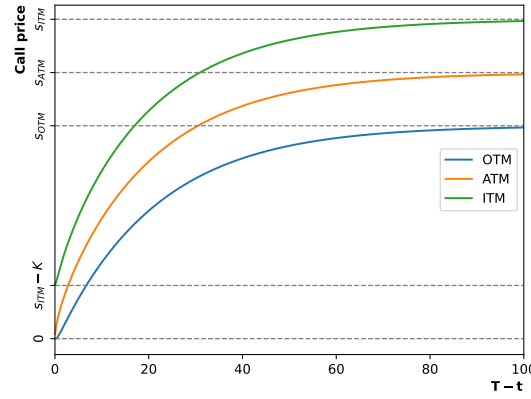


Figure 7.19: Option price as a function of time to maturity. Time is generally measured in years.

7.9.3 Vega

Vega, denoted by $\mathcal{V}$, represents the sensitivity of the option price to volatility¹⁸

$$\mathcal{V}_t = \frac{\partial V}{\partial \sigma}. \quad (7.54)$$

Figure 7.21 shows the behavior of a European call's price as a function of volatility for ITM, ATM, and OTM options. Similarly, Figure 7.22 analyzes the value of a European put as a function of volatility.

As shown, the option price is strictly increasing for both calls and puts, regardless of whether the option is ITM, ATM, or OTM. This implies that $\mathcal{V}_t > 0$.

However, we will later see that this property does not hold for more complex structures. For example, $\mathcal{V}$ can be negative if an increase in volatility reduces the derivative's value, as in strategies such as short straddles.

Table 7.3 gives the expression for $\mathcal{V}$ for the options studied in Section 7.7.

¹⁸In the Black-Scholes model, volatility is assumed constant. Nevertheless, it is useful to study how the option price behaves when σ is treated as an arbitrary parameter.

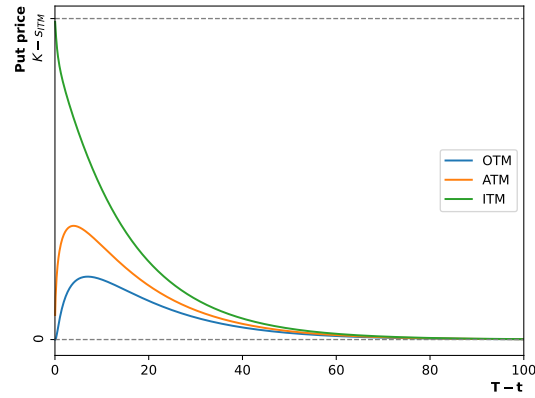


Figure 7.20: Option price as a function of time to maturity. Time is generally measured in years.

Option type	θ_t
European call	$-\frac{s n(d_+) \sigma}{2\sqrt{T-t}} - rK e^{-r(T-t)} \mathcal{N}(d_-)$
European put	$-\frac{s n(d_+) \sigma}{2\sqrt{T-t}} + rK e^{-r(T-t)} \mathcal{N}(-d_-)$
Binary call	$e^{-r(T-t)} \left[r \mathcal{N}(d_-) + n(d_-) \left(\frac{r}{\sigma\sqrt{T-t}} - \frac{d_+}{2(T-t)} \right) \right]$
Binary put	$e^{-r(T-t)} \left[-r \mathcal{N}(-d_-) + n(-d_-) \left(\frac{r}{\sigma\sqrt{T-t}} - \frac{d_+}{2(T-t)} \right) \right]$
Asset-or-nothing call	$-\frac{s n(d_+) \sigma}{2\sqrt{T-t}} - rs \mathcal{N}(d_+)$
Asset-or-nothing put	$-\frac{s n(d_+) \sigma}{2\sqrt{T-t}} + rs \mathcal{N}(-d_+)$

Table 7.2: Expression for θ_t for the options studied in Section 7.7.

7.10 Option pricing from a trader's perspective

7.10.1 Hedging strategies using derivatives

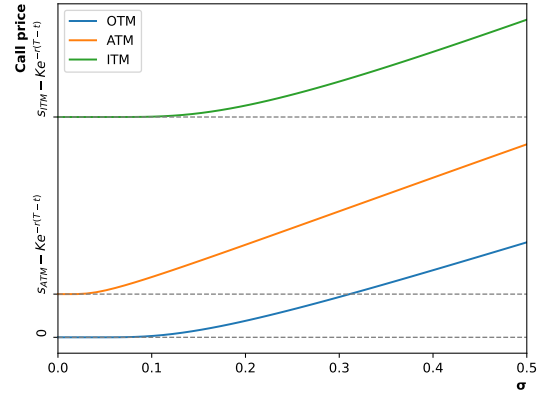


Figure 7.21: Call option price as a function of volatility.

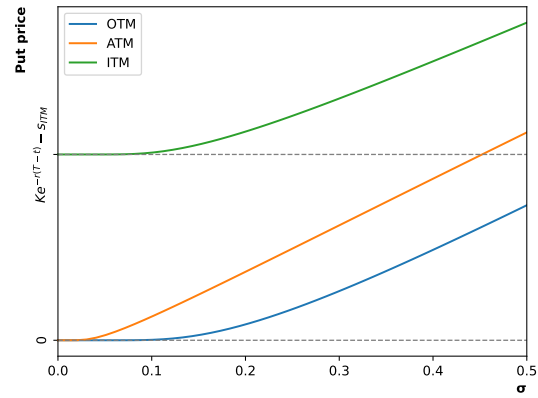


Figure 7.22: Put option price as a function of volatility.

Option type	$\mathcal{V}_t$
European call/put	$s \mathbf{n}(d_+) \sqrt{T-t}$
Binary call	$-e^{-r(T-t)} \mathbf{n}(d_-) \frac{d_+}{\sigma}$
Binary put	$e^{-r(T-t)} \mathbf{n}(d_-) \frac{d_+}{\sigma}$
Asset-or-nothing call	$s \mathbf{n}(d_+) \left(\sqrt{T-t} - \frac{d_+}{\sigma} \right)$
Asset-or-nothing put	$-s \mathbf{n}(d_+) \left(\sqrt{T-t} - \frac{d_+}{\sigma} \right)$

 Table 7.3: Expression for $\mathcal{V}_t$ for the options studied in Section 7.7.

Chapter 8

Local volatility models

Stochastic volatility models are best motivated from the perspective of a trader's needs.

As seen in Section [to be added], the Black-Scholes model has numerous inconsistencies and idealizes the underlying's lognormal dynamics. These limitations might suggest that traders use far more complex models to capture those dynamics adequately. Nevertheless, Black-Scholes remains widely used in practice, especially for managing risk in derivatives portfolios.

An exhaustive search for much more complex models would make sense only if the underlying asset's prices were precisely determined by the model. Financial asset prices, however, are so unpredictable and dynamic that this approach can become impractical. Moreover, traders buy and sell derivatives at actual market prices, rather than model-implied prices. They therefore take a more practical approach, using models to calculate option prices without relying on them completely to predict future market behavior.

A pricing formula does not predict the future of markets; it helps explain how gains and losses on a derivatives position change as different variables evolve. This allows traders to anticipate how market movements affect their P&L without necessarily predicting future prices precisely.

More sophisticated models do more than calculate derivative prices: they explain in detail how different factors affect P&L. Traders use these models to identify and control the associated risks.

Traders therefore face a decision about how to manage the risks associated with their derivatives positions. They can hedge them with other options or derivatives, or retain them in the portfolio and accept the market exposure. The choice depends on their strategy and risk tolerance.

8.1 Which models can describe the underlying?

As usual, we consider a market consisting of a risk-free asset B , serving as numeraire, and a risky asset whose price S is a nonnegative stochastic process defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and adapted to a filtration $\{\mathcal{F}_t : t \in \mathbb{R}_+\}$. We denote the risk-neutral measure by $\mathbb{P}^*$ and assume that it exists and is unique –so that the fundamental theorems of asset pricing from Chapter 7 apply.

Let $V(t, s; T, K, \boldsymbol{\theta})$ be the price of a European option with payoff $\Lambda : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathbb{R}$, where T and K denote maturity and strike, respectively, and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)$ represents

other possible parameters treated as constants, e.g., the risk-free interest rate r , continuous dividends q , etc. When unambiguous, we abbreviate $V(t, s; T, K, \boldsymbol{\theta})$ as $V(t, s)$.

Suppose that we have developed a model to price this option and want to neutralize the portfolio's delta –to delta hedge it. A first step in checking the model is clearly to verify whether $V(T, s) = \Lambda(s)$ for every $s \in \mathbb{R}_+$.

Once this condition is verified, we proceed as follows: (i) sell the option and receive the premium $V(t, s)$; (ii) invest the premium in a zero-coupon bond earning interest at rate r , initially assumed constant; (iii) borrow to neutralize delta by buying the underlying, which may pay dividends represented by q . The initial portfolio value is then zero by construction, and the P&L change over an infinitesimal time interval is

$$dP\&L = \underbrace{-(V(t+dt, s+ds) - V(t, s))}_{(i)} + \underbrace{rV(t, s)dt}_{(ii)} + \underbrace{\Delta(ds - (r-q)sdt)}_{(iii)},$$

where ds is the underlying price change over the interval $(t, t+dt)$.

For sufficiently small dt ¹, we can expand term (i) as

$$V(t+dt, s+ds) - V(t, s) = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial s}ds + \frac{1}{2}\frac{\partial^2 V}{\partial s^2}ds^2 + \frac{1}{2}\frac{\partial^2 V}{\partial t^2}dt^2 + \frac{\partial^2 V}{\partial t \partial s}dt ds + \dots$$

We retain first-order terms in dt and second-order terms² in ds . Delta hedging removes first-order exposure to underlying price changes, so the term linear in ds disappears from P&L by construction. Thus,

$$dP\&L = -\left(\frac{\partial V}{\partial t} - rV + (r-q)s\Delta\right)dt - \left(\frac{\partial V}{\partial s} - \Delta\right)ds - \frac{1}{2}\frac{\partial^2 V}{\partial s^2}ds^2.$$

Maintaining a delta-neutral portfolio necessarily requires setting $\Delta = \partial V/\partial s$, as seen in the definition of the Greeks in Section 7.9. Trading the underlying can neutralize only portfolio delta (the other Greeks must be hedged with derivatives, hence the saying “options are hedged with options”). Finally,

$$dP\&L = -\left(\frac{\partial V}{\partial t} - rV + (r-q)s\frac{\partial V}{\partial s}\right)dt - \frac{1}{2}\frac{\partial^2 V}{\partial s^2}ds^2.$$

Recalling the definitions in Section 7.9, we identify $\Gamma = \frac{\partial^2 V}{\partial s^2}$ and $\Theta = \partial V/\partial t$, where Δ has been set to $\partial V/\partial s$ to eliminate first-order portfolio exposure to underlying price changes. Traders often use an alternative definition of gamma, known as dollar gamma, $\Gamma^D = s^2\Gamma$, which has the same units as V and quantifies the P&L contribution from the position's convexity per unit of relative quadratic variation of the underlying. Therefore,

$$dP\&L = -(\Theta - rV + (r-q)s\Delta)dt - \frac{1}{2}\Gamma^D\left(\frac{ds}{s}\right)^2.$$

¹The typical underlying price change over an interval dt is of order $s\sigma\sqrt{dt}$, where σ is volatility. For assets with moderate volatility, daily delta rebalancing is generally adequate, $dt = 1$, whereas higher volatility requires intraday hedging [29].

²That is, we retain first-order terms in dt , since $\mathbb{E}^*[dtdS_t|\mathcal{F}_t] \approx dt^{3/2}$ and $\mathbb{E}^*[ds_t^2|\mathcal{F}_t] \approx dt$.

We define the functions $A(t, s; T, K, \boldsymbol{\theta}) = (\Theta_t - rV + (r - q)s\Delta_t)$ and $B(t, s; T, K, \boldsymbol{\theta}) = \frac{1}{2}\Gamma_t^D$, where $\Theta_t \equiv \Theta(t, s; T, K, \boldsymbol{\theta})$ and $\Gamma_t^D \equiv \Gamma^D(t, s; T, K, \boldsymbol{\theta})$. When discussing portfolio P&L, the function $A(t, s)$ is often called portfolio theta because it reflects the portfolio's *effective* theta³ (recall that the minus signs arise from the short option position).

We can distinguish the following cases: (i) If $A(t, s) \geq 0$ and $B(t, s) \geq 0$, the trader loses money regardless of ds , so model $V(t, s)$ cannot be used to hedge the option because it underestimates the true value. The initial premium was too low. (ii) If $A(t, s) \leq 0$ and $B(t, s) \leq 0$, the trader earns *free money*, again suggesting mispricing because the model overestimates the true value: the initial premium was too high. (iii) The model is therefore usable (and we call it a market model) only if A and B have opposite signs for any $(t, s) \in \mathbb{R}_+^2$. Moreover, condition $dP\&L \approx 0$ holds only if

$$\left(\frac{ds}{s}\right)^2 \approx -\frac{A(t, s; \boldsymbol{\theta})}{B(t, s; \boldsymbol{\theta})}dt.$$

Although dS_t is not $\mathcal{F}_t$ -measurable, any Itô process satisfies that dS_t^2 is $\mathcal{F}_t$ -measurable by construction. Thus, the model is properly calibrated when expected P&L under the risk-neutral measure is zero⁴.

Specifically, since P&L discounted by the numeraire B is a martingale, $dP\&L$ is driftless and therefore $\mathbb{E}^*[dP\&L | \mathcal{F}_t] = 0$. This condition gives

$$A_t dt = -\frac{1}{2}\Gamma_t^D \mathbb{E}^*\left[\frac{dS_t^2}{S_t^2} \middle| \mathcal{F}_t\right] \Leftrightarrow \mathbb{E}^*\left[\frac{dS_t^2}{S_t^2} \middle| \mathcal{F}_t\right] = -2\frac{A_t}{\Gamma_t^D} dt \equiv \sigma_{\text{break-even}}^2(t, S_t) dt.$$

so a model is valid when a break-even volatility exists such that

$$\Theta_t - rP_t + (r - q)S_t\Delta_t = -\frac{1}{2}\Gamma_t^D \sigma_{\text{break-even}}^2(t, S_t)$$

where $\sigma_{\text{break-even}}(t, S_t)$ is a possibly stochastic process adapted to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$ by construction. Rewriting this relation in terms of option price derivatives gives

$$\frac{\partial P}{\partial t} - rP_t + (r - q)S_t \frac{\partial P}{\partial s} + \frac{\sigma_{\text{break-even}}^2(t, S_t)}{2} S_t^2 \frac{\partial^2 P}{\partial s^2} = 0, \quad (8.1)$$

So far, we have not specified any model for the underlying. The Feynman-Kac theorem now allows us to determine which underlying asset models permit delta hedging of the portfolio and hence correct replication of the option price. That is, we can apply Feynman-Kac to study which models admit equation (8.1).

³Indeed, assuming that P&L is a function of t and s , we can write $dP\&L(t, s) = \frac{\partial P\&L}{\partial t} dt + \frac{\partial P\&L}{\partial s} ds + \frac{1}{2} \frac{\partial^2 P\&L}{\partial s^2} ds^2 = \frac{\partial P\&L}{\partial t} dt + \frac{\partial P\&L}{\partial s} ds + \frac{1}{2} s^2 \frac{\partial^2 P\&L}{\partial s^2} \left(\frac{ds}{s}\right)^2$, where $\frac{\partial P\&L}{\partial s} = 0$ because delta has been neutralized. Thus, A and B correspond to the portfolio's total Θ and Γ^D .

⁴The relationship between P&L and the replicating portfolio value Π_t over an infinitesimal interval $(t, t + dt)$ is $dP\&L(t) = d\Pi_t - r\Pi_t dt$. If the model is correct and replication is perfect, P&L must be zero on average, that is, $\mathbb{E}^*[dP\&L | \mathcal{F}_t] = 0$, implying that the portfolio satisfies $\mathbb{E}^*[d\Pi_t | \mathcal{F}_t] = r\Pi_t dt$ (because Π_t is a martingale under $\mathbb{P}^*$). In general, when local portfolio risk is eliminated by neutralizing delta, $\mathbb{E}^*[dP\&L | \mathcal{F}_t] = dP\&L$, so the preceding condition is equivalent to $dP\&L = 0$ or $d\Pi_t = r\Pi_t dt$.

Theorem 8.1: Feynman-Kac

Given the partial differential equation

$$\frac{\partial}{\partial t}u(t, s) + \mu(t, s)\frac{\partial}{\partial s}u(t, s) + \frac{1}{2}\sigma^2(t, s)\frac{\partial^2}{\partial s^2}u(t, s) - h(t, s)u(t, s) + f(t, s) = 0, \quad (8.2)$$

defined on $(t, s) \in I \times \mathbb{R}$ and subject to the terminal condition

$$u(T, s) = \Lambda(s) \quad (8.3)$$

where μ, σ, Λ, h and f are known functions, T is a parameter, and $u : I \times \mathbb{R} \rightarrow \mathbb{R}$ is the unknown, $u(t, s)$ can be expressed as a conditional expectation under the probability measure $\mathbb{P}^*$ of the form

$$u(t, s) = \mathbb{E}^* \left[e^{-\int_t^T h(v, S_v)dv} \Lambda(S_T) + \int_t^T f(v, S_v) e^{-\int_t^s h(z, S_z)dz} dv \mid S_t = s \right], \quad (8.4)$$

where S is an Itô process satisfying

$$dS_t = \mu(t, S_t)dt + \sigma(t, S_t)dW_t \quad (8.5)$$

with W_t a Wiener process under $\mathbb{P}^*$.

Applying the Feynman-Kac theorem therefore gives the general form of the dynamics of S_t under $\mathbb{P}^*$:

$$dS_t = (r - q)S_t dt + \sigma_{\text{break-even}}^2(t, S_t)S_t dW_t,$$

and consequently S_t must be described by a diffusion process.

The preceding argument yields the following conclusion: **a pricing model must ensure the existence of a break-even volatility for which the infinitesimal P&L change is zero on average. In that case, the underlying asset dynamics must follow a diffusion process.** Thus, the only *valid* pricing model with constant break-even volatility is Black-Scholes. If $\sigma_{\text{break-even}} = \sigma_{\text{break-even}}(t)$, we recover Black-Scholes with a term structure, while $\sigma_{\text{break-even}} = \sigma_{\text{break-even}}(t, s)$ gives a local volatility model.

However, pricing an option is not equivalent to describing the *real* world. In practice, one finds a σ that reproduces observed market prices when inserted into the pricing model. As seen in Proposition 8.2, this is implied volatility, and it is unique for each pair (T, K) . In general, implied volatility differs from realized volatility.

Formally, given $(t, s) \in \mathbb{R}_+^2$, we define $\{V_M(T, K; t, s) : (T, K) \in I_T \times I_K\}$ as the family of market option prices, where I_T and I_K represent the discrete sets of maturities and strikes available in the market. That is, we consider market option prices as a function of strike and expiry for specific t and s , treated as parameters. This family of prices naturally changes with time and the underlying value.

Proposition 8.2

Let $t, s \in \mathbb{R}_+^2$ be fixed and $V_M : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ be defined by $(T, K) \mapsto V_M(T, K; t, s)$ such that $\max\{s - K, 0\} \leq V_M(t, s; K) \leq s$ for every $(T, K) \in \mathbb{R}_+^2$. Then equation $V_M(T, K; t, s) = V_{BS}(t, s; T, K, \sigma, \boldsymbol{\theta})$ has a unique nonnegative solution, denoted by σ_{imp} , called implied volatility.

Proof. As seen in Section 7.9, the map $\sigma \mapsto V_{BS}(t, s; T, K, \sigma, \boldsymbol{\theta})$ is strictly increasing and convex. In particular, $\lim_{\sigma \rightarrow 0} V_{BS}(t, s; T, K, \sigma, \boldsymbol{\theta}) = \max\{s - K, 0\}$ and $\lim_{\sigma \rightarrow +\infty} V_{BS}(t, s; T, K, \sigma, \boldsymbol{\theta}) = s$ hold. The result follows directly by applying the intermediate value theorem to $f(\sigma) = V_M(T, K; t, s) - V_{BS}(t, s; T, K, \sigma, \boldsymbol{\theta})$ on $\mathbb{R}_+$. $\square$

Thus, given $\#(I_T \times I_K)$ market observations for different maturities and strikes, we can obtain $\#(I_T \times I_K)$ implied volatilities for each T and K —with parameters t, s and $\boldsymbol{\theta}$ fixed in advance. Hence, $\sigma_{\text{imp}} = \sigma_{\text{imp}}(T, K; t, s, \boldsymbol{\theta})$, abbreviated as $\sigma_{\text{imp}}(T, K)$, whose image is called the volatility surface. The bounds on V_M always hold in the market; otherwise, an immediate arbitrage opportunity would exist.

Expected P&L can therefore be written as

$$dP\&L(t, s; \boldsymbol{\theta}) = -\frac{1}{2}s^2 \frac{\partial^2 V}{\partial s^2} (\sigma_{\text{break-even}}^2 - \sigma_{\text{imp}}^2) dt, \quad (8.6)$$

and is commonly called gamma/theta P&L or carry P&L in options trading. Carry P&L can thus be analyzed in terms of realized volatility relative to implied volatility:

- If the trader (in this option-selling strategy) is long⁵ gamma (that is, $\Gamma^D > 0$, see Section 7.9), P&L is positive if realized volatility is below implied volatility, and negative otherwise. Generally, a long gamma position has losses limited to the option premium and unlimited gains, owing to the option payoff structure. Being long gamma is therefore a long volatility strategy in the sense that it benefits from large underlying price movements.
- If the trader (in this option-selling strategy) is short gamma (that is, $\Gamma^D < 0$), P&L is positive if realized volatility exceeds implied volatility, and negative otherwise. Unlike the preceding case, selling options limits gains to the premium received, while losses are unlimited. Thus, being short gamma means being short volatility.

Applying this theorem shows that necessarily

$$dS_t = (r - q)S_t dt + \sigma(t, S_t)S_t dW_t, \quad (8.7)$$

which includes Black-Scholes when $\sigma(t, s) = \sigma$ is constant.

⁵In most situations, selling options directly implies a short gamma position. However, there are exceptions in which being short options can imply being long gamma.

8.2 Analysis of the standard deviation of P&L

The value of $dP\&L$ is expressed in monetary units at $t+dt$. To express $dP\&L$ in monetary units at time t , we must discount at a risk-free interest rate r . That is,

$$dP\&L(t, s; \boldsymbol{\theta}) = -\frac{1}{2}e^{-rt}s^2\frac{\partial^2 V}{\partial s^2}\left(\frac{ds^2}{s^2} - \sigma_{\text{imp}}^2 dt\right) \quad (8.8)$$

Continuing the preceding argument, since the discounted P&L of a self-financing strategy is a martingale under $\mathbb{P}^*$, see [17], we have

$$D_t \mathbb{E}^*[P\&L | \mathcal{F}_t] = 0,$$

since $P\&L_0 = 0$. Therefore,

$$-\frac{1}{2}\mathbb{E}^*\left[e^{-rt}S_t^2\frac{\partial^2 V}{\partial s^2}(t, S_t)(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2) \middle| \mathcal{F}_t\right] = 0,$$

which can be rewritten as

$$\mathbb{E}^*\left[e^{-rt}S_t^2\frac{\partial^2 V}{\partial s^2}(t, S_t)\sigma_{\text{real}}^2 \middle| \mathcal{F}_t\right] = \mathbb{E}^*\left[e^{-rt}S_t^2\frac{\partial^2 V}{\partial s^2}(t, S_t)\sigma_{\text{imp}}^2 \middle| \mathcal{F}_t\right]. \quad (8.9)$$

8.3 The need for stochastic volatility models

In practice, delta hedging is insufficient. Although the portfolio is hedged against first-order underlying price movements, the effect of ds^2 is substantial, and gamma can be hedged only through other options, such as vanilla options.

Suppose we buy λ vanilla options on the same underlying, with price $C(t, s; T', K', \boldsymbol{\theta})$. Total P&L changes over an interval $(t, t+dt)$ by

$$dP\&L(t, s; \boldsymbol{\theta}) = -\frac{1}{2}s^2\frac{\partial^2 V}{\partial s^2}\left(\frac{ds^2}{s^2} - \sigma_{\text{imp}}^2 dt\right) + \frac{\lambda}{2}s^2\frac{\partial^2 C}{\partial s^2}\left(\frac{ds^2}{s^2} - \hat{\sigma}_{\text{imp}}^2 dt\right),$$

which can be grouped as

$$dP\&L(t, s; \boldsymbol{\theta}) = -\frac{1}{2}s^2(\lambda\Gamma_C\hat{\sigma}_{\text{imp}}^2 - \Gamma_V\sigma_{\text{imp}}^2)dt + \frac{1}{2}s^2(\lambda\Gamma_C - \Gamma_V)\frac{ds^2}{s^2},$$

where $\hat{\sigma}_{\text{imp}}$ denotes the implied volatility of the vanilla option priced at $C(t, s)$. Initially, we assume that the implied volatilities differ, that is, $\hat{\sigma}_{\text{imp}} \neq \sigma_{\text{imp}}$.

Thus, the P&L term quadratic in ds vanishes if we buy⁶

$$\lambda(t, s; \boldsymbol{\theta}) = \frac{\Gamma_V(t, s; \boldsymbol{\theta})}{\Gamma_C(t, s; \boldsymbol{\theta})} \quad (8.10)$$

units of the vanilla option priced at $C(t, s; \boldsymbol{\theta})$ per unit of option V to gamma hedge the portfolio. In practice, because the gamma profiles of the options can differ substantially,

⁶More precisely, we buy if $\lambda > 0$ and sell if $\lambda < 0$.

the required quantity of vanilla options λ must be rebalanced as time passes and the underlying moves.

Hedging gamma while using different implied volatilities would lead to the contradictory case introduced earlier, where $A(t, s; \boldsymbol{\theta}) \neq 0$ and $B(t, s; \boldsymbol{\theta}) = 0$ imply a systematic loss or gain. We must therefore choose $\hat{\sigma}_{\text{imp}} = \sigma_{\text{imp}}$.

Thus, to first order in dt and second order in ds , we have $dP\&L(t, s; \boldsymbol{\theta}) = 0$ once portfolio delta and gamma have been hedged.

In practice, however, the option price depends not only on time and the underlying, but also on implied volatility levels. We must therefore allow $V = V(t, s, \sigma_{\text{imp}}; T, K, \boldsymbol{\theta})$ and $C = C(t, s, \sigma_{\text{imp}}; T, K, \boldsymbol{\theta})$. In particular, since σ_{imp} depends on the vanilla option prices C through pricing formula (8.1), we allow the price of P to depend on vanilla option prices. Including implied volatility dependence to second order gives the P&L of a position hedged in delta and gamma as

$$\begin{aligned} dP\&L(t, s, \sigma_{\text{imp}}; \boldsymbol{\theta}) = & - \left(\frac{\partial V}{\partial \sigma_{\text{imp}}} - \lambda \frac{\partial C}{\partial \sigma_{\text{imp}}} \right) d\sigma_{\text{imp}} - \frac{1}{2} \left(\frac{\partial^2 V}{\partial \sigma_{\text{imp}}^2} - \lambda \frac{\partial^2 C}{\partial \sigma_{\text{imp}}^2} \right) d\sigma_{\text{imp}}^2 \\ & - \left(\frac{\partial^2 V}{\partial s \partial \sigma_{\text{imp}}} - \lambda \frac{\partial^2 C}{\partial s \partial \sigma_{\text{imp}}} \right) ds d\sigma_{\text{imp}}, \end{aligned} \quad (8.11)$$

which is generally called mark-to-market P&L. Since V and C are determined by the Black-Scholes equation (8.1), we have the relation

$$\mathcal{V} = \Gamma^D \sigma_{\text{imp}} T,$$

introduced in Section 7.9, now replacing constant volatility with the implied volatility function $\sigma_{\text{imp}}(T, K; t, s, \boldsymbol{\theta})$. Gamma and vega are therefore related, so a gamma-hedged position is automatically vega-hedged. For exotic options *in practice*, of course, gamma hedging does not imply vega hedging. Depending on the relative risk magnitudes and the strategy, the trader chooses which to hedge.

Eliminating the term ds^2 removes P&L dependence on realized volatility, but introduces exposure to implied volatility dynamics⁷.

Unlike the case where ds^2 was unhedged, there is now no term in dt to offset $d\sigma_{\text{imp}}^2$ and $ds d\sigma_{\text{imp}}$. Depending on their realized values, the trader may therefore systematically gain or lose money. For carry P&L, implied volatility controlled how gamma and theta offset one another, but we now have no analogous term to offset volga and vanna. The model contains neither an implied volatility for σ_{imp} nor an implied correlation between S and σ_{imp} . This indicates that these options must be priced using a different equation that explicitly includes this parameter, generating additional theta terms to resolve the problem. This is the role of stochastic volatility models, including local volatility.

Thus, acquiring other options as hedging instruments can partially or fully reduce exposure to realized volatility, at the cost of exposure to implied volatility and its evolution, that is, its dynamics.

The task is therefore to model implied volatility dynamics, rather than realized volatility dynamics, since the latter exposure is hedged with options or other instruments. This allows us to determine when a trader will gain or lose money on a hedged position.

⁷In fact, some exposure to realized volatility remains through the covariance term $ds d\sigma_{\text{imp}}$.

For very liquid financial assets, we do not model a single implied volatility σ_{imp} ; instead, we model the dynamics of all implied volatilities $\sigma_{\text{imp}}(T, K)$, where T and K denote vanilla option maturities and strikes. The image of $\sigma_{\text{imp}} : I_T \times I_K \rightarrow \mathbb{R}_+$ is precisely what is known as the volatility surface.

8.4 Introduction to local volatility

Market models treat vanilla option prices observed in the market at $t = 0$ as fundamental model inputs, alongside the underlying price. They do not attempt to derive vanilla prices from theoretical parameters, but incorporate market information directly as initial input. These prices are assumed available and valid (arbitrage-free).

Thus, vanilla option prices are used as hedging instruments rather than explained or derived. The model must reproduce them consistently because they are the system's *initial conditions*.

The simplest extension of Black-Scholes allows instantaneous relative volatility to be a deterministic function of time and the underlying, given the correspondence between Itô and Markov processes. We therefore take $\sigma = \sigma_{\text{loc}}(t, s)$, with the underlying price determined under the risk-neutral measure⁸ by

$$dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(t, S_t)S_t dW_t \quad (8.12)$$

The local volatility model is useful because it can take an arbitrage-free implied volatility surface as a direct input. It therefore fits vanilla option prices at $t = 0$ exactly by construction. Its main distinguishing feature is that, once calibrated to observed market implied volatility, it is completely determined, with no additional free parameters.

This clearly has drawbacks. Knowing the volatility surface at $t = 0$ gives the associated local volatility function $\sigma_{\text{loc}}(t, s)$, which implicitly specifies dynamics for the local volatility surface fixed by $\sigma_{\text{loc}}(t, s)$. The dynamics are therefore induced directly by the model.

By the Feynman-Kac theorem, the associated partial differential equation is

$$\frac{\partial V}{\partial t} - rV + (r - q)s \frac{\partial V}{\partial s} + \frac{\sigma_{\text{loc}}(t, s)^2}{2} s^2 \frac{\partial^2 V}{\partial s^2} = 0, \quad (8.13)$$

so we can ask whether, given a set of market prices, a function $\sigma(t, s)$ can be found such that solving equation (8.13) reproduces those prices.

8.4.1 Local volatility from market prices: Dupire's formula

We assume that, under a risk-neutral probability measure, the underlying asset follows a multiplicative diffusion process $dS_t = (r - q)S_t dt + \sigma_t S_t dW_t$, where σ_t is an arbitrary process⁹. Recall that a European call's price at time t is given by

$$V(t, s) = e^{-r(T-t)} \mathbb{E}^* [(S_T - K)^+ | S_t = s]. \quad (8.14)$$

The option price also depends on the parameters T and K . We seek the local volatility model that exactly reproduces observed market prices, so t and s are now fixed. For

⁸Assumed to exist and be unique.

⁹Generally, this condition means that $\sigma_t \in L^2_{\text{ad}}$.

simplicity, take $t = 0$ and $s = s_0$. We can thus view $V(T, K; 0, s_0, \boldsymbol{\theta})$ as a function of maturity and strike, since market option prices differ across parameter pairs.

Since we consider a one-dimensional stochastic differential equation under $\mathbb{P}^*$ with a unique strong solution, Hörmander's condition holds¹⁰ whenever $\sigma(t, s) \neq 0$. Thus, stochastic differential equation $dS_t = a(t, S_t)dt + b(t, S_t)dW_t$ —and in particular the multiplicative diffusion process considered here—admits a probability density $p(t, s)$ absolutely continuous with respect to Lebesgue measure, such that

$$\mathbb{P}^*(S_t \in A) = \int_A p(t, s)ds. \quad (8.15)$$

We can therefore express the call price as

$$V(T, K) = e^{-(r-q)T} \int_K^{+\infty} (s - K)p(T, s)ds, \quad (8.16)$$

and the fundamental theorem of calculus gives

$$\frac{\partial V}{\partial K} = -e^{-(r-q)T} \int_K^{+\infty} p(T, s)ds, \quad (8.17)$$

and

$$\frac{\partial^2 V}{\partial K^2} = e^{-(r-q)T} p(T, K), \quad (8.18)$$

in agreement with the result obtained in [28].

Since we assume the necessary differentiability conditions, Theorem 4.8 requires the density function $p(t, s)$ to satisfy the Fokker-Planck equation. In this case, it can be written as

$$\frac{\partial p}{\partial t} + \frac{\partial}{\partial s}(a(t, s)p(t, s)) - \frac{1}{2} \frac{\partial^2}{\partial s^2}(b(t, s)^2 p(t, s)) = 0, \quad (8.19)$$

and for $a(t, s) = (r - q)s$ and $b(t, s) = \sigma_{\text{loc}}(t, s)s$, where $\sigma_{\text{loc}}(t, s)$ denotes local volatility, we have

$$\frac{\partial p}{\partial t} + (r - q) \frac{\partial}{\partial s}(s p(t, s)) - \frac{1}{2} \frac{\partial^2}{\partial s^2}(\sigma(t, s)^2 s^2 p(t, s)) = 0. \quad (8.20)$$

First, differentiating expression (8.16) with respect to maturity gives

$$\frac{\partial V}{\partial T} = -r e^{-rT} \int_K^{+\infty} (s - K)p(T, s)ds + e^{-rT} \int_K^{+\infty} (s - K) \frac{\partial p}{\partial T}(T, s)ds, \quad (8.21)$$

which, identifying the first term on the right, can be rewritten as

$$\frac{\partial V}{\partial T} + rV = e^{-rT} \int_K^{+\infty} (s - K) \frac{\partial p}{\partial T}(T, s)ds. \quad (8.22)$$

Using the Fokker-Planck equation (8.20) gives

$$\frac{\partial V}{\partial T} + rV = e^{-rT} \int_K^{+\infty} (s - K) \left(\frac{1}{2} \frac{\partial^2}{\partial s^2}(\sigma(T, s)^2 s^2 p(T, s)) - (r - q) \frac{\partial}{\partial s}(s p(T, s)) \right) ds, \quad (8.23)$$

¹⁰Here, it suffices to check when $b(t, S_t) \frac{\partial}{\partial s}$ generates the tangent space at each point.

which can be integrated by parts. We proceed step by step. First,

$$\int_K^{+\infty} (s - K) \frac{\partial}{\partial s} (s p(T, s)) ds = \lim_{M \rightarrow +\infty} s(s - K) p(T, s) \Big|_{s=K}^{s=M} - \int_K^{+\infty} s p(T, s) ds,$$

and since we assume $\lim_{s \rightarrow +\infty} p(T, s) = 0$,

$$\int_K^{+\infty} (s - K) \frac{\partial}{\partial s} (s p(T, s)) ds = - \int_K^{+\infty} s p(T, s) ds.$$

This term can be expressed in terms of the option price and its derivatives at time T and strike K . In particular, expressions (8.16), (8.17), and (8.18) give

$$e^{-rT} \int_K^{+\infty} s p(T, s) ds = V(T, K) - K \frac{\partial V}{\partial K}(T, K). \quad (8.24)$$

Similarly, integrating by parts twice gives

$$\begin{aligned} \int_K^{+\infty} (s - K) \frac{\partial^2}{\partial s^2} (\sigma_{\text{loc}}^2(T, s) s^2 p(T, s)) ds &= \lim_{M \rightarrow +\infty} (s - K) \frac{\partial}{\partial s} (\sigma_{\text{loc}}^2(T, s) s^2 p(T, s)) \Big|_{s=K}^{s=M} \\ &\quad - \int_K^{+\infty} \frac{\partial}{\partial s} (\sigma_{\text{loc}}^2(T, s) s^2 p(T, s)) ds, \end{aligned}$$

so: (i) since we assume $\lim_{s \rightarrow +\infty} (s - K) p(t, s) = \lim_{s \rightarrow +\infty} (s - K) \frac{\partial p}{\partial s} = 0$, the first term on the right is identically zero; and (ii) relation (8.18) holds. Applying the fundamental theorem of calculus again reduces the full expression to

$$e^{-rT} \int_K^{+\infty} (s - K) \frac{\partial^2}{\partial s^2} (\sigma_{\text{loc}}^2(T, s) s^2 p(T, s)) ds = \sigma_{\text{loc}}^2(T, K) K^2 \frac{\partial^2 V}{\partial K^2}(T, K). \quad (8.25)$$

Combining this result with expression (8.24) finally gives

$$\frac{\partial V}{\partial T} + rV = \frac{1}{2} \sigma_{\text{loc}}^2(T, K) K^2 \frac{\partial^2 V}{\partial K^2} + (r - q)V - (r - q)K \frac{\partial V}{\partial K},$$

which, after simplification, yields $\sigma_{\text{loc}}^2(T, K)$ in the form

$$\boxed{\sigma_{\text{loc}}^2(T, K) = 2 \frac{\frac{\partial V}{\partial T} + qV + (r - q)K \frac{\partial V}{\partial K}}{K^2 \frac{\partial^2 V}{\partial K^2}}}, \quad (8.26)$$

known as Dupire's formula, relating local volatility¹¹ to spot option prices.

Given that market smiles exist, does a local volatility function exist that replicates the observed smile? With data for all strikes and maturities, we could use Dupire's formula to obtain local volatility, but could it be negative? This is resolved by noting that a negative numerator or denominator in Dupire's formula would imply arbitrage opportunities [29]. In particular:

¹¹This derivation can be generalized to an arbitrary Itô process σ_t , giving $\mathbb{E}^* [\sigma_T^2 | S_T = K]$ on the right-hand side of Dupire's formula. Thus, two volatility processes σ_t and σ'_t have the same vanilla smile if their conditional expectations coincide for every $T, K \in I_T \times I_K$ [29].

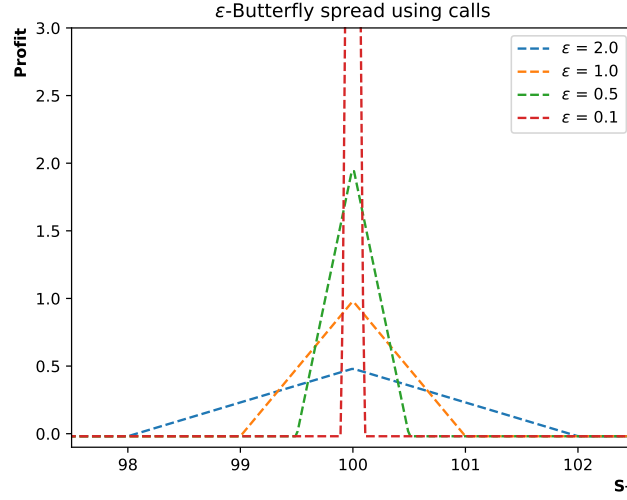


Figure 8.1: Payoff diagram of the ϵ -butterfly spread strategy for different values of ϵ .

- By the definition of a derivative, assuming that $K \mapsto V(T, K)$ is twice continuously differentiable in a neighborhood of K , we can write

$$\frac{\partial^2 V}{\partial K^2} = \lim_{h \rightarrow 0} \frac{V(T, K + h) - 2V(T, K) + V(T, K - h)}{h^2}.$$

The right-hand side can be interpreted as the current price of a combination of calls (assuming European calls): a long position in $\frac{1}{h^2}$ options with strike $K + h$, a short position in $\frac{2}{h^2}$ options with strike K , and another long position in $\frac{1}{h^2}$ options with strike $K - h$. This price combination is exactly the current price (the time-zero market value) of a butterfly spread centered at strike K , scaled by $1/h^2$, as seen in Section 7.8.2.

As shown in Figure 8.1, where we call this strategy an ϵ -butterfly spread, as h tends to zero, the appropriately scaled payoff converges in the distributional sense to a Dirac delta centered at $S_T = K$. The payoff is therefore always nonnegative in this limit. Since the figure shows a nonnegative payoff if $h \rightarrow 0$, the strategy's time-zero price must be nonnegative, recalling that its payoff is the weighted sum of its constituent instruments' payoffs.

If this second derivative were strictly negative, then, since butterfly spreads have positive purchase prices in the market, an arbitrage opportunity would arise, known as strike arbitrage or butterfly spread arbitrage.

Another way to see that $\frac{\partial^2 V}{\partial K^2}$ cannot be negative is through relation (8.18): this would imply a negative diffusion density, which is impossible.

- A negative numerator in expression (8.26) also implies arbitrage opportunities. The numerator can be rewritten as

$$e^{-qT} \frac{d}{dT} [e^{qT} V(T, K e^{(r-q)T})].$$

By Theorem 7.10, assuming in particular a constant effective interest rate $r - q$, we have $\text{For}_S(0, T) \equiv F_T = S_0 e^{(r-q)T}$. Defining the dimensionless quantity $k = K/F_T$, known as moneyness, the numerator can be written as

$$e^{-qT} \frac{d}{dT} [e^{qT} V(T, kF_T)] .$$

The variable k identifies an option as at-the-money forward, ATMF¹², if $k = 1$, OTMF if $k > 1$, and ITMF if $k < 1$.

Suppose that the numerator in (8.26) is negative. Then $e^{qT} V(T, kF_T)$ must decrease with maturity. That is, given $T_1 < T_2$,

$$e^{qT_1} V(T_1, kF_{T_1}) > e^{qT_2} V(T_2, kF_{T_2}). \quad (8.27)$$

We now define the following strategy: (i) buy an option with expiry T_2 and strike kF_{T_2} ; (ii) sell $e^{-q(T_2-T_1)}$ options with expiry T_1 and strike kF_{T_1} , giving a positive initial net cash flow

$$\Pi_0 = e^{-q(T_2-T_1)} V(T_1, kF_{T_1}) - V(T_2, kF_{T_2}) = e^{-qT_2} (e^{qT_1} V(T_1, kF_{T_1}) - e^{qT_2} V(T_2, kF_{T_2})),$$

which is strictly positive by inequality (8.27).

At time T_1 , (iii) take a position Δ in the underlying as follows: $\Delta = 0$ if $S_{T_1} \leq kF_{T_1}$, or $\Delta = -1$ (sell one share of the underlying) if $S_{T_1} > kF_{T_1}$. At maturity T_2 , the strategy's payoff consists of (i), (ii), and (iii). That is,

$$\begin{aligned} \Pi_{T_2} &= (S_{T_2} - kF_{T_2})^+ - e^{r(T_2-T_1)} e^{-q(T_2-T_1)} (S_{T_1} - kF_{T_1})^+ + \Delta (S_{T_2} - e^{(r-q)(T_2-T_1)} S_{T_1}) \\ &= (S_{T_2} - kF_{T_2})^+ - e^{(r-q)(T_2-T_1)} (S_{T_1} - kF_{T_1})^+ - \mathbb{1}_{S_{T_1} > kF_{T_1}} (S_{T_2} - e^{(r-q)(T_2-T_1)} S_{T_1}), \end{aligned}$$

and since $\frac{F_{T_2}}{F_{T_1}} = e^{(r-q)(T_2-T_1)}$,

$$\begin{aligned} \Pi_{T_2} &= (S_{T_2} - kF_{T_2})^+ - \frac{F_{T_2}}{F_{T_1}} (S_{T_1} - kF_{T_1})^+ - \mathbb{1}_{S_{T_1} > kF_{T_1}} \left(S_{T_2} - \frac{F_{T_2}}{F_{T_1}} S_{T_1} \right) \\ &= (S_{T_2} - kF_{T_2})^+ - \left[(\hat{S}_{T_1} - kF_{T_1})^+ + \mathbb{1}_{\hat{S}_{T_1} > kF_{T_2}} (S_{T_2} - \hat{S}_{T_1}) \right], \end{aligned}$$

where $\hat{S}_{T_1} = \frac{F_{T_2}}{F_{T_1}} S_{T_1}$. Taking $f(x) = (x - kF_{T_2})^+$, which is convex by definition, we can write

$$\Pi_{T_2} = f(S_{T_2}) - \left[f(\hat{S}_{T_1}) - \frac{df}{dx}(\hat{S}_{T_1})(S_{T_2} - \hat{S}_{T_1}) \right].$$

Since $f(x)$ is convex on $\mathbb{R}_+$, we have

$$f(x) \geq f(a) + f'(a)(x - a)$$

¹²That is, its strike equals the underlying forward price for the option's maturity. ATM (at-the-money) means that the strike equals the underlying spot price, whereas ATMF (at-the-money forward) means that it equals the forward price. In practice, derivatives traders –especially in equities, FX, rates, and commodities– primarily use ATMF when discussing options, abbreviating it simply as ATM.

for every $x \in \mathbb{R}_+$ with fixed $a > 0$. This directly implies $\Pi_{T_2} \geq 0$.

Assuming a negative numerator in equation (8.26) has therefore allowed us to construct a strategy generating positive cash flow both at inception and at T_2 . We have found an arbitrage opportunity, known as calendar spread arbitrage or maturity arbitrage, contradicting the assumed existence of at least one risk-neutral measure.

Positivity of the numerator in equation (8.26) can also be proved using Jensen's inequality. Moreover, one can show, see [29], that positivity of Dupire's numerator is equivalent to integrated implied variance $\sigma_{\text{imp}}^2(T, K)$ increasing with maturity for any moneyness k . That is,

$$T_1 \sigma_{\text{imp}}^2(T_1, kF_{T_1}) \leq T_2 \sigma_{\text{imp}}^2(T_2, kF_{T_2}) \quad (8.28)$$

Definition 8.3. A volatility surface $(T, K) \mapsto \sigma_{\text{imp}}(T, K)$ is said to be free of static arbitrage¹³ if and only if both conditions hold: (i) there are no calendar spread arbitrage opportunities; and (ii) for fixed T , each curve $K \mapsto \sigma_{\text{imp}}(T, K)$ is free of butterfly spread arbitrage.

In general, in the absence of arbitrage, implied volatility can be defined for any option with a convex payoff. Given a European option with payoff $h(x)$, with $x = S_T/F_T$ and convex h , there exists a unique implied volatility that exactly reproduces market prices for that payoff when inserted into the Black-Scholes equation [30]. Furthermore, absence of arbitrage across maturities T_1 and T_2 implies $T_2 \sigma_{\text{imp}}^2(T_2) \geq T_1 \sigma_{\text{imp}}^2(T_1)$.

8.4.2 Local volatility from implied volatility

Dupire's formula (8.26) gives the local volatility function from market option prices¹⁴. As a first approximation, assume that the underlying pays no discrete dividends –although we allow a continuous dividend yield q . With discrete dividends, option prices no longer follow the unadjusted standard Black-Scholes model¹⁵. In this case, define

$$y = \log \left(\frac{K}{F_t} \right), \quad (8.29)$$

$$w(t, y) = (t - t_0) \sigma_{\text{imp}}^2(t, K), \quad (8.30)$$

¹³More precisely, a set of call prices is free of static arbitrage if: (i) there exists at least one model under which the underlying is a nonnegative local martingale under a probability measure; and (ii) observed prices equal the discounted expected option payoff under that model [34].

¹⁴The derivation here uses calls. As seen in the preceding section, a similar derivation applies to any European option with a convex payoff.

¹⁵Black-Scholes assumes continuous, diffusive, Markovian underlying dynamics. Discrete dividends introduce deterministic jumps that violate continuity and alter the Markov property of the spot price. European option prices with discrete dividends therefore cannot be explained directly by unadjusted Black-Scholes, and implied volatility extracted from those prices no longer represents the instantaneous volatility of a continuous spot process.

where we generally take $t_0 = 0$. Applying Dupire's formula (8.26) to the Black-Scholes formula with implied volatility $\sigma_{\text{imp}}^2(T, K)$ then gives

$$\sigma_{\text{loc}}^2(t, s) = \frac{\frac{\partial w}{\partial t}}{\left(\frac{y}{2w} \frac{\partial w}{\partial y} - 1\right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial y^2} - \frac{1}{4} \left(\frac{1}{4} + \frac{1}{w}\right) \left(\frac{\partial w}{\partial y}\right)^2} \bigg|_{y=\log\left(\frac{s}{F_t}\right)} \quad (8.31)$$

8.5 Calibration of the local volatility surface

As noted in Proposition 8.2, options are not traded continuously across all strikes and maturities; instead, there are discrete sets of each. To obtain the implied volatility surface $\sigma_{\text{imp}}^2(T, K)$, market data must therefore be interpolated –and the surface possibly extrapolated *outside*¹⁶ the available observations– while satisfying the absence of strike arbitrage –prices convex in strike– and maturity arbitrage, as discussed in equation (8.28).

8.5.1 Affine parametrization

8.5.2 The Andreasen & Huge model

8.5.3 The SVI model

The stochastic volatility inspired (SVI) model was first proposed by Jim Gatheral [31, 33]. It parametrizes the volatility smile for each maturity T , ensuring absence of butterfly spread arbitrage through restrictions on the parametrization.

8.5.4 The eSSVI model

8.5.5 Comparison of models

The practical workflow is generally as follows:

1. **Collect market data.** Obtain implied volatilities for different strike and maturity combinations from European option prices.
2. **Fit the implied volatility surface.** Use interpolation models such as the one given in the equation or SVI-type models to construct a smooth, arbitrage-free surface from the discrete data.
3. **Validate the results.** Check that the surface is stable and smooth and produces neither negative values nor numerical inconsistencies.
4. **Apply Dupire's formula (8.31).** Obtain the local volatility surface, enabling simulations for more appropriate pricing of exotic derivatives.

¹⁶The meaning of *outside* a discrete dataset is subtle: in linear programming or convex optimization, a point is *outside* if it does not belong to the convex domain or feasible polyhedron generated by the data.

8.6 Computing theta

As seen above, the theta of a delta-hedged portfolio consisting of a vanilla option and the underlying is

$$\Theta_t = \left(\frac{\partial V}{\partial t} - rV + (r - q)s \frac{\partial V}{\partial s} \right).$$

Thus, it includes not only the option's theta but also the premium financing cost rV and an adjustment for dividends paid by the underlying held in the portfolio. The difficulty arises when calculating $\partial V / \partial t$, since this essentially yields white noise.

One way to calculate theta is to price the option at time $t + \delta t$ using the same implied volatility $\sigma_{\text{imp}}(T, K)$. However, this is problematic: keeping the implied volatility surface unchanged as time passes overestimates the option value, since total variance $T\sigma_{\text{imp}}^2(T, K)$ does not decrease as it should when time to maturity falls.

8.7 Derivatives pricing under the local volatility model

8.8 Implied volatility from a local volatility model

So far, we have obtained a local volatility model from European option market prices or their implied volatility surface. To analyze its dynamics –how the implied volatility surface *reacts* to underlying price changes or the passage of time– we need an efficient way to obtain σ_{imp} from σ_{loc} . Rather than solving equation (8.13), we will express implied volatility as a function of local volatility.

We next show that implied volatility can be viewed as a weighted average of instantaneous volatility. Let S be a multiplicative diffusion process adapted to the natural filtration $\{\mathcal{F}_t\}_{t \in \mathbb{R}^+}$. Under the –unique– risk-neutral measure $\mathbb{P}^*$, the underlying is described by the stochastic differential equation

$$dS_t = (r - q)S_t dt + \sigma_t dW_t, \quad (8.32)$$

where r is the risk-free interest rate and q characterizes continuous dividends paid by the underlying. As noted earlier, we also take $\sigma_t \in L_{\text{ad}}^2$ such that the assumptions of the strong existence and uniqueness theorem hold for equation (8.32).

Now consider a local volatility model described by equation (8.12). We denote the price of a vanilla option with strike K and expiry T by $V_{\text{loc}}(t, s; T, K, \boldsymbol{\theta})$. As usual, we abbreviate the option price as $V_{\text{loc}}(t, s)$ when unambiguous.

Suppose we want to delta hedge a short position in this option using the local volatility model, while actual underlying dynamics follow equation (8.32). Defining the process $Q_t = e^{-rt} V_{\text{loc}}(t, s)$, we can calculate the change in Q_t over an infinitesimal interval $(t, t + dt)$. Applying Itô's formula from 2.17 gives

$$\begin{aligned} dQ_t &= e^{-rt} \left[\left(\frac{\partial V_{\text{loc}}(t, s)}{\partial t} - rV_{\text{loc}}(t, s) \right) dt + \frac{\partial V_{\text{loc}}(t, s)}{\partial s} ds + \frac{1}{2} \frac{\partial^2 V_{\text{loc}}(t, s)}{\partial s^2} ds^2 \right] \\ &= e^{-rt} \left[\left(\frac{\partial V_{\text{loc}}(t, s)}{\partial t} - rV_{\text{loc}}(t, s) + \frac{1}{2} \sigma_t^2 s^2 \frac{\partial^2 V_{\text{loc}}(t, s)}{\partial s^2} \right) dt + \frac{\partial V_{\text{loc}}(t, s)}{\partial s} ds \right], \end{aligned}$$

and therefore its expectation is

$$\begin{aligned}\mathbb{E}^*[dQ_t | S_t = s] &= e^{-rt} \left(\frac{\partial V_{\text{loc}}(t, s)}{\partial t} - rV_{\text{loc}}(t, s) + (r - q)s \frac{\partial V_{\text{loc}}(t, s)}{\partial s} + \frac{1}{2} \sigma_t^2 s^2 \frac{\partial^2 V_{\text{loc}}(t, s)}{\partial s^2} \right) dt \\ &= e^{-rt} \frac{1}{2} s^2 \frac{\partial^2 V_{\text{loc}}(t, s)}{\partial s^2} (\sigma_t^2 - \sigma_{\text{loc}}^2(t, s)) dt,\end{aligned}$$

where we have used relation (8.13), obtained by considering a delta-hedging strategy. Note also that the paths of S_t are generated by model (8.32).

Note that $Q_0 = V_{\text{loc}}(0, s_0)$ and $Q_T = e^{-rT} f(S_T)$, where f denotes the option payoff function. Thus, $\mathbb{E}^*[Q_T] = V(0, s_0)$, where V denotes the option price under model (8.32). Integrating dQ_t over $[0, T]$ and using Fubini's theorem gives

$$\int_0^T \mathbb{E}^*[dQ_t] = \mathbb{E}^* \left[\int_0^T dQ_t \right] = \mathbb{E}^*[Q_T - Q_0] = \mathbb{E}^*[Q_T] - Q_0.$$

Thus, we can write

$$\mathbb{E}^*[Q_T] = Q_0 + \mathbb{E}^* \left[\int_0^T dQ_t \right]$$

that is,

$$\mathbb{E}^*[Q_T] = Q_0 + \mathbb{E}^* \left[\int_0^T e^{-rt} \frac{1}{2} S_t^2 \frac{\partial^2 V_{\text{loc}}(t, S_t)}{\partial s^2} (\sigma_t^2 - \sigma_{\text{loc}}^2(t, S_t)) dt \right].$$

Since $\mathbb{E}^*[Q_T] = V(0, s_0)$ and $Q_0 = V_{\text{loc}}(0, s_0)$,

$$V(0, s_0) = V_{\text{loc}}(0, s_0) + \mathbb{E}^* \left[\int_0^T e^{-rt} \frac{1}{2} S_t^2 \frac{\partial^2 V_{\text{loc}}(t, S_t)}{\partial s^2} (\sigma_t^2 - \sigma_{\text{loc}}^2(t, S_t)) dt \right], \quad (8.33)$$

which implies that the option price under arbitrary underlying dynamics given by model (8.32) equals the base-model price—here, the local volatility price—plus the discounted expectation of the gamma/theta P&L accumulated by hedging with the local volatility model instead of the *actual*, arbitrary underlying model.

Assume that the first model is a local volatility model calibrated using Black-Scholes so that volatility is constant and equal to the implied volatility of the option considered. That is, $\sigma_{\text{loc}}(t, s; T, K, \sigma_{\text{imp}}, \boldsymbol{\theta}) = \sigma_{\text{imp}}(T, K; t, s, \boldsymbol{\theta})$. Then $V_{\text{loc}}(0, s_0) = V(0, s_0) \equiv V_{BS}(0, s_0; T, K, \sigma_{\text{imp}}, \boldsymbol{\theta})$, where V_{BS} is the Black-Scholes option price with implied volatility $\sigma_{\text{imp}}(T, K)$, so equation (8.33) implies

$$\sigma_{\text{imp}}^2(T, K) = \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \frac{1}{2} S_t^2 \frac{\partial^2 V}{\partial s^2} \sigma_t^2 dt \right]}{\mathbb{E}^* \left[\int_0^T e^{-rt} \frac{1}{2} S_t^2 \frac{\partial^2 V}{\partial s^2} dt \right]}, \quad (8.34)$$

which, using the definition of dollar gamma, can be rewritten as

$$\sigma_{\text{imp}}^2(T, K) = \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \frac{1}{2} \Gamma^D(t, s; T, K, \sigma_{\text{imp}}, \boldsymbol{\theta}) \sigma_t^2 dt \right]}{\mathbb{E}^* \left[\int_0^T e^{-rt} \frac{1}{2} \Gamma^D(t, s; T, K, \sigma_{\text{imp}}, \boldsymbol{\theta}) dt \right]}, \quad (8.35)$$

or, more compactly and canceling the factor $1/2$,

$$\sigma_{\text{imp}}^2(T, K) = \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D \sigma_t^2 dt \right]}{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D dt \right]}. \quad (8.36)$$

Thus, implied volatility –specifically, its square– can be interpreted as a weighted average of instantaneous volatility σ_t , with weights given by discounted dollar gamma calculated at constant volatility $\sigma_{\text{imp}}(T, K)$. In particular, when model (8.32) is a local volatility model, implied volatility is

$$\sigma_{\text{imp}}^2(T, K) = \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D \sigma_{\text{loc}}^2(t, S_t) dt \mid S_0 = s_0 \right]}{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D dt \mid S_0 = s_0 \right]}. \quad (8.37)$$

Equations (8.36) and (8.37) are implicit equations for implied volatility because gamma itself depends on implied volatility –hence the explicit dependence in relation (8.35).

To obtain a readily applicable expression, Bergomi [29] approximates volatility as a perturbation of a base model depending only on time, with spot dependence confined to the first-order term. Denoting local variance by $u(t, s) = \sigma_{\text{loc}}(t, s)^2$, we start from the base model $u(t, s) = u_0(t)$. Equation (8.37) then gives

$$\sigma_{\text{imp}}^2(T, K) = \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D u_0(t, S_t) dt \mid S_0 = s_0 \right]}{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D dt \mid S_0 = s_0 \right]}, \quad (8.38)$$

where expectation is taken under the underlying model $dS_t = (r - q)dt + \sqrt{u_0(t)} S_t dW_t$.

Implied volatility is actually a nonlinear functional of u_0 , denoted by $F[u_0] \equiv \sigma_{\text{imp}}^2(T, K)[u_0]$. How does F , the implied variance, change when a spot-dependent correction is added to u_0 ? That is,

$$u_0(t) \mapsto u(t, s) = u_0(t) + \delta u(t, s).$$

To first order (recall that the functional is nonlinear),

$$F[u_0 + \delta u] = F[u_0] + \frac{d}{d\epsilon} F[u_0 + \epsilon \delta u] \Big|_{\epsilon=0},$$

where the second term is the functional derivative of F , denoted by $DF[u_0](\delta u)$. We compute the functional derivative of $F[u] = A[u]/B[u]$. Since functional derivatives satisfy the quotient rule,

$$DF[u_0](\delta u) = \frac{DA[u_0](\delta u)B[u_0] - A[u_0]B[u_0](\delta u)}{B[u_0]^2},$$

with

$$A[u] = \mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D u(t, S_t) dt \mid S_0 = s_0 \right]$$

and

$$B[u] = \mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D dt \mid S_0 = s_0 \right].$$

Thus,

$$DA[u_0](\delta u) = \frac{d}{d\epsilon} \left(\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D(u_0 + \epsilon \delta u) dt \mid S_0 = s_0 \right] \right) \Big|_{\epsilon=0},$$

and differentiating under the expectation (the conditions for differentiation under the Lebesgue integral hold) gives

$$DA[u_0](\delta u) = \frac{d}{d\epsilon} \left(\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D \delta u dt \mid S_0 = s_0 \right] \right) \Big|_{\epsilon=0}.$$

Since $B[u]$ does not depend on ϵ , differentiating under the integral sign gives $DB[u_0](\delta u) = 0$. Thus, $DF[u_0](\delta u) = DA[u_0](\delta u)/B[u_0]$ and

$$\sigma_{\text{imp}}^2(T, K)[u] = \sigma_{\text{imp}}^2(T, K)[u_0] + \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D(t, S_t) \delta u(t, S_t) dt \mid S_0 = s_0 \right]}{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D(t, S_t) dt \mid S_0 = s_0 \right]} \quad (8.39)$$

where $\sigma_{\text{imp}}^2(T, K)[u_0]$ is given by equation (8.38). Note that S_t is modeled by $dS_t = (r - q)dt + \sigma_0(t)S_t dW_t$ with $\sigma_0(t) = \sqrt{u_0(t)}$ —since we used the approximation $\delta u \ll u_0$.

Thus, the effect on implied volatility of the correction $\delta u(t, s)$ to total variance—denoted by $\delta \sigma_{\text{imp}}^2(T, K)$ —is determined by

$$\delta \sigma_{\text{imp}}^2(T, K) = \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D(t, S_t) \delta u(t, S_t) dt \mid S_0 = s_0 \right]}{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D(t, S_t) dt \mid S_0 = s_0 \right]}. \quad (8.40)$$

One can prove that the discounted derivatives of order n of the option price with respect to the logarithm of the underlying are martingales. That is,

$$\frac{d^n V}{dx^n}(t, x_t) = \mathbb{E}^* \left[e^{-r(T-t)} \frac{d^n V}{dx^n}(T, x_T) \mid x_t = x \right]$$

with $x_t = \log S_t$ [29]. Taking the cases $n = 1$ and $n = 2$ immediately shows that discounted dollar delta and dollar gamma are both martingales. Thus, by Fubini's theorem, the denominator in equation (8.40) satisfies

$$\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D(t, S_t) dt \mid S_0 = s_0 \right] = \int_0^T \mathbb{E}^* \left[e^{-rt} \Gamma^D(t, S_t) \mid S_0 = s_0 \right] dt = T \Gamma^D(0, s_0),$$

so

$$\delta \sigma_{\text{imp}}^2(T, K) = \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D(t, S_t) \delta u(t, S_t) dt \mid S_0 = s_0 \right]}{T \Gamma^D(0, s_0)}.$$

We now analyze the numerator. Using Fubini's theorem again, it can be written as

$$\int_0^T dt \int_0^{+\infty} ds \rho(t, s; \sigma_0) e^{-rt} \Gamma^D(t, s) \delta u(t, s),$$

and since $\sigma_0 = \sqrt{u_0}$ is deterministic, the underlying has a lognormal distribution $S_t \sim \text{LogNormal}(\mu_t, \omega_t)$:

$$\rho(t, s; \sigma_0) = \frac{1}{s\sqrt{2\pi\omega_t}} \exp\left(-\frac{(\log s - \mu_t)^2}{2\omega_t}\right)$$

with

$$\begin{aligned}\mu_t &= \log s_0 + \int_0^t \left(r - q - \frac{1}{2}\sigma_0(\tau)^2\right) d\tau, \\ \omega_t &= \int_0^t \sigma_0(\tau)^2 d\tau,\end{aligned}\tag{8.41}$$

where ω_t is called integrated variance. Rearranging μ_t gives

$$\mu_t = \log s_0 + (r - q)T + \frac{1}{2}\omega_T = \log(s_0 e^{(r-q)T}) - \frac{1}{2}\omega_T = \log F_T - \frac{1}{2}\omega_T$$

so the density becomes

$$\rho(t, s; \sigma_0) = \frac{1}{s\sqrt{2\pi\omega_t}} \exp\left(-\frac{(x + \frac{1}{2}\omega_t)^2}{2\omega_t}\right),$$

after defining log-moneyness $x = \log(s/F_t)$.

Dollar gamma can also be calculated from the Black-Scholes equation, giving [29]

$$\Gamma^D(t, s) = s \frac{F_T}{F_t} e^{-r(T-t)} \frac{1}{\sqrt{2\pi(\omega_T - \omega_t)}} \exp\left(-\frac{(x - x_K + \frac{1}{2}(\omega_T - \omega_t))^2}{2(\omega_T - \omega_t)}\right)$$

with $x_K = \log(K/F_T)$.

Thus, equation (8.40) simplifies to

$$\frac{1}{T} \int_0^T dt \int_{-\infty}^{+\infty} dx \sqrt{\frac{\omega_T}{\omega_t}} \frac{\delta u(t, F_t e^x)}{\sqrt{2\pi(\omega_T - \omega_t)}} \exp\left(-\left(\frac{1}{2(\omega_T - \omega_t)} + \frac{1}{2\omega_t}\right) \frac{(\omega_T x - \omega_t x_K)^2}{\omega_T^2}\right),$$

which, under the change of coordinates

$$x = \frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T - \omega_t)\omega_t}}{\sqrt{\omega_T}} y$$

reduces to

$$\delta\sigma_{\text{imp}}^2(T, K) = \frac{1}{T} \int_0^T dt \int_{-\infty}^{+\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \delta u\left(t, F_t e^{\frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T - \omega_t)\omega_t}}{\sqrt{\omega_T}} y}\right).\tag{8.42}$$

Note that $\sigma_{\text{imp}}^2(T, K) = \sigma_{\text{imp},0}^2(T, K) + \delta\sigma_{\text{imp}}^2(T, K)$. Using the new coordinates (t, y) gives

$$\sigma_{\text{imp},0}^2(T, K) = \frac{1}{T} \int_0^T dt \int_{-\infty}^{+\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} u_0(t) = \frac{1}{T} \int_0^T u_0(t) dt.$$

Thus, the implied volatility of the base model, whose volatility has no spot dependence, can be interpreted as the average instantaneous volatility up to maturity T . Clearly, at zeroth order, implied volatility is independent of strike.

Adding the two terms finally gives

$$\sigma_{\text{imp}}^2(T, K) = \frac{1}{T} \int_0^T dt \int_{-\infty}^{+\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} u \left(t, F_t e^{\frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T - \omega_t)\omega_t}}{\sqrt{\omega_T}}} y \right). \quad (8.43)$$

Expansion around constant volatility

Suppose that the base-model volatility –not variance– is constant, that is, $\sigma_0(t) = \sqrt{u_0(t)} := \sigma_0$, and write $\sigma_{\text{loc}}(t, s) = \sigma_0 + \delta\sigma_{\text{loc}}(t, s)$. Then $u = (\sigma_0 + \delta\sigma_{\text{loc}})^2 \approx \sigma_0^2 + 2\sigma_0\delta\sigma_{\text{loc}} = u_0 + 2\sigma_0\delta\sigma_{\text{loc}}$ –discarding the term of order $\delta\sigma_{\text{loc}}^2$ – and hence $\delta u = 2\sigma_0\delta\sigma_{\text{loc}}$ and $\omega_t = t\sigma_0^2$. Substituting into equation (8.43) gives

$$\sigma_{\text{imp}}(T, K) = \frac{1}{T} \int_0^T dt \int_{-\infty}^{+\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \sigma_{\text{loc}} \left(t, F_t e^{\frac{t}{T} x_K + \sigma_0 \frac{\sqrt{(T-t)t}}{\sqrt{T}}} y \right). \quad (8.44)$$

Although expression (8.43) requires fewer approximations than (8.44), we use the latter to study implied volatility dynamics.

For $t \rightarrow 0$, we have $F_t = s_0$, $x_K = y = 0$, so implied volatility depends only on $\sigma(0, s_0)$, as expected. Similarly, for $t \rightarrow T$, $F_t e^{\frac{t}{T} x_K + \sigma_0 \frac{\sqrt{(T-t)t}}{\sqrt{T}}} y = K$, so only $s = K$ contributes to implied volatility.

The factor $e^{-y^2/2}$ gives greater weight to the realization satisfying $y = 0$, that is, the straight line joining $x = \log s_0$ and $x = \log K$. Thus, instead of integrating over the entire real line, one approximation is to consider only the path $y = 0$, giving

$$\sigma_{\text{imp}}(T, K) \approx \frac{1}{T} \int_0^T \sigma_{\text{loc}} \left(t, F_t e^{\frac{t}{T} \log \frac{K}{F_T}} \right) dt. \quad (8.45)$$

Remark 8.4. Although formula (8.43) approximates implied volatility by integrating local volatility along the most likely logarithmic path connecting S_0 and K , it should be understood as a formal simplification based on a first-order expansion in $\delta\sigma_{\text{loc}}(t, s)$. Most Gaussian probability mass is concentrated near $y = 0$, which provides a probabilistic justification for (8.44). However, this expression does not capture the density's dependence on $\sigma_{\text{loc}}(t, s)$ itself, limiting its accuracy for pronounced smiles or demanding quantitative applications.

Nevertheless, approximation (8.43) remains particularly useful near ATMF, where it allows analytical study of local smile sensitivities –such as skew and curvature, introduced later– in terms of spatial derivatives of local volatility. In particular, it has been used in the literature (cf. [29]) as a theoretical tool to analyze how functional perturbations of $\sigma_{\text{loc}}(t, s)$ affect the implied volatility surface, enabling the study of smile dynamics.

However, approximation (8.43) –and in particular (8.44)– has no practical use in trading settings requiring exact, arbitrage-free prices consistent with market observations. By using a fixed probability density –such as the lognormal density of a base model– without accounting for its dynamic dependence on $\sigma_{\text{loc}}(t, s)$, the formula cannot reproduce

observed option prices with sufficient accuracy. It also provides neither consistent Greeks nor guaranteed no-arbitrage conditions, making it unsuitable for hedging, pricing, or risk management in practical models. More sophisticated alternatives attempt to address this deficiency, such as the most-likely-path approximations reviewed in [32], which use short-time heat-kernel expansions of the density $\rho(t, s)$. For realistic equity smiles, however, numerically solving Dupire's equation (8.26) remains more efficient.

These expressions allow us to study how implied volatility varies with strike for a given local volatility function. In particular, implied volatility dynamics near ATMF are fundamental both practically and theoretically. Options in this region have the greatest sensitivity relevant to hedged-position P&L, since gamma and vega reach their maxima there –see Section 7.9. Options close to the forward are also the most liquid, so their implied volatilities reflect aggregate expectations and risks more accurately and stably. A precise description of the time evolution of $\sigma_{\text{imp}}(T, K)$ near the forward therefore captures the dominant risks in carry –represented by theta– and hedge rebalancing.

We therefore use a second-order expansion –capturing slope and convexity– of $\sigma_{\text{loc}}(t, s)$ around ATMF:

$$\sigma_{\text{loc}}(t, s) = \sigma_0 + \alpha(t) \log \left(\frac{s}{F_t} \right) + \frac{\beta(t)}{2} \log \left(\frac{s}{F_t} \right)^2, \quad (8.46)$$

where, defining¹⁷ $x = \log \left(\frac{s}{F_t} \right)$, we can rewrite it as

$$\sigma_{\text{loc}}(t, x) = \sigma_0 + \alpha(t)x + \frac{\beta(t)}{2}x^2, \quad (8.47)$$

where σ_0 represents the level, $\alpha(t)$ the slope (or skew), and $\beta(t)$ the convexity of the local volatility surface near ATMF.

Thus, expression (8.44) gives

$$\sigma_{\text{imp}}(T, K) = \frac{1}{T} \int_0^T \sigma_0 dt + \frac{1}{T} \int_0^T dt \int_{-\infty}^{+\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \left[\alpha(t)X(t, y) + \frac{\beta(t)}{2}X(t, y)^2 \right],$$

with

$$X(t, y) = \frac{t}{T}x_K + \frac{\sqrt{\sigma_0^2(T-t)}}{\sqrt{T}}y.$$

Expanding $\sigma_{\text{imp}}(T, K)$ and integrating over y therefore gives

$$\sigma_{\text{imp}}(T, K) = \frac{1}{T} \int_0^T \left[\sigma_0 + \frac{t}{T}x_K\alpha(t) + \frac{t}{2T} \left(\frac{t}{T}x_K^2 + \sigma_0^2(T-t) \right) \beta(t) \right] dt. \quad (8.48)$$

For a fixed maturity, the implied volatility skew near ATMF –the slope of the implied volatility surface for fixed T – can be calculated by differentiating with respect to log-moneyness¹⁸ k . That is,

$$\text{Skew}(k) = \frac{\partial \sigma_{\text{imp}}(T, K)}{\partial k} = \frac{1}{T} \int_0^T dt \frac{t}{T} \left(\alpha(t) + \frac{t}{T}x_K\beta(t) \right), \quad (8.49)$$

¹⁷This should not be confused with log-moneyness $k = \log(K/F_t)$, although x is often called spot-relative log-moneyness.

¹⁸Differentiating with respect to $\log K$ rather than k is also common. For fixed t , however, the two definitions are equivalent.

which depends on log-moneyness k . Recall that the resulting skew is meaningful only near ATMF. We denote the skew at ATMF for fixed maturity T by $\mathcal{S}_T = \text{Skew}(0)$. In particular,

$$\boxed{\mathcal{S}_T = \frac{1}{T} \int_0^T \frac{t}{T} \alpha(t) dt.} \quad (8.50)$$

Furthermore, if $\alpha(t) = \alpha_0$ is constant, then $\mathcal{S}_T = \alpha_0/2$.

The convexity of implied volatility near ATMF can likewise be calculated by differentiating with respect to log-moneyness k . That is,

$$\text{Conv}(k) = \frac{\partial^2 \sigma_{\text{imp}}(T, K)}{\partial k^2} = \frac{1}{T} \int_0^T \frac{t^2}{T^2} \beta(t) dt \quad (8.51)$$

which is independent of log-moneyness k . In particular, ATMF convexity $\mathcal{C}_T = \text{Conv}(0)$ for fixed maturity T is

$$\boxed{\mathcal{C}_T = \frac{1}{T} \int_0^T \frac{t^2}{T^2} \beta(t) dt,} \quad (8.52)$$

and if $\beta(t) = \beta_0$ is constant, then $\mathcal{C}_T = \beta_0/3$.

Expansion around deterministic volatility

Now take base-model volatility $\sqrt{u_0(t)} = \sigma_0(t)$ and write $\sigma_{\text{loc}}(t, s) = \sigma_0(t) + \delta\sigma_{\text{loc}}(t, s)$. Then $u(t, s) = (\sigma_0(t) + \delta\sigma_{\text{loc}}(t, s))^2 \approx u_0(t) + 2\sigma_0(t)\delta\sigma_{\text{loc}}(t, s)$ —discarding the term of order $\delta\sigma_{\text{loc}}^2$ —and hence $\delta u = 2\sigma_0\delta\sigma_{\text{loc}}$. Starting from equation (8.43),

$$\sigma_{\text{imp}}^2(T, K) = \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} (\sigma_0^2(t) + 2\sigma_0(t)\delta\sigma_{\text{loc}}(t, X(t, y)))$$

where

$$X(t, y) = \frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T - \omega_t)\omega_t}}{\sqrt{\omega_T}} y.$$

Thus, we can write

$$\sigma_{\text{imp}}^2(T, K) = \frac{1}{T} \int_0^T \sigma_0^2(t) dt + \frac{2}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \sigma_0(t) \delta\sigma_{\text{loc}}(t, X(t, y)),$$

so we can consider the first-order expansion of implied volatility around

$$\bar{\sigma}_T^2 = \frac{1}{T} \int_0^T \sigma_0^2(t) dt \quad (8.53)$$

as $\sigma_{\text{imp}}^2(T, K) = \bar{\sigma}_T^2 + 2\bar{\sigma}_T \delta\sigma_{\text{imp}}(T, K)$. Thus,

$$\delta\sigma_{\text{imp}}(T, K) = \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \delta\sigma_{\text{loc}}(t, X(t, y)),$$

and finally

$$\sigma_{\text{imp}}(T, K) = \frac{1}{T} \int_0^T \sigma_0(t) dt + \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \delta\sigma_{\text{loc}} \left(t, F_t e^{\frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T - \omega_t)\omega_t}}{\sqrt{\omega_T}} y} \right), \quad (8.54)$$

which reduces to expression (8.44) when $\sigma_0(t) = \sigma_0$ is constant.

We now calculate the implied volatility skew and convexity for ATMF options using expansion (8.54). Similarly, we choose

$$\sigma_{\text{loc}}(t, x) = \sigma_0(t) + \alpha(t)x + \frac{\beta(t)}{2}x^2, \quad (8.55)$$

where $\sigma_0(t)$ represents the local volatility level –now time dependent– for ATMF options and $\delta\sigma_{\text{loc}}(t, x) = \alpha(t)x + \frac{\beta(t)}{2}x^2$. Thus, implied volatility is

$$\sigma_{\text{imp}}(T, K) = \frac{1}{T} \int_0^T \sigma_0(t) dt + \frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\omega_t}{\omega_T} \left[x_K \alpha(t) + \left(\frac{\omega_t}{\omega_T} x_K^2 + (\omega_T - \omega_t) \right) \frac{\beta(t)}{2} \right], \quad (8.56)$$

which reduces to expression (8.48) when $\sigma_0(t) = \sigma_0$. Differentiating with respect to log-moneyness k gives the skew as

$$\text{Skew}(k) = \frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\omega_t}{\omega_T} \left(\alpha(t) + x_K \frac{\omega_t}{\omega_T} \beta(t) \right), \quad (8.57)$$

and since $\omega_\tau = \tau \bar{\sigma}_\tau^2$, we have

$$\text{Skew}(k) = \frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{t \bar{\sigma}_t^2}{T \bar{\sigma}_T^2} \left(\alpha(t) + x_K \frac{t \bar{\sigma}_t^2}{T \bar{\sigma}_T^2} \beta(t) \right), \quad (8.58)$$

so the skew for ATMF options is

$$\mathcal{S}_T = \frac{1}{T} \int_0^T dt \frac{t}{T} \frac{\sigma_0(t) \bar{\sigma}_t^2}{\bar{\sigma}_T^3} \alpha(t). \quad (8.59)$$

Similarly, convexity as a function of k in a neighborhood of $k = 0$ is

$$\text{Conv}(k) = \frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{t^2 \bar{\sigma}_t^4}{T^2 \bar{\sigma}_T^4} \beta(t), \quad (8.60)$$

which is independent of k because we have used a second-order expansion. Thus, ATMF convexity is

$$\mathcal{C}_T = \frac{1}{T} \int_0^T dt \frac{t^2}{T^2} \frac{\sigma_0(t) \bar{\sigma}_t^4}{\bar{\sigma}_T^5} \beta(t). \quad (8.61)$$

8.9 Volatility surface dynamics

As we can see, once the local volatility function $\sigma(t, s)$ is fixed, implied volatility changes only if t or s changes –assuming the local volatility model. Thus, to characterize the joint dynamics of spot and implied volatility, we need to determine how implied volatility responds to changes in spot. For simplicity, we assume that the zeroth-order local volatility is time dependent. That is, we use (8.55), where implied volatility –to second order around ATMF– is determined by equation (8.56). Recall that $\sigma_{\text{imp}}(T, K) = \sigma_{\text{imp}}(T, K; t, s, \boldsymbol{\theta})$, where $s = s_0$ at $t = 0$. Thus, differentiating (8.54) with respect to $\log s_0$ –to capture relative changes in spot– gives

$$\begin{aligned} \frac{\partial \sigma_{\text{imp}}}{\partial \log s_0} &= \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\partial(\delta\sigma_{\text{loc}}(t, z))}{\partial z} \frac{\partial z(t, s_0)}{\partial \log s_0} \\ &= \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\partial(\delta\sigma_{\text{loc}}(t, z))}{\partial z} \frac{\partial}{\partial \log s_0} \left(F_t e^{\frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T - \omega_t)\omega_t}}{\sqrt{\omega_T}} y} \right) \\ &= \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\partial(\delta\sigma_{\text{loc}}(t, z))}{\partial z} \left(1 - \frac{\omega_t}{\omega_T} \right) z(t, s_0) \\ &= \frac{1}{T} \int_0^T dt \left(1 - \frac{\omega_t}{\omega_T} \right) \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\partial(\delta\sigma_{\text{loc}}(t, z))}{\partial \log z}, \end{aligned}$$

and since

$$\delta\sigma_{\text{loc}}(t, z) = \alpha(t) \log(z/F_t) + \frac{\beta(t)}{2} \log(z/F_t)^2$$

and $z = z(t, y) = F_t \exp \left(\frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T - \omega_t)\omega_t}}{\sqrt{\omega_T}} y \right)$, we have

$$\frac{\partial \sigma_{\text{imp}}}{\partial \log s_0} = \frac{1}{T} \int_0^T dt \left(1 - \frac{\omega_t}{\omega_T} \right) \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \left(\alpha(t) + \frac{\beta(t)}{2} X(t, y) \right),$$

where substituting $X(t, y)$ and integrating over y finally gives

$$\boxed{\frac{\partial \sigma_{\text{imp}}}{\partial \log s_0} = \frac{1}{T} \int_0^T dt \left(1 - \frac{\omega_t}{\omega_T} \right) \frac{\sigma_0(t)}{\bar{\sigma}_T} \left(\alpha(t) + x_K \frac{\beta(t)}{2} \frac{\omega_t}{\omega_T} \right)}. \quad (8.62)$$

In particular, for ATMF implied volatility, $k = 0$ –or equivalently $K = F_T$ –, we obtain

$$\boxed{\left. \frac{\partial \sigma_{\text{imp}}}{\partial \log s_0} \right|_{k=0} = \frac{1}{T} \int_0^T dt \left(1 - \frac{t}{T} \frac{\bar{\sigma}_t^2}{\bar{\sigma}_T^2} \right) \frac{\sigma_0(t)}{\bar{\sigma}_T} \alpha(t)}. \quad (8.63)$$

This expression quantifies the change in implied volatility at a fixed strike equal to the forward, $K = F_T$, when spot changes. Note that equation (8.63) describes how implied volatility changes at a fixed strike, which we subsequently set equal to the forward. However, we could also calculate how ATMF implied volatility moves, taking into account

that the forward also changes when spot changes. Thus, applying the chain rule –and since $K = F_T = F_T(s_0)$ – gives

$$\frac{\partial \sigma_{\text{imp}}(T, F_T; s_0)}{\partial \log s_0} = \left(\frac{\partial \sigma_{\text{imp}}(T, K)}{\partial \log K} \frac{\partial \log K}{\partial \log s_0} + \frac{\partial \sigma_{\text{imp}}(T, K)}{\partial \log s_0} \right) \bigg|_{K=F_T},$$

and hence

$$\boxed{\frac{\partial \sigma_{\text{imp}}(T, F_T)}{\partial \log s_0} = \frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \alpha(t)}, \quad (8.64)$$

which determines the dynamics of ATMF implied volatility. The preceding expression can also be rewritten as

$$\boxed{\frac{\partial \sigma_{\text{imp}}(T, F_T)}{\partial \log s_0} = \mathcal{S}_T + \frac{1}{T} \int_0^T dt \frac{\sigma_0^2(t)}{\bar{\sigma}_t \bar{\sigma}_T} \mathcal{S}_t}. \quad (8.65)$$

In practice, expressing the dynamics of ATMF implied volatility in units of skew is very useful, since it allows volatility movements to be interpreted relative to the local slope of the smile. This normalization provides a natural scale for quantifying relative changes, facilitating comparisons across assets or models and providing a robust measure of the risk of smile deformation. For example, if an asset's ATMF implied volatility increases by 0.5% and the local skew is $\mathcal{S}_T = -0.25$, the ratio $\Delta \sigma_{\text{imp}} / \mathcal{S}_T = 0.5\% / (-0.25) = -2$ indicates that the observed change is equivalent to a moneyness shift of -2% under a fixed smile. If the underlying has not changed, this change cannot be attributed to a simple geometric translation, revealing a rotation or deformation of the implied volatility structure. This interpretation is essential for understanding market behavior and managing risk in option portfolios. We therefore define the skew stickiness ratio (SSR) as

$$\boxed{\mathcal{R}_T = \frac{1}{\mathcal{S}_T} \frac{\partial \sigma_{\text{imp}}(T, F_T)}{\partial \log s_0}}. \quad (8.66)$$

Table 8.1 summarizes the main quantities calculated above, both for the model with time-dependent ATMF local volatility (8.55) and for the model with constant ATMF local volatility (8.46).

Moreover, $\mathcal{R}_T = 2$ if $\alpha(t)$ is constant or if $T \rightarrow 0$ [cf. [29]].

8.10 ATMF skew under a power law

Consider the model described by (8.46). In general, we have seen that the ATMF skew is determined by $\alpha(t)$, whereas convexity is determined by $\beta(t)$. In practice, $\alpha(t)$ is usually assumed to be a power function of t with exponent $\gamma \in \mathbb{R}_+$. That is, $\alpha(t)$ is assumed to decay as $1/t^\gamma$. To prevent $\alpha(t)$ from diverging as $t \rightarrow 0$, a cutoff is introduced for $t < \tau_0$, so we set

$$\alpha(t) = \begin{cases} \alpha_0, & 0 \leq t \leq \tau_0 \\ \alpha_0 \left(\frac{\tau_0}{t}\right)^\gamma, & t > \tau_0 \end{cases} \quad (8.67)$$

Quantity	Model σ_0 (8.46)	Model $\sigma_0(t)$ (8.55)
Skew(k)	$\frac{1}{T} \int_0^T dt \frac{t}{T} (\alpha(t)dt + \frac{t}{T} x_K \beta(t))$	$\frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\sigma_T} \frac{t\sigma_t^2}{T\sigma_T^2} \left(\alpha(t) + x_K \frac{t\sigma_t^2}{T\sigma_T^2} \beta(t) \right)$
$\mathcal{S}_T$	$\frac{1}{T} \int_0^T dt \frac{t}{T} \alpha(t)$	$\frac{1}{T} \int_0^T dt \frac{t}{T} \frac{\sigma_0(t)\sigma_t^2}{\sigma_T^3} \alpha(t)$
Conv(k)	$\frac{1}{T} \int_0^T dt \frac{t^2}{T^2} \beta(t)$	$\frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\sigma_T} \frac{t^2\sigma_t^4}{T^2\sigma_T^4} \beta(t)$
$\mathcal{C}_T$	$\frac{1}{T} \int_0^T dt \frac{t^2}{T^2} \beta(t)$	$\frac{1}{T} \int_0^T dt \frac{t^2}{T^2} \frac{\sigma_0(t)\sigma_t^4}{\sigma_T^5} \beta(t)$
$\frac{\partial \sigma_{\text{imp}}(T, K)}{\partial \log s_0}$	$\frac{1}{T} \int_0^T dt \left(1 - \frac{t}{T}\right) \left(\alpha(t) + x_K \frac{t}{T} \frac{\beta(t)}{2} \right)$	$\frac{1}{T} \int_0^T dt \left(1 - \frac{\omega_t}{\omega_T}\right) \frac{\sigma_0(t)}{\sigma_T} \left(\alpha(t) + x_K \frac{\beta(t)}{2} \frac{\omega_t}{\omega_T} \right)$
$\frac{\partial \sigma_{\text{imp}}(T, K)}{\partial \log s_0} \Big _{k=0}$	$\frac{1}{T} \int_0^T dt \left(1 - \frac{t^2}{T^2}\right) \alpha(t)$	$\frac{1}{T} \int_0^T dt \left(1 - \frac{t^2}{T^2} \frac{\sigma_t^2}{\sigma_T^2}\right) \frac{\sigma_0(t)}{\sigma_T} \alpha(t)$
$\frac{\partial \sigma_{\text{imp}}(T, F_T)}{\partial \log s_0}$	$\mathcal{S}_T + \frac{1}{T} \int_0^T dt \mathcal{S}_t$	$\mathcal{S}_T + \frac{1}{T} \int_0^T dt \frac{\sigma_0^2(t)}{\sigma_t \sigma_T} \mathcal{S}_t$
$\mathcal{R}_T$	$1 + \frac{1}{T} \int_0^T dt \frac{\mathcal{S}_t}{\mathcal{S}_T}$	$1 + \frac{1}{T} \int_0^T dt \frac{\sigma_0^2(t)}{\sigma_t \sigma_T} \frac{\mathcal{S}_t}{\mathcal{S}_T}$

Table 8.1: Summary of implied volatility dynamics under the local volatility models given in (8.46) and (8.55).

In this case, integration gives the skew around ATMF as

$$\mathcal{S}_T = \begin{cases} \frac{\alpha_0}{2}, & 0 \leq T \leq \tau_0 \\ \frac{\alpha_0}{2-\gamma} \left(\frac{\tau_0}{T}\right)^\gamma - \frac{\alpha_0\gamma}{2(2-\gamma)} \left(\frac{\tau_0}{T}\right)^2, & T > \tau_0 \end{cases} \quad (8.68)$$

since generally $\gamma < 2$. Typically, in equity markets, $\gamma \approx 1/2$. Empirically, the term structure of ATMF skew¹⁹ in equity markets follows a power law only for maturities longer than one month, whereas its behavior at shorter maturities differs markedly [35]. In particular, the skew does not diverge as maturity approaches, as some theoretical formulations predict, but tends to stabilize around a typical absolute value close to 1.5. Thus, models with richer structure, such as implied volatility models with 3 or more parameters derived from the Bergomi model –which we will study later– best capture the term structure of ATMF skew in equity markets. Usually, we take $\tau_0 \approx 3$ months.

For very long times to maturity, $T \gg \tau_0$, the term of order 2 in τ_0/T can be neglected, so

$$\mathcal{S}_T \approx \frac{\alpha_0}{2-\gamma} \left(\frac{\tau_0}{T}\right)^\gamma = \frac{1}{2-\gamma} \alpha(T), \quad (8.69)$$

and therefore, for very long maturities, the ATMF skew decays with the same exponent γ as the local volatility skew, rescaled by a factor of $1/(2-\gamma)$. Meanwhile, the SSR is

$$\mathcal{R}_T = \begin{cases} 2, & 0 \leq T \leq \tau_0 \\ \frac{2-\gamma}{1-\gamma} + \frac{\tau_0}{T} \left(1 + \frac{(2-\gamma)}{(1-\gamma)} \frac{1+\gamma(1-\frac{\tau_0}{T})}{\gamma(\frac{\tau_0}{T})^2 - 2(\frac{\tau_0}{T})^\gamma} \right), & T > \tau_0 \end{cases} \quad (8.70)$$

¹⁹Unless we explicitly refer to local volatility skew, we mean implied volatility skew.

In particular, if $T \gg \tau_0$, then

$$\mathcal{R}_\infty = \frac{2 - \gamma}{1 - \gamma},$$

which is independent of maturity T . For $\gamma \approx 1/2$, we have $\mathcal{R}_\infty \approx 3$. Figure ?? shows examples of $\mathcal{R}_T$ as a function of T for different choices of τ_0 . As can be seen, $\mathcal{R}_T$ is highly sensitive to the choice of τ_0 ; that is, the SSR depends strongly on the ATMF skew at short maturities. Furthermore, if $\gamma = 1$, one can verify that $\mathcal{R}_T \sim \log T$ for very long maturities. This follows directly from the form of $\mathcal{S}_T \sim 1/T$ when $\gamma = 1$.

We now consider some examples.

Example 8.5. Example 1.

Example 8.6. Example 2.

8.11 Volatility of implied volatility (vol-of-vol)

As we saw in expression (8.37), implied volatility depends on time and spot and is assumed to be at least $\mathcal{C}^{1,2}$. In particular, the same holds for ATMF implied volatility, so we can write

$$d\sigma_{\text{imp}}(t, s; T, F_T) = \frac{\partial \sigma_{\text{imp}}(t, s; T, F_T)}{\partial t} dt + \frac{\partial \sigma_{\text{imp}}(t, s; T, F_T)}{\partial \log s} d \log s,$$

and, from (8.66),

$$d\sigma_{\text{imp}}(t, s; T, F_T) = \frac{\partial \sigma_{\text{imp}}(t, s; T, F_T)}{\partial t} dt + \mathcal{R}_T \mathcal{S}_T d \log s. \quad (8.71)$$

Thus, the logarithmic variance of $\sigma_{\text{imp}}(t, s; T, F_T)$ can be calculated as

$$\begin{aligned} \text{Var}(d \log \sigma_{\text{imp}}(t, s; T, F_T)) &= \left(\frac{d\sigma_{\text{imp}}(t, s; T, F_T)}{\sigma_{\text{imp}}(t, s; T, F_T)} \right)^2 \\ &= \frac{1}{\sigma_{\text{imp}}(t, s; T, F_T)^2} \left(\frac{\partial \sigma_{\text{imp}}(t, s; T, F_T)}{\partial t} dt + \mathcal{R}_T \mathcal{S}_T d \log s \right)^2, \end{aligned}$$

which, after simplification and neglecting terms of order higher than dt , gives

$$\text{Var}(d \log \sigma_{\text{imp}}(t, s; T, F_T)) = \frac{\mathcal{R}_T^2 \mathcal{S}_T^2}{\sigma_{\text{imp}}(t, s; T, F_T)^2} (d \log s)^2,$$

where $(d \log s)^2$ is the logarithmic variance of spot. At the initial time, with $t = 0$ and $s = s_0$, we have

$$\text{Var}(d \log \sigma_{\text{imp}}(t, s; T, F_T)) = \frac{\mathcal{R}_T^2 \mathcal{S}_T^2 \sigma_{\text{loc}}(0, s_0)^2}{\sigma_{\text{imp}}(t, s; T, F_T)^2}.$$

From expression (8.37), since $\Gamma^D(t, s) \rightarrow \delta(s - K)$ in the sense of distributions as $t \rightarrow 0$, we have $\sigma_{\text{loc}}(0, s_0) = \sigma_{\text{imp}}(0, F_0)$. Thus, the logarithmic volatility of ATMF implied volatility (vol-of-vol), $\text{LVol}_{\text{ATMF}}(\sigma_{\text{imp}}(T, F_T))$, at the initial time is

$$\text{LVol}_{\text{ATMF}}(\sigma_{\text{imp}}(T, F_T)) = \frac{\mathcal{R}_T \mathcal{S}_T \sigma_{\text{imp}}(0, F_0)}{\sigma_{\text{imp}}(T, F_T)}. \quad (8.72)$$

In particular, for very short maturities, i.e., $T \approx 0$, volatility simplifies to

$$\text{LVol}_{ATMF}(\sigma_{\text{imp}}(T, F_T)) = 2\mathcal{S}_T, \quad (8.73)$$

since $\mathcal{R}_T \rightarrow 2$ and $\sigma_{\text{imp}}(T, F_T) \rightarrow \sigma_{\text{imp}}(0, F_0)$.

One of the main limitations of the local volatility model is that the volatility of implied volatility is determined by skew dynamics, leaving no additional degrees of freedom to better fit the observed behavior of volatility.

8.12 Future skews and vol-of-vol

For simplicity, we study the constant-volatility case $\sigma_0(t) = \sigma_0$. Suppose that the local volatility model given in (8.46) has been calibrated to market data. Now consider time $\tau > 0$, with spot price S_τ . What ATMF skew does the local volatility model generate for a time to maturity of θ , starting at τ ?

Recall that the ATMF skew obtained in equation (8.50) is calculated at time $t = 0$. The local volatility function is expressed in terms of $x = \log(s/F_t)$, where $F_t = \text{For}_S(0, t)$ and $S_0 = s$. Thus, if $\tau < t$, we can redefine the variable x as

$$\begin{aligned} x &= \log\left(\frac{s}{\text{For}_S(0, t)}\right) \\ &= \log\left(\frac{s}{\text{For}_S(\tau, t)}\right) + \log\left(\frac{\text{For}_S(\tau, t)}{\text{For}_S(0, t)}\right) \\ &= y + \log\left(\frac{S_\tau e^{(r-q)(t-\tau)}}{se^{(r-q)\tau}}\right) \\ &= y + \log\left(\frac{S_\tau}{F_\tau}\right) \\ &= y + x_\tau, \end{aligned}$$

which allows us to rewrite local volatility as

$$\sigma_{\text{loc}}(t, s) = \sigma_{0,\tau}(t) + \alpha_\tau(t)y + \frac{1}{2}\beta_\tau(t)y^2, \quad (8.74)$$

where the coefficients of this second-order polynomial in y are given by

$$\begin{aligned} \sigma_{0,\tau}(t) &= \sigma_0(t) + \alpha(t)x_\tau + \frac{1}{2}\beta(t)x_\tau^2 \\ \alpha_\tau(t) &= \alpha(t) + \beta(t)x_\tau \\ \beta_\tau(t) &= \beta(t). \end{aligned}$$

Note that if $\beta(t) \neq 0$, since $\alpha_\tau(t)$ depends on x_τ , the ATMF skew at τ depends on the spot price at τ , S_τ . This implies that the local volatility model with $\beta(t) \neq 0$ is not time homogeneous. That is, the forward skew would depend on absolute time rather than time to maturity. We therefore set $\beta(t) = 0$.

We now calculate the ATMF skew at τ for a time to maturity of θ , starting at τ . That is, for maturity $\tau + \theta$. Differentiating implied volatility $\sigma_{\text{imp}}(\tau + \theta, K; \tau, S_\tau)$ with respect to log-moneyness k gives

$$\mathcal{S}_\theta(\tau) = \left. \frac{\partial}{\partial k} \sigma_{\text{imp}}(\tau + \theta, K; \tau, S_\tau) \right|_{k=0} = \frac{1}{\theta} \int_\tau^{\tau+\theta} \frac{t - \tau}{\theta} \alpha(t) dt, \quad (8.75)$$

where $\mathcal{S}_\theta(\tau)$ denotes the forward-start ATMF skew at τ for a time to maturity of θ . Note that the integration interval has been transformed from $[0, T]$ to $[\tau, \tau + \theta]$ through $t \mapsto \frac{\theta}{T}t + \tau$.

Since expression (8.50) gives $\alpha(t) = \frac{d}{dt}(t\mathcal{S}_t) + \mathcal{S}_t$, the ATMF skew at τ for a time to maturity of θ can be rewritten as

$$\mathcal{S}_\theta(\tau) = \mathcal{S}_{\tau+\theta} - \frac{\tau}{\theta} \left(\frac{1}{\theta} \int_\tau^{\tau+\theta} \mathcal{S}_t dt - \mathcal{S}_{\tau+\theta} \right), \quad (8.76)$$

where $\mathcal{S}_t$ denotes the ATMF skew at the initial time for maturity t .

Note that the forward-starting skew, $\mathcal{S}_\theta(\tau)$, uses only the skews in the interval $[\tau, \tau + \theta]$. It therefore need not resemble $\mathcal{S}_\theta$.

Furthermore, when the skew is decreasing (as is typical in equity markets),

$$|\mathcal{S}_\theta(\tau)| \leq |\mathcal{S}_{\tau+\theta}| \ll |\mathcal{S}_\theta|,$$

which implies that the forward skew is smaller in absolute value than the spot skew for the same time to maturity.

Moreover, since ATMF vol-of-vol is related to ATMF skew in the local volatility model, see equation (8.73), such a low forward skew implies that the model predicts very low levels of future volatility of volatility.

8.13 Delta and carry P&L

8.14 How to hedge vega

8.15 Forward-start options

Before introducing stochastic volatility models, we study forward-start options because—as we will see in this section—their analysis provides key insights into which aspects of implied volatility dynamics matter for pricing complex derivatives. These options, whose strike is fixed at a future date, reveal that prices are not tightly determined by the vanilla option volatility smile, as demonstrated by deriving upper and lower price bounds from that smile. This observation highlights the limitations of models calibrated solely to current implied volatilities. In particular, a detailed study of a forward-start call shows how the local volatility model handles the risk associated with the future evolution of the smile (forward-smile risk), allowing us to assess its suitability for these options.

Definition 8.7. Let S_t be the price of the underlying asset at time t , and let $0 < t < T_1 < T_2$. A forward-start call option, abbreviated FSCall, with strike fixing at T_1 and maturity T_2 , is a call option on the ratio S_{T_2}/S_{T_1} whose payoff²⁰ is

$$\text{FSCall}(T_2, S_{T_2}; T_1, T_2, k) = \max \left\{ \frac{S_{T_2}}{S_{T_1}} - k, 0 \right\}. \quad (8.77)$$

Forward-start put options, denoted FSPut, are defined analogously.

Remark 8.8. Note that $k = 1$ defines a forward-start option that is ATM –rather than ATMF– at its fixing time. Furthermore, $k > 1$ indicates that the option is initially OTM, whereas $k < 1$ indicates that it is ITM at $t = T_1$.

Exotic options such as forward-start and cliquet options²¹ are particularly sensitive to forward-smile risk: the risk associated with uncertainty about the future shape of the implied volatility smile. Unlike vanilla options, whose prices can be inferred directly from current implied volatilities for different maturities, forward-start options depend on the behavior of the underlying between two future dates. Thus, their valuation cannot rely solely on current market data; it requires assumptions about the distribution of future returns, that is, how the underlying asset and the implied volatility structure evolve over time. Careful modeling of volatility dynamics is therefore essential for pricing and hedging these options correctly.

To price and hedge these options, we first need a pricing model and a hedging portfolio that neutralizes gamma, theta, and vega risk within that model. We also estimate the cost of continuous reheding, which in practice uses vanilla options. This cost arises because, even if the hedge is calibrated at a given time, market conditions change continuously, and maintaining the hedge requires dynamic trading in other derivatives such as calls and puts. This generates transaction costs and potential hedging errors due to the model's limitations.

In the Black-Scholes model –with constant volatility– volatility has no term structure, i.e., it is independent of expiry T . Thus, since an FS option involves S_{T_1} and S_{T_2} , it is more appropriate to consider a Black-Scholes model with time-dependent volatility. That is, we consider a base model with volatility $\sigma_{\text{loc}} = \sigma_0(t)$ –to which we will subsequently add corrections involving spot, $\delta\sigma_{\text{loc}}(t, s)$. In particular, we study the case (8.55).

Proposition 8.9. *The price of an FS option is independent of S_t for $t < T_1$. In particular, its price depends only on the integrated variance*

$$\bar{\sigma}_{T_1, T_2}^2 = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \sigma^2(t) dt. \quad (8.78)$$

²⁰This payoff is dimensionless, since it is measured in units relative to the value of the underlying at T_1 . Equivalently, some authors use the dimensional version $(S_{T_2} - kS_{T_1})^+$, which corresponds to the same derivative expressed in monetary units. Both expressions are financially equivalent and are related by a factor of S_{T_1} . The dimensionless version is particularly convenient for defining forward implied volatility, since its value does not involve the factor S_{T_1} . When using the dimensional form $(S_{T_2} - kS_{T_1})^+$, this factor must be removed from the price before obtaining the corresponding implied volatility.

²¹A series of forward-start options is called a cliquet or ratchet option [36].

Proof. Under the risk-neutral measure $\mathbb{P}^*$, the price of an FSCall option for $t < T_1$ is²²

$$\text{FSCall}(t, s; T_1, T_2, k) = \mathbb{E}^* \left[e^{-r(T_1-t)} \left(\frac{S_{T_2}}{S_{T_1}} - k \right)^+ \middle| \mathcal{F}_t \right],$$

and since $\sigma_t = \sigma(t)$ is a deterministic volatility process, spot follows a lognormal distribution such that

$$\log S_{T_i} = \log S_t + \int_t^{T_i} \left(r - q - \frac{1}{2} \sigma^2(s) \right) ds + \int_t^{T_i} \sigma(s) dW_s,$$

for $i = 1, 2$. Thus,

$$\log \frac{S_{T_2}}{S_{T_1}} = \int_{T_1}^{T_2} \left(r - q - \frac{1}{2} \sigma^2(s) \right) ds + \int_{T_1}^{T_2} \sigma(s) dW_s,$$

so S_{T_2}/S_{T_1} is a lognormal random variable with mean

$$\left(r - q - \frac{1}{2} \bar{\sigma}_{T_1, T_2}^2 \right) (T_2 - T_1),$$

and variance

$$(T_2 - T_1) \bar{\sigma}_{T_1, T_2}^2.$$

That is, $\mathbb{E}^* \left[(S_{T_2}/S_{T_1} - k)^+ \middle| \mathcal{F}_t \right]$ is a function only of the integrated variance, which we denote by $G = G(\bar{\sigma}_{T_1, T_2})$. We therefore denote the price of an FSCall (and, respectively, of an FSPut) simply by

$$\text{FSCall}(t; T_1, T_2, k) = e^{-r(T_1-t)} G(\bar{\sigma}_{T_1, T_2}). \quad (8.79)$$

where

$$G(\bar{\sigma}_{T_1, T_2}) = e^{-(r-q)(T_2-T_1)} \text{BSCall}(1, k; T_2 - T_1, \bar{\sigma}_{T_1, T_2}) \quad (8.80)$$

with BSCall denoting the vanilla call option price given by expression (7.28). $\square$

Remark 8.10. Note that

$$\bar{\sigma}_{T_1, T_2}^2 = \frac{(T_2 - t) \bar{\sigma}_{T_2}^2 - (T_1 - t) \bar{\sigma}_{T_1}^2}{T_2 - T_1}, \quad (8.81)$$

where $\bar{\sigma}_{T_i}^2$ is defined in equation (8.53). That is, the price of an FSCall depends only on the integrated variances $\bar{\sigma}_{T_1}^2$ and $\bar{\sigma}_{T_2}^2$.

Given the preceding result, we need to analyze: (i) how to hedge this FS option; and (ii) which payoffs with maturities T_1 and T_2 should be used to calibrate $\bar{\sigma}_{T_1}$ and $\bar{\sigma}_{T_2}$. As we saw earlier, a unique strictly positive implied volatility can be associated with any convex payoff, so we now also refer to both quantities as implied volatilities. Recall that these payoffs must satisfy the condition for the absence of calendar arbitrage. That is,

$$(T_2 - t) \bar{\sigma}_{T_2}^2 \geq (T_1 - t) \bar{\sigma}_{T_1}^2.$$

²²Again, for $T_1 < t < T_2$, the price is determined by the price of a standard call option.

Remark 8.11. For $t < T_1$, the delta, gamma, and vanna of an FS option are identically zero by Proposition 8.9. If $T_1 < t < T_2$, the option becomes a standard call or put, and its Greeks are calculated in the usual way.

One possibility would be to hedge this instrument with call or put options –and hedge their delta with spot. However, we have seen that their implied volatilities depend on spot, and the gamma of a portfolio of calls and puts need not vanish. Thus, we need to find a European derivative –on spot– such that a portfolio of these derivatives with maturities T_1 and T_2 has spot-independent vega.

8.15.1 How to hedge the risks of a forward-start option

Suppose we construct a portfolio, Π , of European call and put options. Since the option price can be calculated using the Black-Scholes formula with volatility set equal to implied volatility, the price –in the case of a call– is

$$V(t, s; T, K, r, q, \sigma_{\text{imp}}) = s\mathcal{N}(d_+(s/K, T-t)) - Ke^{-r(T-t)}\mathcal{N}(d_-(s/K, T-t)),$$

which is homogeneous of degree 1 in the pair (s, K) . That is, $V(t, \lambda s; T, \lambda K, r, q, \sigma_{\text{imp}}) = \lambda V(t, s; T, K, r, q, \sigma_{\text{imp}})$. Thus, simplifying the notation to $V = V(s, K)$ gives

$$V(s, K) = V\left(K \frac{s}{K}, K\right) = K \cdot V\left(\frac{s}{K}, 1\right) = K f\left(\frac{s}{K}\right),$$

where $f(s/K)$ is also a function of implied volatility. That is, $f = f(s/K; \sigma_{\text{imp}})$. Defining

$$\varphi(s/K) = \frac{\partial f}{\partial \sigma_{\text{imp}}}\left(\frac{s}{K}; \sigma_{\text{imp}}\right),$$

the vega of this option is given by

$$\mathcal{V}(s, K) = K\varphi\left(\frac{s}{K}\right).$$

Let $\rho(K)$ be the distribution specifying the quantity of options with strike $K \in [K_{\min}, K_{\max}]$ held in portfolio Π . The portfolio's total vega is then given by

$$\mathcal{V}_{\Pi}(t, s) = \int_{K_{\min}}^{K_{\max}} dK \rho(K) K \varphi\left(\frac{s}{K}\right), \quad (8.82)$$

The question is then: which strike distribution $\rho(K)$ should we choose for the portfolio to make its vega independent of spot –as the FS option's vega is? With the change of variable $u = \frac{s}{K}$, we have

$$\mathcal{V}_{\Pi}(t, s) = \int_{s/K_{\max}}^{s/K_{\min}} du \frac{1}{u^3} s^2 \rho\left(\frac{s}{u}\right) \varphi(u).$$

Thus, choosing $\rho(s/u) \propto u^2/s^2$ eliminates the dependence of total vega on s –when taking the limits $K_{\min} \rightarrow 0$ and $K_{\max} \rightarrow +\infty$. Since $s/u = K$, this choice is equivalent to setting

$$\rho(K) \propto \frac{1}{K^2}. \quad (8.83)$$

Thus, a portfolio of European options with density proportional to K^{-2} has spot-independent vega. The question now is: what is the payoff of this portfolio?

Applying the Carr-Madan (or Breeden-Litzenberger) formula [28, 38], any payoff $\Lambda(S_T)$ of class $\mathcal{C}^2(\mathbb{R}_{++})$ can be rewritten as

$$\Lambda(S_T) = \Lambda(K_0) + \Lambda'(K_0)(S_T - K_0) + \int_0^{K_0} (K - S_T)^+ \Lambda''(K) dK + \int_{K_0}^{+\infty} (S_T - K)^+ \Lambda''(K) dK, \quad (8.84)$$

and setting $\Lambda(s) = \log s$ gives

$$\log S_T = \log K_0 + \frac{1}{K_0}(S_T - K_0) - \int_0^{K_0} \frac{1}{K^2} (K - S_T)^+ dK - \int_0^{K_0} \frac{1}{K^2} (S_T - K)^+ dK.$$

That is, up to an affine function of S_T given by $\log K_0 + (S_T - K_0)/S_T$, the payoff can be replicated with a portfolio of European call and/or put options. If $K_0 \rightarrow 0$, the portfolio consists of calls at all strikes. If $K_0 \rightarrow +\infty$, it consists of puts at all strikes. Equivalently, we can define a portfolio of calls and puts with density $\rho(K) \propto K^{-2}$ and integrate twice with respect to strike.

In fact, for reasons to be introduced later in connection with variance swaps, it is preferable to work with a payoff equal to $-2 \log S_T$. In this case, the price of a log contract is

$$\text{LC}(t, s) = -2e^{-(r-q)(T-t)} \mathbb{E}^* [\log S_T | \mathcal{F}_t], \quad (8.85)$$

and, assuming that spot is described –under the risk-neutral measure $\mathbb{P}^*$ – by

$$dS_t = (r - q)S_t dt + \sigma_0(t)S_t dW_t$$

we have

$$d \log S_t = \left(r - q - \frac{1}{2} \sigma_0(t)^2 \right) dt + \sigma_0(t) dW_t.$$

Integrating between t and T gives

$$\log S_T - \log S_t = (r - q)(T - t) - \frac{1}{2} \int_t^T \sigma_0(u)^2 du + \int_t^T \sigma_0(u) dW_u,$$

so taking expectations –and applying Theorem 2.14– finally gives

$$\mathbb{E}^* [\log S_T | \mathcal{F}_t] = \log s + (r - q)(T - t) - \frac{1}{2} \int_t^T \sigma_0(u)^2 du. \quad (8.86)$$

Recalling definition (8.53) and using the payoff expression (8.85), we have

$$\boxed{\text{LC}(t, s) = -2e^{-r(T-t)} \left(\log s + (r - q)(T - t) - \frac{1}{2} \bar{\sigma}_T^2(T - t) \right)}. \quad (8.87)$$

Remark 8.12. The delta of a log contract at time t is given by $\Delta(t, s) = -2e^{-(r-q)(T-t)} s^{-1}$, and its dollar gamma by $\Gamma^D(t, s) = 2e^{-(r-q)(T-t)}$. In particular, a delta-hedged log contract is essentially a derivative on volatility [37]. Since dollar gamma is constant, the log contract provides stable gamma exposure across all spot levels. Recall that our aim was to hedge the vega of forward-start options.

In particular, the vega of a log contract is

$$\mathcal{V}(t) = 2e^{-r(T-t)}(T-t)\bar{\sigma}_T. \quad (8.88)$$

Let us return to the question of how to hedge the vega of a forward-start option using log contracts. In particular, the sensitivity of the FSCall price to volatilities $\bar{\sigma}_{T_1}$ and $\bar{\sigma}_{T_2}$ is given by

$$\begin{aligned} \frac{\partial \text{FSCall}(t; T_1, T_2, \bar{\sigma}_{T_1}, \bar{\sigma}_{T_2})}{\partial \bar{\sigma}_{T_1}} &= -e^{-r(T_1-t)}(T_1-t)\bar{\sigma}_{T_1}N \\ \frac{\partial \text{FSCall}(t; T_1, T_2, \bar{\sigma}_{T_2}, \bar{\sigma}_{T_2})}{\partial \bar{\sigma}_{T_2}} &= e^{-r(T_1-t)}(T_2-t)\bar{\sigma}_{T_2}N, \end{aligned}$$

where

$$N = \frac{1}{(T_2 - T_1)\bar{\sigma}_{T_1, T_2}} \frac{\partial G}{\partial \bar{\sigma}_{T_1, T_2}}.$$

It is therefore straightforward to verify that the portfolio

$$\begin{aligned} \Pi &= -\text{FSCall}(t; T_1, T_2, \bar{\sigma}_{T_1}, \bar{\sigma}_{T_2}) \\ &\quad + \frac{N}{2} [e^{r(T_2-T_1)}\text{LC}(t; T_1, \bar{\sigma}_{T_1}) - \text{LC}(t; T_2, \bar{\sigma}_{T_2})], \end{aligned} \quad (8.89)$$

is vega hedged. That is, it has no sensitivity to either $\bar{\sigma}_{T_1}$ or $\bar{\sigma}_{T_2}$.

Remark 8.13. Note that although the dollar gamma of a log contract is not identically zero, it is constant. Thus, the total gamma of this portfolio is

$$\Gamma_{\Pi}^D = -\Gamma_{\text{FSCall}}^D + \frac{N}{2} [e^{r(T_2-T_1)}\Gamma_{\text{LC}(T_1)}^D - \Gamma_{\text{LC}(T_2)}^D],$$

and, as we saw earlier, $\Gamma_{\text{FSCall}}^D = 0$. Hence,

$$\Gamma_{\Pi}^D = \frac{N}{2} [e^{r(T_2-T_1)} \cdot 2e^{r(T_1-t)} - 2e^{r(T_2-t)}] = 0.$$

Thus, when hedging the vega of an FSCall with log contracts, the portfolio's total gamma is also zero because of the required combination of log contracts. Moreover, under the assumption that our volatility model is $\sigma_t = \sigma_0(t)$, it is easy to verify that the portfolio's theta also vanishes. Thus, this portfolio is theta, delta, gamma, and vega neutral under a deterministic volatility model²³. We have therefore constructed a static hedge²⁴ that perfectly hedges first-order sensitivities to $\bar{\sigma}_{T_1, T_2}$.

Market implied volatilities depend not only on maturity T but also on strike. How should we define $\bar{\sigma}_T$? Following Bergomi's argument [29], we take $\bar{\sigma}_T$ as the implied volatility²⁵, since

$$\bar{\sigma}_T = \left(e^{r(T-t)}\text{LC}_{\text{Mkt}} + \frac{2}{T-t} \log s + 2(r-q) \right)^{1/2}. \quad (8.90)$$

²³This neutrality holds only under the assumption that instantaneous volatility is a deterministic function of time, i.e., $\sigma_t = \sigma_0(t)$. If volatility is stochastic or depends on spot, this hedge can generate residual P&L of second order in forward volatility changes.

²⁴A hedge is static if it does not require rebalancing as time or spot changes.

²⁵Since the log contract payoff is $\Lambda(S_T) = -2 \log S_T$, which is convex in S_T , formula (8.87) can be inverted to obtain implied volatility by equating it to the observed market price, denoted by LC_{Mkt} . That is, $\text{LC}(t, s; \bar{\sigma}_T, \theta) = \text{LC}_{\text{Mkt}}$, yielding a unique solution.

Remark 8.14. From now on, we use $\sigma_{\text{imp,LC}}(T) = \bar{\sigma}_T$ to refer to the implied volatility of a log contract with maturity T . The log contract's implied volatility thus has a clear interpretation: it is the square root of the time-weighted average of future instantaneous variance under the risk-neutral measure [cf. Eq. (8.53)]. As we will see later, this quantity coincides with the variance swap rate (adjusted for the payoff factor) and is obtained by inverting the market price of the log contract.

We will see later that, in the absence of dividends, the implied volatility of log contracts can be expressed as an average of vanilla option implied volatilities.

8.15.2 Additional P&L terms

Which terms affect $t < T_1$? We now analyze the P&L changes²⁶ of the preceding portfolio over an infinitesimal time interval. Thus,

$$\begin{aligned} dP\&L &= \Pi(t + dt, s + ds, \bar{\sigma}_{T_1, T_2} + d\bar{\sigma}_{T_1, T_2}, \boldsymbol{\theta}) - (1 + rdt)\Pi(t, s, \bar{\sigma}_{T_1, T_2}, \boldsymbol{\theta}) \\ &= e^{-r(T_1 - (t + dt))} G(\bar{\sigma}_{T_1, T_2} + d\bar{\sigma}_{T_1, T_2}) - (1 + rdt)e^{-r(T_1 - t)} G(\bar{\sigma}_{T_1, T_2}) \\ &\quad + \frac{N}{2} (\text{LC}(t + dt, s + ds, \sigma_{\text{imp}}(T_2) + d\sigma_{\text{imp}}(T_2), \boldsymbol{\theta}) - \text{LC}(t, s, \sigma_{\text{imp}}(T_2), \boldsymbol{\theta})) e^{r(T_2 - T_1)} \\ &\quad - \frac{N}{2} (\text{LC}(t + dt, s + ds, \sigma_{\text{imp}}(T_1) + d\sigma_{\text{imp}}(T_1), \boldsymbol{\theta}) - \text{LC}(t, s, \sigma_{\text{imp}}(T_1), \boldsymbol{\theta})). \end{aligned}$$

Expanding to first order in dt and second order in ds –but not in $d\bar{\sigma}_{T_1, T_2}$ – gives [29]:

$$dP\&L = e^{-r(T_1 - t)} \left[- (G(\bar{\sigma}_{T_1, T_2} + d\bar{\sigma}_{T_1, T_2}) - G(\bar{\sigma}_{T_1, T_2})) + \frac{\partial G}{\partial \bar{\sigma}_{T_1, T_2}^2} d\bar{\sigma}_{T_1, T_2}^2 \right]. \quad (8.91)$$

Remark 8.15. Expression (8.91) should not be interpreted as a truncated second-order Taylor expansion around $\bar{\sigma}_{T_1, T_2}$: it represents the exact P&L induced by a finite shift in forward variance $\bar{\sigma}_{T_1, T_2}^2$ –since risks of first order in dt and second order in ds are hedged by construction, i.e., this portfolio is theta, delta, and gamma neutral. Since the hedge was constructed to cancel the first-order derivative with respect to this variance (i.e., the portfolio is vega neutral), the linear term vanishes by construction. The residual is purely quadratic.

Although the model may allow independent perturbations in $\bar{\sigma}_{T_1}$ and $\bar{\sigma}_{T_2}$, the only combination that affects P&L is their joint contribution to forward variance $\bar{\sigma}_{T_1, T_2}^2$. Indeed, even if $d\bar{\sigma}_{T_1}^2$ and $d\bar{\sigma}_{T_2}^2$ appear explicitly in the intermediate expression, they do so only through:

$$d(\bar{\sigma}_{T_1, T_2}^2) = \frac{T_2 d\bar{\sigma}_{T_2}^2 - T_1 d\bar{\sigma}_{T_1}^2}{T_2 - T_1}.$$

Thus, the portfolio has no individual exposure to $\bar{\sigma}_{T_1}$ or $\bar{\sigma}_{T_2}$, but only to the shape of the implied volatility curve between these points.

²⁶We define P&L as the difference between the portfolio value at $t + dt$ and the value it would have reached had it grown at the risk-free rate r . That is, we compare $\Pi(t + dt)$ with $(1 + rdt)\Pi(t)$, which is equivalent to writing $dP\&L = d\Pi - r\Pi dt$ to first order.

Remark 8.16. The entire derivation has been carried out in terms of implied variances rather than volatilities. This is natural both analytically and operationally, since log contract prices—and hence the hedge—are linear functions of $\bar{\sigma}_T^2$. Working with $\bar{\sigma}^2$ rather than $\bar{\sigma}$ is analogous to using S (rather than $\sqrt{S}$) when calculating the P&L of a delta-hedged option: it simplifies the analysis and better reflects the system's actual linear dependencies.

In particular, if G is affine in $\bar{\sigma}_{T_1, T_2}^2$, the hedge is perfect and P&L remains exactly zero until time T_1 . This would hold, for example, for a derivative whose payoff is a linear combination of log contracts, such as $\log\left(\frac{S_{T_2}}{S_{T_1}}\right)$, although this product has no commercial appeal. A more relevant example is a forward variance swap, which pays the realized variance over $[T_1, T_2]$.

In practice, G is generally not affine in $\bar{\sigma}_{T_1, T_2}^2$, so P&L contains second-order terms. For an at-the-money forward call, we have approximately

$$G(\bar{\sigma}_{T_1, T_2}) \simeq \frac{\bar{\sigma}_{T_1, T_2}}{\sqrt{2\pi}} \sqrt{T_2 - T_1},$$

and the P&L generated by a finite change in forward variance is given by:

$$dP\&L \simeq e^{-r(T_1-t)} \frac{\sqrt{T_2 - T_1}}{\sqrt{2\pi}} \cdot \frac{1}{2\bar{\sigma}_{T_1, T_2}} (d\bar{\sigma}_{T_1, T_2})^2. \quad (8.92)$$

This result highlights *vol-of-vol risk*: P&L is nonzero even without directional volatility exposure, and must be anticipated and incorporated into the derivative's price at $t = 0$ as a premium associated with the curvature of G .

What happens at $t = T_1$? The price of the forward-start option at $t = T_1$ can be analyzed in two different ways:

- (i) At $t = T_1$, the forward-start option becomes a European vanilla option with maturity T_2 . As we saw in equation (8.84), any option with a European payoff can be replicated with a portfolio of vanilla options. Since their prices are determined by their implied volatilities for maturity T_2 observed at T_1 , the forward-start option's price at T_1 is also a function of these vanilla implied volatilities. For example, if at T_1 the FS option is an at-the-money call with implied volatility $\sigma_{\text{imp}}^{(T_1)}(T_2, K)$, where the superscript (T_1) denotes the implied volatility observed at T_1 , then the forward-start option's price at T_1 is

$$V\left(T_1, S_{T_1}; T_2, K = S_{T_1}, \sigma_{\text{imp}}^{(T_1)}(T_2, K = S_{T_1})\right), \quad (8.93)$$

where, recall, $V(t, s; T, K, \sigma)$ denotes the price of a European call option at t , with spot s , implied volatility σ , maturity T , and strike K .

- (ii) In portfolio Π , constructed in (8.89), both the forward-start option and its hedge—a combination of log contracts—depended on implied volatilities $\bar{\sigma}_{T_1}$ and $\bar{\sigma}_{T_2}$ through $\bar{\sigma}_{T_1, T_2}$ in equation (8.81). In particular, at $t = T_1$, we have $\bar{\sigma}_{T_1, T_2} = \bar{\sigma}_{T_2}$. Thus, the forward-start option's price at T_1 is

$$V(T_1, S_{T_1}; T_2, K = S_{T_1}, \bar{\sigma}_{T_2}^{(T_1)}) \quad (8.94)$$

where the superscript (T_1) again denotes the implied volatility –of the log contract– observed at T_1 .

Note that the prices in (8.93) and (8.94) generally differ, since they are calculated using different volatilities: the market implied volatility of an ATM call, $\sigma_{\text{imp}}^{(T_1)}(T_2, K = S_{T_1})$, and that of the log contract, $\bar{\sigma}_{T_2}^{(T_1)}$. This difference reflects future smile risk, justifying a price correction when valuing the derivative at $t < T_1$.

The risk arising from uncertainty about the smile observed at T_1 for maturity T_2 –given the log contract’s implied volatility– is called forward-smile risk. Thus, for a forward-start option with payoff $\Lambda = \Lambda(S_{T_2}/S_{T_1})$, we must add a correction to the initial price to cover the difference between the market price of that payoff at time $t = T_1$ and the price calculated using Black-Scholes with implied volatility $\bar{\sigma}_{T_2}^{(T_1)}$.

If the forward-start option’s payoff depends only weakly on the log contract’s volatility (as in a digital forward-start option), hedging vega with log contracts is not meaningful, and forward-smile risk completely dominates its price –the option becomes an instrument exposed purely to the forward smile. In this case, vol-of-vol risk disappears, but the risk associated with the future shape of the smile remains and cannot be hedged with log contracts.

Remark 8.17. One might ask: why not calculate forward implied volatility $\bar{\sigma}_{T_1, T_2}$ directly, which would suffice to price a forward-start option correctly? In practice, this is impossible for a fundamental reason: forward-start options are not directly quoted in financial markets, so their prices cannot be observed to invert formula (8.79) and extract their implied volatility in the strict sense. However, it is possible to observe (or accurately replicate) log contract prices at different maturities, whose expectation under the risk-neutral measure allows total implied variance up to each maturity to be recovered. These prices define an average forward variance over $[T_1, T_2]$, denoted $\bar{\sigma}_{T_1, T_2}^2$, as a linear combination of the variances of log contracts maturing at T_1 and T_2 ; see equation (8.81). This quantity should not be confused with the market implied volatility of the forward-start option, since it does not fully incorporate effects such as the volatility smile, skew, or vol-of-vol risk. The price obtained from $\bar{\sigma}_{T_1, T_2}$ therefore requires corrections, typically of second order, that capture its sensitivity to changes in forward variance.

8.15.3 Forward-start options and their pricing

We choose to price forward-start options –and assess their risks– assuming time-dependent volatility, recalibrated daily to the implied volatility of log contracts, $\sigma_{\text{imp,LC}}(T)$. We can then hedge the option’s delta and gamma as above and make the two necessary price adjustments: (i) $V_{\text{vol-of-vol}}$ to cover implied volatility-of-volatility risk over $[0, T_1]$, as shown in equation (8.92); and (ii) $V_{\text{forward-smile}}$ to cover future smile risk arising from the difference between expressions (8.93) and (8.94).

Thus, the price of the forward-start option at $t = 0$ is

$$\boxed{\text{FSCall}(t; T_1, T_2, k) = e^{-rT_1} G(\bar{\sigma}_{T_1, T_2}(0)) + V_{\text{vol-of-vol}} + V_{\text{forward-smile}},} \quad (8.95)$$

but how can $V_{\text{vol-of-vol}}$ and $V_{\text{forward-smile}}$ be estimated?

- (i) If there is no market for vol-of-vol and the forward smile:
- $V_{\text{vol-of-vol}}$: Historical data on log contract implied volatilities are used to estimate conservatively²⁷ the expected realized volatility of $\bar{\sigma}_{T_1, T_2}$ over the interval $[0, T_1]$.
 - $V_{\text{forward-smile}}$: Historical market smile data for European vanilla options (calls and puts) with time to maturity $T_2 - T_1$ are used. At different historical dates, these options allow the implied volatility curve as a function of strike to be constructed for times to maturity equal to $T_2 - T_1$, i.e., the same remaining maturity as the forward-start option at T_1 . Comparing the implied volatility at the strike relevant to the payoff with that of the log contract in each case yields an empirical distribution of differences. A conservative value is estimated from this distribution to ensure, with high probability, that the estimate prudently covers the risk of forward-smile miscalibration.
- (ii) If a liquid market exists in which the risks of the forward-start option can be hedged –for example, by trading other forward-start products²⁸– historical estimates can be replaced by implied values. Under this approach, $V_{\text{vol-of-vol}}$ and $V_{\text{forward-smile}}$ are not calculated separately; their sum is obtained directly as a total adjustment implied by the price.

As we approach T_1 , vol-of-vol risk disappears because past volatility behavior is already known, so $V_{\text{vol-of-vol}}$ must tend to zero. Forward-smile risk, by contrast, is gradually resolved, and at T_1 we can observe the exact difference between our option's implied volatility and that of a standard contract. Therefore, $V_{\text{forward-smile}}$ must adjust to this actual difference.

For a model to be appropriate, $V_{\text{vol-of-vol}}(t)$ must tend to zero as $t \rightarrow T_1$. The model must also ensure that $V_{\text{forward-smile}}(t)$ converges, as $t \rightarrow T_1$, to the smile for maturity T_2 .

Remark 8.18. $V_{\text{forward-smile}}$ accounts for most of the volatility risk in options primarily exposed to forward-smile risk, such as forward-start ATM call spreads or digital payoffs.

In these cases, G has very little sensitivity to forward volatility $\bar{\sigma}_{T_1, T_2}$, so $V_{\text{vol-of-vol}} \approx 0$.

Remark 8.19. Although local volatility models reproduce the current market smile, they provide no explicit control over future smiles. They are therefore unsuitable for correctly pricing forward-start products sensitive to the forward smile (such as digital options). We return to this point later when studying forward variance models [cf. [29, chapters 7 and 12]], which allow more precise control over the adjustments $V_{\text{vol-of-vol}}$ and $V_{\text{forward-smile}}$.

²⁷A conservative estimate constructs an estimator $\bar{v}$ of the realized volatility of $\bar{\sigma}_{T_1, T_2}(t)$ over $[0, T_1]$ such that $\mathbb{P}(\bar{v} \geq v_{\text{real}}) \geq 1 - \varepsilon$ for a sufficiently small confidence level ε . In practice, this is implemented using high percentiles of the empirical distribution or by adding a multiple of the standard deviation to the sample mean.

²⁸For example, consider a digital forward-start option that starts in $T_1 = 1$ year and matures in $T_2 = 2$ years, with an at-the-money strike fixed at T_1 . Its payoff is 1 if $S_{T_2}/S_{T_1} > k = 1$, and 0 otherwise. This option is highly sensitive to the future shape of the smile, so its price depends critically on both the evolution of implied volatility and the smile profile at T_1 for maturity T_2 . Instead of estimating the vol-of-vol and smile risk adjustments separately, a model calibrated to the market allows $V_{\text{vol-of-vol}} + V_{\text{forward-smile}}$ to be calculated directly.

Although pricing models are commonly calibrated to the vanilla option smile in practice, this strategy is neither sufficient nor appropriate for capturing the risks of forward-start options. In particular, vanilla option prices cannot robustly determine either the forward smile or volatility-of-volatility risk, since these dependencies cannot be locked in²⁹ through hedging strategies using observable instruments. Thus, calibrating a model exclusively to the spot smile in the hope that this determines $V_{\text{vol-of-vol}}$ and $V_{\text{forward-smile}}$ is not only incorrect but potentially dangerous. Rigorous pricing of these products requires models that explicitly address these dynamic dependencies, using log contract implied volatilities as better-founded hedging benchmarks. Moreover, vanilla option prices provide only loose bounds on forward-start option prices, demonstrating that calibration to the vanilla smile is insufficient to determine $V_{\text{forward-smile}}$ [cf. [29, chapter 3.1.7]].

8.16 Forward-start options with absolute payoffs

What happens if, instead of analyzing a payoff that depends on the ratio of underlying prices at T_1 and T_2 , we consider a payoff involving their difference in absolute terms?

Definition 8.20. Let S_t be the price of the underlying asset at time t , and let $0 < t < T_1 < T_2$. An absolute forward-start call option³⁰, abbreviated FSACall, with strike fixing at T_1 and maturity T_2 , is a call option whose strike is defined as a multiple of the price T_1 , i.e., $K = kS_{T_1}$, and whose payoff is

$$\text{FSACall}(T_2, S_{T_2}; T_1, T_2, k) = \max \{S_{T_2} - kS_{T_1}, 0\}. \quad (8.96)$$

As with relative forward-start options, the value of an FSACall can be shown to depend on forward volatility $\bar{\sigma}_{T_1, T_2}$.

Proposition 8.21. *The price of an FSACall option is given by*

$$\text{FSACall}(t, s; T_1, T_2, k) = se^{-q(T_1-t)}G(\bar{\sigma}_{T_1, T_2}). \quad (8.97)$$

Proof. Under the risk-neutral measure $\mathbb{P}^*$, the price of an FSACall option for $t < T_1$ is³¹

$$\text{FSACall}(t, S_t) = \mathbb{E}^* \left[e^{-r(T_2-t)} (S_{T_2} - kS_{T_1})^+ \mid \mathcal{F}_t \right].$$

Given the lognormal structure of S_{T_2} , we can write $S_{T_2} = S_{T_1} \cdot \frac{S_{T_2}}{S_{T_1}}$, with $\frac{S_{T_2}}{S_{T_1}}$ independent of S_t for $t < T_1$. Thus:

$$\text{FSACall}(t, S_t) = \mathbb{E}^* \left[e^{-r(T_1-t)} \cdot e^{-r(T_2-T_1)} S_{T_1} \cdot \left(\frac{S_{T_2}}{S_{T_1}} - k \right)^+ \mid \mathcal{F}_t \right].$$

²⁹Since forward-smile and vol-of-vol risk cannot be replicated with vanilla instruments, their effect on price cannot be eliminated by hedging and must be modeled exogenously. Thus, calibrating a model only to the vanilla smile does not robustly determine forward-start product prices, and any dependence of $V_{\text{forward-smile}}$ on the spot smile is essentially a modeling assumption rather than a market implication.

³⁰This is commonly known as a forward volatility agreement (FVA) in FX markets.

³¹Again, for $T_1 < t < T_2$, the price is determined by the price of a standard call option.

That is, we reduce the problem to an option with payoff proportional to S_{T_1} :

$$\text{FSACall}(t, S_t) = e^{-r(T_1-t)} \cdot \mathbb{E}^* \left[S_{T_1} \cdot e^{-r(T_2-T_1)} \cdot \left(\frac{S_{T_2}}{S_{T_1}} - k \right)^+ \middle| \mathcal{F}_t \right].$$

Since S_{T_1} –which depends only on the path over $[0, T_1]$ – and S_{T_2}/S_{T_1} –which depends on the path over $[T_1, T_2]$ – are independent, and $\mathbb{E}^*[S_{T_1} | \mathcal{F}_t] = S_t e^{(r-q)(T_1-t)}$, we have:

$$\text{FSACall}(t, s) = s \cdot e^{-q(T_1-t)} \cdot G(\bar{\sigma}_{T_1, T_2}),$$

where G is the function of forward volatility obtained in the proof of Proposition 8.9. $\square$

Remark 8.22. Note that the FSA option's delta is now nonzero. However, gamma and higher-order derivatives with respect to spot remain identically zero, as for FS options.

As for forward-start options with relative payoffs, we need to determine how to hedge this product's risks and what its price should be. That is, (i) which vanilla options can be used to hedge its risks? (ii) Which additional P&L terms should be taken into account? We consider each question separately.

8.16.1 How to hedge the risks of a forward-start option with an absolute payoff

Note that the FSA option's vega is

$$\mathcal{V}(s, K) = s e^{-q(T_1-t)} \frac{\partial G(\bar{\sigma}_{T_1, T_2})}{\partial \bar{\sigma}_{T_1, T_2}},$$

so $\mathcal{V}$ is linear in spot. To hedge vega, we therefore need a vanilla option (or a set of vanilla options) that provides a vega position linear in spot.

Following the reasoning in Section 8.15.1, we construct a portfolio of vanilla options that yields a linear position. Since the price now includes a prefactor s , the portfolio's vega is

$$\mathcal{V}(\Pi) = \int_0^{+\infty} du \frac{1}{u^3} s \rho\left(\frac{s}{u}\right) \varphi(u),$$

so we must choose an option density as a function of strike of $\rho(K) \propto K^{-1}$. From expression (8.84), the payoff of the vanilla option portfolio –consisting, for example, of calls at all strikes– is $\Lambda(S_T) = S_T \log S_T$. If we regard this portfolio as a single contract, it is called an $s \log s$ contract.

We can price an $s \log s$ contract –abbreviated SLC– in the usual way as the discounted expected value of its payoff. That is,

$$\text{SLC}(t, s) = e^{-r(T-t)} \mathbb{E}^* [S_T \log S_T | \mathcal{F}_t]. \quad (8.98)$$

Again, since we assume time-dependent volatility,

$$S_T = S_t e^{(r-q-\frac{1}{2}\bar{\sigma}_T^2)(T-t) + \bar{\sigma}_T dW_s},$$

where $\bar{\sigma}_T^2$ is defined in equation (8.53) as³²

$$\bar{\sigma}_T = \frac{1}{T-t} \int_t^T \sigma_0(u)^2 du.$$

Thus, with the change of variable $X = \log S_T$, we have

$$\mathbb{E}^* [S_T \log S_T | \mathcal{F}_t] = \mathbb{E}^* [e^X X | \mathcal{F}_t],$$

where $X \sim N(\mu, \sigma^2)$, with $\mu = \log S_t + (r - q - \frac{1}{2}\bar{\sigma}_T^2)(T - t)$ and $\sigma^2 = \bar{\sigma}_T^2(T - t)$. Thus,

$$\mathbb{E}^* [S_T \log S_T | \mathcal{F}_t] = \int_{-\infty}^{+\infty} dx x e^x \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

Combining the exponents and completing the square gives

$$e^{\mu + \frac{1}{2}\sigma^2} \int_{-\infty}^{+\infty} dx x \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - (\mu + \sigma^2))^2}{2\sigma^2}\right),$$

and the change of variable $y = x - (\mu + \sigma^2)$ gives

$$e^{\mu + \frac{1}{2}\sigma^2} \int_{-\infty}^{+\infty} dy (y + \mu + \sigma^2) \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{y^2}{2\sigma^2}\right),$$

where $Y = X - (\mu + \sigma^2)$ follows a normal distribution with zero mean and variance σ^2 . Separating the preceding integral into a sum of terms shows that

$$\mathbb{E}^* [S_T \log S_T | \mathcal{F}_t] = e^{\mu + \frac{1}{2}\sigma^2} (\mu + \sigma^2),$$

and substituting the preceding values and discounting to the present finally gives the price of an $s \log s$ contract as

$$\boxed{\text{SLC}(t, s; T, \bar{\sigma}_T) = s e^{-q(T-t)} \left(\log s + \left(r - q + \frac{1}{2}\bar{\sigma}_T^2 \right) (T - t) \right)}. \quad (8.99)$$

Having obtained the contract's price, we calculate its sensitivity –vega– to $\bar{\sigma}_T$. In particular,

$$\frac{\partial \text{SLC}}{\partial \bar{\sigma}_T} = s e^{-q(T-t)} (T - t) \bar{\sigma}_T. \quad (8.100)$$

Thus, using the dependence of the price of a forward-start option with an absolute payoff on forward volatilities $\bar{\sigma}_{T_1}$ and $\bar{\sigma}_{T_2}$ –rather than on forward volatility $\bar{\sigma}_{T_1, T_2}$ –, we obtain

$$\begin{aligned} \frac{\partial \text{FSACall}}{\partial \bar{\sigma}_{T_1}} &= -s e^{-q(T_1-t)} (T_1 - t) \bar{\sigma}_{T_1} N, \\ \frac{\partial \text{FSACall}}{\partial \bar{\sigma}_{T_2}} &= s e^{-q(T_1-t)} (T_2 - t) \bar{\sigma}_{T_2} N, \end{aligned} \quad (8.101)$$

³²Note that in expression (8.53) we set $t = 0$.

with

$$N = \frac{1}{(T_2 - T_1)\bar{\sigma}_{T_1, T_2}} \frac{\partial G}{\partial \bar{\sigma}_{T_1, T_2}}.$$

Thus, the portfolio

$$\begin{aligned} \Pi &= -\text{FSACall}(t, s; T_1, T_2, \bar{\sigma}_{T_1}, \bar{\sigma}_{T_2}) \\ &\quad + N \left[e^{q(T_2 - T_1)} \text{SLC}(t; T_2, \bar{\sigma}_{T_2}) - \text{SLC}(t; T_1, \bar{\sigma}_{T_1}) \right], \end{aligned} \quad (8.102)$$

has no sensitivity to either $\bar{\sigma}_{T_1}$ or $\bar{\sigma}_{T_2}$.

Remark 8.23. Let us examine the choice of hedge ratio N . The portfolio's sensitivity to $\bar{\sigma}_{T_1}$ is given by:

$$\frac{\partial \Pi}{\partial \bar{\sigma}_{T_1}} = -\frac{\partial \text{FSACall}}{\partial \bar{\sigma}_{T_1}} + \frac{\partial N}{\partial \bar{\sigma}_{T_1}} \left(e^{q(T_2 - T_1)} \text{SLC}(t; T_2, \bar{\sigma}_{T_2}) - \text{SLC}(t; T_1, \bar{\sigma}_{T_1}) \right) - N \frac{\partial \text{SLC}(t; T_1, \bar{\sigma}_{T_1})}{\partial \bar{\sigma}_{T_1}},$$

which expands to

$$\begin{aligned} \frac{\partial \Pi}{\partial \bar{\sigma}_{T_1}} &= -s e^{-q(T_1 - t)} \frac{\partial G}{\partial \bar{\sigma}_{T_1, T_2}} \frac{\partial \bar{\sigma}_{T_1, T_2}}{\partial \bar{\sigma}_{T_1}} \\ &\quad + \frac{\partial N}{\partial \bar{\sigma}_{T_1}} \left(e^{q(T_2 - T_1)} \text{SLC}(t; T_2, \bar{\sigma}_{T_2}) - \text{SLC}(t; T_1, \bar{\sigma}_{T_1}) \right) \\ &\quad - s N e^{-q(T_1 - t)} (T_1 - t) \bar{\sigma}_{T_1}. \end{aligned}$$

An analogous expression is obtained for $\partial \Pi / \partial \bar{\sigma}_{T_2}$. Note that all three terms are linear in s (since the prices of $s \log s$ contracts are proportional to spot), so if we eliminate vega for given values of $\bar{\sigma}_{T_1}$ and $\bar{\sigma}_{T_2}$, vega remains zero even if spot changes.

If the hedge ratio proposed in [29] is chosen, the portfolio is constructed so that the preceding derivatives vanish exactly at the point where $\bar{\sigma}_{T_1}$ and $\bar{\sigma}_{T_2}$ are evaluated, i.e.,

$$\left. \frac{\partial \Pi}{\partial \bar{\sigma}_{T_1}} \right|_{\text{Mkt}} = 0, \quad \left. \frac{\partial \Pi}{\partial \bar{\sigma}_{T_2}} \right|_{\text{Mkt}} = 0.$$

Although N is functionally dependent on average volatility $\bar{\sigma}_{T_1, T_2}$, in practice it is fixed as a constant when the portfolio is constructed using observed market volatilities. Thus, the sensitivity of Π to these volatilities vanishes only at that point. This implies that second-order derivatives (volga and cross-volga) generally remain nonzero and contribute to second-order P&L. However, since eliminating vega leaves it at zero even when spot changes, there is no spot/vol correlation.

Moreover, the hedge ratios³³ are independent of time and spot. The hedge is therefore static –it remains stable as time advances and spot changes. In particular, it depends only on $\bar{\sigma}_{T_1, T_2}$. Thus, rebalancing is required only when this forward volatility $\bar{\sigma}_{T_1, T_2}$ changes.

Remark 8.24. Although the FSACall's gamma is zero, the portfolio's gamma must be checked. Differentiating twice with respect to spot confirms that the portfolio's gamma is zero. Similarly, differentiating with respect to time shows that its theta vanishes.

Thus, portfolio (8.102) is theta, delta, gamma, and vega neutral, with vega neutrality understood to be invariant under changes in $\bar{\sigma}_{T_1}$ or $\bar{\sigma}_{T_2}$.

³³The hedge ratios in this strategy are $N e^{q(T_2 - T_1)}$ against $\text{SLC}(t; T_2, \bar{\sigma}_{T_2})$ and $-N$ against $\text{SLC}(t; T_1, \bar{\sigma}_{T_1})$.

Remark 8.25. Note that the delta of portfolio (8.102) is nonzero. Differentiating Π with respect to s gives

$$\frac{\partial \Pi}{\partial s} = -e^{-q(T_1-t)} \left[G(\bar{\sigma}_{T_1, T_2}) + N \left(r - q + \frac{1}{2} \bar{\sigma}_{T_1, T_2}^2 \right) (T_2 - T_1) \right],$$

which is independent of s —as expected, since the portfolio’s gamma is zero.

Thus, the new portfolio

$$\begin{aligned} \Pi_{\Delta\text{-hedged}} &= -\text{FSACall}(t, s; T_1, T_2, \bar{\sigma}_{T_1}, \bar{\sigma}_{T_2}) \\ &\quad + N \left[e^{q(T_2-T_1)} \text{SLC}(t; T_2, \bar{\sigma}_{T_2}) - \text{SLC}(t; T_1, \bar{\sigma}_{T_1}) \right] \\ &\quad + \Delta(t)s, \end{aligned} \tag{8.103}$$

with

$$\Delta(t) = -e^{-q(T_1-t)} \left[G(\bar{\sigma}_{T_1, T_2}) + N \left(r - q + \frac{1}{2} \bar{\sigma}_{T_1, T_2}^2 \right) (T_2 - T_1) \right],$$

is theta, delta, gamma, and vega neutral. Note that $\Delta(t) < 0$ for all $t \in \mathbb{R}_+$. Delta is therefore hedged by taking a short spot position, which must be rehedge as time advances.

Again, since the payoffs of $s \log s$ contracts are convex, their implied volatilities are well defined. The implied volatility of $s \log s$ contracts is $\bar{\sigma}_T$, obtained directly by inverting their pricing formula after equating it to the market price. That is,

$$\bar{\sigma}_T = \frac{2}{(T-t)s} \left(e^{q(T-t)} \text{SLC}_{\text{Mkt}} + s (\log s + (r-q)(T-t)) \right), \tag{8.104}$$

is the implied volatility of the $s \log s$ contract associated with price SLC_{Mkt} .

As with log contracts, the implied volatility of $s \log s$ contracts can be expressed as an average of vanilla option implied volatilities in the absence of dividends.

8.16.2 Additional P&L terms

Consider portfolio $\Pi_{\Delta\text{-hedged}}$ and expand P&L to first order in t and second order in s , $\bar{\sigma}_{T_1}$, and $\bar{\sigma}_{T_2}$. As we saw earlier, the portfolio is theta, delta, gamma, and vega neutral, so there are no contributions from ds , ds^2 , dt , $d\bar{\sigma}_{T_1}$, and $d\bar{\sigma}_{T_2}$. But what happens to the cross terms $\frac{\partial \Pi_{\Delta\text{-hedged}}}{\partial s \partial \bar{\sigma}_{T_i}}$ for $i = 1, 2$? And to the second-order terms $d\bar{\sigma}_{T_i}^2$?

As we saw in the preceding section, the cross terms involving the correlation between spot and implied volatility are zero, $\frac{\partial \Pi_{\Delta\text{-hedged}}}{\partial s \partial \bar{\sigma}_{T_i}} = 0$ for $i = 1, 2$, since the portfolio’s vega is linear in s —so once vega is hedged, the hedge remains effective even if spot changes.

Furthermore, expressions (8.100) and (8.101) give

$$\begin{aligned} \frac{\partial^2 \text{SLC}}{\partial (\bar{\sigma}_T)^2} &= s e^{-q(T-t)} (T-t), \\ \frac{\partial^2 \text{FSA}}{\partial (\bar{\sigma}_{T_1})^2} &= -s e^{-q(T_1-t)} (T_1-t) \left(N + \bar{\sigma}_{T_1} \frac{\partial N}{\partial \bar{\sigma}_{T_1}} \right), \\ \frac{\partial^2 \text{FSA}}{\partial (\bar{\sigma}_{T_2})^2} &= s e^{-q(T_1-t)} (T_2-t) \left(N + \bar{\sigma}_{T_2} \frac{\partial N}{\partial \bar{\sigma}_{T_2}} \right). \end{aligned}$$

so expanding the derivatives and applying the chain rule to calculate the second derivative of $\Pi_{\Delta\text{-hedged}}$ with respect to forward volatility $\bar{\sigma}_{T_1, T_2}$ gives

$$dP\&L = -\frac{s}{2}e^{-q(T_1-t)}\frac{\partial^2 G}{\partial(\bar{\sigma}_{T_1, T_2})^2}d(\bar{\sigma}_{T_1, T_2}^2)^2. \quad (8.105)$$

The construction of portfolio Π simultaneously hedges first-order sensitivities to implied volatility, delta, and time, providing an effective vega, gamma, and theta hedge for the forward-start option. This is possible because we use European instruments with payoffs of the form $s \log s$, whose implied volatilities serve as benchmarks. Although a superficial analysis might suggest that the correlation risk between spot, s , and forward volatility is relevant, the model shows that this risk is offset by the appropriate choice of hedging instruments, eliminating the need to account for spot/vol correlation.

In this setting, as for forward-start options with relative payoffs, vol-of-vol risk and forward-smile risk must be priced. Specifically, using the preceding notation, $V_{\text{vol-of-vol}}$ represents the estimated P&L of the hedged portfolio between times $[0, T_1]$, as given in formula (8.105). Meanwhile, $V_{\text{forward-smile}}$ captures the difference between the market price at T_1 of a vanilla option with strike kS_{T_1} and maturity T_2 and its theoretical model price, calculated using the implied volatility of an $s \log s$ contract maturing at T_2 . These two corrections reconstruct the price of a forward-start option with an absolute payoff from European instruments observable in the market.

8.17 Forward-start options in the local volatility model

We now examine how to price vol-of-vol risk and forward-smile risk using the local volatility model for forward-start options with relative payoffs. As we saw in expression (8.90), we can obtain a log contract's implied volatility if it is quoted in the market. But what if log contracts are not quoted instruments? How can we then obtain their implied volatility?

We assume that European call and put options on the underlying S_t are traded. We can then calibrate the implied volatility surface and use Dupire's formula—which, recall, applies only to this class of options—to obtain the local volatility that models the underlying. That is, standard call/put options yield $\sigma_{\text{loc}}(t, s)$ such that, under the risk-neutral measure,

$$dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(t, S_t)S_t dW_t.$$

Starting from expression (8.86), we know that the expected payoff of a log contract with maturity T can be rewritten, assuming a constant volatility that reproduces the market price—denoted $\sigma_{\text{imp, LC}}$ —as

$$\mathbb{E}^*[\log S_T | S_t = s] = \log(se^{(r-q)(T-t)}) - \frac{(T-t)}{2}\sigma_{\text{imp, LC}}^2,$$

where the first term on the right-hand side is exactly $\log \text{For}_S(t, T)$. Setting $t = 0$, the time at which we calibrated the implied volatility surface—and hence the local volatility function—, gives $\text{For}_S(t, T) = F_T$, so

$$\mathbb{E}^*\left[\log\left(\frac{S_T}{F_T}\right) \middle| S_0 = s\right] = -\frac{T}{2}\sigma_{\text{imp, LC}}^2.$$

Furthermore, evaluating the expectation assuming that the underlying follows the local volatility model gives

$$\mathbb{E}^* \left[\log \left(\frac{S_T}{F_T} \right) \middle| S_0 = s \right] = -\frac{1}{2} \int_0^T \mathbb{E}^* \left[\sigma_{\text{loc}}^2(t, S_t) \middle| S_0 = s \right] dt,$$

and equating the two expressions gives

$$\sigma_{\text{imp,LC}}^2 = \frac{1}{T} \int_0^T \mathbb{E}^* \left[\sigma_{\text{loc}}^2(t, S_t) \middle| S_0 = s \right]. \quad (8.106)$$

Remark 8.26. In the special case $\sigma_{\text{loc}}(t, s) = \sigma_0(t)$, we recover the earlier expression for implied volatility, $\sigma_{\text{imp,LC}} = \bar{\sigma}_T$, where $\bar{\sigma}_T$ is given by (8.53).

Note that expression (8.36) can be obtained for any convex European payoff. Thus, for log contracts, we can rewrite it as

$$\sigma_{\text{imp,LC}}^2(T) = \frac{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D \sigma_{\text{loc}}^2(t, S_t) dt \middle| S_0 = s \right]}{\mathbb{E}^* \left[\int_0^T e^{-rt} \Gamma^D dt \middle| S_0 = s \right]}. \quad (8.107)$$

As in the analysis of volatility surface dynamics, we expand around a time-dependent volatility. That is, we start from the base model $u_0(t) = \sigma_{\text{loc}}(t, s)^2$. Introducing the modification $u_0(t) \mapsto u(t, s) = u_0(t) + \delta u(t, s)$ produces an implied volatility correction given by (8.40). The main difference is that, for log contracts, dollar gamma is independent of spot, simplifying the calculation of the expectation in the numerator.

Thus, the numerator is calculated as

$$\int_0^T dt e^{-rt} \Gamma^D(t) \int_0^{+\infty} ds \rho(t, s; \sigma_0) \delta u(t, s),$$

and, again, since $\sigma_0(t) = \sqrt{u_0(t)}$ is deterministic, the underlying follows a lognormal process with density

$$\rho(t, s; \sigma_0) = \frac{1}{s\sqrt{2\pi\omega_t}} \exp \left(-\frac{(\log(s/F_t) + \frac{1}{2}\omega_t)^2}{2\omega_t} \right),$$

where ω_t is defined in equation (8.41).

The log contract's dollar gamma is

$$\Gamma^D(t, s) = s^2 \frac{\partial^2 \text{LC}(t, s)}{\partial s^2} = 2e^{-r(T-t)}$$

so the change of variable $x = \log(s/F_t)$ —with $dx = s^{-1}ds$ — gives

$$2e^{-rT} \int_0^T dt \int_{-\infty}^{+\infty} dx \frac{1}{\sqrt{2\pi\omega_t}} \exp \left(-\frac{(x + \frac{1}{2}\omega_t)^2}{2\omega_t} \right) \delta u(t, F_t e^x).$$

Using the change of variable $y = \frac{1}{\sqrt{\omega_t}} \left(x + \frac{1}{2}\omega_t \right)$, the numerator can be rewritten as

$$2e^{-rT} \int_0^T dt \int_{-\infty}^{+\infty} dy \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \delta u(t, F_t e^{-\frac{1}{2}\omega_t + \sqrt{\omega_t}y}). \quad (8.108)$$

The denominator is $2e^{-rT}T$, so

$$\delta\sigma_{\text{imp,L C}}^2 = \frac{1}{T} \int_0^T dt \int_{-\infty}^{+\infty} dy \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \delta u \left(t, F_t e^{-\frac{1}{2}\omega_t + \sqrt{\omega_t}y} \right). \quad (8.109)$$

Expanding around a constant volatility, $\sigma_0(t) = \sigma_0$, recovers the expression given in [29]

$$\delta\sigma_{\text{imp,L C}}^2 = \frac{1}{T} \int_0^T dt \int_{-\infty}^{+\infty} dy \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \delta u \left(t, F_t e^{-\frac{\sigma_0^2 t}{2} + \sigma_0 \sqrt{t}y} \right). \quad (8.110)$$

Thus, following the same reasoning as for the implied variance of European call/put options gives³⁴

$$\sigma_{\text{imp,L C}}(T) = \frac{1}{T} \int_0^T \sigma_0(t) dt + \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \delta\sigma_{\text{loc}} \left(t, F_t e^{-\frac{1}{2}\omega_t + \sqrt{\omega_t}y} \right). \quad (8.111)$$

Again, using the second-order expansion of $\sigma_{\text{loc}}(t, s)$ around ATMF given in expression (8.55), i.e.,

$$\sigma_{\text{loc}}(t, x) = \sigma_0(t) + \alpha(t)x + \frac{\beta(t)}{2}x^2, \quad (8.112)$$

where $x = \log(s/F_t)$ and $\sigma_0(t)$ is the ATMF local volatility –since $s = F_t$ implies $x = 0-$. In this case, $\delta\sigma_{\text{loc}} = \alpha(t)x + \frac{\beta(t)}{2}x^2$, giving

$$\sigma_{\text{imp,L C}}(T) = \frac{1}{T} \int_0^T \sigma_0(t) dt - \frac{1}{2T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \omega_t \left[\alpha(t) - \beta(t) \left(1 + \frac{\omega_t}{4} \right) \right]. \quad (8.113)$$

As we saw in expression (8.59), the term proportional to $\alpha(t)$ in (8.113) can be written in terms of the ATMF skew, $\mathcal{S}_T$. That is, since

$$\mathcal{S}_T = \frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\omega_t}{\omega_T} \alpha(t),$$

the first term in the integral becomes

$$\frac{1}{2T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \omega_t \alpha(t) = \frac{\omega_T}{2} \mathcal{S}_T,$$

so finally

$$\sigma_{\text{imp,L C}}(T) = \frac{1}{T} \int_0^T \sigma_0(t) dt - \frac{\omega_T}{2} \mathcal{S}_T + \frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \omega_t \left(1 + \frac{\omega_t}{4} \right) \frac{\beta(t)}{2}. \quad (8.114)$$

³⁴Do not confuse $\bar{\sigma}_T$ in expression (8.56) with the log contract's implied volatility.

Let us now analyze its dynamics: what happens to the log contract's implied volatility when spot changes? Differentiating expression (8.113) with respect to $\log s$ –to express the change in units of s , i.e., as a percentage change– gives

$$\begin{aligned}
 \frac{\partial \sigma_{\text{imp,L C}}(T; s_0)}{\partial \log s_0} &= \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\partial(\delta \sigma_{\text{loc}}(t, z))}{\partial z} \frac{\partial z(t, s_0)}{\partial \log s_0} \\
 &= \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\partial(\delta \sigma_{\text{loc}}(t, z))}{\partial z} \frac{\partial}{\partial \log s_0} \left(F_t e^{-\frac{1}{2}\omega_t + \sqrt{\omega_t} y} \right) \\
 &= \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\partial(\delta \sigma_{\text{loc}}(t, z))}{\partial z} z(t, y) \\
 &= \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \frac{\partial(\delta \sigma_{\text{loc}}(t, z))}{\partial \log z},
 \end{aligned}$$

and since

$$\delta \sigma_{\text{loc}}(t, z) = \alpha(t) \log(z/F_t) + \frac{\beta(t)}{2} \log(z/F_t)^2$$

with $z = z(t, y) = F_t \exp(-\frac{1}{2}\omega_t + \sqrt{\omega_t} y)$, we have

$$\frac{\partial(\delta \sigma_{\text{loc}}(t, z))}{\partial \log z} = \alpha(t) + \beta(t) \log(z/F_t),$$

and hence

$$\frac{\partial \sigma_{\text{imp,L C}}(T; s_0)}{\partial \log s_0} = \frac{1}{T} \int_0^T dt \int_{-\infty}^{\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \frac{\sigma_0(t)}{\bar{\sigma}_T} \left[\alpha(t) + \left(-\frac{1}{2}\omega_t + \sqrt{\omega_t} y \right) \beta(t) \right].$$

Integrating with respect to y finally gives

$$\boxed{\frac{\partial \sigma_{\text{imp,L C}}(T; s_0)}{\partial \log s_0} = \frac{1}{T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \left(\alpha(t) - \frac{1}{2}\omega_t \beta(t) \right)}. \quad (8.115)$$

If ATM volatility is constant, i.e., $\sigma_0(t) = \sigma_0$, the preceding expression reduces to

$$\frac{\partial \sigma_{\text{imp,L C}}(T; s_0)}{\partial \log s_0} = \frac{1}{T} \int_0^T \alpha(t) dt - \frac{\sigma_0^2}{2} \int_0^T \frac{t}{T} \beta(t) dt,$$

which agrees with the expression presented in [29].

Remark 8.27. Note that in the first-order approximation around ATM –setting $\beta(t) = 0$ – expression (8.113) reduces to

$$\sigma_{\text{imp,L C}}(T) = \frac{1}{T} \int_0^T \sigma_0(t) dt - \frac{\omega_T}{2} \mathcal{S}_T.$$

Moreover, taking constant ATM volatility $\sigma_0(t) = \sigma_0$, the implied volatility of vanilla ATM options reduces to

$$\sigma_{\text{imp}}(T, F_T) = \sigma_0$$

since $x_K = 0$ in this case; see equation (8.56). Thus,

$$\omega_T = T\sigma_0^2 = T\sigma_{\text{imp}}(T, F_T)^2,$$

giving the following first approximation to a log contract's implied volatility:

$$\sigma_{\text{imp,LC}}(T) = \sigma_{\text{imp}}(T, F_T) - \frac{T}{2}\sigma_{\text{imp}}(T, F_T)^2\mathcal{S}_T, \quad (8.116)$$

which provides an approximation to the log contract price in terms of vanilla ATMF option prices and their skew.

Moreover, from expression (8.64), the term proportional to $\alpha(t)$ coincides exactly with the change in vanilla ATMF implied volatility. Thus,

$$\frac{\partial\sigma_{\text{imp,LC}}(T; s_0)}{\partial\log s_0} = \frac{\partial\sigma_{\text{imp}}(T, F_T; s_0)}{\partial\log s_0} - \frac{1}{2T} \int_0^T dt \frac{\sigma_0(t)}{\bar{\sigma}_T} \omega_t \beta(t).$$

and, in particular, if $\beta(t) = 0$, then

$$\frac{\partial\sigma_{\text{imp,LC}}(T; s_0)}{\partial\log s_0} = \frac{\partial\sigma_{\text{imp}}(T, F_T; s_0)}{\partial\log s_0}. \quad (8.117)$$

8.17.1 Does the convexity term matter for log contract implied volatility dynamics?

Note that equations (8.116) and (8.117) are obtained for local volatility expanded to first order around ATMF, given by

$$\sigma_{\text{loc}}(t, s) = \sigma_0(t) + \alpha(t) \log\left(\frac{s}{F_t}\right),$$

and their interpretation is clear:

- (i) Since the slope of the equity smile at ATMF is usually negative, $\mathcal{S}_T < 0$, log contract implied volatility is higher than ATMF call/put implied volatility.
- (ii) The rate of change of log contract implied volatility coincides with the implied volatility dynamics of ATMF calls/puts.

However, as shown in [29], this approximation –which gives log contract implied volatility using only ATMF call/put implied volatilities– is not useful. This means that the difference $\sigma_{\text{imp,LC}}(T) - \sigma_{\text{imp}}(T, F_T)$ is not determined by ATMF skew, a local property of the smile around ATMF.

Thus, a first-order local volatility approximation is systematically less accurate at capturing log contract implied volatility than call/put implied volatility. Formula (8.116) is valid only when the smile's slope is not too steep [cf. [29, page 123]].

Moreover, plotting the ratio $\frac{\partial\sigma_{\text{imp,LC}}(T; s_0)}{\partial\log s_0} / \frac{\partial\sigma_{\text{imp}}(T, F_T; s_0)}{\partial\log s_0}$ using actual market data shows that it is not equal to 1, as approximation (8.117) would suggest. Nevertheless, it does give the correct order of magnitude.

8.17.2 Estimating vol-of-vol risk

Recall that the integrated variance of forward-start options can be expressed in terms of log contract implied volatilities as

$$\bar{\sigma}_{T_1, T_2}^2 = \frac{(T_2 - t)\bar{\sigma}_{T_2}^2 - (T_1 - t)\bar{\sigma}_{T_1}^2}{T_2 - T_1},$$

so $\bar{\sigma}_{T_1, T_2} = \bar{\sigma}_{T_1, T_2}(\bar{\sigma}_{T_1}, \bar{\sigma}_{T_2})$. Taking the differential of $\bar{\sigma}_{T_1, T_2}^2$ and applying the chain rule gives

$$d\bar{\sigma}_{T_1, T_2} = \frac{T_2 - t}{T_2 - T_1} \frac{\bar{\sigma}_{T_2}}{\bar{\sigma}_{T_1, T_2}} d\bar{\sigma}_{T_1} - \frac{T_1 - t}{T_2 - T_1} \frac{\bar{\sigma}_{T_1}}{\bar{\sigma}_{T_1, T_2}} d\bar{\sigma}_{T_2}$$

In particular, if the term structure of log contract implied volatility is flat, i.e., $\bar{\sigma}_{T_1} \approx \bar{\sigma}_{T_2}$, then $\bar{\sigma}_{T_1, T_2} \approx \bar{\sigma}_{T_1} \approx \bar{\sigma}_{T_2}$. Moreover, since

$$d\bar{\sigma}_T = \left(\frac{1}{T - t} \int_t^T \alpha(u) du \right) d \log S_t,$$

we have

$$d\bar{\sigma}_{T_1, T_2} \approx \left(\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \alpha(t) dt \right) d \log S_t,$$

where $\alpha(t) = \frac{\partial \sigma_{\text{loc}}(t, S_t)}{\partial \log S_t}$ is the local volatility skew. Note that if local volatility skew is zero, implied volatility remains constant at all times.

Let us now calculate the variance of the forward variance $d\bar{\sigma}_{T_1, T_2}$. Since

$$\text{Var} [d \log S_t | \mathcal{F}_t] = \sigma_{\text{loc}}(t, S_t)^2 dt,$$

we have

$$\text{Var} [d\bar{\sigma}_{T_1, T_2} | \mathcal{F}_t] \approx \left(\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \alpha(t) dt \right)^2 \sigma_{\text{loc}}(t, S_t)^2 dt,$$

so vol-of-vol is

$$\text{Vol} [d\bar{\sigma}_{T_1, T_2} | \mathcal{F}_t] \approx \left(\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \alpha(t) dt \right) \sigma_{\text{loc}}(t, S_t).$$

Thus, at $t \in [0, T_1]$, vol-of-vol is determined by the local volatility skew between T_1 and T_2 , which need not agree with actual market observations.

Furthermore, $d\bar{\sigma}_{T_1, T_2}$ is not a function of $T_1 - t$, so the model is not stationary. This means that price dynamics and the risks associated with forward-start options depend not only on the time remaining until T_1 , but also on the calendar date at which the model is applied, which is unrealistic.

To address this problem, stochastic volatility models –introduced in the next chapter– allow volatility smile dynamics to be modeled by fitting observed market data.

Conclusion 8.28. The vol-of-vol of forward-start options generated by the local volatility model depends exclusively on local volatility skew $\alpha(t)$ at the time the local volatility surface is calibrated. It can therefore yield values above or below historical or implied levels of the volatility of implied volatility. Thus, $V_{\text{vol-of-vol}}$ will be systematically incorrect if calculated using the local volatility model.

8.17.3 Estimating forward-smile risk

Using approximation (8.116) for log contract implied volatility, at T_1 we have

$$\sigma_{\text{imp,LC}}(T_2; T_1, S_{T_1}) - \sigma_{\text{imp}}(T_2, \text{For}_S(T_1, T_2); T_1, S_{T_1}) = -\frac{T}{2} \sigma_{\text{imp}}(T_2, \text{For}_S(T_1, T_2); T_1, S_{T_1})^2 \mathcal{S}_{T_2-T_1}(T_1; S_{T_1})$$

where, following the notation introduced in Section 8.12, $\mathcal{S}_{T_2-T_1}(T_1; S_{T_1})$ denotes the forward-starting skew at T_1 for maturity T_2 –equivalently, for remaining maturity $T_2 - T_1$.

Writing $\theta = T_2 - T_1$, we have

$$\mathcal{S}_\theta(T_1; S_{T_1}) = \frac{1}{\theta} \int_{T_1}^{T_1+\theta} \frac{t - T_1}{\theta} \alpha(t) dt,$$

and using the change of variable

$$t = T_1 + u\theta$$

with $u \in [0, 1]$, the preceding integral can be rewritten as

$$\mathcal{S}_\theta(T_1; S_{T_1}) = \frac{1}{\theta} \int_0^1 \alpha(T_1 + \theta u) u du,$$

which we can compare with the spot-starting ATMF skew for the same time to maturity θ ,

$$\mathcal{S}_\theta(0; S_0) = \int_0^1 \alpha(u\theta) u du.$$

For typical equity smiles, where $\alpha(t)$ is taken to decay as a power law with exponent γ , both skews can be calculated explicitly as

$$\mathcal{S}_\theta(T_1; S_{T_1}) \propto T_1^{-\gamma} \quad \mathcal{S}_\theta(0; S_0) \propto \theta^{-\gamma}.$$

Thus, for short remaining maturities, $\theta \ll T_1$ (i.e., for forward-start options whose life is much shorter than the time until their future start date), the ATMF skew generated by the local volatility model at T_1 decays as $1/T_1^\gamma$, which is much smaller than the ATMF skew observed at $t = 0$ for the same remaining maturity.

Conclusion 8.29. When buying or, especially, selling a forward-start call option, at time T_1 we are exposed to risk arising from the difference between the implied volatilities of the log contract and the ATMF call maturing at T_2 . If T_1 is far in the future (relative to the remaining maturity), the local volatility model generates a smile at T_1 for maturity T_2 that is much flatter than the smile likely to be observed in the market for these options. Thus, $V_{\text{forward-risk}}$ will also be incorrectly determined if calculated using a local volatility model.

In the local volatility model, future vol-of-vol is completely determined by ATMF skew and its evolution over time, which are themselves deterministic functions fixed by the initial calibration. As a result, the model generates vol-of-vol values systematically below those observed in the market, where the volatility smile evolves stochastically.

8.17.4 How to hedge forward-start vega under the local volatility model

8.18 Toward a stochastic volatility model

Once the implied volatility surface has been obtained by inverting the Black-Scholes equation against market prices, we might expect to know how vanilla option prices will evolve. However, in the market, vanilla option prices themselves are observed to be stochastic processes –potentially correlated with the underlying price– whose evolution depends on external factors. An initial approach might therefore be to model the underlying price, S_t , and vanilla option price, $C_t(T, K)$, jointly as a two-dimensional Itô process such that, under the risk-neutral measure $\mathbb{P}^*$,

$$dS_t = (r - q)S_t dt + \sigma_t S_t dW_t^1 \quad (8.118)$$

$$dC_t = rC_t dt + \Lambda_t dW_t^2, \quad (8.119)$$

where W_t^1 and W_t^2 are Wiener processes with correlation ρ , Λ is the volatility of C_t , and σ is the underlying's volatility.

The Itô processes (8.118) and (8.119) have initial conditions $S_0 = s$ –where s is the spot price at $t = 0$ – and $C_0 = C_{\text{Mkt}}$ –where C_{Mkt} is the market option price at $t = 0$. Equation (8.119) also has a boundary condition, since the option price at T must equal the payoff. Thus, $C_T(T, K) = (S_T - K)^+$.

As we have seen throughout this chapter, there is a bijection between option prices and implied volatility through $C_t(T, K) = V(t, s, \sigma_{\text{imp}}; T, K, r, q)$. We can therefore model implied volatilities $\sigma_{\text{imp}}(t; T, K)$. Thus,

$$dS_t = (r - q)S_t dt + \sigma_t S_t dW_t^1 \quad (8.120)$$

$$d\sigma_{\text{imp}}(t) = \mu_t dt + \lambda_t dW_t^2. \quad (8.121)$$

The function $\mu_t(T, K)$ is determined by applying the two-dimensional Itô formula to the relation $C_t(T, K) = V(t, s, \sigma_{\text{imp}}; T, K, r, q)$

$$dC_t = \frac{\partial V}{\partial t} dt + \frac{1}{2} \frac{\partial^2 V}{\partial s^2} d[S]_t + \frac{1}{2} \frac{\partial^2 V}{\partial \sigma_{\text{imp}}^2} d[\sigma_{\text{imp}}]_t + \frac{\partial^2 V}{\partial s \partial \sigma_{\text{imp}}} d[S, \sigma_{\text{imp}}]_t + \frac{\partial V}{\partial s} dS_t + \frac{\partial V}{\partial \sigma_{\text{imp}}} d\sigma_{\text{imp}}(t),$$

where $d[S]_t = \sigma_t^2 S_t^2 dt$ and $d[\sigma_{\text{imp}}]_t = \lambda_t^2 dt$ are the quadratic variations of processes S_t and $\sigma_{\text{imp}}(t)$, respectively, and $d[S, \sigma_{\text{imp}}]_t = \sigma_t \lambda_t S_t \sigma_{\text{imp}}(t) \rho dt$ determines their covariation, with ρ denoting the correlation between Wiener processes W^1 and W^2 .

Using equations (8.120) and (8.121) to obtain the drift and equating it to $(r - q)C_t dt$ gives $\mu_t(T, K)$. Calculating the derivatives of $V(t, s, \sigma_{\text{imp}}; T, K, r, q)$ analytically and simplifying yields

$$\mu_t = \frac{1}{\sigma_{\text{imp}}} \left(\frac{\sigma_{\text{imp}}^2 - \sigma_t^2}{2(T - t)} - \frac{1}{2} d_+ d_- \lambda_t^2 + \frac{d_2}{\sqrt{T - t}} \sigma_t \lambda_t \rho \right). \quad (8.122)$$

Thus, if instantaneous volatility σ_t were known, the joint evolution of price S_t and implied volatility $\sigma_{\text{imp}}(t; T, K)$ would be fixed –given λ_t . However, σ_t and $\sigma_{\text{imp}}(t; T, K)$ must

satisfy the no-arbitrage conditions. For example, at very short maturities, ATMF option implied volatility must tend to instantaneous volatility, i.e., $\lim_{T \rightarrow t} \sigma_{\text{imp}}(t) = \sigma_t$. In practice, specifying σ_t in $[0, T]$ already determines the prices (and hence implied volatilities) of all options with maturity $\tau < T$; in other words, the initial smile must be incorporated into process σ_t itself. Attempting to model the implied volatility surface directly as a separate process is therefore usually unrealistic.

8.18.1 Modeling the dynamics of the local volatility function

Recall that, given a volatility surface satisfying the no-arbitrage conditions in Section 8.4.1, Dupire's formula (8.31) provides a bijection between that surface and a local volatility function $\sigma_{\text{loc}}(t, s)$. An advantage of using the local volatility function is that the no-arbitrage conditions translate into positivity of this function.

Remark 8.30. Note that we are not using the local volatility model; we are simply using the local volatility function as an alternative representation of implied volatilities at (8.121). The underlying's relative volatility is a process σ_t adapted to the filtration $\{\mathcal{F}_t\}_{t \in I}$.

This one-to-one relation between implied volatility and the local volatility function allows us to specify implied volatility dynamics –and hence option price dynamics– through dynamics for the local volatility function, denoted $\sigma_{\text{loc}}(t; \tau, s)$. The notation $\sigma_{\text{loc}}(t; \tau, s)$ refers to the local volatility function associated with the implied volatility surface at t . Thus, for fixed τ and s , $\sigma_{\text{loc}}(t; \tau, s)$ is a process defined for $t \leq \tau$.

Note that the local volatility model is recovered by setting $\sigma_t = \sigma_{\text{loc}}(t_0; t, s)$, indicating that $\sigma_{\text{loc}}(t_0; t, s)$ is the local volatility function obtained from an implied volatility surface calibrated at t_0 . In a model with a dynamic local volatility function, however, the underlying's volatility is determined by $\sigma_t = \sigma_{\text{loc}}(t; t, s)$.

As shown in Section 8.4.1, the underlying's instantaneous volatility can be related to the local volatility function through

$$\sigma_{\text{loc}}^2(t; \tau, s) = \mathbb{E}^* [\sigma_t^2 | S_\tau = s] = 2 \frac{\frac{\partial V}{\partial T} + qV + (r - q)K \frac{\partial V}{\partial K}}{K^2 \frac{\partial^2 V}{\partial K^2}} \bigg|_{\substack{T=\tau \\ K=s}},$$

which relates local volatility to the underlying's density function. This density is a function of t and s and a functional of the local volatility function (specifically, of its square). For simplicity, we denote it by $p(t, s, \sigma_{\text{loc}}^2)$. Calculating the differential of p allows us to recalculate μ_t in equation (8.121), obtaining –for the case $r = q = 0$ [cf. [29, chapter 2.4]]– the drift

$$\begin{aligned} \mu_t(\tau, s) = & \frac{\lambda_t(\tau, s)}{p(\tau, s, \sigma_{\text{loc}}^2)} \left(s \sigma_{\text{loc}}(t, s) \frac{\partial p(\tau, s, \sigma_{\text{loc}}^2)}{\partial s} \frac{d[W^1, W^2]_t}{dt} \right. \\ & \left. + \frac{1}{2} \int_t^\tau du \int_0^{+\infty} dx p(u, x, \sigma_{\text{loc}}^2) \lambda_t(\tau, s) x^2 \frac{\partial^2 p(u, x, \sigma_{\text{loc}}^2)}{\partial x^2} \frac{d[W^2, W^2]_t}{dt} \right) \end{aligned} \quad (8.123)$$

Analyzing this expression, Bergomi concludes that this approach is also infeasible for modeling implied volatility dynamics. This is because implied volatilities are indirectly

linked to option prices, making their drift nonlocal and dependent on the entire local volatility structure, so specifying direct, nontrivial dynamics becomes infeasible.

Recall, however, that implied volatility can be defined for any European payoff that is concave or convex. Is there a particular payoff whose implied volatility is easier to model?

8.18.2 Implied volatilities of power-payoff options

Consider a European payoff of the form $\Lambda(S_T) = S_T^p$, where p is an arbitrary parameter. These are called power payoffs, and we denote the prices of these options by³⁵ $\text{Pow}_p(t, s)$. Thus,

$$\text{Pow}_p(t, s) = e^{-r(T-t)} \text{For}_S(t, T)^p e^{\frac{p(p-1)}{2}(T-t)\bar{\sigma}_T^2}, \quad (8.124)$$

where $\bar{\sigma}_T^2$ is the implied volatility obtained by equating the expression to market prices. This implied volatility is well defined for $p \in \mathbb{R} \setminus \{0, 1\}$, since the payoffs are concave if $p \in (0, 1)$ and convex otherwise. We denote this volatility –observed at time t for maturity T – by $\sigma_{\text{imp}, \text{Pow}_p}(t, T)$.

The implied volatilities of European power-payoff options with exponent p , denoted $\sigma_{\text{imp}, \text{Pow}_p}(T)$, can be obtained by inverting equation (8.124) against market prices $\text{Pow}_p^{\text{Mkt}}$. For $p \in (0, 1)$, power-payoff option prices are bounded by an affine function of s . However, for $p < 0$ or $p > 1$, market prices may not be finite.

Since

$$\text{Pow}_p(t, s) = \mathbb{E}^* [e^{-r(T-t)} S_T^p | \mathcal{F}_t],$$

dividing by $\text{For}_S(t, T)^p = S_t e^{r(T-t)}$, which is nonrandom given the information available up to t , gives

$$e^{r(T-t)} \frac{\text{Pow}_p(t, s)}{\text{For}_S(t, T)^p} = \mathbb{E}^* \left[\left(\frac{S_T}{\text{For}_S(t, T)} \right)^p | \mathcal{F}_t \right] = \mathbb{E}^* [\exp(px) | \mathcal{F}_t],$$

where $x = x(T) = \log \frac{S_T}{\text{For}_S(t, T)}$. Since $p(t, s)$ denotes the density function of S_T , we can calculate the conditional expectation as

$$\mathbb{E}^* [\exp(px) | \mathcal{F}_t] = \int_0^{+\infty} du p(t, u) e^{p \log \frac{u}{\text{For}_S(t, T)}},$$

with $\text{For}_S(t, T) = s e^{r(T-t)}$ –since we are essentially conditioning on $S_t = s$ – and, using equation (8.18), rewrite the integral as

$$\mathbb{E}^* [\exp(px) | \mathcal{F}_t] = \int_0^{+\infty} dK e^{r(T-t)} \frac{\partial^2 V(T, K)}{\partial K^2} e^{p \log \frac{K}{\text{For}_S(t, T)}},$$

where $V(T, K)$ is the European vanilla option price with strike K and maturity T given in (7.28). Taking the change of variable

$$z(K) = \frac{\log \left(\frac{\text{For}_S(t, T)}{K} \right)}{\sigma_{\text{imp}}(T, K; t) \sqrt{T}} - \frac{\sigma_{\text{imp}}(T, K; t) \sqrt{T}}{2},$$

³⁵Since the payoff depends only on T and S_T , these option prices depend only on t and $S_t = s$; see [4, chapter 6.3].

Details of the change of variable $z(K)$ and the result...

$$\mathbb{E}^* [\exp(px) | \mathcal{F}_t] = \int_{-\infty}^{+\infty} dz \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} e^{-p\left(\frac{\omega^2(z)}{2} + z\omega(z)\right)} \left(1 + p \frac{d\omega(z)}{dz}\right) dz,$$

where $\omega(z) = \sigma_{\text{imp}}(T, K(z))\sqrt{T}$. Taking a further change of variable $y = z + p\omega(z)$ such that

$$y(K) = \frac{\log\left(\frac{\text{For}_S(t, T)}{K}\right)}{\sigma_{\text{imp}}(T, K; t)\sqrt{T}} + \left(p - \frac{1}{2}\right) \sigma_{\text{imp}}(T, K; t)\sqrt{T}.$$

Thus, the implied volatility of European power-payoff options with exponent p is

$$e^{-\frac{u^2}{2}} e^{\frac{p(p-1)}{2} \sigma_{\text{imp}, \text{Pow}_p}^2(T)} = \int_{-\infty}^{+\infty} dy \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} e^{\frac{p(p-1)}{2} \sigma_{\text{imp}}^2(T, K(y, p); t)} dy, \quad (8.125)$$

for $p \in (0, 1)$, where $\sigma_{\text{imp}}(T, K(y, p))$ is the implied volatility of vanilla call options at “moneyness” y . That is, at a strike K such that y and K are related through expression $y(K)$.

The term $\sigma_{\text{imp}}(T, K(y, p))$ can be obtained numerically in a straightforward way: choose a set of strikes K_i . For each K_i , calculate the corresponding value y_i using the expression for $y(K)$, yielding tuples $(y_i, \sigma_{\text{imp}}(T, K_i))$. Interpolating these tuples, for example with a spline, produces the function $\sigma_{\text{imp}}(T, K(y, p))$.

Note that the limits $p \rightarrow 0$ and $p \rightarrow 1$ lead to two types of options introduced earlier in this chapter: log contracts and $s \log s$ contracts.

- (i) If $p \rightarrow 0$, since $s^p \approx 1 + p \log s$ in a neighborhood of $p = 0$, we have $\lim_{p \rightarrow 0} \sigma_{\text{imp}, \text{Pow}_p}(T) = \sigma_{\text{imp}, \text{LC}}(T)$. As we will see in this chapter, these options are closely related to variance swaps. Since $p \approx 0$, we can expand the exponential on both sides of equation (8.125) to obtain

$$1 - \frac{p}{2} \sigma_{\text{imp}, \text{LC}}^2(T) = \int_{-\infty}^{+\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \left(1 - \frac{p}{2} \sigma_{\text{imp}}^2(T, K(y, 0); t) T\right).$$

Simplifying this expression finally gives

$$\sigma_{\text{imp}, \text{LC}}^2(T) = \int_{-\infty}^{+\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \sigma_{\text{imp}}^2(T, K(y); t), \quad (8.126)$$

where $y(K)$ is given by

$$y(K) = \frac{\log\left(\frac{\text{For}_S(t, T)}{K}\right)}{\sigma_{\text{imp}}(T, K; t)\sqrt{T}} - \frac{\sigma_{\text{imp}}(T, K; t)\sqrt{T}}{2}. \quad (8.127)$$

- (ii) Now setting $p = 1 - \varepsilon$ and taking the limit $\varepsilon \rightarrow 0$, since $s^p \approx s - \varepsilon s \log s$ for small ε , we have $\lim_{\varepsilon \rightarrow 0} \sigma_{\text{imp}, \text{Pow}_p}(T) = \sigma_{\text{imp}, \text{SLC}}(T)$ (notation $\sigma_{\text{imp}, \text{SLC}}(T)$ in Section 8.16.1. Equation (8.125) then becomes

$$\sigma_{\text{imp}, \text{SLC}}^2(T) = \int_{-\infty}^{+\infty} dy \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \sigma_{\text{imp}}^2(T, K(y); t), \quad (8.128)$$

with $y(K)$ given by

$$y(K) = \frac{\log\left(\frac{\text{For}_S(t, T)}{K}\right)}{\sigma_{\text{imp}}(T, K; t)\sqrt{T}} + \frac{\sigma_{\text{imp}}(T, K; t)\sqrt{T}}{2}. \quad (8.129)$$

8.18.3 Forward variance of power-payoff options

Power payoffs have the form $h(x)$ with $x = S_T/F_T$, where h is convex if $p < 0$ or $p > 1$, and concave if $p \in (0, 1)$. Thus, as we saw in Section 8.4.1, an associated implied volatility can be defined that also satisfies the convexity condition. That is,

$$(T_2 - t)\sigma_{\text{imp}, \text{Pow}_p}^2(t, T_2) \geq (T_1 - t)\sigma_{\text{imp}, \text{Pow}_p}^2(t, T_1),$$

where $T_1 < T_2$. We can therefore define an associated forward variance that is always positive. Its discrete definition is

$$\xi_t^{(p)}(T_1, T_2) = \frac{(T_2 - t)\sigma_{\text{imp}, \text{Pow}_p}^2(t, T_2) - (T_1 - t)\sigma_{\text{imp}, \text{Pow}_p}^2(t, T_1)}{T_2 - T_1}, \quad (8.130)$$

and its continuous definition is

$$\xi_t^{(p)}(T) = \frac{\partial}{\partial T} \left((T - t)\sigma_{\text{imp}, \text{Pow}_p}^2(t, T) \right). \quad (8.131)$$

The set of forward variances $\xi^{(p)}(T)$ for all T , $\{\xi^{(p)}(T) : T \in [t, M]\}$ —where t is the present time and M is determined by the longest maturity available in market data—is called the variance curve for index p . These are the variables we will model.

Remark 8.31. $\xi^{(p)}$ is a random field $(t, T) \mapsto \xi^{(p)}(t, T)$. For fixed T , $t \mapsto \xi_t^{(p)}(T)$ is an Itô process: it evolves stochastically in calendar time. For fixed t , $T \mapsto \xi^{(p)}(t, T)$ is a random maturity curve, not a process in the Itô sense. Moreover, forward variance as defined in (8.131) can be inverted by integrating between t and T as

$$\sigma_{\text{imp}, \text{Pow}_p}^2(t, T) = \frac{1}{T - t} \int_t^T \xi_t^{(p)}(u) du. \quad (8.132)$$

8.18.4 Forward variance dynamics

We model forward variance as an Itô process. That is,

$$dS_t = (r - q)S_t dt + \sigma_t S_t dW_t^S \quad (8.133)$$

$$d\xi_t^{(p)}(T) = \mu_t^{(p)}(T)dt + \lambda_t^{(p)}(T)dW_t^T, \quad (8.134)$$

where the initial conditions S_0 and ξ_0 are fixed by the market, with $\xi_0(\cdot)$ denoting the market curve for $T \in [0, M]$ and M the longest observable market maturity.

Note that, for each T , we have a stochastic process given by equation (8.134) with its corresponding Wiener process. These Wiener processes may be correlated across maturities and with the underlying's Wiener process W^S . For two fixed maturities u, v , let $\rho_t(u, v) = \frac{d[W^u, W^v]_t}{dt}$ denote the correlation between their Wiener processes and

$\rho_t(S, u) = \frac{d[W^S, W^u]_t}{dt}$ the correlation between the Wiener process for maturity u and W^S . Since we consider an uncountable continuum of maturities $T > t$, we have an infinite-dimensional Itô process; see Appendix ??.

Rewriting the power-payoff option price, $\text{Pow}_p(t, S_t, \bar{\sigma}_T^2)$, in terms of forward variance gives

$$\text{Pow}_p(t, s, \xi^{(p)}; T, r, q, p) = e^{-r(T-t)} \text{For}_S(t, T)^p e^{\frac{p(p-1)}{2} \int_t^T \xi_t^{(p)}(u) du}, \quad (8.135)$$

with $S_t = s$. Since the discounted price of $\text{Pow}_p(t, s)$ is a martingale, its drift must equal $r\text{Pow}_p(t, s)dt$. Thus,

$$\frac{d\text{Pow}_p(t, s, \xi^{(p)})}{\text{Pow}_p(t, s, \xi^{(p)})} = rdt + (\text{noise}),$$

so, using the two-dimensional Itô formula³⁶, we can calculate $\mu_t^{(p)}(T)$ by imposing the preceding condition.

First, since

$$\text{For}_S(t, T) = S_t e^{(r-q)(T-t)},$$

we have

$$\text{Pow}_p(t, S_t, \xi_t^{(p)}) = S_t^p e^{-r(T-t) + p(r-q)(T-t) + \frac{p(p-1)}{2} \int_t^T \xi_t^{(p)}(u) du}.$$

Note that Pow_p is a function of t and S_t and a functional of curve $u \mapsto \xi^{(p)}(t, u)$ at $[t, T]$. For each fixed T , $\xi_t^{(p)}(T)$ is an Itô process in t for $t \in [0, T]$; for fixed t , $T \mapsto \xi_t^{(p)}(T)$ is a random curve (not a process in T).

The classical Itô formula therefore cannot be applied directly here. The state includes the entire curve $u \mapsto \xi^{(p)}(t, u)$ (an infinite-dimensional object), so $\text{Pow}_p(t, S_t, \xi_t^{(p)}(\cdot))$ is not a $C^{1,2}$ function of a semimartingale taking values in $\mathbb{R}^n$, as required by the classical Itô formula introduced in Theorem 2.17. Instead of partial derivatives with respect to a scalar ξ , functional derivatives with respect to $\phi = \xi_t^{(p)}(\cdot)$ in a function space are required. In particular, since, for each fixed u , $\xi_t^{(p)}(u)$ belongs to³⁷ $L_{\text{ad}}^2([0, M] \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mu \times \mathbb{P}^*)$, $\phi(u) = \xi_t^{(p)}(u) \mathbb{1}_{[t, T]}(u)$ is also square integrable, so $\phi \in L^2([t, M])$ for fixed t . Since it is also adapted to $\mathcal{F}_t$ by definition of ϕ , we have $\phi \in L_{\text{ad}}^2([t, M] \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mu \times \mathbb{P}^*)$.

Thus, the two-dimensional Itô formula can be applied as follows³⁸:

$$\begin{aligned} d\text{Pow}_p(t, s)[\phi] &= \frac{\partial \text{Pow}_p}{\partial t} dt + \frac{\partial \text{Pow}_p}{\partial s} dS_t + \frac{1}{2} \frac{\partial^2 \text{Pow}_p}{\partial s^2} d[S]_t \\ &\quad + \langle D_\phi \text{Pow}_p, d\phi_t \rangle + \frac{1}{2} \langle \langle D_\phi^2 \text{Pow}_p, d[\phi]_t \rangle \rangle \\ &\quad + \left\langle D_\phi \left(\frac{\partial \text{Pow}_p}{\partial s} \right), d[\phi, S]_t \right\rangle. \end{aligned} \quad (8.136)$$

where $\phi_t(\cdot) = \xi_t^{(p)}(\cdot)$. When differentiating with respect to t , we must account for its appearance in the lower limit of the forward variance integral. Moreover, the functional

³⁶By two dimensions, we mean that Pow_p depends on a time variable, t , and two other random variables, S_t and $\xi_t^{(p)}$.

³⁷Again, M denotes the longest observable market maturity.

³⁸I use the notation $d\text{Pow}_p(t, s)[\xi^{(p)}]$ to emphasize that Pow_p is a function of t and s and a functional of $\xi^{(p)}$.

derivatives act on a function h such that

$$\langle D_\phi \text{Pow}_p[\phi], h \rangle = \left\langle \frac{\delta \text{Pow}_p}{\delta \phi(u)}, h \right\rangle = \int_t^T du \frac{\delta \text{Pow}_p}{\delta \phi(u)} h(u) \quad (8.137)$$

$$\langle \langle D_\phi^2 \text{Pow}_p[\phi], h \rangle \rangle = \int_t^T \int_t^T dudv \frac{\delta \text{Pow}_p}{\delta \phi(u) \delta \phi(v)} h(u, v) \quad (8.138)$$

$$\left\langle D_\phi \left(\frac{\partial \text{Pow}_p}{\partial s} \right), h \right\rangle = \int_t^T du \frac{\delta}{\delta \phi(u)} \left(\frac{\partial \text{Pow}_p}{\partial s} \right) h(u). \quad (8.139)$$

Note that $D_\phi \text{Pow}_p[\phi]$ belongs to the dual of $L^2([t, M])$, so $D_\phi \text{Pow}_p[\phi] \in L^2([t, M])^*$. Since $L^2([t, M])$ is a Hilbert space, however, $L^2([t, M]) \simeq L^2([t, M])^*$. Furthermore, $D_\phi^2 \text{Pow}_p[\phi]$ is a continuous bilinear form on $L^2([t, M]) \times L^2([t, M])$, i.e., a bounded linear operator $L^2([t, M]) \rightarrow L^2([t, M])^*$.

Let us calculate the derivatives step by step:

(i) The derivatives with respect to t and s are straightforward, giving

$$\begin{aligned} \frac{\partial \text{Pow}_p}{\partial t} dt &= \left(r - p(r - q) - \frac{p(p-1)}{2} \xi_t^{(p)} \right) \text{Pow}_p dt \\ \frac{\partial \text{Pow}_p}{\partial s} dS_t &= p(r - q) \text{Pow}_p dt + p \sigma_t \text{Pow}_p dW_t^S \\ \frac{1}{2} \frac{\partial^2 \text{Pow}_p}{\partial s^2} d[S]_t &= \frac{p(p-1)}{2} \sigma_t^2 \text{Pow}_p dt. \end{aligned}$$

(ii) Writing the power-payoff option price in terms of ϕ gives

$$\text{Pow}_p(t, s)[\phi] = s^p e^{-r(T-t) + p(r-q)(T-t) + \frac{p(p-1)}{2} \int_t^T \phi(u) du}, \quad (8.140)$$

so, using the Euler-Lagrange formula (and noting that the functional does not depend on derivatives of ϕ), we obtain

$$D_\phi \text{Pow}_p[\phi] = \frac{p(p-1)}{2} \text{Pow}_p(t, s)[\phi],$$

since the derivative is 1 at every point of the domain $u \in (t, T)$. Applying this distribution to $d\phi_t$ gives

$$\begin{aligned} \langle D_\phi \text{Pow}_p[\phi], d\phi_t \rangle &= \frac{p(p-1)}{2} \text{Pow}_p(t, s)[\phi] \int_t^T du d\phi_t(u) \\ &= \frac{p(p-1)}{2} \text{Pow}_p(t, s)[\phi] \int_t^T du d\xi_t^{(p)}(u) \\ &= \frac{p(p-1)}{2} \text{Pow}_p(t, s)[\phi] \int_t^T du d\xi_t^{(p)}(u) \\ &= \frac{p(p-1)}{2} \text{Pow}_p(t, s)[\phi] \left[\left(\int_t^T \mu_t^{(p)}(u) du \right) dt + \left(\int_t^T \lambda_t^{(p)}(u) dW_t^u \right) \right]. \end{aligned}$$

(iii) Similarly, the second derivative is

$$D_\phi^2 \text{Pow}_p[\phi] = \frac{p(p-1)}{2} D_\phi \text{Pow}_p(t, s)[\phi] = \frac{p^2(p-1)^2}{4} \text{Pow}_p(t, s)[\phi],$$

and $d[\phi]_t = d[\phi(u), \phi(v)]_t = \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) d[W^u, W^v]$. From the definition of ρ_t , we have $d[\phi]_t = \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) \rho_t(u, v) dt$, so

$$\begin{aligned} \frac{1}{2} \langle \langle D_\phi^2 \text{Pow}_p[\phi], d\phi_t \rangle \rangle &= \frac{p^2(p-1)^2}{8} \text{Pow}_p(t, s)[\phi] \int_t^T \int_t^T dudv d[\phi]_t(u, v) \\ &= \left(\frac{p^2(p-1)^2}{8} \text{Pow}_p(t, s)[\phi] \int_t^T \int_t^T dudv \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) \rho_t(u, v) \right) dt. \end{aligned}$$

(iv) Finally, since $\frac{\partial \text{Pow}_p}{\partial s} = \frac{p}{s} \text{Pow}_p$, the cross derivative is given by

$$D_\phi \left(\frac{\partial \text{Pow}_p}{\partial s} \right) = \frac{p}{s} D_\phi \text{Pow}_p(t, s)[\phi] = \frac{1}{s} \frac{p^2(p-1)}{2} \text{Pow}_p(t, s)[\phi],$$

so applying it to $d[\phi, S]_t$ gives

$$\begin{aligned} \left\langle D_\phi \left(\frac{\partial \text{Pow}_p}{\partial s} \right), d[\phi, S]_t \right\rangle &= \frac{1}{s} \frac{p^2(p-1)}{2} \text{Pow}_p \int_t^T du d[\phi, S]_t \\ &= \left(\frac{p^2(p-1)}{2} \int_t^T du \lambda_t^{(p)}(u) \sigma_t \rho_t(S, u) \right) \text{Pow}_p dt \\ &= \left(\frac{p^2(p-1)}{2} \sigma_t \int_t^T du \lambda_t^{(p)}(u) \rho_t(S, u) \right) \text{Pow}_p dt \end{aligned}$$

Summing all terms yields the stochastic differential equation modeling the power-payoff option price. Dividing by $\text{Pow}_p(t, s)[\phi]$ and simplifying gives

$$\begin{aligned} \frac{d\text{Pow}_p(t, s)[\phi]}{\text{Pow}_p(t, s)[\phi]} &= \left[r + \frac{p(p-1)}{2} (\sigma_t^2 - \xi_t^{(p)}) + \frac{p(p-1)}{2} \int_t^T \mu_t^{(p)}(u) du \right. \\ &\quad + \frac{p^2(p-1)^2}{8} \int_t^T \int_t^T dudv \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) \rho_t(u, v) \\ &\quad \left. + \frac{p^2(p-1)}{2} \sigma_t \int_t^T du \lambda_t^{(p)}(u) \rho_t(S, u) \right] dt \\ &\quad + p \sigma_t dW_t^S + \frac{p(p-1)}{2} \int_t^T \lambda_t^{(p)}(u) dW_t^u. \end{aligned} \tag{8.141}$$

Equating the preceding expression to $r dt$ gives

$$\begin{aligned} 0 &= \frac{p(p-1)}{2} (\sigma_t^2 - \xi_t^{(p)}) + \frac{p(p-1)}{2} \int_t^T \mu_t^{(p)}(u) du \\ &\quad + \frac{p^2(p-1)^2}{8} \int_t^T \int_t^T dudv \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) \rho_t(u, v) \\ &\quad + \frac{p^2(p-1)}{2} \sigma_t \int_t^T du \lambda_t^{(p)}(u) \rho_t(S, u), \end{aligned} \tag{8.142}$$

and differentiating with respect to T gives the drift $\mu_t^{(p)}$ of the variance curve such that

$$\frac{p(p-1)}{2}\mu_t^{(p)}(T) + \frac{p^2(p-1)}{2}\sigma_t\lambda_t^{(p)}(T)\rho_t(S, T) + \frac{p^2(p-1)^2}{4}\int_t^T du \lambda_t^{(p)}(T)\lambda_t^{(p)}(u)\rho_t(T, u) = 0, \quad (8.143)$$

hence

$$\mu_t^{(p)}(T) = -\left(p\sigma_t\lambda_t^{(p)}(T)\rho_t(S, T) + \frac{p(p-1)}{2}\int_t^T du \lambda_t^{(p)}(T)\lambda_t^{(p)}(u)\rho_t(T, u)\right). \quad (8.144)$$

Taking the limit $T \rightarrow t$ in expression (8.141) leaves only the nonzero term

$$\left[r + \frac{p(p-1)}{2}\left(\sigma_t^2 - \xi_t^{(p)}(t)\right)\right] dt,$$

so equating this to $r dt$ gives the condition

$$\boxed{\xi_t^{(p)}(t) = \sigma_t^2}. \quad (8.145)$$

Thus, the short end of the variance curve equals the underlying's instantaneous volatility S for any value of p . Finally, the system of stochastic differential equations for S and $\xi^{(p)}(T)$ is

$$dS_t = (r - q)S_t dt + \sqrt{\xi_t^{(p)}(t)}S_t dW_t^S \quad (8.146)$$

$$d\xi_t^{(p)}(T) = -\left(p\sqrt{\xi_t^{(p)}(t)}\lambda_t^{(p)}(T)\rho_t(S, T) + \frac{p(p-1)}{2}\int_t^T du \lambda_t^{(p)}(T)\lambda_t^{(p)}(u)\rho_t(T, u)\right) dt + \lambda_t^{(p)}(T)dW_t^T. \quad (8.147)$$

8.18.5 Markov representation of the variance curve

Suppose we want to simulate system (8.146) and (8.147) by Monte Carlo for fixed T . The problem is that forward variance depends on different values of $T \in [t, M]$, requiring each forward variance to be evolved individually, which is impractical. One way to overcome this obstacle would be to rewrite $\xi^{(p)}(T)$ as a Markov process, in which $\xi^{(p)}(T)$ depends on a finite number of state variables that generate the entire variance curve at time t . We will study a similar procedure when introducing the HJM framework for fixed-income models.

Suppose that the covariance structure ρ_t allows system (8.146) and (8.147) to be rewritten as

$$dS_t = (r - q)S_t dt + \sqrt{\xi_t^{(p)}(t)}S_t dW_t^S \quad (8.148)$$

$$d\xi_t^{(p)}(T) = \mu_t(T)dt + \sum_{i=0}^n \alpha^i(T)\beta_t^i dW_t^i, \quad (8.149)$$

where W^i are correlated Wiener processes for $i = 0, 1, \dots, n$, $\alpha^i(T)$ are functions of T , and β_t^i are stochastic processes adapted to $\mathcal{F}_t$. Similarly, $\rho_t(i, j)$ denotes the correlation between W_t^i and W_t^j , and $\rho_t(S, i)$ the correlation between W_t^S and W_t^i .

The chosen form of the volatility structure in this SDE is reasonable and captures the properties we wish to model.

Example 8.32. If we set $\beta_t^i = e^{k_i t}$ and $\alpha^i(T) = \alpha_i e^{-k_i T}$, then

$$d\xi_t^{(p)}(T) = \mu_t(T)dt + \sum_{i=0}^n \alpha_i e^{-k_i(T-t)} dW_t^i,$$

so the dynamics of $\xi^{(p)}(T)$ are time homogeneous –since the volatilities depend only on $T - t$, and the correlations of $\xi^{(p)}(T_1)$ and $\xi^{(p)}(T_2)$ depend on $T_1 - t$ and $T_2 - t$.

Integrating equation (8.149) gives (add derivation)

$$\begin{aligned} \xi_t^{(p)}(T) &= \xi_0^{(p)}(T) - p \sum_{i=0}^n \int_0^t \alpha^i(T) \beta_u^i \rho_u(S, i) \sqrt{\xi_u^{(p)}(u)} du \\ &\quad - \frac{p(p-1)}{2} \int_0^t \left(\int_u^T \sum_{i=0}^n \rho_u(i, j) \alpha^i(T) \beta_u^i \alpha^j(v) \beta_u^j dv \right) du \\ &\quad + \sum_{i=0}^n \int_0^t \alpha^i(T) \beta_u^i dW_u^i \\ &= \xi_0^{(p)}(T) + \sum_{i=0}^n \alpha^i(T) \left[-p \rho_t(S, i) \int_0^t \beta_u^i \sqrt{\xi_u^{(p)}(u)} du \right. \\ &\quad \left. - \frac{p(p-1)}{2} \sum_{j=0}^n \int_0^t \rho_u(i, j) \beta_u^i \beta_u^j (A^j(T) - A^j(u)) du + \int_0^t \beta_u^i dW_u^i \right], \end{aligned}$$

where we have introduced the functions $A^j(u) = \int_0^u \alpha^j(v) dv$. Defining the process $B_t(i, j) = \rho_t(i, j) \int_0^t \beta_u^i \beta_u^j du$ then gives

$$\begin{aligned} \xi_t^{(p)}(T) &= \xi_0^{(p)}(T) + \sum_{i=0}^n \alpha^i(T) \left[-\frac{p(p-1)}{2} \sum_{j=0}^n B_T(i, j) (A^j(T) - A^j(t)) \right. \\ &\quad \left. + \int_0^t \left(-p \rho_u(S, i) \beta_u^i \sqrt{\xi_u^{(p)}(u)} - \frac{p(p-1)}{2} \sum_{j=0}^n B_u(i, j) \alpha^j(u) du + \beta_u^i dW_u^i \right) \right]. \end{aligned}$$

We can simplify the preceding expression further by defining the integrand as an Itô process dx_t^i , with initial condition $x_0^i = 0$, for each i . Thus,

$$\begin{aligned} x_0^i &= 0 \\ dx_t^i &= -p \rho_t(S, i) \beta_t^i \sqrt{\xi_t^{(p)}(t)} - \frac{p(p-1)}{2} \sum_{j=0}^n B_t(i, j) \alpha^j(t) dt + \beta_t^i dW_t^i, \end{aligned}$$

so the variance curve at time t is given by

$$\boxed{\xi_t^{(p)}(T) = \xi_0^{(p)}(T) + \sum_{i=0}^n \alpha^i(T) \left[-\frac{p(p-1)}{2} \sum_{j=0}^n B_t(i, j) (A^j(T) - A^j(t) + x_t^i) \right]} \quad (8.150)$$

Note that if $T \rightarrow t$, the short end of the curve—which equals the underlying's instantaneous variance S_t —tends to

$$\xi_t^{(p)}(t) = \xi_0^{(p)}(t) + \sum_{i=0}^n \alpha^i(t) x_t^i. \quad (8.151)$$

Thus, a Monte Carlo simulation requires only the evolution of processes x_t^i , $B_t(i, j)$, and the underlying S_t , for $i = 0, 1, \dots, n$. There are $2n + 3$ Itô processes in total, with the system of stochastic differential equations

$$dx_t^i = -p\rho_t(S, i)\beta_t^i \sqrt{\xi_t^{(p)}(t)} - \frac{p(p-1)}{2} \sum_{j=0}^n B_t(i, j)\alpha^j(t)dt + \beta_t^i dW_t^i, \quad (8.152)$$

$$dB_t(i, j) = \rho_t(i, j)\beta_t^i\beta_t^j dt, \quad (8.153)$$

$$dS_t = (r - q)S_t dt + \sqrt{\sum_{i=0}^n \alpha^i(t)x_t^i S_t} dW_t^S, \quad (8.154)$$

and initial conditions $x_0^i = 0$, $B_0(i, j) = 0$, and $S_0 = s$.

Once these processes have been simulated, the variance curve is obtained directly by substitution into equation (8.150).

In system (8.152), the volatilities β_t^i may depend arbitrarily on processes x_t^i and correlations $B_t(i, j)$. Thus, even if the forward variance curve $\xi_t^{(p)}(T)$ can be represented with finitely many factors, the resulting dynamics may still depend on the entire curve in t . This is equivalent to applying a local volatility model to $\xi_t^p(T)$ and provides no real structural simplification.

Moreover, unlike forward rates in HJM, instantaneous variance $V_t = \xi_t^{(p)}(t)$ cannot be negative. Under the ansatz proposed in equation (8.151), there is no general guarantee that this expression remains nonnegative for all t , since fluctuations in x_t^i and the signs of $\alpha^i(t)$ can produce $\xi_t^{(p)}(t) < 0$.

Bergomi suggests imposing lognormal dynamics on each $\xi_t^{(p)}(T)$, of the form

$$\xi_t^{(p)}(T) = \xi_t^{(p)}(T) \left(\mu_t^{(p)}(T)dt + \sum_{i=0}^n \alpha^i(T) \beta_t^i dW_t^i \right),$$

which would in principle preserve positivity. However, a Markov representation consistent with this form can be constructed only for $p = 0$ (log contracts) and $p = 1$ ($s \log s$ contracts), failing for general values of p .

Moreover, there is no clear procedure for directly modeling and consistently calibrating the joint dynamics of several variance curves $\xi_t^{(p)}(T)$. In practice, only one curve for a given p is modeled explicitly, and the others are derived without being able to fix their initial conditions.

Thus, the proposed ansatz for a low-dimensional Markov representation cannot be usefully extended to forward variances associated with power payoffs for $p \neq 0, 1$.

8.18.6 Log contracts as a solution for modeling the variance curve

Let ζ_t^T denote the forward variance curve of log contracts (recovered by taking $p \rightarrow 0$) with maturity T . Then

$$\zeta_t^T = \xi_t^{(0)}(T) = \frac{d}{dT} \left((T - t) \sigma_{\text{imp,LC}}^2(t, T) \right),$$

where, as we saw, $\sigma_{\text{imp,LC}}(t, T)$ is always well defined because log contract prices are finite, so implied volatility is well defined whenever the no-arbitrage conditions hold. The main advantage of log contracts becomes apparent from equation (8.147), which models forward variance curve evolution: setting $p \rightarrow 0$ eliminates the drift and yields the system

$$dS_t = \sqrt{\zeta_t^T} S_t dW_t^S, \quad (8.155)$$

$$d\zeta_t^T = \lambda_t(T) dW_t^T, \quad (8.156)$$

which allows the curve to be modeled simply.

As we saw in Section 8.15.1, log contracts can be replicated with a portfolio containing vanilla options over a continuous density of strikes, which does not exist in practice. Log contracts are therefore not available as traded market instruments. Moreover, this analysis does not apply to fixed cash dividends –rather than proportional dividends with q – since the property required to prove the convexity condition then fails.

Fortunately, closely related instruments known as variance swaps are traded in the market. Equation (8.156) remains valid, with ζ_t^T replaced by variance swap forward variances, which we study in the next chapter.

Chapter 9

Variance swaps

Variance swaps (VS) were introduced in the 1990s and are considered a better way to gain exposure to an underlying's volatility than using options. Options have the disadvantage that their vega and gamma exposure (essentially the main forms of volatility exposure) is concentrated around the strike –owing to the functional form of these Greeks; see Section 7.9. Thus, if the underlying moves far from the strike, spot movements become irrelevant to P&L.

Consider an example: a trader believes that implied volatility is cheap and bets on an increase in future realized volatility by buying a call option. Recall that gamma/theta P&L over the interval $(t, t + dt)$ is given by

$$dP\&L = \frac{1}{2}\Gamma^D(t, S_t)(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)dt,$$

so total P&L up to maturity T is

$$P\&L = \frac{1}{2} \int_0^T \Gamma^D(t, S_t)(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)dt.$$

Thus, if realized volatility exceeds implied volatility while the option has gamma (which occurs near the strike), the trader makes money.

However, suppose the underlying moves far from the strike with very little volatility (i.e., it moves away from the strike with $\sigma_{\text{real}} < \sigma_{\text{imp}}$ until time t^*). Up to that point, the trader loses money.

Even if the underlying becomes extremely volatile from t^* until T (such that $\sigma_{\text{real}} > \sigma_{\text{imp}}$) while far from the strike, P&L is weighted by dollar gamma –which is approximately zero far from the strike–, so P&L from t^* onward is zero. Total P&L therefore decomposes as

$$\begin{aligned} P\&L &= \frac{1}{2} \int_0^{t^*} \Gamma^D(t, S_t)(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)dt + \frac{1}{2} \int_{t^*}^T \Gamma^D(t, S_t)(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)dt \\ &= \frac{1}{2} \int_0^{t^*} \Gamma^D(t, S_t)(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)dt \\ &< 0, \end{aligned}$$

meaning that the trader loses money.

Thus, although the trader correctly predicted that realized volatility would exceed implied volatility, the strategy loses money. It is not enough for realized volatility to exceed implied volatility on average; this must happen while the option has gamma.

Volatility swaps and variance swaps were developed to avoid this consequence of obtaining volatility exposure through options, allowing exposure to volatility without exposure to the underlying price. We focus on variance swaps because their linear structure in variance facilitates analysis, simplifies replication with option portfolios, and makes them more liquid and more commonly traded in OTC markets than volatility swaps.

Definition 9.1. A variance swap (VS) is a financial derivative with a structure analogous to a futures contract, whose underlying is the realized variance of a financial asset (generally equity, FX, etc.). Its payoff at maturity T is therefore

$$\text{VS}(T, T) = N \left(\text{RV}_T - \sigma_{\text{VS}}^2(t, T) \right), \quad (9.1)$$

where N is the notional,

$$\text{RV}_T = \frac{252}{\#(\text{trading days})} \sum_{i=0}^{N-1} \log^2 \left(\frac{S_{i+1}}{S_i} \right) \quad (9.2)$$

is the realized variance¹ of the underlying asset –with S_i denoting the closing asset price on the dates specified in the contract (generally at the end of each day)–, and $\sigma_{\text{VS}}^2(t, T)$ is the variance strike fixed at the initial date t so that the contract has zero initial value.

Remark 9.2. A variance swap’s zero value at inception does not imply that its price remains zero throughout its life. For example, a VS entered into at $t = 0$ with strike $\sigma_{\text{VS}}(T; 0)$ has zero initial value by definition. At $t = 1$, however, the same contract reflects accumulated information and variance expectations for the remaining period, so its value may be nonzero. A new VS entered into at $t = 1$ with strike $\sigma_{\text{VS}}(T; 1)$ has zero initial value at $t = 1$, so the two contracts do not have the same price at that time.

We will in fact use the convention

$$\text{RV}_T = \frac{1}{T - t} \sum_{i=0}^{N-1} \log^2 \left(\frac{S_{i+1}}{S_i} \right). \quad (9.3)$$

9.1 Forward variances of variance swaps

Suppose that at time t we construct portfolio Π_t consisting of: (i) a long position with notional $N = T_2 - t$ in $\text{VS}(t, T_2)$ and (ii) a short position with notional $N = (T_1 - t)e^{-r(T_2 - T_1)}$ in $\text{VS}(t, T_1)$, with $T_2 > T_1$. At date T_2 , the portfolio value –after accruing

¹The factor 252 is the number of days on which trading is generally possible in a year. For a discussion of the relevance of this factor, see [39, chapter 20].

interest on the swap maturing at T_1 – is

$$\begin{aligned}
 \Pi_{T_2} &= (T_2 - t)\text{VS}(T_2) - e^{r(T_2-T_1)}(T_1 - t)e^{-r(T_2-T_1)}\text{VS}(T_1) \\
 &= (T_2 - t)\text{VS}(t, T_2) - (T_1 - t)\text{VS}(t, T_1) \\
 &= \sum_{j=1}^m \log^2 \left(\frac{S_{j+1}}{S_j} \right) - [(T_2 - t)\sigma_{\text{VS}}^2(t, T_2) - (T_1 - t)\sigma_{\text{VS}}^2(t, T_1)] \\
 &= \sum_{j=0}^m \log^2 \left(\frac{S_{j+1}}{S_j} \right) - (T_2 - T_1)\xi_t^{\text{VS}}(T_1, T_2),
 \end{aligned}$$

where $j = 0$ represents time T_1 and $j = m$ time T_2 . Moreover, $\xi_t^{\text{VS}}(T_1, T_2)$ denotes the discrete VS forward variance between T_1 and T_2 , similarly to expression (8.130). That is,

$$\xi_t^{\text{VS}}(T_1, T_2) = \frac{(T_2 - t)\sigma_{\text{VS}}^2(t, T_2) - (T_1 - t)\sigma_{\text{VS}}^2(t, T_1)}{T_2 - T_1}. \quad (9.4)$$

Note that $\xi_t^{\text{VS}}(T_1, T_2) > 0$ for any t by construction, since the convexity condition holds. Now suppose we unwind our position at $t' > t$. We construct portfolio Π'_t : (i) a short position with notional $N = T_2 - t'$ in $\text{VS}(t', T_2)$ and (ii) a long position with notional $N = (T_1 - t')e^{-r(T_2-T_1)}$ in $\text{VS}(t', T_1)$. At date T_2 , the portfolio value is

$$\begin{aligned}
 \Pi'_{T_2} &= e^{r(T_2-T_1)}(T_1 - t')e^{-r(T_2-T_1)}\text{VS}(t', T_1) - (T_2 - t')\text{VS}(t', T_2) \\
 &= (T_1 - t')\text{VS}(t', T_1) - (T_2 - t')\text{VS}(t', T_2) \\
 &= [(T_2 - t')\sigma_{\text{VS}}^2(t', T_2) - (T_1 - t')\sigma_{\text{VS}}^2(t', T_1)] - \sum_{j=1}^m \log^2 \left(\frac{S_{j+1}}{S_j} \right) \\
 &= (T_2 - T_1)\xi_{t'}^{\text{VS}}(T_1, T_2) - \sum_{j=0}^m \log^2 \left(\frac{S_{j+1}}{S_j} \right),
 \end{aligned}$$

so the combined portfolio is $\Pi_{T_2} + \Pi'_{T_2}$, given by

$$(T_2 - T_1) [\xi_{t'}^{\text{VS}}(T_1, T_2) - \xi_t^{\text{VS}}(T_1, T_2)], \quad (9.5)$$

Thus, strategy $\Pi_{T_2} + \Pi'_{T_2}$ has no exposure to the underlying's realized variance over $[T_1, T_2]$; it depends exclusively on the change in VS forward variance over the interval $[t, t']$.

What is the initial cost of this strategy? Since a VS is defined to have zero cost at inception, the strategy's initial cost is zero. Thus, we have constructed a payoff linear in forward variance $\xi_t^{\text{VS}}(T_1, T_2)$ over $[t, t']$ at zero initial cost.

Since the VS price at inception t is zero, its discounted expected value at t must be zero. Thus,

$$\text{VS}(t, T) = e^{-r(T-t)}\mathbb{E}^* [N (\text{RV}_T - \sigma_{\text{VS}}^2(t, T)) \mid \mathcal{F}_t],$$

and since $\sigma_{\text{VS}}^2(t, T)$ is a constant fixed at t ,

$$\sigma_{\text{VS}}^2(t, T) = \mathbb{E}^* [\text{RV}_T \mid \mathcal{F}_t].$$

Is $\sigma_{\text{VS}}^2(t, T)$ a martingale under $\mathbb{P}^*$? Taking its expectation conditional on filtration $\mathcal{F}_s$, with $s < t$, and applying property (i) of Proposition A.27 –since $\mathcal{F}_s \subset \mathcal{F}_t$ – gives

$$\mathbb{E}^* [\sigma_{\text{VS}}^2(t, T) \mid \mathcal{F}_s] = \mathbb{E}^* \left[\mathbb{E}^* [\text{RV}_T \mid \mathcal{F}_t] \mid \mathcal{F}_s \right] = \mathbb{E}^* [\text{RV}_T \mid \mathcal{F}_s] = \sigma_{\text{VS}}^2(s, T),$$

so $\sigma_{\text{VS}}^2(t, T)$ is a $\mathbb{P}^*$ -martingale. By the martingale representation theorem (to be added), there exists an adapted stochastic process $\{\phi_t(T) : t \leq T_1\}$ such that

$$d\sigma_{\text{VS}}^2(t, T) = \phi_t(T) dW_t^T, \quad (9.6)$$

where W^T is a Wiener process under $\mathbb{P}^*$.

Similarly, we can define the continuous forward variance of a VS, denoted $\xi_t^{\text{VS}}(T)$, as

$$\xi_t^{\text{VS}}(T) = \frac{\partial}{\partial T} ((T - t)\sigma_{\text{VS}}^2(t, T)), \quad (9.7)$$

which can also be expressed as a driftless stochastic differential equation given by

$$d\xi_t^{\text{VS}}(T) = \phi_t(T) dW_t^T. \quad (9.8)$$

Note that equation (9.8) has the same structure as the evolution of log contract forward variance in equation (8.156). But is there any other relation between variance swaps and log contracts?

9.2 Relation between variance swaps and log contracts

For simplicity, assume that the price observations $(S_0, S_1, \dots, S_N)$ are equally spaced. That is, $t_{i+1} - t_i = \delta t$ for all $i = 0, 1, \dots, N - 1$. For a small change in $\delta S_i = S_{i+1} - S_i$, approximating the logarithm to second order and discounting to $t = 0$ gives

$$e^{-r(T-t)} \sum_{i=0}^{N-1} \left(\frac{\delta S_i}{S_i} \right)^2 = \sum_{i=0}^{N-1} e^{-r(T-t_i)} e^{-r(t_i)} \left(\frac{\delta S_i}{S_i} \right)^2,$$

where taking the limit $\delta t \rightarrow 0$ gives the discounted variance swap payoff—setting $\sigma_{\text{VS}}^2(t, T) = 0$ —as

$$\int_t^T du e^{-r(T-u)} e^{-ru} \left(\frac{dS_u}{S_u} \right)^2,$$

or, similarly, the P&L differential generated by this position is

$$dP\&L(t, S_t) = e^{-r(T-t)} e^{-rt} \left(\frac{dS_t}{S_t} \right)^2,$$

which resembles the theta/gamma P&L differential of a portfolio consisting of a delta-hedged European option, as seen in (8.8). In particular, comparing the two expressions shows that necessarily $\sigma_{\text{imp}}^2 = 0$, but we have $e^{-r(T-t)}$ instead of $\frac{1}{2}s^2 \frac{\partial^2 V}{\partial s^2}$. Is there a European option whose dollar gamma at zero implied volatility satisfies the following relation?

$$\frac{1}{2}\Gamma^D = \frac{1}{2}s^2 \frac{\partial^2 V}{\partial s^2} = e^{-r(T-t)}.$$

Yes: as we saw in Remark 8.12, a log contract's dollar gamma equals $e^{-r(T-t)}$.

Thus, the P&L differential generated by a variance swap coincides –to second order in δS_i – with the theta/gamma P&L generated by a log contract at zero implied volatility, denoted $LC(t, s; \sigma_{\text{imp}, LC}(T) = 0)$.

What value of $\sigma_{VS}^2(t, T)$ makes the initial VS price at t zero? If the log contract's market price at the initial time t were $LC(t, s; \sigma_{\text{imp}, LC}(T) = 0)$, then $\sigma_{VS}^2(t, T)$ would have to be zero. In practice, however, the log contract has market price LC_{Mkt} . Buying it therefore generates mark-to-market P&L of $-(LC_{\text{Mkt}} - LC(t, s; \sigma_{\text{imp}, LC}(T) = 0))$ for us. We must consequently pass this cost on to the client as a VS premium. This premium is $(T - t)LC_{\text{Mkt}}$, payable at maturity T . Thus, $\sigma_{VS}^2(t, T)$ must be

$$\sigma_{VS}^2(t, T) = \frac{1}{T - t} (LC_{\text{Mkt}} - LC(t, s; \sigma_{\text{imp}, LC}(T) = 0)),$$

which, using expression (8.87), expresses $\sigma_{VS}^2(t, T)$ in terms of the log contract's implied volatility. We finally obtain

$$\sigma_{VS}(t, T) = \sigma_{\text{imp}, LC}(T) \quad (9.9)$$

$$\xi_t^{\text{VS}}(T) = \zeta_t^T. \quad (9.10)$$

Thus, log contract and VS forward variances are identical. Moreover, since a log contract can be replicated with a portfolio of vanilla options over an infinite set of strikes, we can calculate $\sigma_{VS}^2(t, T)$ in terms of vanilla option prices as

$$\sigma_{VS}^2(t, T) = \frac{e^{r(T-t)}}{T - t} \int_0^{+\infty} \frac{2}{K^2} (V_{\text{Mkt}}(T, K) - V(T, K; \sigma_{\text{imp}} = 0)) dK, \quad (9.11)$$

where

$$V(T, K; \sigma_{\text{imp}} = 0) = e^{-r(T-t)} (Se^{(r-q)(T-t)} - K)^+,$$

with V denoting the value of put or call options (the difference is the same by put-call parity).

Thus, $\sigma_{VS}^2(t, T)$ can be written as an integral of vanilla option prices weighted by K^{-2} . This integral is well defined whenever the smile is arbitrage free, but places substantial weight on low strikes (deep OTM puts). Since market quotes are available only over a finite strike range, the smile must be extrapolated beyond that range, and the VS value depends strongly on this choice. For highly liquid underlyings (indices), many market makers therefore reverse the procedure: they use the quoted VS price to infer implied volatilities at low strikes (anchoring the tail with the VS). For a very long maturity T , the variance strike σ_{VS}^2 must be adjusted for interest rate volatility.

From Remark 8.12, we saw that the log contract's delta and gamma do not depend on implied volatility. Its risk can therefore be managed using any σ_{imp} . The natural choice is $\sigma_{\text{imp}} = \sigma_{\text{imp}, LC}(T)$ (the implied volatility consistent with the log contract/VS itself): with this choice, delta hedging the log contract generates gamma P&L and theta P&L that exactly match the two components of decomposition (9.1).

Setting implied volatility to zero in the replication is particularly useful for analyzing payoffs in which realized variance is weighted by functions of spot, such as conditional variance swaps, or in the presence of discrete dividends. We examine this later.

In the absence of discrete dividends and jumps, σ_{VS}^2 can be calculated as the log contract's implied volatility, obtained in expression (8.126).

9.3 Weighted variance swaps

In the standard variance swap introduced in equation (9.3), all terms in the sum have weights $\frac{1}{T-t}$. Defining another weight function w , potentially dependent on the underlying, $w(S_i)$, gives different payoffs. These derivatives are called weighted variance swaps (WVS), with payoff

$$\text{WVS}(t, T; w) = N \left[\frac{1}{T-t} \sum_{i=0}^{N-1} w(S_i) \text{RV}_T - \sigma_{w\text{VS}}^2(t, T) \right], \quad (9.12)$$

or, in continuous and more general form, w can be taken as a process adapted to the filtration $\mathcal{F}_t$ such that

$$\text{WVS}(t, T; w) = N \left[\frac{1}{T-t} \int_t^T w_u d[\log S]_u - \sigma_{w\text{VS}}^2(t, T) \right]. \quad (9.13)$$

If $t = 0$ denotes the current date, the WVS is called a spot-starting weighted variance swap (SSWVS); if $t > 0$, it is called a forward-starting weighted variance swap (FSWVS). The most common cases are:

- (i) Standard variance swap (VS), introduced above. It is recovered by setting the weights to $\omega_t = 1$.
- (ii) Gamma swap (GS). It is obtained with weights $\omega_t = S_t$.
- (iii) Arithmetic variance swap (AVS). It results from choosing weights $\omega_t = S_t^2$.
- (iv) Corridor variance swap (CVS). Here the weights are the indicator function of a connected interval C . That is, $\omega_t = \mathbb{1}_C$. Setting $C = [0, H]$ gives a special case of CVS called a down variance swap (DVS). Setting $C = [L, \infty)$ gives a special case called an up variance swap (UVS).

9.3.1 How to replicate a WVS

We now study how a WVS can be replicated and which instruments are required [40], analogously to the VS case. In particular, we analyze cases (ii)–(iv) introduced above.

9.4 Other issues concerning variance swaps

Suppose the underlying S is modeled by a jump-diffusion process such that

$$dS_t = (r - q)S_t dt + \sigma_t S_t dW_t^S + S_{t-} (J dN_t - \lambda \bar{J} dt), \quad (9.14)$$

where J is the relative jump size, with jumps independently and identically distributed (iid) with mean $\bar{J} = \mathbb{E}^*[J]$, and N_t is a Poisson process counting the jumps. J and N_t are assumed independent, and $\mathbb{E}^*[dN_t] = \lambda dt$. (a jump models chapter will be added in the future)

In this case, the difference between σ_{VS}^2 and $\sigma_{\text{imp,LC}}$ is given by

$$\sigma_{\text{VS}}^2(T) - \sigma_{\text{imp,LC}}^2(T) = \lambda \mathbb{E}^* [\log^2(1+J) + 2\log(1+J) - 2J], \quad (9.15)$$

which can be written in terms of ATMF skew $\mathcal{S}_T$ –to first order in $\mathcal{S}_T$ – as

$$\sigma_{\text{VS}}(T) \approx \sigma_{\text{imp,LC}}(T) (1 - \sigma_{\text{imp,LC}}(T) \mathcal{S}_T), \quad (9.16)$$

so the difference between σ_{VS} and $\sigma_{\text{imp,LC}}$ is independent of maturity T only if ATMF skew, $\mathcal{S}_T$, decays as $1/T$. Moreover, the difference between σ_{VS} and $\sigma_{\text{imp,LC}}$ can be obtained asymptotically –independently of the chosen model– as

$$\frac{\sigma_{\text{VS}}(T)}{\sigma_{\text{imp,LC}}(T)} - 1 \approx -\frac{s_{\Delta t}}{6} \sigma_{\text{imp,LC}}(T) \sqrt{\Delta t},$$

where $s_{\Delta t}$ is the ATMF skew of daily returns. However, inferring this return skewness over a short period is impractical and depends on the chosen model. Realized daily skewness is so small that the variance swap adjustment is practically zero.

Among variance payoffs, only payoffs quadratic in price –i.e., arithmetic variance swaps, AVS– are replicated exactly by delta hedging. All others (including log variances or volatility swaps) require additional assumptions or approximations.

In theory, the variance swap replication formula uses a continuous integral over strikes, but in practice only discrete strikes are quoted, making replication imperfect. This adds another source of error to the difference between the market variance strike and the strike implied by model-free theory.

For practical purposes, on highly liquid indices, variance swap implied volatility and log contract volatility can be assumed equal. Their forward variances are therefore equal, and the model is simulated using the system of stochastic differential equations

$$dS_t = (r - q)S_t dt + \sqrt{\xi_t^{\text{VS}}(t)} S_t dW_t^S, \quad (9.17)$$

$$d\xi_t^{\text{VS}}(T) = \lambda_t(T) dW_t^T. \quad (9.18)$$

To match market prices more closely, an adjustment to realized variance can be introduced using free parameters λ and $\bar{J}$ such that

$$\log^2 \left(\frac{S_{i+1}}{S_i} \right) \mapsto \log^2 \left(\frac{S_{i+1}}{S_i} \right) + (\lambda \Delta) \mathbb{E}^* [\log^2(1+J) + 2\log(1+J) - 2J],$$

and forward variance is adjusted as

$$\zeta_t^T = \xi_t^{\text{VS}}(T) + \lambda \mathbb{E}^* [\log^2(1+J) + 2\log(1+J) - 2J].$$

Dividends do not affect a variance swap's payoff when adjusted returns are used, and their impact on indices is negligible. They do, however, affect replication: fixed cash dividends introduce deterministic jumps in the underlying price, breaking standard log contract replication. To correct this, the replication strategy must be supplemented with additional European payoffs maturing on the dividend dates.

9.5 How to price a VS using PDEs

Chapter 10

Trading volatility

10.1 Trading volatility with options

As we have seen, an option provides exposure to the underlying asset. For example, an investor who expects an asset's value to rise can buy a call option to profit while limiting risk exposure. However, we have also seen that the risk associated with movements in the underlying price can be neutralized through delta hedging.

This is the mechanism used by volatility traders. It gives them exposure to volatility rather than movements in the equity price. More precisely, they are exposed to all Greeks other than delta. Nevertheless, (i) volatility traders generally do not neutralize gamma, but use it to guide delta reheding¹; (ii) theta represents the loss of value over time (time decay) and is accepted as a cost of trading volatility; (iii) rho is less influential, since interest rate changes affect option prices but generally have a smaller impact than vega, gamma, and theta; (iv) the remaining Greeks may matter for more advanced strategies, especially in high-volatility environments. The key to volatility trading is to optimize vega exposure without incurring excessive costs to hedge the other Greeks.

10.1.1 Risks of volatility trading

Volatility traders face risks beyond volatility itself. Dividends and financing costs can affect their positions: a dividend reduces the underlying price and changes option valuations, while high borrowing costs can make the short selling required for delta hedging more expensive. Moreover, near expiry, traders' hedging activity may cause a stock to remain close to certain strike prices, a phenomenon known as pinning.

When delta hedging an option, ideally a forward with the same maturity should be used, rather than shares or forwards with a different maturity. Otherwise, additional risk is introduced.

A forward with a different maturity incorporates a different term structure into its price, potentially creating hedging mismatches.

Using the stock directly, on the other hand, exposes the trader to dividends, since the underlying price falls when dividends are paid, affecting the hedge.

¹Continually hedging delta is called gamma scalping.

Chapter 11

Fixed-income models

Fixed-income models play a fundamental role in quantitative finance, providing mathematical tools for pricing and managing debt-based financial instruments. From government bonds to structured products, these models allow the analysis of price and interest rate dynamics and the risks associated with different investment strategies.

This chapter explores the relation between interest rate models and fixed-income asset pricing, covering both deterministic and stochastic models. It presents methods for constructing yield curves, measuring bond price sensitivity to interest rate changes (duration and convexity), and hedging and managing risk. It also examines practical applications in portfolio management and the structuring of fixed-income products in financial markets.

For simplicity, I follow the notation used in [19], which I will later adapt to the notation used in earlier chapters of these notes.

11.1 Introduction to interest rates

An interest rate determines the amount a borrower promises to pay a lender. Many different interest rates exist for each currency: mortgage rates, deposit rates, personal loan rates, and so on.

Interest rates generally depend on the borrower's credit risk, i.e., the probability of defaulting on the amount promised to the lender. The higher the credit risk, the higher the interest rate. Lending to someone more likely to default is more costly in terms of risk, so the lender charges a higher rate to compensate.

As we saw in Section 7.7, the Black-Scholes model uses a risk-free interest rate. This assumes that the borrower has zero probability of default.

Which rates are risk free? A first approximation is the interest rates on developed countries' government debt. These are rates associated with a country's Treasury securities (for example, Spanish Treasury bills, medium-term bonds, and long-term bonds, depending on the contract's term). Governments are generally assumed always able to meet their commitments (although numerous sovereign defaults have occurred historically), so these rates are commonly used as proxies for –or even called– risk-free interest rates.

In practice, Treasury rates are not used as risk-free rates; rates associated with more complex structures, such as overnight indexed swaps (OIS), are used instead. We study these later.

Why do Treasury rates differ from risk-free rates? Among other reasons, government regulations in some countries require financial institutions to hold Treasury securities on their balance sheets. This increases bond demand, raising prices and lowering yields.

11.2 Short rates, zero-coupon bonds, and spot interest rates

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, consider an d -dimensional Brownian motion $W_t = (W_t^1, W_t^2, \dots, W_t^d)$ defined for $t \in [0, T]$. As in Chapter 2, assume that $\{\mathcal{F}_t : t \in [0, T]\}$ is the filtration generated by W_t . We therefore work on the filtered space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$.

Let R_t be a stochastic process adapted to the filtration $\{\mathcal{F}_t\}_{t \in [0, T]}$. We define B_t as the stochastic process describing the value of a money market account, whose dynamics are modeled by the (now stochastic) differential equation

$$dB_t = R_t dt, \quad (11.1)$$

with initial condition $M_0 = 1$. Solving equation (7.8) therefore gives

$$B_t = \exp \left(\int_0^t R_s ds \right). \quad (11.2)$$

This definition gives the value of the money market account at time t , where R_t is the instantaneous interest rate. In particular, if $R_t = r$ is constant, we recover expression (7.8).

Process B_t is important for relating amounts of currency available at different times. Specifically, what is the value at time t of one unit of currency at time T ?

If we deposit A units of a currency in the money market account at time $t = 0$, then at time t we have $A \cdot B_t$ units. In particular, to have exactly one unit in the account at time T , we require

$$A \cdot B_T = 1,$$

so we should have invested $A = B_T^{-1}$ units of currency. The account value at time t , given by $A \cdot B_t$, is therefore B_t/B_T .

Definition 11.1. The stochastic discount factor between times t and T , denoted $D(t, T)$, is the amount of currency that must be invested at time t in a money market account to be equivalent to one unit of that currency at time T . That is,

$$D(t, T) = \frac{B_t}{B_T} = \exp \left(- \int_t^T R_s ds \right). \quad (11.3)$$

The qualifier “stochastic” is meaningful only when the short rate is described by a stochastic process. For convenience, following the notation in [4], we write the stochastic discount factor as $D(t, T) = \frac{D_T}{D_t} = \frac{D(0, T)}{D(0, t)}$. Thus, $D_t = B_t^{-1}$ is the stochastic discount factor between times 0 and t .

If $R_t = r$ is constant, the discount factor coincides with that introduced in Chapter 7. The short rate r then coincides with the interest rate used, for example, in the Black-Scholes model. This assumption may be reasonable when pricing derivatives such as

equity or currency options, since fluctuations in the underlying asset dominate interest rate fluctuations for pricing these products. However, it is inappropriate for interest rate products [19]. These notes introduce different models (stochastic differential equations) for interest rate evolution, following examples in [4].

Definition 11.2. A zero-coupon bond with maturity T (also known as a T -maturity zero-coupon bond, pure discount bond, or simply a T -bond) is a contract guaranteeing its holder payment of one unit of an agreed currency at time T , with no intermediate payments (hence the name zero coupon). We denote its price at time t by $P(t, T)$.

By construction, $P(T, T) = 1$. One might expect the bond's value at time t to equal the stochastic discount factor $D(t, T)$. However, this holds only if the interest rate is deterministic: when the short rate is stochastic, $D(t, T)$ is a random variable by construction. A zero-coupon bond's price at time t , however, cannot be random, since it must be fixed in the market.

Assuming a unique risk-neutral measure $\mathbb{P}^*$ exists, $D_t P(t, T)$ is a martingale under that measure (by definition of a risk-neutral measure). That is,

$$D_t P(t, T) = \mathbb{E}^* [D_T P(T, T) | \mathcal{F}_t] = \mathbb{E}^* [D_T | \mathcal{F}_t] \quad (11.4)$$

and the zero-coupon bond price at time t , given by $P(t, T)$, can be calculated as the expected discount factor under this measure. That is,

$$P(t, T) = D_t^{-1} \mathbb{E}^* [D_T | \mathcal{F}_t] = \mathbb{E}^* [D(t, T) | \mathcal{F}_t] = \mathbb{E}^* \left[\exp \left(- \int_t^T R_s ds \right) \middle| \mathcal{F}_t \right]. \quad (11.5)$$

Note that $P(t, T)$ defines an $\mathcal{F}_t$ -measurable stochastic process. Thus, although at time t a zero-coupon bond's future price, $P(t', T)$ with $t < t' \leq T$, is a random variable for each t' , its current price is known.

When considering interest rate evolution, the short rate R_t gives a pointwise view: it is the instantaneous interest rate at time t . For a zero-coupon bond's return, however, we are interested in the return accumulated over a period, which we call the spot rate.

Definition 11.3. The continuously compounded spot rate, $R(t, T)$, at time t for maturity T is the constant interest rate at which an investment of $P(t, T)$ monetary units at time t yields one monetary unit at time T . That is,

$$R(t, T) = -\frac{1}{T-t} \ln \mathbb{E}^* \left[\exp \left(- \int_t^T R_s ds \right) \middle| \mathcal{F}_t \right] \quad (11.6)$$

Definition (11.6) allows the zero-coupon bond price at time t to be written as

$$P(t, T) = e^{-R(t, T)(T-t)}, \quad (11.7)$$

where $R(T, T) = 0$ by construction. The spot rate is thus the effective interest rate observable today for discounting a single payment at a future date.

Definition 11.4. The simply compounded spot rate, $L(t, T)$, at time t for maturity T is the constant interest rate at which an investment of $P(t, T)$ monetary units at time t yields one monetary unit at time T when accrual is proportional to the investment period. Thus,

$$L(t, T) = \frac{1}{T-t} \left(\frac{1}{P(t, T)} - 1 \right). \quad (11.8)$$

This definition is motivated by requiring linear discounting to satisfy

$$P(t, T) = \frac{1}{1 + L(t, T)(T - t)}, \quad (11.9)$$

and again $L(T, T) = 0$.

Definition 11.5. The annually compounded spot rate, $Y(t, T)$, at time t for maturity T is the constant interest rate at which an investment of $P(t, T)$ monetary units at time t yields one monetary unit at time T when compounding (with earnings reinvested) occurs once a year. That is,

$$Y(t, T) = \frac{1}{P(t, T)^{1/(T-t)}} - 1. \quad (11.10)$$

Thus, the zero-coupon bond price at time t is

$$P(t, T) = \frac{1}{(1 + Y(t, T))^{(T-t)}}, \quad (11.11)$$

where, recall, $P(t, T)$ satisfies equation (11.5), from which each interest rate defined above can be obtained.

Definition 11.6. The spot rate compounded k times annually, $Y^k(t, T)$, at time t for maturity T is the constant interest rate at which an investment of $P(t, T)$ monetary units at time t yields one monetary unit at time T when compounding (with earnings reinvested) occurs k times a year. That is,

$$Y^k(t, T) = \frac{k}{P(t, T)^{1/(k(T-t))}} - k. \quad (11.12)$$

The preceding definition directly gives the zero-coupon bond price as

$$P(t, T) = \left(1 + \frac{Y^k(t, T)}{k}\right)^{-k(T-t)}. \quad (11.13)$$

In particular, for fixed $Y^k(t, T) = Y$, note that if $k \rightarrow +\infty$, then

$$\lim_{k \rightarrow +\infty} P(t, T) = e^{-Y(T-t)},$$

which coincides with the continuously compounded zero-coupon bond price.

All interest rates defined above are equivalent over infinitesimal intervals and coincide with the short rate in the limit $T \rightarrow t^+$.

11.3 Yield curve

The zero-coupon curve gives zero-coupon bond yields across time horizons, reflecting market expectations of future interest rates and their term structure.

Definition 11.7. The zero-coupon curve (yield curve or term structure of interest rates) at time t is defined by the function

$$T \rightarrow \begin{cases} L(t, T) & t < T < t + 1, \\ Y(t, T) & T \geq t + 1, \end{cases} \quad (11.14)$$

where time is measured in years.

The yield curve is at least of class $\mathcal{C}^0(0, T)$, since $L(t, t + 1) = Y(t, t + 1)$ always holds by definition. Figure 11.1 shows the Spanish yield curve as an example.

The term yield curve is often used for several different curves derived from interest rates, which can cause confusion. An alternative definition of the zero-coupon curve is the graph of the function $t \rightarrow R(t, T)$ for $T > t$.

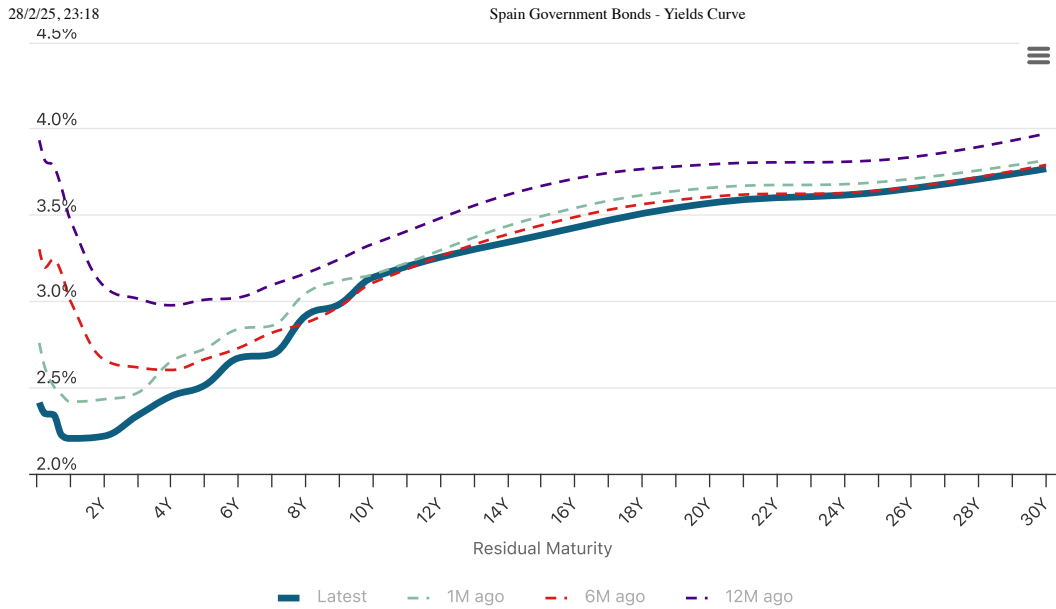


Figure 11.1: Spanish government bond yield curve as of 28 February 2025. Obtained from [World Government Bonds](#).

11.4 Forward rates

Forward interest rates are inferred from current rates at time t and indicate financing costs or expected returns for a future period, between expiry date $T > t$ and maturity date $S > T$. In other words, they indicate the interest expected to be paid or received in the future based on today's rates. For example, using current interest rates, we can calculate the yield of a three-month Treasury bill to be issued in six months.

Thus, knowing current interest rates at different maturities allows us to calculate the rate for a specific future period that is consistent with rates available today. These rates are fundamental to pricing fixed-income instruments and designing hedging strategies in financial markets, since they allow future costs and returns to be estimated from current information.

An investor who expects future interest rates to differ from current forward rates may find various investment strategies attractive. One involves a contract known as a forward rate agreement (FRA).

A forward rate agreement is a contract (specifically, an interest rate derivative) defined by three dates: (i) the current time t , (ii) an expiry date $T > t$, and (iii) a maturity date $S > T$. It allows its holder to receive (or make) a payment based on the difference between a fixed rate K agreed in the contract and the spot rate $L(T, S)$. The contract thus locks in the interest rate between dates T and S at a desired value K . In practice, the rates used in the contract are simply compounded, so we use $L(T, S)$.

By construction, the contract buyer receives $NK(S - T)$ units of currency, where K is the contractual fixed rate and N is the notional. The contract value at S is therefore

$$N(S - T)(K - L(T, S)), \quad (11.15)$$

where rewriting the spot rate in terms of the zero-coupon bond price in equation (11.8) gives

$$N \left((S - T)K - \frac{1}{P(T, S)} + 1 \right). \quad (11.16)$$

Consider each term separately. The term $P(T, S)^{-1}$ represents the number of monetary units at S obtained per unit invested at T in a zero-coupon bond maturing at S . Thus, $P(t, T)$ represents the amount of currency at t that grows to 1 monetary unit at T . Hence, $P(T, S)^{-1}$ at S is equivalent to $P(t, T)$ at t .

Meanwhile, the amount $B = (S - T)K + 1$ at S is worth $P(t, S)B$ at t . That is,

$$P(t, S)B = P(t, S)(S - T)K + P(t, S).$$

Thus, the FRA value at time t is

$$\text{FRA}(t, T, S, N, K) = N(P(t, S)(S - T)K - P(t, T) + P(t, S)). \quad (11.17)$$

An FRA must have zero initial value because the fixed rate K is set so that the net present value of future cash flows (the difference between payments and receipts) is zero. This follows from the no-arbitrage condition, which prevents either party from making an immediate riskless gain. Both parties thus start on equal terms at inception.

Equating equation (11.17) to zero gives the contractual fixed rate K , which defines the forward rate.

Definition 11.8. The simply compounded forward rate prevailing at t with expiry T and maturity S is denoted $F(t, T, S)$ and given by

$$F(t, T, S) = \frac{1}{S - T} \left(\frac{P(t, T)}{P(t, S)} - 1 \right). \quad (11.18)$$

We will later see that $F(t, T, S)$ is the expectation of $L(T, S)$ at time t under a particular probability measure.

Using definition 11.8, we can rewrite the FRA price at time t as

$$\boxed{\text{FRA}(t, T, S, N, K) = NP(t, S)(S - T)(K - F(t, T, S))}, \quad (11.19)$$

where K is the forward rate fixed when the contract is entered into, with expiry T and maturity S . Thus, if inception is at $t = 0$, then $K = F(0, T, S)$, so clearly $\text{FRA}(0, T, S, N, K) = 0$.

Taking the limit $S \rightarrow T^+$ in definition 11.8 defines the instantaneous forward rate. We have

$$\lim_{S \rightarrow T^+} F(t, T, S) = \lim_{S \rightarrow T^+} -\frac{1}{P(t, S)} \frac{P(t, S) - P(t, T)}{S - T}$$

and the definition of the derivative motivates the following definition.

Definition 11.9. The instantaneous forward rate prevailing at t for maturity T is denoted $f(t, T)$ and given by

$$f(t, T) = -\frac{\partial \ln P(t, T)}{\partial T}, \quad (11.20)$$

so

$$P(t, T) = \exp \left(- \int_t^T f(u, T) du \right). \quad (11.21)$$

Thus, the instantaneous forward rate is the rate agreed today for a loan that starts at T and lasts an instant.

It is important to distinguish this from the short rate, the instantaneous interest rate at t . Thus, R_t is an interest rate observed or determined at a specific time, whereas $f(t, T)$ is the rate implied by today's market for a future infinitesimal period. Clearly, $f(t, t) = R_t$.

For the instantaneous forward rate to be well defined, the zero-coupon bond price function is assumed differentiable.

11.5 Numéraire

A numéraire is an asset or financial instrument chosen as the reference unit for valuing other assets. It serves as a kind of “currency” in which other assets' relative prices are expressed within a financial model. A particular country's currency is commonly used as the numéraire, and one can change numéraire by using another country's currency.

For example, if a discount bond is chosen as numéraire, all other asset prices are expressed in units of that bond. This facilitates calculations and simplifies modeling.

Definition 11.10. A *numéraire* is any non-dividend-paying reference asset with strictly positive value.

Consider m underlying assets whose prices are modeled by the system of stochastic differential equations

$$dS_t^i = \alpha_t^i dt + S_t^i \sum_{j=1}^d \sigma_t^{ij} dW_t^j. \quad (11.22)$$

Assuming a unique risk-neutral measure exists, both fundamental theorems of risk-neutral asset pricing apply; see Theorems 7.4 and 7.7. The risk-neutral measure is constructed by applying the multidimensional Girsanov theorem. In particular, the second

fundamental theorem of asset pricing ensures that any derivative can be replicated by trading only the underlying assets and the money market account.

Under the risk-neutral measure, $\mathbb{P}^*$, discounted asset prices, $D_t S_t^i$, are martingales. That is,

$$\mathbb{E}^* [D_T S_T^i | \mathcal{F}_t] = D_t S_t^i. \quad (11.23)$$

The price of the i th asset in units of the money market account is $S_t/B_t = D_t S_t^i$. Thus, the i th underlying asset is worth $D_t S_t^i$ “shares” of the money market account. Moreover, its price in these units is a martingale. We therefore call $\mathbb{P}^*$ a risk-neutral measure for the money market account numéraire.

Changing numéraire to express the i th underlying asset’s value in terms of another money market account, for example N_t , gives the value of the i th asset in units of N_t , i.e., S_t^i/N_t , which is no longer a martingale under $\mathbb{P}^*$. We therefore need to change measure to preserve risk neutrality.

In principle, any asset whose price is strictly positive at all times can be used as numéraire, with other asset prices expressed in its units. This is equivalent to changing the units in which prices are measured. From now on, $\mathbb{P}^*$ denotes the risk-neutral measure associated with the domestic-currency money market account numéraire.

Remark 11.11. Any asset price divided by a numéraire is a martingale under the measure associated with that numéraire.

Since domestic currency can be invested in the money market account, its value in nominal monetary units follows dynamics incorporating a growth rate (the risk-free interest rate). Domestic currency viewed as a financial asset can therefore be interpreted as paying dividends in the form of interest when invested.

We henceforth focus on non-dividend-paying assets, since choosing a numéraire without dividend cash flows simplifies relative-price modeling and construction of the corresponding risk-neutral measure. If the numéraire paid dividends, the dynamics of relative asset prices in its units would require additional adjustment terms, complicating both the change of measure and the no-arbitrage analysis.

We mainly study the following numéraires:

- Domestic money market account. Its associated risk-neutral measure is denoted $\tilde{\mathbb{P}}$.
- Foreign money market account. Its associated risk-neutral measure is denoted $\tilde{\mathbb{P}}^f$.
- A zero-coupon bond¹ maturing at T . Its associated risk-neutral measure is denoted $\tilde{\mathbb{P}}^T$.

11.6 Interest rate swaps

Interest rate swaps (IRS) generalize forward rate agreements. An interest rate swap is a financial contract in which two parties agree to exchange interest payments based on different rates. These payment streams (called legs) are typically linked to (i) a fixed

¹A zero-coupon bond promises to pay a fixed amount, called its face value, at a future time T , without intermediate payments. To simplify notation, we take its face value to be 1.

rate and (ii) a variable or floating rate. For example, one party pays fixed and receives floating, while the other does the reverse.

A forward-start swap is a swap whose payments begin at a predetermined future date rather than immediately. Thus, the contract is entered into today, but cash-flow exchanges begin in the future.

Plain vanilla IRS (standard swaps without exotic features) fall into two categories: (i) payer swaps, in which a party pays fixed and receives floating, and (ii) receiver swaps, in which a party pays floating and receives fixed. In any IRS, one party necessarily holds a payer swap and the other a receiver swap.

11.6.1 Payer/Receiver Forward-start Interest Rate Swaps

A payer forward-start interest rate swap (PFS) starts at a predetermined future time rather than immediately, and its holder takes the payer position, paying a fixed rate and receiving a floating rate.

Thus, at each time T_i in a predefined schedule $\mathcal{T} = \{T_\alpha, T_{\alpha+1}, \dots, T_\beta\}$, the fixed-leg payer (the holder of the payer swap) pays

$$N(T_i - T_{i-1})K,$$

for $i > \alpha$, where K is the fixed interest rate and T_α is the contract start date. The floating-leg payer (the swap seller) pays²

$$N(T_i - T_{i-1})L(T_{i-1}, T_i).$$

The floating rates are determined on dates $T_\alpha, T_{\alpha+1}, \dots, T_{\beta-1}$ and paid on dates $T_{\alpha+1}, T_{\alpha+2}, \dots, T_\beta$, respectively.

Thus, the PFS payoff discounted to time t is

$$\sum_{i=\alpha+1}^{\beta} D(t, T_i) N(T_i - T_{i-1}) (L(T_i, T_{i-1}) - K),$$

so the PFS price at time t is given by

$$\text{PFS}(t, \mathcal{T}, N, K) = \sum_{i=\alpha+1}^{\beta} \mathbb{E}^* [D(t, T_i) N(T_i - T_{i-1}) (L(T_i, T_{i-1}) - K) | \mathcal{F}_t], \quad (11.24)$$

where $\mathbb{E}^*$ denotes expectation under the risk-neutral measure and we have used linearity of expectation.

Similarly, the value of a receiver forward-start interest rate swap (RFS) is

$$\text{RFS}(t, \mathcal{T}, N, K) = \sum_{i=\alpha+1}^{\beta} \mathbb{E}^* [D(t, T_i) N(T_i - T_{i-1}) (K - L(T_i, T_{i-1})) | \mathcal{F}_t], \quad (11.25)$$

In particular, if $T_\alpha = t$, the contract names simplify to payer IRS, denoted PS, and receiver IRS, denoted RS.

²In most plain vanilla IRS, the floating leg pays a simply compounded floating spot rate, so we use $L(T_{i-1}, T_i)$.

11.6.2 Coupon-bearing bonds and floating-rate notes

A coupon-bearing bond pays interest periodically on specified dates $T_{\alpha+1}, T_{\alpha+2}, \dots, T_\beta$ through coupons $c_i = N(T_i - T_{i-1})K$ for $\alpha < i < \beta$ and $c_\beta = N(T_\beta - T_{\beta-1})K + N$, where N is the bond's face value and K is the fixed interest rate. Its price is therefore

$$\boxed{\text{CP}(t, \mathcal{T}, c) = \sum_{i=\alpha+1}^{\beta} c_i P(t, T_i),} \quad (11.26)$$

where $c = \{c_{\alpha+1}, c_{\alpha+2}, \dots, c_\beta\}$.

A floating-rate note, on the other hand...

11.6.3 Caplets/Floorlets

11.6.4 Forward-start Caps/Floors

A cap is similar to a payer IRS, but payments occur only when positive, i.e., when $L(T_i, T_{i-1}) > K$. A cap can therefore be viewed as a collection of caplets.

$$\text{FSCap}(t, \mathcal{T}, N, K) = \sum_{i=\alpha+1}^{\beta} \mathbb{E}^* [D(t, T_i) N(T_i - T_{i-1}) (L(T_i, T_{i-1}) - K)^+ | \mathcal{F}_t] \quad (11.27)$$

where the usual definition of an interest rate cap is recovered by setting $T_\alpha = t$.

Similarly, a forward-start floor consists of a series of floorlets. Its price is given by

$$\text{FSFlr}(t, \mathcal{T}, N, K) = \sum_{i=\alpha+1}^{\beta} \mathbb{E}^* [D(t, T_i) N(T_i - T_{i-1}) (K - L(T_i, T_{i-1}))^+ | \mathcal{F}_t] \quad (11.28)$$

In particular, using Girsanov's theorem and the relation $\mathbb{E}^*[D(t, T)X_T | \mathcal{F}_t] = P(t, T)\mathbb{E}^{T_i}[X_T | \mathcal{F}_t]$, where expectation is taken under the T_i -forward measure and $P(t, T)$ is the zero-coupon bond price at time t , the forward-start interest rate cap price is given by

$$\boxed{\text{FSCap}(t, \mathcal{T}, N, K) = \sum_{i=\alpha+1}^{\beta} P(t, T_i) \mathbb{E}^{T_i} [N(T_i - T_{i-1}) (K - L(T_i, T_{i-1}))^+ | \mathcal{F}_t]} \quad (11.29)$$

and the forward-start interest rate floor price by

$$\boxed{\text{FSFlr}(t, \mathcal{T}, N, K) = \sum_{i=\alpha+1}^{\beta} P(t, T_i) \mathbb{E}^{T_i} [N(T_i - T_{i-1}) (K - L(T_i, T_{i-1}))^+ | \mathcal{F}_t]} \quad (11.30)$$

11.6.5 Swaptions

11.7 Vasicek model

The Vasicek model is one of the earliest stochastic approaches to describing interest rate evolution.

Interest rates are influenced by monetary policy decisions, inflation, economic growth, and other macroeconomic variables. In practice, central banks adjust rates to maintain economic stability, preventing them from remaining too high or too low for extended periods. Excessively high rates may slow the economy, prompting the central bank to lower them. Conversely, excessively low rates may generate inflation, prompting the central bank to raise them. Mean reversion is therefore a central feature of the Vasicek model.

Empirically, interest rates do not follow a purely random movement but tend to fluctuate around a long-term average. Although they can deviate significantly during certain periods (such as economic crises or inflationary cycles), they generally return to equilibrium values determined by structural economic factors.

A model without mean reversion, such as pure Brownian motion, would allow interest rates to grow indefinitely or become negative, which does not reflect observed financial market behavior.

Specifically, the Vasicek model describes short-rate dynamics through the stochastic differential equation

$$dR_t = \kappa(\theta - R_t)dt - \sigma dW_t, \quad (11.31)$$

where κ is the speed at which the interest rate tends to revert to its long-term mean, denoted θ , and σ represents volatility.

The Vasicek equation describes an Ornstein-Uhlenbeck process, analyzed in Section 3.1.1. Interest rates can therefore be negative, which rarely occurs in practice (although it has occurred historically).

The model also assumes constant volatility, σ . When rates are high, however, central banks tend to intervene more frequently to adjust the economy, generating greater interest rate fluctuations. In practice, volatility is therefore not constant but depends on the level of rates.

Market yield curves can have various shapes, but, as we will see later, the Vasicek model generates only relatively simple yield curves. Moreover, although it can model yield curve evolution, it does not generally guarantee the absence of arbitrage.

For these reasons, the Vasicek model is not widely used in practical applications. A considerable advantage, however, is its closed-form solution, given by

$$R_t = R_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-s)} dW_s, \quad (11.32)$$

where the final term is the Itô integral of the deterministic function $\exp(-\kappa(t-s))$. As we saw in Chapter 2, a deterministic integrand implies that the integral is a normal random variable with zero mean and variance $\frac{\sigma^2}{2\kappa}(1 - e^{-\kappa t})$.

11.8 CIR model

The Cox-Ingersoll-Ross (CIR) model extends the Vasicek model by introducing volatility proportional to the square root of the interest rate. That is,

$$dR_t = \kappa(\theta - R_t)dt + \sigma\sqrt{R_t}dW_t, \quad (11.33)$$

where κ is the speed of mean reversion, θ is the long-term interest rate level, and σ is volatility.

Unlike the Vasicek model, CIR volatility depends on the square root of the interest rate, ensuring nonnegativity. Moreover, if the Feller condition, $2\kappa\theta \geq \sigma^2$, holds, the interest rate can be shown to remain strictly positive.

Despite the coefficient $\sqrt{R_t}$, a unique solution can be shown to exist [20], given by

$$R_t = R_s e^{-\kappa(t-s)} + \theta (1 - e^{-\kappa(t-s)}) + \sigma \int_s^t e^{-\kappa(t-u)} \sqrt{R_u} dW_u, \quad (11.34)$$

whose expectation conditional on knowing the interest rate at time s , i.e., $R_s = r_s$, is

$$\mathbb{E}^*[R_t | R_s = r_s] = r_s e^{-\kappa(t-s)} + \theta (1 - e^{-\kappa(t-s)}), \quad (11.35)$$

since the Itô integral has zero expectation; see Theorem 2.14.

The CIR model's main drawbacks are: (i) with only one source of risk, it cannot fully capture yield curve complexity; (ii) since volatility is proportional to $\sqrt{R_t}$, it tends to underestimate volatility when interest rates are very low; (iii) parameter estimation is nontrivial and often requires advanced numerical methods; (iv) it is unsuitable for negative-rate environments, an issue particularly relevant over the past decade with negative interest rates in Europe and Japan.

The fundamental difference from the Vasicek model is that the Itô integral no longer has a deterministic integrand, making its solution harder to analyze. Later, however, we use numerical methods developed in Chapter 5 to study paths and their statistical properties.

The solution at time T of the CIR model in equation (11.33), conditional on the interest rate at time t , can be shown to follow a noncentral chi-squared distribution, allowing exact sampling without discretizing the differential equation [23]. Specifically,

$$R_t | R_s = r_s \sim \frac{\sigma^2 (1 - e^{-\kappa(t-s)})}{4\kappa} \chi_\nu^2(\lambda), \quad (11.36)$$

where

$$\nu = \frac{4\kappa\theta}{\sigma^2} \quad (11.37)$$

denotes the number of degrees of freedom and

$$\lambda = \frac{4\kappa e^{-\kappa(t-s)} r_s}{\sigma^2 (1 - e^{-\kappa(t-s)})} \quad (11.38)$$

is the noncentrality parameter. Thus, we can calculate

$$\mathbb{E}^*[R_t | R_s = r_s] = \frac{\sigma^2 (1 - e^{-\kappa(t-s)})}{4\kappa} (\nu + \lambda), \quad (11.39)$$

which agrees with expression (11.35).

CIR paths can also be simulated by discretization, using numerical schemes such as the Taylor methods (with reflection [24]) introduced in Chapter 5 or other methods, such as Alfonsi's [21], that preserve positivity of the solutions.

In particular, we analyze a method that preserves the exact solution's mean-reversion property and whose convergence has been established in both the strong and weak senses. We also study a method that exactly captures the CIR process's first and second asymptotic moments, improving the representation of its long-term behavior [22].

11.8.1 Zero-coupon bond price

11.8.2 Parameter estimation

11.8.3 Numerical simulation

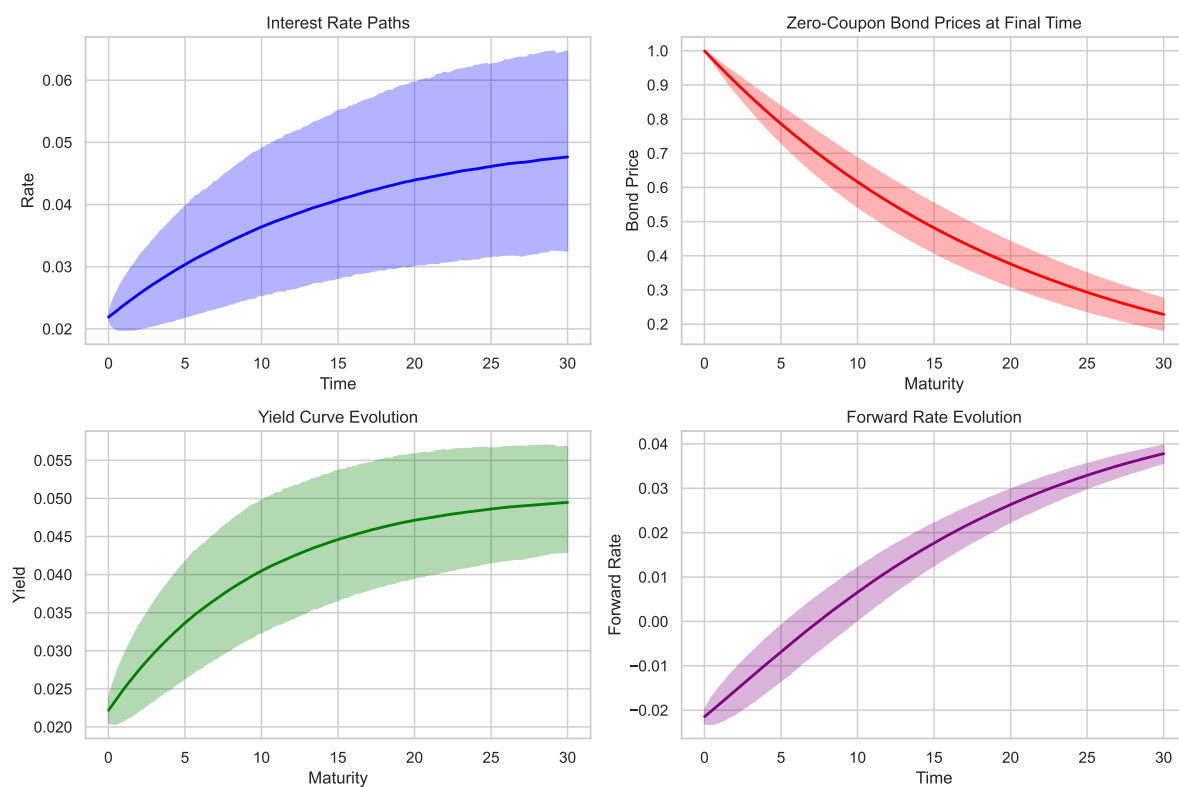


Figure 11.2: Evolution of short rates (top left), zero-coupon bond prices at maturity (top right), the yield curve (bottom left), and forward rates (bottom right). The dark line shows the mean. The 95% confidence intervals are included.

Appendix A

Preliminaries

A.1 σ -algebras

In probability, the set Ω represents the possible outcomes of an experiment. For example, rolling a 6-sided die once might give $\Omega = \{1, 2, \dots, 6\}$. For a coin toss, $\Omega = \{\text{heads}, \text{tails}\}$, whereas measuring the time until a telephone exchange receives a call gives $\Omega =]0, \infty[$. Thus, Ω need not consist of numbers.

Basic probability courses assume that all *information* about the system is available, so the event that occurred in an experiment can be fully determined. This need not hold in more general settings. It is therefore important to define precisely what *information* is available about experimental outcomes.

Definition A.1. Let Ω be a nonempty set. A σ -algebra on Ω is a collection $\mathcal{F}$ of subsets of Ω satisfying:

- (a) $\emptyset \in \mathcal{F}$.
- (b) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
- (c) If $(A_n)_{n=1}^\infty \subset \mathcal{F}$, then $\cup_{n=1}^\infty A_n \in \mathcal{F}$. In particular, if $A, B \in \mathcal{F}$, then $A \cup B \in \mathcal{F}$.

The definition directly implies that a σ -algebra also satisfies:

- (d) $\Omega \in \mathcal{F}$.
- (e) If $(A_n)_{n=1}^\infty \subset \mathcal{F}$, then $\cap_{n=1}^\infty A_n \in \mathcal{F}$. In particular, if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.
- (f) If $A, B \in \mathcal{F}$ and $A \subset B$, then $B \setminus A \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a measurable space, and the elements of $\mathcal{F}$ are said to be $\mathcal{F}$ -measurable. Thus, the elements of $\mathcal{F}$ are events whose occurrence can be determined in a realization of the experiment.

Example A.2. For a die roll, $\Omega = \{1, 2, \dots, 6\}$, the collection

$$\mathcal{F}_1 = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}$$

is a σ -algebra on Ω . If we roll a die once and can observe only whether the result is even or odd, we do not have all the *information* about possible outcomes; we can determine

(measure) only the events in $\mathcal{F}_1$. Thus, with the σ -algebra $\mathcal{F}_1$, we cannot determine, for example, whether the result is 2, since $X^{-1}(2) = \{2\} \notin \mathcal{F}_1$ (see Definition A.6).

Continuing the example, knowing whether a die roll is odd also tells us whether it is even. Observed events therefore convey information beyond themselves. It is useful to define mathematically the information generated by a single event or a family of events.

Definition A.3. Let $\mathcal{A} = \{A_i : i \in I\}$ be a collection of events, with I not necessarily countable. The σ -algebra generated by $\mathcal{A}$ on Ω is the smallest σ -algebra containing all events A_i . We denote it by $\sigma(\mathcal{A})$.

Example A.4. An important example is the Borel σ -algebra on $\mathbb{R}$. Let $\mathcal{Y} \subset \mathcal{P}(\mathbb{R})$ denote the collection of all intervals in $\mathbb{R}$. The Borel σ -algebra on $\mathbb{R}$ is defined as $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{Y})$. Although counterintuitive, there are elements of $\mathcal{P}(\mathbb{R})$ that do not belong to $\mathcal{B}(\mathbb{R})$. Thus, $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{P}(\mathbb{R})$. The elements of $\mathcal{B}(\mathbb{R})$ are called *Borel sets*.

For our later study of stochastic processes, it is important to define a *hierarchy* of information about a system.

Definition A.5. Let Ω be nonempty and $\mathcal{F}$ a σ -algebra on Ω . A sub- σ -algebra of $\mathcal{F}$ is a collection $\mathcal{G}$ of events such that $A \in \mathcal{G}$ implies $A \in \mathcal{F}$, where $\mathcal{G}$ itself defines a σ -algebra on Ω . We write $\mathcal{G} \subset \mathcal{F}$.

A.2 Random variables

Definition A.6. Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be measurable spaces. A random variable on $(\Omega, \mathcal{F})$ is a map $X : \Omega \rightarrow \Omega'$ satisfying

$$\{\omega \in \Omega : X(\omega) \in F'\} \in \mathcal{F}, \quad \forall F' \in \mathcal{F}' \quad (\text{A.1})$$

that is, $X^{-1}(F') \in \mathcal{F}$ for all $F' \in \mathcal{F}'$.

These notes focus on the case $\Omega' = \mathbb{R}$ and $\mathcal{F}' = \mathcal{B}(\mathbb{R}^{d \times m})$. We also say that X is $\mathcal{F}$ -measurable to emphasize that *knowing* the information in $\mathcal{F}$ means *knowing* the value of X .

Remark A.7. Condition (A.1) is equivalent to $\{\omega \in \Omega : X(\omega) \leq t\} \in \mathcal{F}, \forall t \in \mathbb{R}$ when $\Omega' = \mathbb{R}$ and $\mathcal{F}' = \mathcal{B}(\mathbb{R}^{d \times m})$.

Since probability often involves random variables that are functions of other random variables, the following proposition is useful.

Proposition A.8. If X is an $\mathcal{F}$ -measurable random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel (or Borel-measurable¹) function, then $Y = g(X)$ is also an $\mathcal{F}$ -measurable random variable².

Just as we defined the information generated by a family of events, it is useful to define the information contained in a random variable.

¹That is, g satisfies condition (A.1) on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}^{d \times m}))$.

²The function $g(X)$ is well defined since $g(X) = g \circ X : \Omega \rightarrow \mathbb{R}$.

Definition A.9. Let $\{X_i, i \in I\}$ be a collection of random variables on a measurable space $(\Omega, \mathcal{F})$. The σ -algebra generated by $X_i, i \in I$ is the smallest σ -algebra on Ω , $\mathcal{G}$, such that all random variables X_i are $\mathcal{G}$ -measurable. We denote it by $\sigma(X_i, i \in I)$.

This definition applies to Proposition A.8 by taking $\mathcal{F}$ to be $\sigma(X)$. In particular, when $g(X) = X$, X is a $\sigma(X)$ -measurable random variable.

A.3 Probability measures

Definition A.10. Let $(\Omega, \mathcal{F})$ be a measurable space. A map $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure if:

- (a) $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
- (b) (Additivity). If $(A_n)_{n=1}^\infty \subset \mathcal{F}$ is a family of disjoint subsets, then $\mathbb{P}(\cup_{n=1}^\infty A_n) = \sum_{n=1}^\infty \mathbb{P}(A_n)$. In particular, if $A, B \in \mathcal{F}$ are disjoint, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability space³. The notion of $\mathcal{F}$ -measurability now becomes clearer, since we want to *measure* the probability that X belongs to a subset of $\mathbb{R}$. The preceding conditions imply:

- (c) If $(A_n)_{n=1}^\infty \subset \mathcal{F}$, then $\mathbb{P}(\cup_{n=1}^\infty A_n) \leq \sum_{n=1}^\infty \mathbb{P}(A_n)$.
- (d) If $A, B \in \mathcal{F}$ satisfy $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$. In particular, $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- (e) Given $A, B \in \mathcal{F}$, we have $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- (f) If $(A_n)_{n=1}^\infty \subset \mathcal{F}$ is a family of subsets such that $A_n \subset A_{n+1} \forall n \in \mathbb{N}$, then $\mathbb{P}(\cup_{n=1}^\infty A_n) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$.
- (g) If $(A_n)_{n=1}^\infty \subset \mathcal{F}$ is a family of subsets such that $A_n \supset A_{n+1} \forall n \in \mathbb{N}$, then $\mathbb{P}(\cap_{n=1}^\infty A_n) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$.

Distribution of a random variable

Definition A.11. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X a random variable defined on it. The *probability distribution* of X is the map $\mu_X : \mathcal{B}(\mathbb{R}^{d \times m}) \rightarrow [0, 1]$ defined by

$$\mu_X(B) = \mathbb{P}(\{X^{-1}(B)\}) = \mathbb{P}(\{\omega : X(\omega) \in B\}) = \mathbb{P}(\{X \in B\}). \quad (\text{A.2})$$

The preceding definition also yields the cumulative distribution function.

Definition A.12. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X a random variable defined on it. The *cumulative distribution function* (or simply *distribution function*) of X is the map $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(t) = \mu_X([-\infty, t]) = \mathbb{P}(\{X \leq t\}), \quad t \in \mathbb{R} \quad (\text{A.3})$$

³Clearly, this is a special case of a measure space $(\Omega, \mathcal{F}, \mu)$.

Note that F_X is well defined, since $] - \infty, t] \in \mathcal{B}(\mathbb{R}^{d \times m})$ for all $t \in \mathbb{R}$.

Remark A.13. It is easy to verify that the distribution function satisfies:

- (i) $\lim_{t \rightarrow -\infty} F_X(t) = 0$ and $\lim_{t \rightarrow \infty} F_X(t) = 1$,
- (ii) $F_X(s) \leq F_X(t)$ for all $s < t$,
- (iii) F_X is right continuous.

Definition A.14. A random variable X is *discrete* if it takes values in a finite countable subset of $\mathbb{R}$.

Definition A.15. A random variable X is *continuous* if $\mathbb{P}(\{X \in B\}) = 0$ for every $B \in \mathcal{B}(\mathbb{R}^{d \times m})$ such that $\lambda(B) = 0$, where λ denotes Lebesgue measure.

Since $\lambda(\{x\}) = 0$, we have $\mathbb{P}(X = x) = 0$ for all $x \in \mathbb{R}$.

Proposition A.16. If X is a continuous random variable, there exists a differentiable function $f_X : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) \geq 0$ for all $x \in \mathbb{R}$, $\int_{\mathbb{R}} f_X(x) dx = 1$, and

$$\mu_X(B) = \int_B f_X(x) dx \quad \forall B \in \mathcal{B}(\mathbb{R}^{d \times m}).$$

This is called the density function and satisfies $\frac{d}{dt} F_X(t) = f_X(t)$.

A.4 Independence

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We now study independence of events, σ --algebras, and random variables.

Definition A.17. A collection of events $\{A_i\}_{i \in I} \subset \mathcal{F}$ is independent if

$$\mathbb{P}(\cap_{i \in I} A_i) = \prod_{i \in I} \mathbb{P}(A_i) \tag{A.4}$$

Definition A.18. Let $\{\mathcal{G}_1, \dots, \mathcal{G}_n\}$ be a collection of sub- σ -algebras of $\mathcal{F}$. They are independent if

$$\mathbb{P}(A_1 \cap \dots \cap A_n) = \mathbb{P}(A_1) \dots \mathbb{P}(A_n) \tag{A.5}$$

for all $A_i \in \mathcal{G}_i$, $i = 1, 2, \dots, n$.

Definition A.19. Let $\{X_1, \dots, X_n\}$ be a collection of random variables. They are independent if and only if

$$\mathbb{P}(\{X_1 \leq t_1, \dots, X_n \leq t_n\}) = \mathbb{P}(\{X_1 \leq t_1\}) \dots \mathbb{P}(\{X_n \leq t_n\}) \tag{A.6}$$

for all $t_i \in \mathbb{R}$, $i = 1, \dots, n$.

A.5 Expectation

The scope of these notes precludes a detailed theoretical development of the expectation of a random variable $\mathbb{E}[X]$.

Definition A.20. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X a random variable defined on it. Its expectation is defined as

$$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P} \quad (\text{A.7})$$

where the integral is the Lebesgue integral with respect to measure $\mathbb{P}$.

We now consider two useful special cases. Given a random variable X on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a Borel function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $\mathbb{E}[|g(X)|] < \infty$:

(i) If X is a discrete random variable with support S_X , then

$$\mathbb{E}[g(X)] = \sum_{x \in S_X} g(x) \mathbb{P}(\{X = x\}) \quad (\text{A.8})$$

(ii) If X is a continuous random variable with density f_X , then

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f_X(x) dx \quad (\text{A.9})$$

Now define the space of random variables

$$\mathcal{L}^p = \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P}) = \{X : X \text{ is a random variable on } (\Omega, \mathcal{F}) \text{ and } \mathbb{E}[|X|^p] < \infty\} \quad (\text{A.10})$$

and, given the equivalence relation $X \sim Y = \{X, Y \in \mathcal{L}^p : X = Y \text{ c.t.p.}\}$ ⁴, let L^p be the set of equivalence classes in $\mathcal{L}^p$ under $\sim$. If we equip L^p with the norm

$$\|X\|_p = (\mathbb{E}[|X|^p])^{1/p} \quad (\text{A.11})$$

then L^p is a Banach space under this norm. In particular, L^2 is a Hilbert space with inner product

$$\langle X, Y \rangle = \mathbb{E}[XY] \quad (\text{A.12})$$

for $X, Y \in L^2$. Elements of L^2 are also called square-integrable random variables.

Furthermore, if $X \in L^1$, then $|\mathbb{E}[X]| \leq \mathbb{E}[|X|]$,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y] \quad (\text{A.13})$$

and, for $X \leq Y$,

$$\mathbb{E}[X] \leq \mathbb{E}[Y] \quad (\text{A.14})$$

Thus, $\mathbb{E}$ is a linear and monotone functional on L^1 . Standard functional analysis results give the following inequalities:

⁴The abbreviation c.t.p. means that equality holds everywhere except on a set of measure zero.

(i) Hölder's inequality

$$|\langle X, Y \rangle| \leq \|X\|_p \|Y\|_q \quad (\text{A.15})$$

where $p > 1$, with $1/p + 1/q = 1$ and $X, Y \in L^p$. If⁵ $p = 2$, then (A.15) can be rewritten as $\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[X^2]} \sqrt{\mathbb{E}[Y^2]}$.

(ii) Minkowski's inequality (or the triangle inequality in L^p)

$$\|X + Y\|_p \leq \|X\|_p + \|Y\|_p \quad (\text{A.16})$$

where $p \geq 1$ and $X, Y \in L^p$.

(iii) Chebyshev's inequality

$$\mathbb{P}[\{|X| \geq c\}] \leq c^{-p} \mathbb{E}[|X|^p] \quad (\text{A.17})$$

Variance and covariance

Definition A.21. Let $X, Y \in L^2$. The *variance* of X is defined as

$$\text{Var}[X] = \mathbb{E}[X - \mathbb{E}[X]]^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2 \geq 0 \quad (\text{A.18})$$

and the covariance of X and Y as

$$\text{Cov}(X, Y) = \mathbb{E}[X - \mathbb{E}[X]] \mathbb{E}[Y - \mathbb{E}[Y]] = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] \quad (\text{A.19})$$

If $\text{Cov}(X, Y) = 0$, random variables X and Y are said to be uncorrelated. If $\mathbb{E}[X] = 0$ or $\mathbb{E}[Y] = 0$, the preceding expression gives covariance as $\mathbb{E}[XY]$; in this case, we use that expression to denote covariance.

Proposition A.22. Let $X, Y \in L^2$ be independent random variables (in the sense of Definition A.19). Then

(i) $\text{Var}[cX + Y] = c^2 \text{Var}[X] + \text{Var}[Y]$ for all $c \in \mathbb{R}$.

(ii) $\text{Cov}(X, Y) = 0$, i.e., X and Y are uncorrelated⁶.

A.6 Convergence of random variables

For a sequence of random variables $\{X_n\}_{n \in \mathbb{N}}$ defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$, several notions of convergence are available.

(a) If there exists a null set $F \in \mathcal{F}$, i.e., $\mathbb{P}(F) = 0$, such that for every $\omega \notin F$ the sequence $X_n(\omega) \in \mathbb{R}^d$ converges in the usual sense to $X(\omega) \in \mathbb{R}^d$, we say that $\{X_n\}_{n \in \mathbb{N}}$ converges to X **with probability 1**, and write

$$\text{ac} \lim_{n \rightarrow \infty} X_n = X$$

⁵For $p = 2$, this is also known as the Schwarz inequality.

⁶The converse is false.

(b) If, for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \epsilon\}) = 0$$

we say that $\{X_n\}_{n \in \mathbb{N}}$ converges **in probability** to X , and write

$$\text{st} \lim_{n \rightarrow \infty} X_n = X$$

(c) If $X_n \in L^p(\Omega, \mathcal{F}, \mathbb{P})$ for all $n \in \mathbb{N}$ and $X \in L^p(\Omega, \mathcal{F}, \mathbb{P})$, we say that $\{X_n\}_{n \in \mathbb{N}}$ converges to X **in pth mean**. If $p = 1$, it converges in mean, and if $p = 2$, it converges **in mean square**, denoted by

$$\text{qm} \lim_{n \rightarrow \infty} X_n = X$$

(d) If F_n is the distribution function of X_n and F that of X , we say that $\{X_n\}_{n \in \mathbb{N}}$ converges to X **in distribution** if

$$\lim_{n \rightarrow \infty} F_n = F$$

These notions are related as follows: convergence with probability 1 implies convergence in probability, which implies convergence in distribution; convergence in q th mean implies convergence in p th mean if $p \leq q$, which in turn implies convergence in probability. Moreover, a sufficient condition for $X_n \rightarrow X$ with probability 1 is

$$\sum_{n=1}^{\infty} \mathbb{E}[|X_n - X|^p] < \infty \quad (\text{A.20})$$

for some $p > 0$.

A.7 Random vectors

Random vectors generalize random variables to an d -dimensional range, with $1 < d < \infty$. We always take the target space to be the measurable space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, where $\mathcal{B}(\mathbb{R}^d)$ is the Borel σ -algebra defined⁷ on $\mathbb{R}^d$ by

$$\mathcal{B}(\mathbb{R}^d) = \sigma((a_1, b_1) \times \cdots \times (a_d, b_d) : a_i, b_i \in \mathbb{R} \text{ with } a_i < b_i)$$

Thus, a random vector is a function

$$\begin{aligned} X &: (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)) \\ \omega &\mapsto X(\omega) = (X_1(\omega), \dots, X_d(\omega)) \end{aligned}$$

such that

$$\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$$

for all $B \in \mathcal{B}(\mathbb{R}^d)$. This condition can be shown to be equivalent to

$$\{\omega \in \Omega : X_1(\omega) \leq t_1, \dots, X_d(\omega) \leq t_d\} \in \mathcal{F}$$

for all $t_1, \dots, t_n \in \mathbb{R}$.

⁷For $d = 1$, this recovers the earlier definition in Example A.4.

Definition A.23. Given a random vector X , its joint distribution function is defined as

$$F(t_1, \dots, t_d) = \mathbb{P}(\{\omega \in \Omega : X_1(\omega) \leq t_1, \dots, X_d(\omega) \leq t_d\}) \equiv \mathbb{P}(X \leq t)$$

For $d = 1$, $F(x)$ coincides with the definition of $F_X(t)$ in A.12. Moreover, $F(x_1, \dots, x_d)$ satisfies the same requirements as in the one-dimensional case, so F is indeed a distribution function. The marginal distributions associated with each component can then be defined as

$$F_{X_k}(t_k) = \mathbb{P}(X_k \leq t_k) = F(\infty, \dots, \infty, t_k, \infty, \dots, \infty)$$

In practice, random variables are specified by their distribution functions. Indeed, given a distribution function F , one can always construct a probability space and a random variable X with distribution function F . For example, take $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mathbb{P}_F)$ and $X(\omega) = \omega$, where $\mathbb{P}_F$ is the probability measure defined by F on $\mathcal{B}(\mathbb{R}^d)$. If P_F is absolutely continuous with respect to Lebesgue measure⁸, there exists a nonnegative integrable function such that

$$F(x_1, \dots, x_d) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_d} f(y_1, \dots, y_d) dy_1 \dots dy_d$$

where f is called the joint density function. In this case, F is absolutely continuous and hence differentiable almost everywhere in $\omega \in \Omega$, with

$$\frac{\partial^d F}{\partial x_1 \dots \partial x_d} = f(x_1, \dots, x_d)$$

We can also define the expectation and covariance matrix of a random vector by

$$\mathbb{E}[X] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_d])$$

and

$$\text{Cov}[X_i, X_j] = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i] \mathbb{E}[X_j]$$

The covariance matrix is easily shown to be symmetric and positive semidefinite.

A random vector X belongs to $L^p(\Omega, \mathcal{F}, \mathbb{P})$ if and only if $\mathbb{E}[|X|] < \infty$, where $|X| = \text{tr}(XX')$. Similarly, $G \in M_{n \times n}(\mathbb{R})$ belongs to this space if and only if $\mathbb{E}[|G|] < \infty$, where $|G| = \text{tr}(GG')$.

A.8 Conditional expectation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X and Y finite-valued random variables defined on it. It is natural to define the conditional probability of X given $Y = y_i$, with $y_i \in Y(\Omega)$, as the frequency of obtaining x_i in a realization of X among outcomes for which $Y = y_i$ occurs, i.e.,

$$\mathbb{P}(\{X = x_i\} | \{Y = y_i\}) = \frac{\mathbb{P}(\{(X, Y) = (x_i, y_i)\})}{\mathbb{P}(\{Y = y_i\})}$$

for $1 \leq i \leq n$ and $1 \leq j \leq m$, with $n = |X(\Omega)|$ and $m = |Y(\Omega)|$.

⁸That is, $P_F(N) = 0$ for every $N \in \mathcal{B}(\mathbb{R}^d)$ of Lebesgue measure zero.

Note that $\mathbb{P}^{Y=y_j} := \mathbb{P}(\{X = x_i\}|\{Y = y_j\})$ defines a new probability measure on $(\Omega, \mathcal{F})$. The expectation of X under this measure is therefore naturally defined, analogously to (A.8), as

$$\mathbb{E}[X|Y = y_j] := \sum_{i=1}^n x_i \mathbb{P}(\{X = x_i\}|\{Y = y_j\})$$

for $j = 1, \dots, m$. We now seek a more general definition of a random variable $\xi := \mathbb{E}[X|Y]$ satisfying $\xi(\omega) = \mathbb{E}[X|Y = Y(\omega)]$ for every $\omega \in \Omega$, called the conditional expectation of X given Y . Note that ξ is completely determined by the values of Y , i.e., it is a $\sigma(Y)$ -measurable random variable. Consequently, $\xi = \varphi(Y)$ for some function φ .

This construction extends to continuous random variables through joint and marginal density functions. In both cases,

$$\langle X - \mathbb{E}[X|Y], \varphi(Y) \rangle_2 = 0$$

for every $\varphi : \mathbb{R} \rightarrow \mathbb{R}$, where $\langle \cdot, \cdot \rangle_2$ denotes the inner product on $L^2(\Omega, \mathcal{F}, \mathbb{P})$ described in A.5. Thus, $X - \mathbb{E}[X|Y]$ is orthogonal to the linear subspace of $\sigma(Y)$ -measurable functions.

Geometrically, $\mathbb{E}[X|Y]$ can therefore be interpreted as the projection of X onto the linear subspace of all $\sigma(Y)$ -measurable random variables. In other words, $\mathbb{E}[X|Y]$ is the best approximation to X by a function of Y .

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{G}$ a sub- σ -algebra of $\mathcal{F}$. Following this motivation, we introduce conditional expectation more abstractly as the orthogonal projection under the inner product in L^2 , i.e.,

$$\mathbb{E}[X|\mathcal{G}] := \operatorname{argmin} \{ \|X - Y\|_2 : Y \in \mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P}) \}$$

for $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. This definition is possible because $L^2(\Omega, \mathcal{F}, \mathbb{P})$ is a Hilbert space with norm $\|\cdot\|_2$.

Lemma A.24. *The orthogonal projection operator $\mathbb{E}[X|\mathcal{G}]$ is well defined on $L^2(\Omega, \mathcal{G}, \mathbb{P})$ and satisfies*

$$\mathbb{E}[X 1_G] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}] 1_G] \quad (\text{A.21})$$

for every $G \in \mathcal{G}$ and $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Moreover:

- (a) If $X \geq 0$ almost everywhere, then $\mathbb{E}[X|\mathcal{G}] \geq 0$ almost everywhere.
- (b) $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]] = \mathbb{E}[X]$

Theorem A.25. *For every random variable $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$, there exists a random variable Z such that*

- (a) Z is $\mathcal{F}$ -measurable.
- (b) $\mathbb{E}[|Z|] < \infty$
- (c) For every $F \in \mathcal{F}$, $\mathbb{E}[X 1_F] = \mathbb{E}[Z 1_F]$

Moreover, if another random variable $\tilde{Z}$ satisfies all these conditions, then $Z = \tilde{Z}$ almost everywhere.

Definition A.26. The random variable Z described in Theorem A.25 is called the conditional expectation of X given $\mathcal{F}$, denoted $\mathbb{E}[X|\mathcal{F}]$, and is unique in the sense of that theorem.

For simplicity, if $\mathcal{F} = \sigma(Y_1, \dots, Y_n)$, we write $\mathbb{E}[X|Y_1, \dots, Y_n]$. We now examine some useful properties of conditional expectation.

Proposition A.27. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, X and Y random variables defined on it, and $\mathcal{G}$ a sub- σ -algebra. Conditional expectation as defined in A.26 satisfies:

- (a) If $\mathcal{G} = \{\emptyset, \Omega\}$, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$.
- (b) If $X \geq 0$, then $\mathbb{E}[X|\mathcal{G}] \geq 0$.
- (c) If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X|\mathcal{G}] = X$.
- (d) If $X = a$ is constant, then $\mathbb{E}[X|\mathcal{G}] = a$.
- (e) If $X, Y \in L^1(\Omega, \mathcal{F}, \mathbb{P})$, then $\mathbb{E}[aX + bY|\mathcal{G}] = a\mathbb{E}[X|\mathcal{G}] + b\mathbb{E}[Y|\mathcal{G}]$, with $a, b \in \mathbb{R}$.
- (f) If $X \leq Y$, then $\mathbb{E}[X|\mathcal{G}] \leq \mathbb{E}[Y|\mathcal{G}]$.
- (g) If X is $\mathcal{G}$ -measurable and $X, XY \in L^1(\Omega, \mathcal{F}, \mathbb{P})$, then $\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}]$. In particular, $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]Y|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}]\mathbb{E}[Y|\mathcal{G}]$.
- (h) If $\sigma(X)$ and $\mathcal{G}$ are independent, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$.
- (i) Let $\mathcal{G}_1 \subset \mathcal{G}_2$ be two sub- σ -algebras. Then

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]|\mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}_1]|\mathcal{G}_2] = \mathbb{E}[X|\mathcal{G}_1]$$

where all preceding equalities and inequalities hold almost everywhere.

It is then straightforward to define the conditional probability of an event A given $\mathcal{G} \subset \mathcal{F}$ as

$$\mathbb{P}(A|\mathcal{G}) = \mathbb{E}[\mathbb{1}_A|\mathcal{G}]$$

where $\mathbb{1}_A$ denotes the indicator function

$$\mathbb{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

A.9 Change of measure

This section is particularly relevant when moving from the real-world (or physical) measure, which describes actual asset price dynamics, to a risk-neutral measure (*risk-neutral measure*; see Chapter 7), used for pricing derivatives and other financial instruments. Under this measure, the discounted expected value of future cash flows equals the asset's current price.

Theorem A.28. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and Z a random variable nonnegative with probability 1⁹, with $\mathbb{E}[Z] = 1$. Then, for $A \in \mathcal{F}$,

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega), \quad (\text{A.22})$$

defines a probability measure. Moreover, if X is a nonnegative random variable, then

$$\tilde{\mathbb{E}}[X] = \mathbb{E}[XZ]. \quad (\text{A.23})$$

Finally, if Z is strictly positive with probability 1, then

$$\mathbb{E}[X] = \tilde{\mathbb{E}}\left[\frac{X}{Z}\right]. \quad (\text{A.24})$$

Note that $\tilde{\mathbb{E}}$ denotes the expectation defined in (A.7) using probability measure $\tilde{\mathbb{P}}$. That is,

$$\tilde{\mathbb{E}}[X] = \int_{\Omega} X(\omega) d\tilde{\mathbb{P}}(\omega) \quad (\text{A.25})$$

Definition A.29. Let $(\Omega, \mathcal{F})$ be a measurable space. Two probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$ defined on it are equivalent if every subset $A \in \mathcal{F}$ of measure zero under $\mathbb{P}$ also has measure zero under $\tilde{\mathbb{P}}$, and conversely.

Thus, equivalent probability measures agree on *impossible* events (of zero probability). They may, however, assign different probabilities to other possible events (of nonzero probability).

Under the assumptions of Theorem A.28, probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$ can be shown to be equivalent; see p. 34 in [4]. Thus, it is unnecessary to specify the measure when referring to events occurring with probability 1.

Definition A.30. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\tilde{\mathbb{P}}$ another probability measure equivalent to $\mathbb{P}$, and Z a random variable strictly positive with probability 1 that relates the two measures through expression (A.22). Then Z is called the Radon-Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$, written

$$Z = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}. \quad (\text{A.26})$$

As shown in Chapter 7, the existence of a risk-neutral measure is related to the absence of arbitrage. We are therefore interested in the existence of risk-neutral measures.

Theorem A.31. Let $\mathbb{P}$ and $\tilde{\mathbb{P}}$ be equivalent measures on a measurable space $(\Omega, \mathcal{F})$. Then there exists a random variable Z , strictly positive with probability 1, such that $\mathbb{E}[Z] = 1$ and

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega), \quad (\text{A.27})$$

for every $A \in \mathcal{F}$.

⁹That is, $\mathbb{P}(\{\omega \in \Omega : Z(\omega) \geq 0\}) = 1$.

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